Metasol Meta Solution

MC/TOR

Contactors and Overload Relays



LSIS



Metasol Contactors and Overload Relays

New generation of Contactors from LSIS



Metasol Contactors

Metasol Contactors



■ Designed to show superior technology: The Metasol series

Sealed structure has improved its safety features so that no arc flash is exposed. In order to emphasize its durability as an industrial device, Metasol series adopt simple design form and sophisticated shape applying diamond cut concepts on product surface.

LSIS META SOLUTION CONTACTORS

Contents :

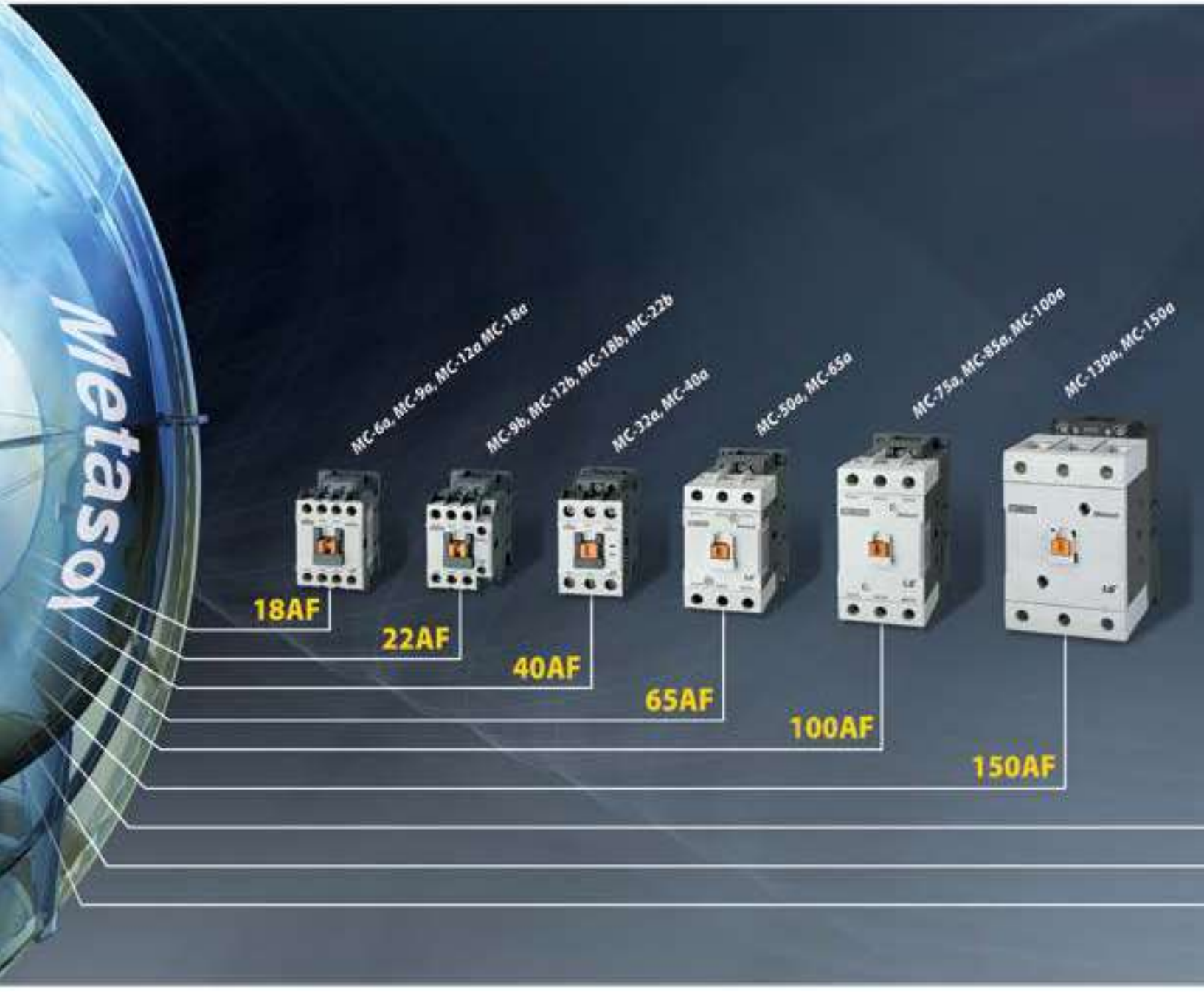


Table of specifications ····· 10
Type numbering system ····· 18
Environment ····· 21
Connections ····· 22
Control coil characteristics ····· 24
Marking ····· 28
Internal structure ····· 29
3-pole Contactors ····· 30
Motor starters ····· 52
Reversing contactors ····· 56
LED voltage monitoring contactor ····· 58
Contactor relays ····· 59
Latching contactors ····· 62
4-pole Contactors ····· 66
Enclosed motor starters, Mold enclosure ····· 78
Enclosed motor starters, Steel enclosure ····· 79
Capacitor contactors ····· 80
2-pole AC Contactors, Definite purpose ····· 81
3-pole AC Contactors, Definite purpose ····· 82
Single-pole AC Contactors, Definite purpose ····· 83
2-pole DC Contactors ····· 84
Accessory ····· 85
Overload relays ····· 96
- Overload relay specifications ····· 98
- Overload relay operation ····· 102
- Accessories for overload relay ····· 103
- Overload relay ordering types ····· 108
- Overload relay trip curves ····· 114

Type 2 Coordination (380/415V) ····· 122
Type 2 Coordination (440V) ····· 123
Type C Coordination (240V) ····· 127
Type C Coordination (480V) ····· 129
Type C Coordination (600V) ····· 130
Type D Coordination (240V) ····· 131
Type D Coordination (480V) ····· 132
Type D Coordination (600V) ····· 133
Rated UL 508 combination motor controller Type-F ····· 134
Ready for IE3 ····· 137
Dimension ····· 139
Delayed Opening Device ····· 175
Electronic timer unit ····· 176
Contact arrangement ····· 177
Comparison table ····· 180
Certifications ····· 186

>>>

Metasol Series completion



More Safe, More Valuable



Contactors & Overload Relays



- *Advanced technology & Expert solution*
- *Offering economical solution with compact size & easy connection*
- *Ensure reliability in the system: Type II Coordination*
- *Perfect system with peripheral devices.*
- *Environment friendly product: complies with International environment standard RoHS.*
- *25Models in 9 Frames*

Motor Starters

- *Compact design for space saving*
- *DIN rail and screw mountable (upto 150AF)*
- *Overload relays directly mountable to contactors separately*
- *Finger-proof design*
- *Easy to combine with manual motor starters*
- *Broad range of accessories*
- *World class products conforming to IEC and UL standards*



18AF	22AF	40AF	65AF	100AF	150AF
Contactor	**Contactor**	**Contactor**	**Contactor**	**Contactor**	**Contactor**
MC-6a, 9a, 12a, 18a	**MC-9b, 12b, 18b, 22b**	**MC-32a, 40a**	**MC-50a, 65a**	**MC-75a, 85a, 100a**	**MC-130a, 150a**
Rated current: 6, 9, 12, 18A	Rated current: 9, 12, 18, 22A	Rated current: 32, 40A	Rated current: 50, 65A	Rated current: 75, 85, 100A	Rated current: 130, 150A
Rated insulation voltage: 690V	Rated insulation voltage: 690V	Rated insulation voltage: 1000V	Rated insulation voltage: 1000V	Rated insulation voltage: 1000V	Rated insulation voltage: 1000V
Overload relay MT-12	**Overload relay MT-32**	**Overload relay MT-32**	**Overload relay MT-63**	**Overload relay MT-95**	**Overload relay MT-150**
Current range: 0.1~18A	Current range: 0.1~40A	Current range: 0.1~40A	Current range: 4~65A	Current range: 7~100A	Current range: 34~150A
Trip class: Class10A, 20	Trip class: Class10A, 20	Trip class: Class10A, 20	Trip class: Class10A, 20	Trip class: Class10A, 20	Trip class: Class10A, 20



225AF

Contactor
MC-185a, 225a
Rated current:
185, 225A
Rated insulation voltage:
1000V

Overload relay MT-225
Current range: 65~240A
Trip class: Class10A, 20

400AF

Contactor
MC-265a, 330a, 400a
Rated current:
265, 330, 400A
Rated insulation voltage:
1000V

Overload relay MT-400
Current range: 85~400A
Trip class: Class10A, 20

800AF

Contactor
MC-500a, 630a, 800a
Rated current:
500, 630, 800A
Rated insulation voltage:
1000V

Overload relay MT-800
Current range: 200~800A
Trip class: Class10A, 20

1260AF

Contactor
MC-1260a
Rated current:
1260A
Rated insulation voltage:
1000V

2100AF

Contactor
MC-1400a, 1700a, 2100a
Rated current:
1400, 1700, 2100A
Rated insulation voltage:
1000V



MC/TOR
Accessories

Perfect combination of peripheral devices and various accessories

Offering economical solution with compact size & easy connection

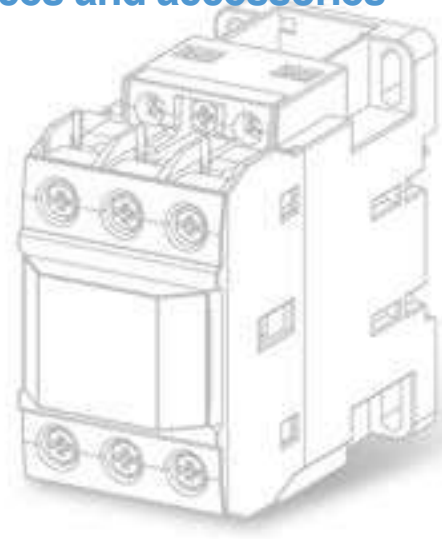


Perfect system with other devices and accessories

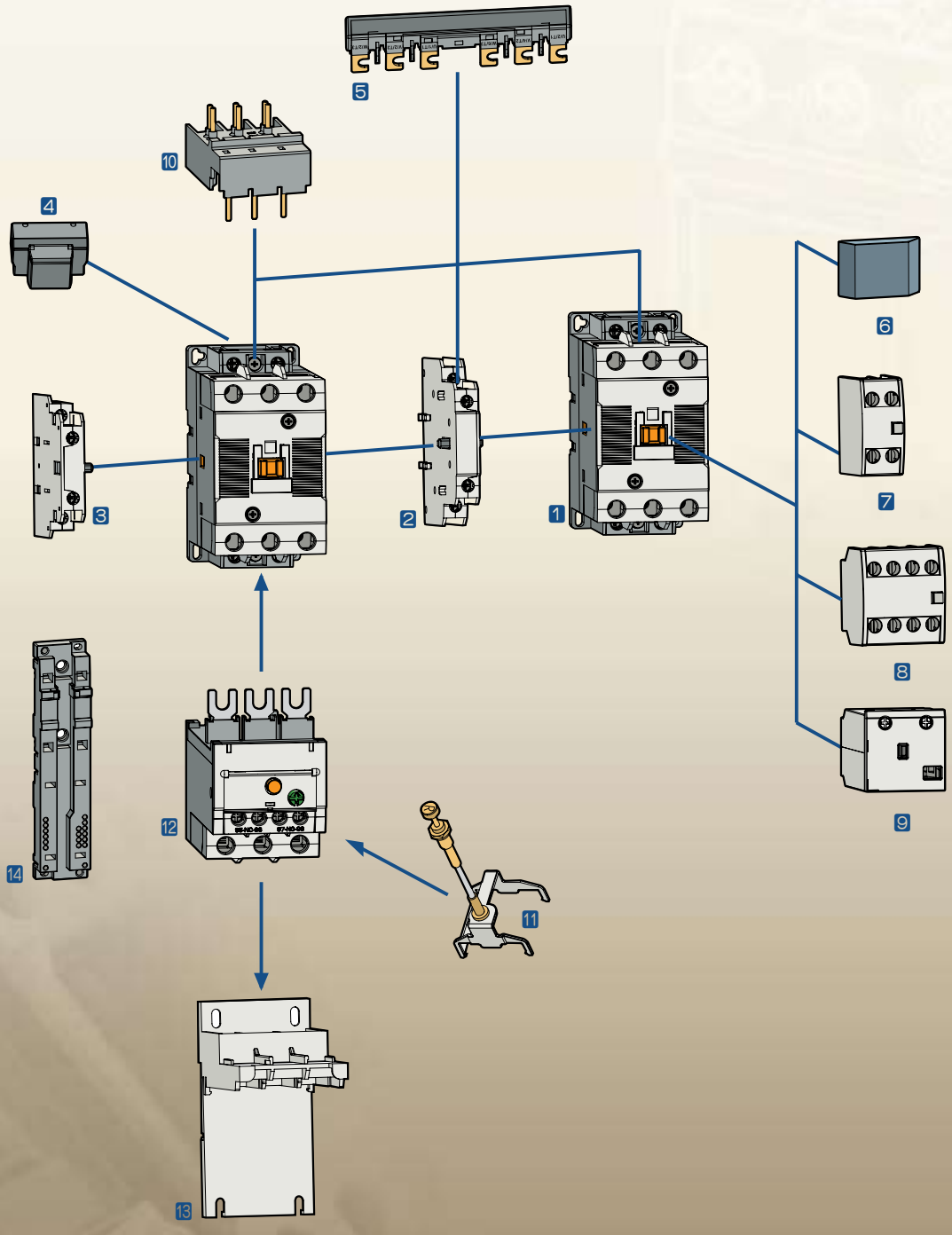
■ One-touch connection
- Din Rail connection without any special tools
- Direct combine a surge unit

■ Simple wiring
- Mounting unit, Adapter
- Reversing wire set

■ Easy Maintenance
- 4-terminal type: Easy wiring of operation coil
- Module type of Screw terminal
- Interlock unit with built-in auxiliary contacts(2NO built-in)



Advanced accessories for Metasol contactors



1 Contactor
2 Interlock unit
3 Auxiliary contact unit
- side mount
4 Surge unit
5 Wire kit
6 Safety cover (Option)
7 Auxiliary contact unit
- front mount(2p)
8 Auxiliary contact unit
- front mount(4p)
9 Mechanical latch unit
10 Direct adaptor to MMS
11 Remote reset unit
12 Overload relay
13 Separate mounting unit
14 Mounting unit for use with MMS

Table of specifications

MC type
Magnetic Contactors





Frame size				18AF				22AF			
Type				MC-6a	MC-9a	MC-12a	MC-18a	MC-9b	MC-12b	MC-18b	MC-22b
	screws clamp terminals			●	●	●	●	●	●	●	●
	Lug clamp terminals										
Number of poles				3pole				3pole			
Rated operational voltage, Ue				690V				690V			
Rated insulation voltage, Ui				690V				690V			
Rated frequency				50/60Hz				50/60Hz			
Rated impulse withstand voltage, Uimp				6kV				6kV			
Maximum operating rate in operating cycles per hour(AC3)				1800 operations per hour				1800 operations per hour			
Durability	Mechanical			15 mil. operations				15 mil. operations			
	Electrical			2.5 mil. operations				2.5 mil. operations			
Current	AC-1, Thermal current		A	25	25	25	32	27	27	32	45
and	AC-3	200/240V	kW	2.2	2.5	3.5	4.5	2.5	3.5	4.5	5.5
power			A	9	11	13	18	11	13	18	22
		380/440V	kW	3	4	5.5	7.5	4	5.5	7.5	11
			A	7	9	12	18	9	12	18	22
		500/550V	kW	3	4	7.5	7.5	4	7.5	7.5	15
			A	6	7	12	13	7	12	13	20
		690V	kW	3	4	7.5	7.5	4	7.5	7.5	15
			A	4	5	9	9	6	9	9	18
		1000V	kW	-	-	-	-	-	-	-	-
			A	-	-	-	-	-	-	-	-
Rated Short-time		1s	A	210	250	280	300	250	280	300	400
withstand current		10s	A	105	110	120	130	110	120	154	186
(IEC 60947)		30s	A	70	70	80	85	70	80	100	130
		1min	A	61	61	61	70	61	61	84	90
		3min	A	40	45	47	50	45	50	60	60
		10min	A	30	30	30	40	30	30	40	50
		≥15min	A	25	26	28	40	26	28	40	45
UL rating	Continuous current		A	25	25	25	32	27	27	32	45
(50/60Hz)	Single	110~120V	HP	0.5	0.5	0.75	1	0.5	0.75	1	2
	phase	220~240V	HP	1.5	1.5	2	3	1.5	2	3	3
		200~208V	HP	2	2	3	7.5	2	3	7.5	7.5
	Three	220~240V	HP	3	3	5	7.5	3	5	7.5	10
	phase	440~480V	HP	5	5	7.5	10	5	7.5	10	15
		550~600V	HP	7.5	7.5	10	15	7.5	10	15	20
	NEMA size			00	00	0	1	00	0	1	
Size	AC	Weight	kg	0.33				0.34			
and	control	Size(W×H×D)	mm	45×73.5×80.4				45×73.5×87.4			
weight	DC	Weight	kg	0.4				0.41			
	control	Size(W×H×D)	mm	45×73.5×96.6				45×73.5×103.6			
Auxiliary(standard)				1a or 1b				1a1b			
Auxiliary	Side mount			UA-1				UA-1			
	Front mount			UA-2, UA-4				UA-2, UA-4			

Note) Minimum conduct current of Auxiliary contactor is DC 17V 5mA.

MT type
Thermal Overload Relays





Type				MT-12/□	MT-32/□
	screws clamp terminals			●	●
	Lug clamp terminals				
Rated operational voltage, Ue				690V	690V
Rated insulation voltage, Ui				690V	690V
Rated impulse withstand voltage, Uimp				6kV	6kV
Trip class				10A, 20	10A, 20
Setting range				0.1~18A	0.1~40A
Size		Weight	kg	0.1	0.17
and weight		Size(W×H×D)	mm	45×73.2×63.7	45×75×90

* The safety cover of magnetic contactor and thermal overload relay is optional.







40AF		65AF		100AF		
MC-32a	MC-40a	MC-50a	MC-65a	MC-75a	MC-85a	MC-100a
●	●	●	●	●	●	●
		●	●	●	●	●
3pole		3pole		3pole		
1000V		1000V		1000V		
1000V		1000V		1000V		
50/60Hz		50/60Hz		50/60Hz		
8kV		8kV		8kV		
1800 operations per hour		1200 operations per hour		1200 operations per hour		
12 mil. operations		12 mil. operations		12 mil. operations		
2 mil. operations		2 mil. operations		2 mil. operations		
55	60	100	115	125	135	160
7.5	11	15	18.5	22	25	30
32	40	55	65	75	85	105
15	18.5	22	30	37	45	55
32	40	50	65	75	85	105
18.5	22	30	33	37	45	55
28	32	43	60	64	75	85
18.5	22	30	33	37	45	55
20	23	28	35	42	45	65
22	22	30	30	37	37	37
17	17	23	23	28	28	28
600	700	1000	1050	1100	1200	1320
260	300	550	700	750	800	900
160	160	330	380	400	450	500
100	100	250	270	300	350	400
70	70	150	200	220	270	270
55	55	90	120	140	170	180
50	50	87	100	114	150	160
55	60	100	115	125	135	160
2	2	3	5	5	7.5	10
5	5	10	15	15	15	20
7.5	15	20	25	25	30	30
10	15	25	30	30	40	40
20	30	40	50	50	60	75
25	30	50	60	60	75	75
1P	2				3	
0.55		1.05		1.93		
69×83×90		79×106×119		94×140×135.8		
0.77		1.3		2.8		
69×83×117.1		79×106×146.4		94×140×172.3		
2a2b		**2a2b**		**2a2b**		
UA-1		UA-1		UA-1		
UA-2, UA-4		UA-2, UA-4		UA-2, UA-4		







MT-32/□	MT-63/□	MT-95/□
●	●	●
	●	●
690V	690V	690V
690V	690V	690V
6kV	6kV	6kV
10A, 20	10A, 20	10A, 20
0.1~40A	4~65A	7~100A
0.17	0.31/0.33	0.48/0.5
45×75×90	55×81×100	70×97×110

Table of specifications

MC type
Magnetic Contactors

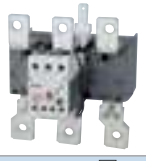




Frame size				150AF		225AF		400AF		
Type				MC-130a	MC-150a	MC-185a	MC-225a	MC-265a	MC-330a	MC-400a
	screws clamp terminals			●	●	●	●	●	●	●
	Lug clamp terminals			●	●	●	●			
Number of poles				3pole		3pole		3pole		
Rated operational voltage, Ue				1000V		1000V		1000V		
Rated insulation voltage, Ui				1000V		1000V		1000V		
Rated frequency				50/60Hz		50/60Hz		50/60Hz		
Rated impulse withstand voltage, Uimp				8kV		8kV		8kV		
Maximum operating rate in operating cycles per hour(AC3)				1200 operations per hour		1200 operations per hour		1200 operations per hour		
Durability	Mechanical			5 mil. operations		5 mil. operations		5 mil. operations		2.5 mil. operations
	Electrical			1 mil. operations		1 mil. operations		1 mil. operations		0.5 mil. operations
Current and power	AC-1, Thermal current		A	200	250	300	350	400	500	520
	AC-3	200/240V	kW	37	45	55	75	80	90	125
			A	130	150	185	225	265	330	400
		380/440V	kW	60	75	90	132	147	160	200
			A	130	150	185	225	265	330	400
		500/550V	kW	60	70	110	132	147	160	225
			A	90	100	180	200	225	250	350
		690V	kW	55	55	110	140	160	200	250
			A	60	60	120	150	185	220	300
		1000V	kW	75	75	132	132	147	147	147
			A	53	53	90	90	105	105	105
Rated Short-time withstand current (IEC 60947)		1s	A	1350	1800	2000	2500	3500	4000	4600
		10s	A	950	1200	1500	1700	2400	3000	4400
		30s	A	700	800	1000	1200	1500	2500	2974
		1min	A	550	600	800	1000	1100	1700	1846
		3min	A	350	450	520	700	800	1000	1313
		10min	A	200	300	350	500	600	620	760
		≥15min	A	175	280	320	400	500	553	699
UL rating (50/60Hz)	Continuous current		A	200	250	300	350	400	500	520
	Single phase	110~120V	HP	10	15	15	15	-	-	-
		220~240V	HP	20	25	30	40	-	-	-
		200~208V	HP	40	40	60	60	75	100	125
	Three phase	220~240V	HP	40	50	60	75	100	100	150
		440~480V	HP	75	100	125	150	200	200	300
		550~600V	HP	75	75	125	150	200	200	300
	NEMA size				4			5		
Size and weight 	AC control	Weight	kg	2.4		5.4		9.2		
		Size(W×H×D)	mm	119×158×130.3		138×203×185.1		163×243×204.4		
	DC control	Weight	kg							
		Size(W×H×D)	mm							
Auxiliary(standard)				2a2b		2a2b		2a2b		
Auxiliary	Side mount			UA-1		AU-100, AU-100E (Max.4a4b)		AU-100, AU-100E (Max.4a4b)		
	Front mount			UA-2, UA-4		-		-		

MT type
Thermal Overload Relays





Type				MT150/□	MT225/□	MT400/□
	screws clamp terminals			●	●	●
	Lug clamp terminals			●	●	
Rated operational voltage, Ue				690V	690V	690V
Rated insulation voltage, Ui				690V	690V	690V
Rated impulse withstand voltage, Uimp				6kV	6kV	6kV
Trip class				10A, 20	10A, 20	10A, 20
Setting range				34~150A	65~240A	85~400A
Size and weight		Weight	kg	0.67	2.5	2.6
		Size(W×H×D)	mm	95×109×113	147×141×184	151×171×198

* The safety cover of magnetic contactor and thermal overload relay is optional.







800AF		
MC-500a	MC-630a	MC-800a
●	●	●
3pole		
1000V		
1000V		
50/60Hz		
8kV		
1200 operations per hour		
2.5 mil. operations		
0.5 mil. operations		
700	900	1050
147	190	220
500	630	800
265	330	440
500	630	800
265	330	500
400	500	720
300	400	500
380	420	630
280	280	280
220	220	220
6000	7000	7500
5050	6400	7000
4400	4500	4900
3400	3500	3800
2000	2200	2500
1400	1550	1550
1100	1300	1300
700	900	1050
-	-	-
-	-	-
150	200	200
200	250	300
400	500	600
400	500	600
6		7
22.4 285×312×245.3		
2a2b		
AU-100, AU-100E (Max.4a4b)		
-		

1260AF
MC-1260a
●
3pole
1000V
1000V
50/60Hz
8kV
300 operations per hour
0.5 mil. operations
0.5 mil. operations
1260
-
-
-
-
-
-
-
-
-
-
8000
7200
5200
4000
2300
3000
1500
1260
-
-
23.5 285×352×246
2a2b
AU-100, AU-100E (Max.4a4b)
-

2100AF		
MC-1400a	MC-1700a	MC-2100a
●	●	●
3pole		
1000V		
1040V		
50/60Hz		
8kV		
300 operations per hour		
0.5 mil. operations		
0.05 mil. operations		
1400	1600	2100
275	320	-
860	1050	-
550	700	900
860	1050	1450
-	-	-
-	-	-
860	950	-
860	950	-
-	-	-
-	-	-
-	-	-
8000	10000	10000
-	-	-
4500	5500	5500
-	-	-
2600	3000	3000
-	-	-
1400	1700	2100
-	-	-
-	-	-
33.8 431×380×246		
2a2b		
AU-100, AU-100E (Max.4a4b)		
-		



MT-800/□
●
690V
690V
6kV
10A, 20
200~800A
11.5
360×530×212

Table of specifications

MC type
Magnetic Contactors (4pole)



Frame size				18AF			
Type				MC-6a/4	MC-9a/4	MC-12a/4	MC-18a/4
	Screw clamp terminal			●			
Number of poles				4pole			
Rated operational voltage (Ue)				690V			
Rated insulation voltage (Ui)				690V			
Rated frequency				50/60Hz			
Rated impulse withstand voltage, Uimp				6kV			
Maximum operating rate in operating cycles per hour(AC1)				1800 operations per hour			
Durability	Mechanical			15 mil. Operations			
	Electrical			0.5 mil. Operations		0.8 mil. Operations	
Current and Power	AC-1	Thermal current	A	25	25	25	40
		200/240V	kW	9	9	9	15
			A	25	25	25	40
		380/400V	kW	17	17	17	27
			A	25	25	25	40
		500/550V	kW	21	21	21	35
			A	25	25	25	40
		690V	kW	27	27	27	44
			A	25	25	25	40
UL rating (50/60Hz)	Continuous current		A	25	25	25	40
	Single Phase	110~120V	HP	0.5	0.5	0.75	1
		220~240V	HP	1.5	1.5	2	3
		200~208V	HP	2	2	3	5
	Three Phase	220~240V	HP	3	3	5	7.5
		440~480V	HP	5	5	7.5	10
		550~600V	HP	7.5	7.5	10	15
Size and weight 	AC	Weight	kg	0.33			
	Control	Size(W×H×D)	mm	45×73.5× 79			
	DC	Weight	kg	0.4			
	Control	Size(W×H×D)	mm	45×73.5×96.6			
Auxiliary(standard)				-			
Auxiliary	Side Mount			UA-1			
	Front Mount			UA-2, UA-4			





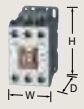


22AF	40AF		85AF			
MC-22a/4	MC-32a/4	MC-40a/4	MC-50a/4	MC-65a/4	MC-75a/4	MC-85a/4
●	●		●			
4pole	4pole		4pole			
690V	690V		690V			
690V	690V		1000V			
50/60Hz	50/60Hz		50/60Hz			
6kV	6kV		8kV			
1800 operations	1800 operations per hour		1800 operations per hour			
15 mil. Operations	15 mil. Operations		12 mil. Operations			
1 mil. Operations	1 mil. Operations		1 mil. Operations			
40	50	60	80	100	110	135
15	18	22	30	37	41	51
40	50	60	80	100	110	135
27	35	42	56	70	76	95
40	50	60	80	100	110	135
35	43	52	70	88	97	120
40	50	60	80	100	110	135
44	55	66	88	110	120	150
40	50	60	80	100	110	135
40	50	60	80	100	110	135
2	2	3	3	5	5	7.5
3	5	5	7.5	10	15	15
7.5	7.5	10	10	15	20	25
7.5	10	10	15	20	25	30
10	20	25	30	40	50	50
15	20	25	30	40	50	50
0.4	0.59		1.2			
47.2×80×86.8	59×83.5×94.5		91×123.5×117.8			
0.5	0.7		1.29			
47.2×80×113.2	59×83.5×121		91×123.5×117.8			
-	-		-			
AU-1	AU-1		AU-1			
UA-2, UA-4	UA-2, UA-4		UA-2, UA-4			

Table of specifications

MC type
Magnetic Contactors (4pole)



Frame size				225AF				
Type				MC-100a/4	MC-130a/4	MC-150a/4	MC-185a/4	MC-225a/4
	Screw clamp terminal					●		
Number of poles						4pole		
Rated operational voltage (Ue)						690V		
Rated insulation voltage (Ui)						1000V		
Rated frequency						50/60Hz		
Rated impulse withstand voltage, Uimp						8kV		
Maximum operating rate in operating cycles per hour(AC1)						1200 operations per hour		
Durability	Mechanical					5 mil. Operations		
	Electrical					0.8 mil. Operations		
Current and Power	AC-1	Thermal current	A	200	250	275	300	350
		200/240V	kW	57	60	76	87	100
			A	150	155	200	230	260
		380/400V	kW	106	110	142	165	185
			A	150	155	200	230	260
		500/550V	kW	132	137	180	205	230
			A	150	155	200	230	260
		690V	kW	165	170	225	255	290
			A	150	155	200	230	260
UL rating (50/60Hz)	Continuous current		A	200	250	275	300	350
	Single Phase	110~120V	HP	7.5	10	15	15	15
		220~240V	HP	15	20	25	30	40
		200~208V	HP	30	40	40	60	60
	Three Phase	220~240V	HP	30	40	50	60	75
		440~480V	HP	60	75	100	125	150
		550~600V	HP	60	75	100	125	150
Size and weight 	AC Control	Weight	kg			5.6		
		Size(W×H×D)	mm			175×203×185		
	DC Control	Weight	kg					
		Size(W×H×D)	mm					
Auxiliary(standard)				2NO2NC				
Auxiliary	Side Mount					AU-100, AU-100E		
	Front Mount					-		

* - FLA = 722 A, LRA = 5618 A
** - FLA = 566 A, LRA = 4495 A



400AF		
MC-265a/4	MC-330a/4	MC-400a/4
	●	
	4pole	
	690V	
	1000V	
	50/60Hz	
	8kV	
	1200 operations per hour	
	2.5 mil. Operations	
	0.5 mil. Operations	
400	500	520
115	135	160
300	350	420
215	250	300
300	350	420
265	315	375
300	350	420
335	390	470
300	350	420
400	500	520
-	-	-
-	-	-
75	100	125
100	125	150
200	250	300
200	250	300
	9.9	
	206×243×205	
2NO2NC		
	AU-100, AU-100E	
	-	



800AF		
MC-500a/4	MC-630a/4	MC-800a/4
	●	
	4pole	
	690V	
	1000V	
	50/60Hz	
	8kV	
	1200 operations per hour	
	2.5 mil. Operations	
	0.5 mil. Operations	
700	900	1050
245	255	310
630	660	800
450	470	570
630	660	800
560	590	710
630	660	800
710	740	900
630	660	800
700	900	1050
-	-	-
-	-	-
150	200	200
200	250	300
400	500	600 *
400	500	600 **
	26.3	
	346×310 ×244	
2NO2NC		
	AU-100, AU-100E	
	-	

Type numbering system

Contactors

MC65A - **40** - **22** - **K7** - **L**

Product (6A~2100A)	
MC6A • • • MC2100A	MC-6a • • • MC-2100a
MC9B • • • MC22B	MC-9b • • • MC-22b

Main contactor	
30	3NO 0NC
40	4NO 0NC
22	2NO 2NC

Auxiliary contact	
00	None
11	Normal 1NO Normal 1NC
22	Normal 2NO Normal 2NC
11L	Low Current 1NO Low current 1NC
22L	Low Current 2NO Low current 2NC
11H	Hoist 1NO Normal 1NC
22H	Hoist 2NO Normal 2NC

Coil voltage
See Coil code Table

*note) See page 23 for details

Terminal	
S	Screw
L	Lug

- A : Metasol A Type
- B : Metasol B Type

ex)

MC65A-30-00-Q7-S-E	MC-65a AC380V 50/60Hz SCREW (Metasol) EXP
MC75A-30-22-H7-L-E	MC-75a AC100V 50/60Hz LUG 2a2b (Metasol) EXP
MC185A-30-22-FV-B-E	MC-185a AC100-240V 50/60Hz, DC100-220V 2a2b (Metasol) EXP
MC85A-40-00-M7-S-E	MC-85a AC220V 50/60Hz 4P (Metasol) EXP
MC800A-40-22-EV-B-E	MC-800a AC/DC200V 2a2b 4P (Metasol) EXP

Contactor relays

MR4 - **22L** - **B7**

Product	
MR4	MR-4
MR6	MR-6
MR8	MR-8

Main contact	
00	None
13	Normal 1NO Normal 3NC
22	Normal 2NO Normal 2NC
22L	Low Current 2NO Low current 2NC
44	Normal 4NO Normal 4NC
44L	Low Current 2NO Low current 2NC

Coil voltage
See Coil code Table

*note) See page 23 for details

ex)

MR4-22-B7-E	MR-4 AC24V 50/60Hz 2a2b (Metasol) EXP
MR6-42-JD-E	MR-6 DC12V 4a2b (Metasol) EXP
MR8-44-MD-E	MR-8 DC220V 4a4b (Metasol) EXP

Thermal Overload Relay

MT-12 - **3H** **0.1**

Rated current	
12	12AF
32	32AF
63	63AF
95	95AF
⋮	⋮
800	800AF

Relay version	
3H	Non-differential (3-heater)
3K	Differential
3D	Class 20

Relay setting current	
0.1	0.1A
⋮	⋮
800	800A

Enclosed Motor starters

MW22B - **M** - **B**

Product	
MW6A ⋮ MW100A	MW-6a ⋮ MW-100a
MW9B ⋮ MW22B	MW-9b ⋮ MW-22b

Meterial	
-	Iron
M	Mold

None/Botton	
-	None
B	Button

ex)

MW32aB-30-22-M7-21.5-3H-S-E	MW-32aB AC220V 50/60Hz SCREW 2a2b 21.5A 3H (Metasol) EXP
MW32A-30-22-N5-34-3H-S-E	MW-32a AC415V 50Hz SCREW 2a2b 34A 3H (Metasol) EXP
MW22BMB-11-Q7-19-SH-S-E	MW-22bMB AC380V 50/60Hz SCREW 1a1b 19A 3H (Metasol) EXP

Main Coil Voltage

AC						DC		Semi-electric (AC)		Electric (AC/DC)	
50Hz		60Hz		50/60Hz				50/60Hz			
Coil Voltage(V)	code	Coil Voltage(V)	code	Coil Voltage(V)	code	Coil Voltage(V)	code	Coil Voltage(V)	code	Coil Voltage(V)	code
24V	B5	24V	B6	24V	B7	12V	JD	100~120	HF	24 (AC24~25 50/60Hz, DC24)	AV
48V	E5	48V	E6	48V	E7	24V	BD	220~240	MU	48 (AC48~50 50/60Hz, DC24)	BV
110V	F5	110V	F6	110V	F7	48V	ED	265~347	OQ	24/48 (AC24~50 50/60Hz, DC100~110)	CV
220V	M5	220V	M6	220V	M7	110V	FD	380~440	OV	100/200 (AC100~240 50/60Hz, DC100~220)	FV
380V	Q5	380V	Q6	380V	Q7	125V	GD	440~575	S7	100/200 (AC100~240 50/60Hz, DC100~220)	DV
415V	N5	415V	N6	415V	N7	220V	MD				EV
440V	R5	440V	R6	440V	R7	240V	OD				GV

* See page 23 for details

Type numbering system

Optional Accessories

Auxiliary contact unit

UA1 — 11

Version	
UA-1	Side mount
AU-100	
AU-100E	
UA-2	Front mount(2P)
UA-4	Front mount(4P)

Note) UA-2: 2NO, 1NO1NC, 2NC
UA-4: 4NO, 3NO1NC, 4NC, 2NO2NC, 1NO3NC,

Contact composition	
11	1NO+1NC
20	2NO
02	2NC
40	4NO
31	3NO+1NC
22	2NO+2NC
13	1NO+3NC
04	4NC

Surge absorber

US - 11

Product Composition
Varistor+RC
Varistor+RC
Varistor+RC
Varistor+RC
Varistor+RC
Varistor+RC
Varistor
Varistor
Varisto
Varistor
RC

Voltage	
1	AC 24~48V
2	AC 100~125V
3	AC 200~240V
4	DC 24~48V
5	DC 100~125V
6	DC 200~220V
11	AC/DC 24~48V
12	AC/DC 100~125V
13	AC/DC 200~240V
14	AC/DC 380~440V
22	AC 100~125V

Note) See page 82 for details.

Interlock unit

UR - 02

Product

Contact composition	
02	2NC
00	None

Wire kit for Interlocking

UW - 32

Product

Rated current	
18	18AF
22	22AF
32	40AF
63	65AF
95	100AF

Separate mounting unit (For relay)

UZ - 32

Product

Rated current	
12	12AF
32	32AF
63	63AF
95	95AF
150	150AF

Remote reset unit (For relay)

UM - 4R

Product

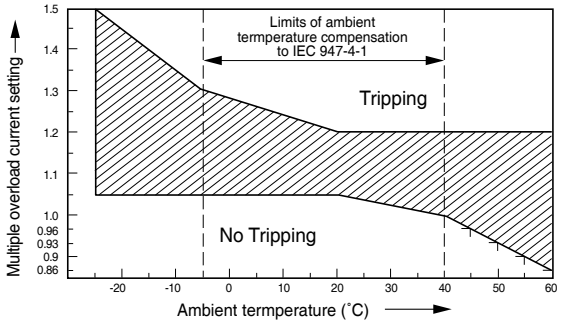
Cable Length	
12	12AF
32	32AF
63	63AF

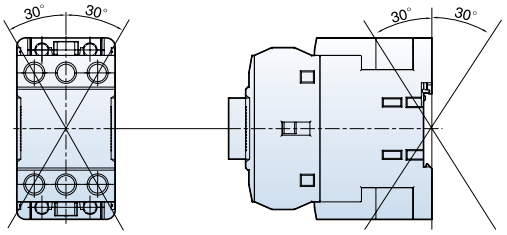
Environment

Environment



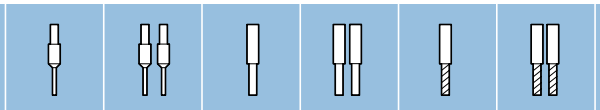
Standards	IEC/EN 60947-1, IEC/EN 60947-4-1, UL508, K60947
Certifications	CE, UL, CCC
Rated insulation voltage (Ui)	1000V (6A~22A: 690V, 32A~800A: 1000V)
Rated impulse withstand voltage (Uimp)	8kV (6A~22A: 6kV, 32A~800A: 8kV)
Degree of protection	IP20 (Conforming to IEC60529)
Ambient air temperature	Storage: -30˚C ~ +65˚C
	Operation: -5˚C ~ +55˚C (AC Type) -5˚C ~ +40˚C (MT, DC Type) -25˚C ~ +60˚C (MT refer to Graph)
Operating altitude	3000m (8900ft)
Operating positions	
Shock resistance	Opened: 8G
(1/2 sine wave =11ms)	Closed: 10G
Vibration resistance (Conforming to IEC68-2-6)	Opened: 2G
(5...300 Hz)	Closed: 4G
Flame resistance	Conforming to UL 94: V0
	Conforming to IEC 695-2-1: 960˚C





Connections

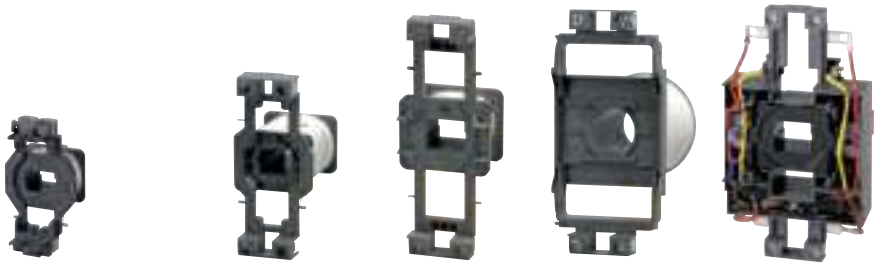
Connections

Wire type / Frame		Main Terminal Size	 (AWG / mm²)			 mm(max)	Torque [lb-in]	Torque [Nm]	Torque [kgf · cm]
MC-6a/9a		M3.5	18~10 / 1~6	18~10 / 1~6	18~10 / 1~6	8.3	10	1.13	12
MC-12a			16~10 / 1.5~10	16~10 / 1.5~10	16~10 / 1.5~10				
MC-18a			16~8 / 1.5~10	16~8 / 1.5~10	16~8 / 1.5~10				
MC-9b		M4	18~10 / 1~6	18~10 / 1~6	18~10 / 1~6	9.6	20	2.25	23
MC-12b			18~10 / 1~6	16~10 / 1~6	16~10 / 1~6				
MC-18b			18~10 / 1~6	16~8 / 1.5~10	16~8 / 1.5~10				
MC-22b			18~10 / 1~6	14~8 / 2.5~10	14~8 / 2.5~10				
MC-32a		M5	18~10 / 1~6	12~8 / 2.5~10	12~8 / 2.5~10	12.8	35	4	41
MC-40a			18~10 / 1~6	8~6 / 10~16	8~6 / 10~16				
MC-50a		M6	-	10~4 / 6~25	10~4 / 6~25	14	35	4	41
MC-65a			-	8~3 / 10~35	8~3 / 10~35				
MC-75a		M8	-	8~2 / 10~35	8~2 / 10~35	17	45	5.1	52
MC-85a			-	8~1/0 / 10~50	8~1/0 / 10~50				
MC-100a			-	8~2/0 / 10~70	8~2/0 / 10~70				
MC-130a		M8	-	3~2/0 / 35~70	3~2/0 / 35~70	24.5	80	9.1	93
MC-150a			-	3~4/0 / 35~95	3~4/0 / 35~95				
MC-6~150a Aux./Coil		M4	20~14 / 0.5~2.5	18~12 / 0.75~2.5	18~12 / 0.75~2.5	7.6	10	1.13	12
MC-185a		M10	-	1~4/0 / 50~95	1~4/0 / 50~95	25	130	14.7	150
MC-225a			-	1/0~300 / 50~150	1/0~300 / 50~150				
MC-265a		M12	-	3/0~500 / 95~240	3/0~500 / 95~240	30	200	22.6	230
MC-330a			-	4/0~500 / 95~240	4/0~500 / 95~240				
MC-400a			-	350~700 / 185~185×2	350~700 / 185~185×2				
MC-500a		M16	-	350~800 / 185~240×2	350~800 / 185~240×2	40	500	56.5	576
MC-630a			-	600~2000	-				
MC-800a			-	1700~Busbar	-				
MC-185~800a Aux/Coil		M4	16~10 / 1.25~5.5	16~10 / 1.25~5.5	16~10 / 1.25~5.5	7.6	15	1.75	18
MC-1260a		M12	-	5mm×100mm×2ea	-	-	200	22.6	230
MC-1400a		M12	-	5mm×100mm×2ea	-	-	200	22.6	230
MC-1700a				5mm×100mm×3ea					
MC-2100a				5mm×100mm×4ea					

Coil code table

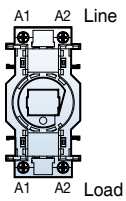
AC						DC		Semi-electric (AC)		Electric (AC/DC)	
50Hz		60Hz		50/60Hz		–		50/60Hz		–	
Coil voltage(V)	code	Coil voltage(V)	code	Coil voltage(V)	code	Coil voltage(V)	code	Coil voltage(V)	code	Coil voltage(V)	code
24	B5	24	B6	24	B7	12	JD	100~120	HF	24 (AC24~25 50/60Hz, DC24)	AV
32	C5	32	-	32	-	20	AD	220~240	MU	48 (AC48~50 50/60Hz, DC24)	BV
36	CB5	36	-	36	CB7	24	BD	300 (265~247)	OQ	24/48 (AC24~50 50/60Hz, DC24~48)	CV
42	D5	42	-	42	D7	30	PD	380~440	OV	100/100 (AC100~127 50/60Hz, DC100~110)	DV
48	E5	48	E6	48	E7	36	CD	500 (440~575)	S7	100/200 (AC100~240 50/60Hz, DC100~220)	FV
65	GE5	65	GE6	65	GE7	42	ID	100~120 (LED)	HFL	200/200 (AC200~240 50/60Hz, DC200~220)	EV
80	G5	80	-	80	-	48	ED	220~240 (LED)	MUL	300/400 (AC265~450 50/60Hz)	GV
100	H5	100	-	100	H7	60	ND	380~440 (LED)	QVL	AC 300 (AC265~347 50/60Hz)	HV
110	F5	110	F6	110	F7	72	SD			AC 400 (AC380~450 50/60Hz)	IV
100/110	-	100/110	-	100/110	HF7	80	KS			AC 500 (AC440~575 50/60Hz)	JV
115	FE5	115	FE6	115	FE7	100	HD				
110/120	-	110/120	-	110/120	HK7	110	FD				
120	K5	120	K6	120	K7	120	KD				
127	KB5	127	KB6	127	KB7	125	GD				
175	LC5	175	LC6	175	LC7	200	LD				
200	L5	200	L6	200	L7	220	MD				
200/208	-	200/208	-	200/208	LJ7	240	OD				
208	LB5	208	LB6	208	LB7	250	UD				
220	M5	220	M6	220	M7						
200/220	-	200/220	-	200/220	LM7						
220/230	-	220/230	-	220/230	MP7						
220/240	-	220/240	-	220/240	MU7						
220L	M5L	220L	M6L	220L	M7L						
230	P5	230	-	230	P7						
240	U5	240	U6	240	U7						
240W	U5W	240W	-	240W	-						
230/240	PU5	230/240	PU6	230/240	PU7						
240/277	-	240/277	-	240/277	UO7						
256	A5	256	A6	256	A7						
277	O5	277	O6	277	O7						
380	Q5	380	Q6	380	Q7						
400	V5	400	-	400	V7						
380/400	-	380/400	-	380/400	QV7						
415	N5	415	N6	415	N7						
440	R5	440	R6	440	R7						
400/440	-	400/440	-	400/440	VR7						
460	-	460	-	460	RW7						
480	W5	480	W6	480	-						
500	S5	500	-	500	S7						
550	X5	550	-	550	X7						
600	T5	600	T6	600	-						

Control coil characteristics (3pole)



Frame size			18AF	22AF	40AF	65AF	100AF	150AF(A)
Type			MC-6a MC-9a MC-12a MC-18a MR-4, 6, 8	MC-9b MC-12b MC-18b MC-22b	MC-32a MC-40a	MC-50a MC-65a MD-30a	MC-75a MC-85a MC-100a MD-60a MD-100a	MC-130a MC-150a
AC coil								
Control voltage		[Uc]						48, 100~240
	50Hz	[V]	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V					110~120, 220~240,
	60Hz	[V]	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V					300, 380~440, 500V
Voltage limit		[Uc]	85 ~110%					
Coil consumption at 20°C								
AC 220V	Inrush	[VA]	70	70	90	110	240	110
60Hz	Holding	[VA]	9	9	9	13	17	18
Heat dissipation		[W]	2.3	2.3	2.7	2.8	5.4	2.7
Operating time								
	Closing	[ms]	15 ... 30	15 ... 30	15 ... 30	15 ... 30	15 ... 30	20 ... 40
	Opening	[ms]	4 ... 19	4 ... 19	4 ... 19	4 ... 19	10 ... 30	50 ... 70
DC coil								
Control voltage		[Uc]	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V					24, 48, 70~110, 100~220, 110, 220V
Voltage limit		[Uc]	85~110%					
Coil consumption at 20°C								
DC 110V	Inrush	[W]	6	6	9	9	18	213
	Holding	[W]	6	6	9	9	18	7.5
Time constant (L/R)		[ms]	28	28	28	65	75	
Operating time								
	Closing	[ms]	40 ... 60	40 ... 60	50 ... 65	50 ... 65	100 ... 120	70 ... 80
	Opening	[ms]	4 ... 19	4 ... 19	4 ... 19	4 ... 19	10 ... 25	60 ... 70

Note) Either semi-electronic coil(A) or electronic coil(B) can be chosen for MC-130a and MC-150a



Note) A1(A2) in line side is internally connected to A1(A2) in load side.



Frame size			150AF(B)	225AF	400AF	800AF	1260AF	2100AF
Type			MC-130a MC-150a	MC-185a MC-225a	MC-265a MC-330a MC-400a	MC-500a MC-630a MC-800a	MC-1260a	MC-1400a MC-1700a MC-2100a
AC/DC common coil								
Control voltage		[Uc]						
	AC/DC		AC 24	24/24	DC 24	-	-	-
	AC/DC		24~48/24~48	48/48	-	100/100	-	-
	AC/DC		100~240/100~220	100~240/100~220	100~240/100~220	200/200	100~240/100~220	100~240/100~220
	AC		-	300	300	300	-	-
	AC		-	400	400	400	-	-
	AC		-	500	500	500	-	-
Voltage limit		[Uc]	85~110% Note 1)					
Coil consumption at 20°C Note 2)								
AC 220V	Inrush	[VA]	320	380	571	1000	1500	2500
50/60Hz	Holding	[VA]	10	11.6	14	29	24	50
Heat dissipation		[W]	4.5	4.7	5	7.8		8.7
Operating time								
	Closing	[ms]	70	70	55	75	75	80
	Opening	[ms]	70	70	55	75	75	70

Note) 1. Voltage limit for DC 24V coil is 75~110%.
2. Operating time for DC 24V coil is 90ms.
3. Either semi-electronic coil(A) or electronic coil(B) can be chosen for MC-130a and MC-150a

Control coil characteristics (4pole)






Frame size			18AF	22AF	40AF	85AF
Type			MC-6a/4P MC-9a/4P MC-12a/4P MC-18a/4P	MC-22a/4P	MC-32a/4P MC-40a/4P	MC-50a/4P MC-65a/4P MC-75a/4P MC-85a/4P
AC coil						
Control voltage		[Uc]				
	50Hz	[V]	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550	24, 32, 42, 48, 65, 80, 100, 110, 220, 230, 240, 380, 400, 415, 500, 550, 240(wide)		24, 48, 100, 110, 200, 220, 230, 240, 380, 400, 415, 440, 500, 550, 240(wide)
	60Hz	[V]	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600	24, 48, 110, 120, 208, 220, 240, 277, 380, 440, 480, 600		24, 48, 110, 120, 208, 220, 230, 240, 277, 380, 440, 480, 600
	50/60Hz	[V]	24, 48, 100, 110, 120, 200, 220, 230, 20, 380, 415, 440, 500, 550	24, 48, 100, 110, 120, 200, 220, 230, 240, 380, 415, 440, 500, 550		24, 48, 110, 120, 220, 230, 240, 380, 415, 440, 500, 550
Voltage limit			85~110%			
Coil consumption at 20°C						
AC 220V	Inrush	[VA]	70	95	95	220
60Hz	Holding	[VA]	9	9	9	17
Heat dissipation		[W]	2.3	2	2	5
Operating time	Closing	[ms]	15...30	10~17	10~17	16~25
	Opening	[ms]	4...9	6~9	6~9	8~15
DC coil						
Control voltage		[Uc]	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250			
Voltage limit			85~110%			
Coil consumption at 20°C						
DC 100V	Inrush	[W]	6	9	9	460
(18AF: DC110V)	Holding	[W]	6	9	9	6
Time constant (L/R)		[ms]	28	50	50	-
Operating time	Closing	[ms]	40...60	45~55	45~55	20~30
	Opening	[ms]	4...19	8~15	8~15	13~20



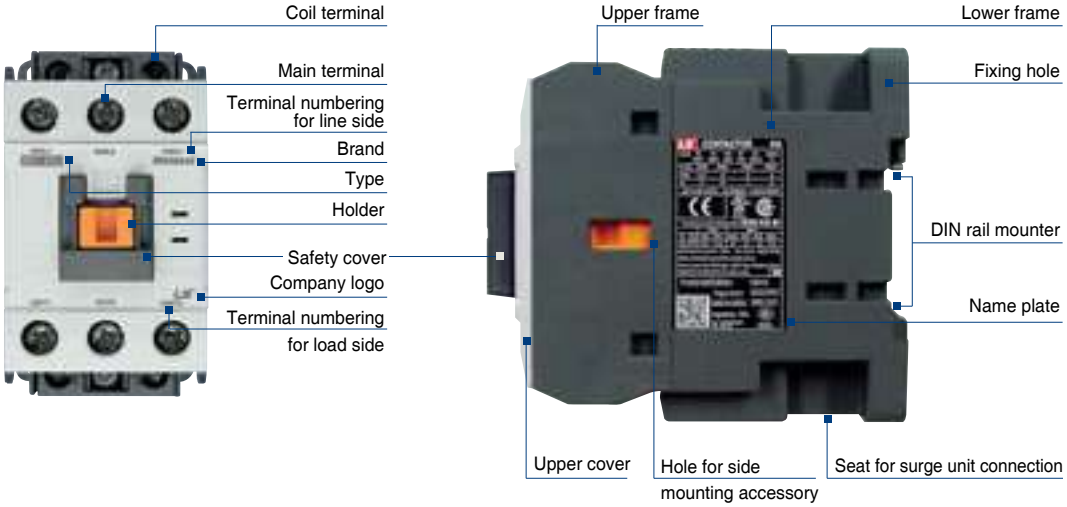




Frame size			225AF	400AF	800AF
Type			MC-100a/4P MC-130a/4P MC-150a/4P MC-185a/4P MC-225a/4P	MC-265a/4P MC-330a/4P MC-400a/4P	MC-500a/4P MC-630a/4P MC-800a/4P
AC/DC common coil					
Control voltage		[Uc]			
		[AC/DC]	24/24	24(DC Only)	-
		[AC/DC]	48/48	-	100/100
		[AC/DC]	100~240/100~220	100~240/100~220	200/200
		[AC]	300	300	300
		[AC]	400	400	400
		[AC]	500	500	500
Voltage limit				85~110%[Noet1)]	
Coil consumption at 20°C[Noet2)]					
AC 220V	Inrush	[VA]	380	571	1000
50/60Hz	Holding	[VA]	11.6	14	29
Heat dissipation		[W]	4.7	5	7.8
Operating time	Closing	[ms]	70	55	75
	Opening	[ms]	70	55	75

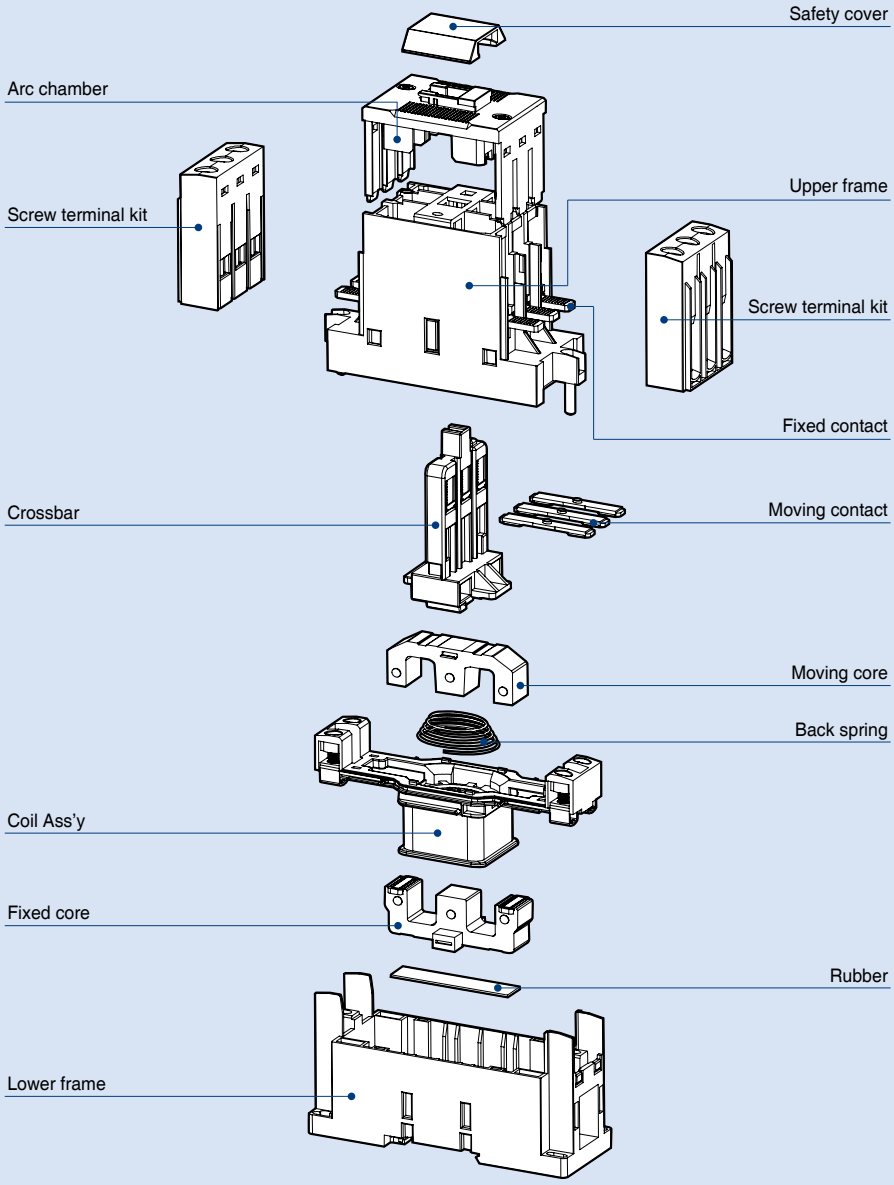
Note) 1. Voltage limit for DC 24V coil is 75~110%.
2. Operating time for DC 24V coil is 90ms.

Marking


Coil terminal
Main terminal
Terminal numbering for line side
Brand
Type
Holder
Safety cover
Company logo
Terminal numbering for load side
Upper frame
Lower frame
Fixing hole
DIN rail mounter
Name plate
Upper cover
Hole for side mounting accessory
Seat for surge unit connection


Company logo
CE marking
UL size
Wire size
Country of origin
Bar code
CONTACTOR 9A
AC3 IEC 60947 EN 60947 VDE 0660 BS 5424 GB 14048.4 K60947 -4-1
V AC 240 440 550 690
kW 2.5 4 4 4
A 11 9 7 5
AC1=Ith=25A Ui=690V Uimp=6kV
This Device is a Direct Replacement to NEMA SIZE 00
1PH 3PH
V AC 120 240 208 240 480 600
UL HP 1/2 1.5 2 3 5 7.5
Continuous current : 25A Aux Cont.: A600 P600
When protected by 20A time delay fuses.
Short circuit 5kA RMS Sym. 600V Max. Break all lines
AWG18-10 60/75°C Cu-wire only
AB
P0104D21005EC9DEMJ01
120110
Torque: 20 lb.in
MADE IN KOREA
Inspected by Y.S.Kim
HOIST-a
Type
Product name
Rating
UL, CSA rating
A, B : Metasol MC
S : Susol MC
Date of manufacture
Certification
(Domestic use only)

Internal structure



Contactors (18AF)

Description

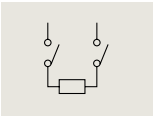
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- AC or DC control in different physical size
- Front/side mountable accessories available
- Direct mountable overload relay available Rating
- 1NO or 1NC Auxiliary contact built-in as standard.
- Degree of protection: IP20



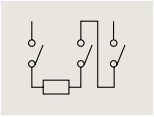
MC-18a

Rating

Contactor type			MC-6a		MC-9a		MC-12a		MC-18a	
			kW	A	kW	A	kW	A	kW	A
AC duty	AC3	200/240V	2.2	9	2.2	11	3.5	13	4.5	18
		380/440V	3	7	4	9	5.5	12	7.5	18
		500/550V	3	6	4	7	7.5	12	7.5	13
		690V	3	4	4	5	7.5	9	7.5	9
	AC4	200/240V	1.5	7	1.5	8	2.2	11	3.7	16
		380/440V	2.2	5	2.2	6	4	9	4	11
	AC1		-	25	-	25	-	25	-	32
DC1 duty	2-pole	24V	-	10	-	10	-	12	-	18
(L/R=1ms)	in series	48V	-	10	-	10	-	12	-	18
		110V	-	6	-	6	-	10	-	13
	3-pole	24V	-	10	-	10	-	12	-	18
	in series	48V	-	10	-	10	-	12	-	18
		110V	-	8	-	8	-	12	-	18
DC2, 4 duty	2-pole	24V	-	8	-	8	-	12	-	12
(L/R=15ms)	in series	48V	-	4	-	4	-	6	-	6
		110V	-	2.5	-	2.5	-	4	-	4
	3-pole	24V	-	8	-	8	-	12	-	12
	in series	48V	-	6	-	6	-	10	-	10
		110V	-	4	-	4	-	8	-	8
Directly mountable Overload relay			MT-12							
Conductor size (solid, stranded)		AWG	18~10		16~10		16~8		14~8	
(the max. number of conducts: 2)		mm²	1~4		1.5~4		1.5~10		2.5~10	
Conductor type			65/75℃ Cu-wire only							



2pole for DC contact



3pole for DC contact



Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

* Please be cautious for wiring control power cable, especially " + " (Positive) and " - " (Negative) polarity should be correctly considered.

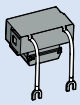
Certification

- CE(IEC)
- UL508
- CCC

Accessories

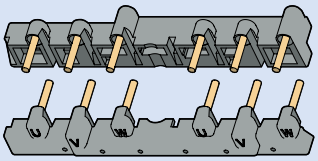


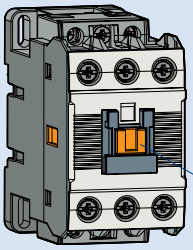
Delayed opening device, AD-9
▶85 page

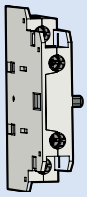


Surge unit, US
connecting to coil terminals
▶88 page

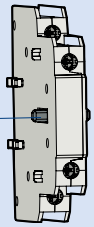
Wire kit, UW
for use with Interlock unit
▶86 page



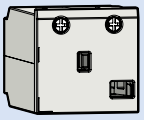




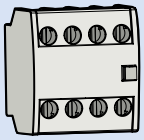
Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶85 page



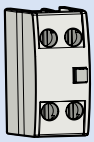
Interlock unit, UR
side mount
▶86 page



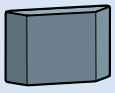
Mechanical latch unit, ML-65
▶84 page



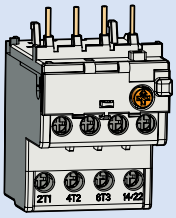
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Safety cover, A7
▶91 page



Overload relay, MT
directly mountable to MC type contactor
▶94 page

Contactors (22AF)

Description

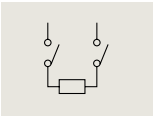
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- AC or DC control in different physical size
- Front/side mountable accessories available
- Direct mountable overload relay available Rating
- 1NO1NC Auxiliary contact built-in as standard.
- UA-1 is available on only left side
- Degree of protection: IP20



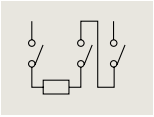
MC-22b

Rating

Contactor type			MC-9b		MC-12b		MC-18b		MC-22b	
			kW	A	kW	A	kW	A	kW	A
AC duty	AC3	200/240V	2.2	11	3.5	13	4.5	18	5.5	22
		380/440V	4	9	5.5	12	7.5	18	11	22
		500/550V	4	7	7.5	12	7.5	13	15	20
		690V	4	6	7.5	9	7.5	9	15	18
	AC4	200/240V	1.5	8	2.2	11	3.7	16	3.7	18
		380/440V	2.2	6	4	9	4	11	5.5	13
	AC1		-	27	-	27	-	32	-	45
DC1 duty	2-pole	24V	-	10	-	12	-	18	-	20
(L/R=1ms)	in series	48V	-	10	-	12	-	18	-	20
		110V	-	6	-	10	-	13	-	15
	3-pole	24V	-	10	-	12	-	18	-	20
	in series	48V	-	10	-	12	-	18	-	20
		110V	-	8	-	12	-	18	-	20
DC 2,4 duty	2-pole	24V	-	8	-	12	-	12	-	20
(L/R=15ms)	in series	48V	-	4	-	6	-	6	-	15
		110V	-	2.5	-	4	-	4	-	8
	3-pole	24V	-	8	-	12	-	12	-	20
	in series	48V	-	6	-	10	-	10	-	20
		110V	-	4	-	8		8	-	15
Directly mountable Overload relay			MT-32							
Conductor size (solid, stranded)		AWG	16~10		16~8		14~8		14~8	
(the max. number of conducts: 2)		mm²	1.5~4		1.5~10		2.5~10		2.5~10	
Conductor type			65/75℃ Cu-wire only							



2pole for DC contact



3pole for DC contact

Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

* Please be cautious for wiring control power cable, especially " + " (Positive) and " - " (Negative) polarity should be correctly considered.

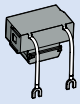
Certification

- CE(IEC)
- UL508
- CCC

Accessories

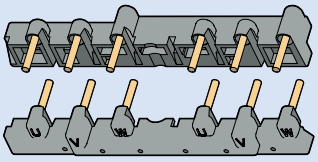


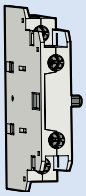
Delayed opening device, AD-9
▶85 page



Surge unit, US
connecting to coil terminals
▶88 page

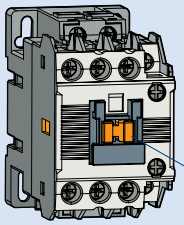
Wire kit, UW
for use with Interlock unit
▶86 page

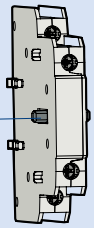




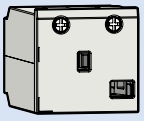
Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶85 page

* It is not possible to attach "UA-1" on the right side of product

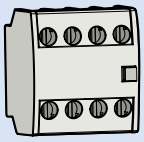




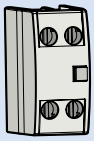
Interlock unit, UR
side mount
▶86 page



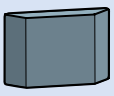
Mechanical latch unit, ML-65
▶84 page



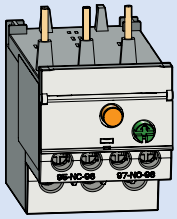
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Safety cover, A7
▶91 page



Overload relay, MT
directly mountable to MC type contactor
▶94 page

Contactors (40AF)

Description

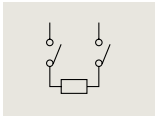
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- AC or DC control in different physical size
- Front/side mountable accessories available
- Direct mountable overload relay available Rating
- 2NO2NC Auxiliary contact built-in is possible as standard
- Degree of protection: IP20



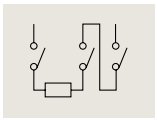
MC-40a

Rating

Contactor type			MC-32a		MC-40a	
			kW	A	kW	A
AC duty	AC3	200/240V	7.5	32	11	40
		380/440V	15	32	18.5	40
		500/550V	18.5	28	22	32
		690V	18.5	20	22	23
		1000V	22	17	22	17
	AC4	200/240V	4.5	20	5.5	25
		380/440V	7.5	17	11	24
	AC1		-	55	-	60
DC1	2-pole	24V	-	25	-	35
(L/R=1ms)	in series	48V	-	25	-	35
		110V	-	25	-	25
	3-pole	24V	-	25	-	35
	in series	48V	-	25	-	35
		110V	-	25	-	35
DC2,4	2-pole	24V	-	25	-	35
(L/R=15ms)	in series	48V	-	20	-	20
		110V	-	10	-	10
	3-pole	24V	-	25	-	35
	in series	48V	-	25	-	30
		110V	-	20	-	20
Directly mountable Overload relay			MT-32			
Conductor size (solid, stranded)		AWG	12~8		8~6	
(the max. number of conducts: 2)		mm²	2.5~10		10~16	
Conductor type			65/75℃ Cu-wire only			



2pole for DC contact



3pole for DC contact



Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

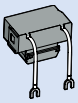
Certification

- CE(IEC)
- UL508
- CCC

Accessories

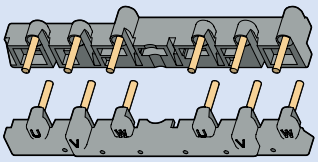


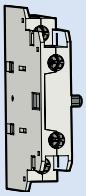
Delayed opening device, AD-9
▶85 page



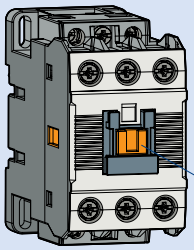
Surge unit, US
connecting to coil terminals
▶88 page

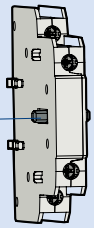
Wire kit, UW
for use with Interlock unit
▶86 page



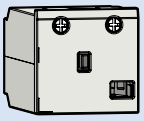


Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶85 page

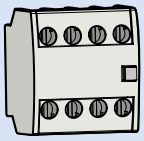




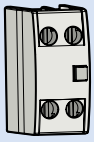
Interlock unit, UR
side mount
▶86 page



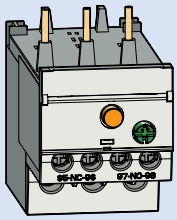
Mechanical latch unit, ML-65
▶84 page



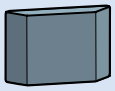
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Overload relay, MT
directly mountable to MC type contactor
▶95 page



Safety cover, A7
▶95 page

Contactors (65AF)

Description

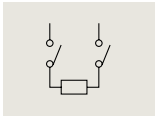
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- Both screw and Box terminal (lug) type are available
- AC or DC control in different physical size
- Front/side mountable accessories available
- Direct mountable overload relay available rating
- 2NO2NC Auxiliary contact built-in is possible as standard
- Degree of protection: IP20



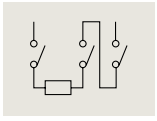
MC-65a

Rating

Contactor type			MC-50a		MC-65a	
			kW	A	kW	A
AC duty	AC3	200/240V	15	55	18.5	65
		380/440V	22	50	30	65
		500/550V	30	43	33	60
		690V	30	28	33	35
		1000V	30	23	30	23
	AC4	200/240V	7.5	35	11	50
		380/440V	15	32	22	47
	AC1		-	100	-	115
DC1 duty	2-pole	24V	-	50	-	50
(L/R=1ms)	in series	48V	-	40	-	40
		110V	-	35	-	35
	3-pole	24V	-	50	-	65
	in series	48V	-	50	-	65
		110V	-	50	-	65
DC2, 4 duty	2-pole	24V	-	45	-	45
(L/R=15ms)	in series	48V	-	25	-	25
		110V	-	15	-	15
	3-pole	24V	-	50	-	50
	in series	48V	-	35	-	35
		110V	-	30	-	30
Directly mountable Overload relay			MT-63			
Conductor size (solid, stranded)		AWG	10~4		8~3	
(the max. number of conducts: 2)		mm²	6~25		10~35	
Conductor type			65/75℃ Cu-wire only			



2pole for DC contact



3pole for DC contact



Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

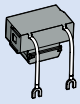
Certification

- CE(IEC)
- UL508
- CCC

Accessories

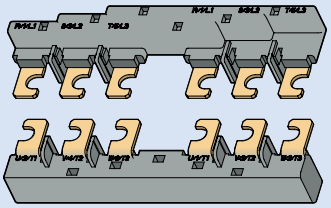


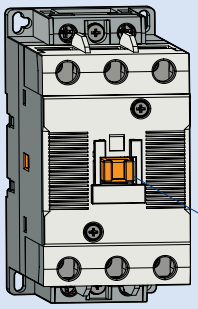
Delayed opening device, AD-50
▶85 page

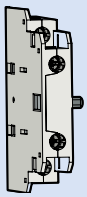


Surge unit, US
connecting to coil terminals
▶88 page

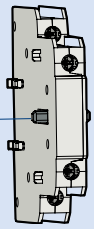
Wire kit, UW
for use with Interlock unit
▶86 page



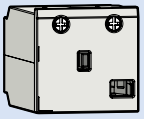




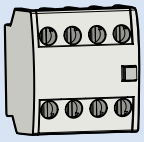
Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶77 page



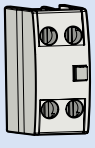
Interlock unit, UR
side mount
▶86 page



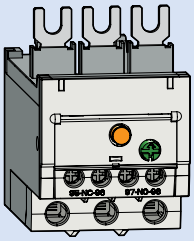
Mechanical latch unit, ML-65
▶84 page



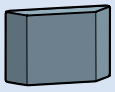
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Overload relay, MT
directly mountable to MC type contactor
▶91 page



Safety cover, A7
▶87 page

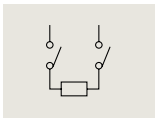
Contactors (100AF)

Description

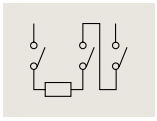
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- Both screw and Box terminal (lug) type are available
- AC or DC control in different physical size
- Front/side mountable accessories available
- Direct mountable overload relay available Rating
- 2NO2NC Auxiliary contact built-in is possible as standard
- Degree of protection: IP20



MC-100a



2pole for DC contact



3pole for DC contact

Rating

Contactor type			MC-75a		MC-85a		MC-100a	
			kW	A	kW	A	kW	A
AC duty	AC3	200/240V	22	75	25	85	30	105
		380/440V	37	75	45	85	55	105
		500/550V	37	64	45	75	55	85
		690V	37	42	45	45	45	65
		1000V	37	28	37	28	37	28
	AC4	200/240V	13	55	15	65	19	80
		380/440V	25	52	30	62	37	75
	AC1		-	125	-	135	-	160
DC1 duty	2-pole	24V	-	75	-	80	-	100
(L/R=1ms)	in series	48V	-	65	-	65	-	100
		110V	-	50	-	50	-	80
	3-pole	24V	-	75	-	80	-	100
	in series	48V	-	75	-	80	-	100
		110V	-	75	-	80	-	100
DC2, 4 duty	2-pole	24V	-	65	-	65	-	100
(L/R=15ms)	in series	48V	-	40	-	40	-	60
		110V	-	20	-	20	-	40
	3-pole	24V	-	80	-	80	-	100
	in series	48V	-	60	-	60	-	90
		110V	-	50	-	50	-	80
Directly mountable Overload relay			MT-95					
Conductor size (solid, stranded)		AWG	8~2		8~0		8~00	
(the max. number of conducts: 2)		mm²	10~35		10~50		10~70	
Conductor type			65/75℃ Cu-wire only					



Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

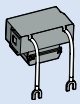
Certification

- CE(IEC)
- UL508
- CCC

Accessories

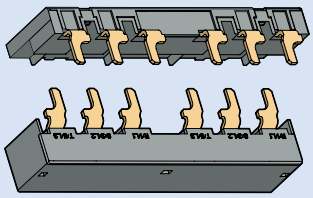


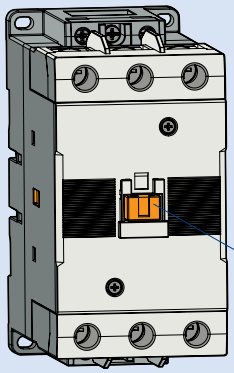
Delayed opening device, AD-50
▶85 page

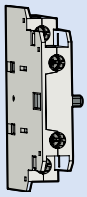


Surge unit, US
connecting to coil terminals
▶88 page

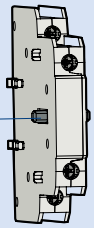
Wire kit, UW
for use with Interlock unit
▶86 page



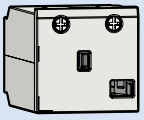




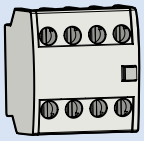
Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶85 page



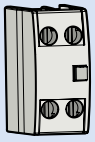
Interlock unit, UR
side mount
▶86 page



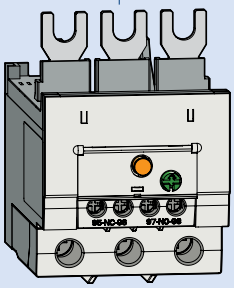
Mechanical latch unit, ML-150
▶84 page



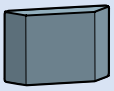
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Overload relay, MT
directly mountable to MC type contactor
▶95 page



Safety cover, A7
▶91 page

Contactors (150AF)

Description

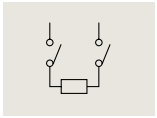
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- Both screw and Box terminal (lug) type are available
- AC or DC control in the same physical size
- Front/side mountable accessories available
- Direct mountable overload relay available
- 2NO2NC Auxiliary contact built-in is possible as standard
- Degree of protection: IP20



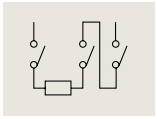
MC-150a

Rating

Contactor type			MC-130a		MC-150a	
			kW	A	kW	A
AC duty	AC3	200/240V	37	130	45	150
		380/440V	60	130	75	150
		500/550V	60	90	70	100
		690V	55	60	55	60
		1000V	75	53	75	53
	AC4	200/240V	22	93	30	120
		380/440V	45	90	55	110
	AC1		-	200	-	250
DC1 duty (L/R=1ms)	2-pole in series	24V	-	120	-	150
		48V	-	100	-	120
		110V	-	80	-	100
	3-pole in series	24V	-	120	-	150
		48V	-	120	-	150
		110V	-	100	-	150
DC2,4 duty (L/R=15ms)	2-pole in series	24V	-	120	-	150
		48V	-	60	-	100
		110V	-	40	-	80
	3-pole in series	24V	-	120	-	150
		48V	-	90	-	130
		110V	-	80	-	120
Directly mountable Overload relay			MT-150			
Conductor size (solid, stranded)		AWG	3~00		3~4/0	
(the max. number of conducts: 2)		mm²	35~70		35~95	
Conductor type			65/75℃ Cu-wire only			



2pole for DC contact



3pole for DC contact



Coil voltage

AC/DC		100~240 / 100~220 / 24~48
AC	50/60Hz	24, 48, 100~240, 110~120, 220~240, 300, 380~440, 500V
DC	DC	24, 48, 70~110, 100~220, 110, 220V

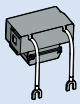
Certification

- CE(IEC)
- UL508
- CCC

Accessories



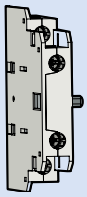
Delayed opening device, AD-100
▶85 page



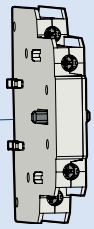
Surge unit, US
connecting to coil terminals
▶88 page

Busbar

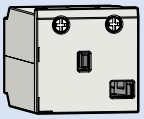
Insulation barrier
▶85 page



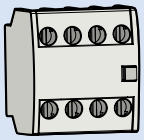
Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶81 page



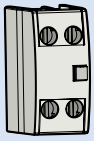
Interlock unit, UR
side mount
▶86 page



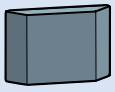
Mechanical latch unit, ML-150
▶84 page



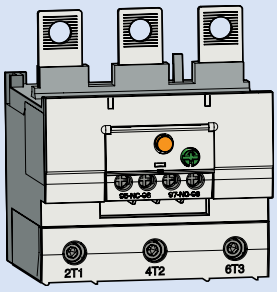
Auxiliary contact unit, UA-4
front mount, 4P
▶81 page



Auxiliary contact unit, UA-2
front mount, 2P
▶81 page



Safety cover, A7
▶91 page



Overload relay, MT
directly mountable to MC type contactor
▶96 page

Contactors (225AF)

Description

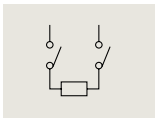
- 3-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available
- Direct mountable overload relays available
- Degree of protection: IP20



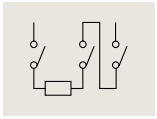
MC-225a

Rating

Contactor type			MC-185a		MC-225a	
			kW	A	kW	A
AC duty	AC3	200/240V	55	185	75	225
		380/440V	90	185	132	225
		500/550V	110	180	132	200
		690V	110	120	140	150
		1000V	132	90	132	90
	AC4	200/240V	37	150	45	180
		380/440V	75	150	90	180
	AC1		-	300	-	350
DC1 duty	2-pole	24V	-	180	-	220
(L/R=1ms)	in series	48V	-	180	-	180
		110V	-	150	-	150
	3-pole	24V	-	180	-	220
	in series	48V	-	180	-	220
		110V	-	180	-	220
DC2, 4 duty	2-pole	24V	-	180	-	220
(L/R=15ms)	in series	48V	-	150	-	150
		110V	-	120	-	120
	3-pole	24V	-	180	-	220
	in series	48V	-	180	-	220
		110V	-	150	-	150
Directly mountable Overload relay			MT-225			
Conductor size (solid, stranded)		AWG	1~4/0		0~300	
(the max. number of conducts: 2)		mm²	50~95		50~150	
Conductor type			75℃ Cu stranded wire			



2pole for DC contact



3pole for DC contact



Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
24V	24 - 25V	24V
48V	48 - 50V	48V
100/200V	100 - 240V	100 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

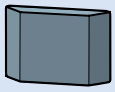
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-100
Delayed opening device
▶85 page



Safety cover, A7
▶91 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶82 page



AP-220
Terminal cover
▶85 page



AR-180
Interlock unit
side mount
▶87 page



AI-180
Insulation barrier
▶85 page



MT-225
Overload relay
directly mountable to the contactor
Current range 65~240A
▶97 page

Contactors (400AF)

Description

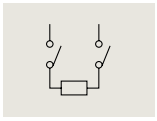
- 3-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available
- Direct mountable overload relays available
- Degree of protection: IP20



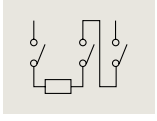
MC-400a

Rating

Contactor type			MC-265a		MC-330a		MC-400a	
			kW	A	kW	A	kW	A
AC duty	AC3	200/240V	80	265	90	330	125	400
		380/440V	147	265	160	330	200	400
		500/550V	147	225	160	250	225	350
		690V	160	185	200	220	250	300
		1000V	147	105	147	105	147	105
	AC4	200/240V	50	200	55	220	75	300
		380/440V	102	200	110	220	150	300
	AC1		-	400	-	500	-	520
DC1 duty	2-pole	24V	-	220	-	300	-	400
(L/R=1ms)	in series	48V	-	180	-	240	-	240
		110V	-	150	-	200	-	200
	3-pole	24V	-	220	-	300	-	400
	in series	48V	-	220	-	300	-	400
		110V	-	220	-	300	-	400
DC2, 4 duty	2-pole	24V	-	220	-	300	-	400
(L/R=15ms)	in series	48V	-	150	-	200	-	200
		110V	-	120	-	150	-	150
	3-pole	24V	-	220	-	300	-	400
	in series	48V	-	220	-	280	-	280
		110V	-	150	-	200	-	200
Directly mountable Overload relay			MT-400					
Conductor size (solid, stranded)		AWG	3/0~500		4/0~500		350~700	
(the max. number of conducts: 2)		mm²	95~240		95~240		185~370	
Conductor type			75℃ Cu stranded wire					



2pole for DC contact



3pole for DC contact

Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
100/200V	100 - 240V	100 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

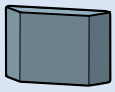
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-300
Delayed opening device
▶85 page



Safety cover, A7
▶91 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶82 page



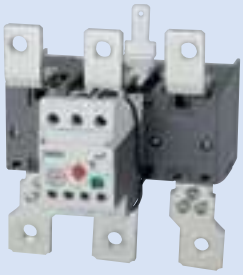
AP-400
Terminal cover
▶83 page



AR-180
Interlock unit
side mount
▶87 page



AI-180
Insulation barrier
▶85 page



MT-400
Overload relay
directly mountable to the contactor
Current range 85~400A
▶97 page

Contactors (800AF)

Description

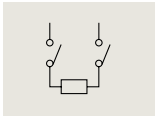
- 3-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available
- Overload relays available
- Degree of protection: IP20



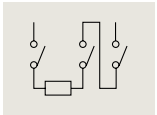
MC-800a

Rating

Contactor type			MC-500a		MC-630a		MC-800a	
			kW	A	kW	A	kW	A
AC duty	AC3	200/240V	147	500	190	630	220	800
		380/440V	265	500	330	630	440	800
		500/550V	265	400	330	500	500	720
		690V	300	380	400	420	500	630
		1000V	280	220	280	220	280	220
	AC4	200/240V	90	350	110	400	160	630
		380/440V	176	350	200	400	300	630
	AC1		-	700	-	900	-	1050
DC1 duty	2-pole	24V	-	400	-	630	-	800
(L/R=1ms)	in series	48V	-	240	-	630	-	800
		110V	-	200	-	630	-	630
	3-pole	24V	-	400	-	630	-	800
	in series	48V	-	400	-	630	-	800
		110V	-	400	-	630	-	800
DC2, 4 duty	2-pole	24V	-	400	-	630	-	800
(L/R=15ms)	in series	48V	-	200	-	630	-	630
		110V	-	150	-	630	-	630
	3-pole	24V	-	400	-	630	-	800
	in series	48V	-	280	-	630	-	630
		110V	-	200	-	630	-	630
Directly mountable Overload relay			MT-800					
Conductor size (solid, stranded)		AWG	350~800		600~2000		1700~Busbar	
(the max. number of conducts: 2)		mm²	185~480		-		-	
Conductor type			75℃ Cu stranded wire					



2pole for DC contact



3pole for DC contact



Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
100V	100 - 127V	100 - 110V
200V	200 - 240V	200 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

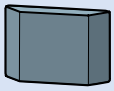
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-600
Delayed opening device
▶85 page



Safety cover, A7
▶91 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶82 page



AP-800
Terminal cover
▶85 page



AR-600
Interlock unit
side mount
▶87 page



AI-600
Insulation barrier
▶85 page



MT-800
Overload relay
directly mountable to the contactor
Current range 200~800A
▶97 page

Contactors (1260AF)

Description

- 3-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available
- Overload relays available
- Degree of protection: IP20



MC-1260a

Rating

Contactor type			MC-1260a	
			kW	A
AC duty	AC3	200/240V	-	-
		380/440V	-	-
		500/550V	-	-
		690V	-	-
		1000V	-	-
	AC1		-	1260
Conductor size (solid, stranded)		Bubar	5mm×100mm×2ea	
Conductor type			Busbar	

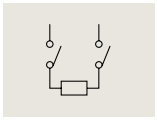


Coil voltage, AC/DC common

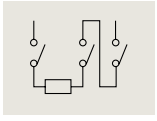
Nominal(ordering) voltage	AC 50/60Hz	DC
100V / 200V	100 - 240V	100 - 220V

Certification

- CE(IEC)
- UL, UL/cUL



2pole for DC contact



3pole for DC contact

Accessories





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶82 page



AI-800
Insulation barrier
▶85 page

Contactors (2100AF)

Description

- 3-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available
- Overload relays available
- Degree of protection: IP20



MC-2100a

Rating

Contactor type			MC-1400a		MC-1700a		MC-2100a	
			kW	A	kW	A	kW	A
AC duty	AC3	200/240V	290	860	310	1050	-	-
		380/440V	550	860	700	1050	900	1450
		500/550V	-	-	-	-	-	-
		690V	800	800	1000	950	-	-
		1000V	-	-	-	-	-	-
	AC1		-	1400	-	1600	-	2100
Conductor size (solid, stranded)		Busbar	5mm×100mm×2ea		5mm×100mm×2ea		5mm×100mm×2ea	
Conductor type			Busbar					



Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
100V / 200V	100 - 240V	100 - 220V

Certification

- CE(IEC)
- UL60947-4-1
- CCC

Accessories





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶82 page



AI-2100
Insulation barrier
▶85 page

Motor starters

Description

- Opened direct motor starter
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- AC or DC control contactor selectable
- Front/side mountable accessories available
- Overload relay is directly mounted to the contactor
- Degree of protection: IP20



Rating

Starter type			18AF				22AF			
			MS-6a	MS-9a	MS-12a	MS-18a	MS-9b	MS-12b	MS-18b	MS-22b
Number of poles			3				3			
Rated operational voltage (Ue)			690V				690V			
Degree of pollution (IEC60529)			IP20				IP20			
Rated operational current Ie (θ≤40℃)										
AC-1	Ue max 690V	A	25	25	25	32	27	27	32	45
AC-3	200/240V	A	9	11	13	18	11	13	18	22
	380/400V	A	7	9	12	18	9	12	18	22
	415V	A	7	9	12	18	9	12	18	22
	440V	A	7	9	12	18	9	12	18	22
	500V	A	6	7	12	13	7	12	13	20
	690V	A	4	5	9	9	6	9	9	18
	1000V	A	-	-	-	-	-	-	-	-
Rated operational power (θ≤40℃)										
AC-3	200/240V	kW	2.2	2.5	3.5	4.5	2.5	3.5	4.5	5.5
	380/400V	kW	3	4	5.5	7.5	4	5.5	7.5	11
	415V	kW	3	4	5.5	7.5	4	5.5	7.5	11
	440V	kW	3	4	5.5	7.5	4	5.5	7.5	11
	500V	kW	3	4	7.5	7.5	4	7.5	7.5	15
	690V	kW	3	4	7.5	7.5	4	7.5	7.5	15
	1000V	kW	-	-	-	-	-	-	-	-
According to IEC60947-4-1										
Rated making capacity AC-3			70	90	120	180	90	120	180	220
Rated breaking capacity AC-3			56	72	96	144	72	96	144	176
Electrical life for Ie/AC-3 Max										
in millions of operating cycles			2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Maximum operating rate										
in operating cycles per hour			1800	1800	1800	1800	1800	1800	1800	1800

Devices usable in the starter combination

Contactors										
with AC or DC control coil			MC-6a	MC-9a	MC-12a	MC-18a	MC-9b	MC-12b	MC-18b	MC-22b
with coil in common use for AC and DC			-	-	-	-	-	-	-	-
Thermal Overload Relays										
Class 10A	Differential type		MT-12/3K				MT-32/3K			
	Non-differential type		MT-12/3H				MT-32/3H			
	Non-differential type(2-heater)		MT-12/2H				MT-32/2H			
Class 20	Differential type		MT-12/3D				MT-32/3D			
	Setting range	A	0.1~18A				0.1~40A			



40AF		65AF		100AF		
MS-32a	MS-40a	MS-50a	MS-65a	MS-75a	MS-85a	MS-100a
3		3		3		
690V		690V		690V		
IP20		IP20		IP20		
55	60	100	115	125	135	160
32	40	55	65	75	85	105
32	40	50	65	75	85	105
32	40	50	65	75	85	105
32	40	50	65	75	85	105
28	32	43	60	64	75	85
20	23	28	35	42	45	65
17	17	23	23	28	28	28
7.5	11	15	18.5	22	25	30
15	18.5	22	30	37	45	55
15	18.5	22	30	37	45	55
15	18.5	22	30	37	45	55
18.5	22	30	33	37	45	55
18.5	22	30	33	37	45	55
22	22	30	30	37	37	37
320	400	500	650	750	850	1050
256	320	400	520	600	650	840
2	2	2	2	2	2	2
1800	1800	1800	1800	1800	1800	1800

MC-32a	MC-40a	MC-50a	MC-65a	MC-75a	MC-85a	MC-100a
-	-	-	-	-	-	-
MT-32/3K		MT-63/3K		MT-95/3K		
MT-32/3H		MT-63/3H		MT-95/3H		
MT-32/2H		MT-63/2H		MT-95/2H		
MT-32/3D		MT-63/3D		MT-95/3D		
0.1~40A		4~65A		7~100A		

Motor starters

Description

- Opened direct motor starter
- 3-pole(NO) main contacts
- Screw mountable
- AC/DC common use control contactor used (from 225AF)
- Additional accessories available
- Overload relay is directly mounted to the contactor
- Degree of protection: IP20



Rating

Starter type				150AF		225AF	
				MS-130a	MS-150a	MS-185a	MS-225a
Number of poles				3		3	
Rated operational voltage (Ue)				690V		690V	
Degree of pollution (IEC60529)				IP20		IP20	
Rated operational current Ie (θ≤40℃)							
AC-1	Ue max 690V		A	200	250	300	350
AC-3	200/240V		A	130	150	185	225
	380/400V		A	130	150	185	225
	415V		A	130	150	185	225
	440V		A	130	150	185	225
	500V		A	90	100	180	200
	690V		A	60	60	120	150
	1000V		A	53	65	90	90
Rated operational power (θ≤40℃)							
AC-3	200/240V		kW	37	45	55	75
	380/400V		kW	60	75	90	132
	415V		kW	60	75	90	132
	440V		kW	60	75	90	132
	500V		kW	60	70	110	132
	690V		kW	55	55	110	140
	1000V		kW	75	90	132	132
According to IEC60947-4-1							
Rated making capacity AC-3				1200	1500	1850	2250
Rated breaking capacity AC-3				960	1200	1480	1800
Electrical life for Ie/AC-3 Max in millions of operating cycles				1	1	1	1
Maximum operating rate in operating cycles per hour				1200	1200	1200	1200

Devices usable in the starter combination

Contactors							
with AC or DC control coil				MC-130a	MC-150a	-	-
with coil in common use for AC and DC				-	-	MC-185a	MC-225a
Thermal Overload Relays							
Class 10A	Differential type			MT-150/3K		MT-225/3K	
	Non-differential type			MT-150/3H		MT-225/3H	
	Non-differential type(2-heater)			MT-150/2H		MT-225/2H	
Class 20	Differential type			MT-150/3D		MT-225/3D	
	Setting range		A	34~150A		65~240A	



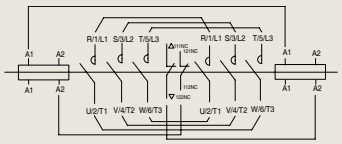
400AF			800AF		
MS-265a	MS-330a	MS-400a	MS-500a	MS-630a	MS-800a
	3			3	
	690V			690V	
	IP20			IP20	
400	500	520	700	900	1050
265	330	400	500	630	800
265	330	400	500	630	800
265	330	400	500	630	800
265	330	400	500	630	800
225	250	350	400	500	720
185	220	300	380	420	630
105	105	105	220	220	220
80	90	125	147	190	220
147	160	200	265	330	440
147	160	200	265	330	440
147	160	200	265	330	440
147	160	225	265	330	500
160	200	250	300	400	500
147	147	147	280	280	280
2650	3300	4000	5000	6300	8000
2120	2640	3200	4000	5040	6400
1	1	0.5	0.5	0.5	0.5
1200	1200	1200	1200	1200	1200
-	-	-	-	-	-
MC-265a	MC-330a	MC-400a	MC-500a	MC-630a	MC-800a
	MT-400/3K			MT-800/3K	
	MT-400/3H			MT-800/3H	
	MT-400/2H			MT-800/2H	
	MT-400/3D			MT-800/3D	
	85~400A			200~800A	

Reversing contactors

Description

- Two AC or DC control contactors are interlocked mechanically and electrically
- 3-pole(NO) main contact in each contactor
- Finger proof design
- DIN rail or screw mountable
- Front/side mountable accessories available
- Overload relay can be directly mounted to the contactor if necessary
- Please use interval time of over 25ms
- Degree of protection: IP20

Circuit diagram





MC-40a/R



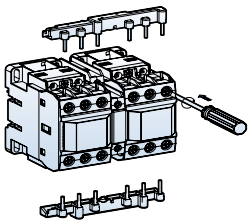
MC-65a/R



MC-100a/R

Rating

Type	AC-3 duty							Current (AC3)	Weight (AC control)
	200 ~ 240V	380 ~ 400V	415V	440V	500V	690V	1000V	440V	
	kW	kW	kW	kW	kW	kW	kW	A	
MC-6a/R	2.2	3	3	3	3	3	-	7	0.63kg
MC-9a/R	2.5	4	4	4	4	4	-	9	0.63kg
MC-12a/R	3.5	5.5	5.5	5.5	7.5	7.5	-	12	0.63kg
MC-18a/R	4.5	7.5	7.5	7.5	7.5	7.5	-	18	0.63kg
MC-9b/R	2.5	4	4	4	4	4	-	9	0.75kg
MC-12b/R	3.5	5.5	5.5	5.5	7.5	7.5	-	12	0.75kg
MC-18b/R	4.5	7.5	7.5	7.5	7.5	7.5	-	18	0.75kg
MC-22b/R	5.5	11	11	11	15	15	-	22	0.75kg
MC-32a/R	7.5	18.5	18.5	18.5	22	22	22	32	0.95kg
MC-40a/R	11	18.5	18.5	18.5	22	22	22	40	0.95kg
MC-50a/R	15	22	22	22	30	30	30	50	1.7kg
MC-65a/R	18.5	30	30	30	33	33	30	65	1.7kg
MC-75a/R	22	37	37	37	37	37	37	75	3.3kg
MC-85a/R	25	45	45	45	45	45	37	85	3.3kg
MC-100a/R	30	55	55	55	55	55	37	105	3.3kg



Coil voltage

AC	50Hz	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

Certification

- CE(IEC)
- UL508
- CCC

Description

- 3-pole main contact
- Mechanically interlocked
- AC/DC common use coil built (from MC-185a/R)
- Wide coil operation voltage
- Screw mountable
- 4NO4NC Auxiliary contacts built-in
- Top/side mountable accessories available
- Please use interval time of over 50ms
- Degree of protection: IP20



Rating

Type	AC-3 duty							Current (AC3)	Weight (AC control)
	200 ~ 240V	380 ~ 400V	415V	440V	500V	690V	1000V	440V	
	kW	kW	kW	kW	kW	kW	kW	A	
MC-130a/R	37	60	60	60	60	55	75	130	4.3kg
MC-150a/R	45	75	75	75	75	55	75	150	4.3kg
MC-185a/R	55	90	90	90	110	110	132	185	13kg
MC-225a/R	75	132	132	132	132	140	132	225	13kg
MC-265a/R	80	147	147	147	147	160	147	265	21.4kg
MC-330a/R	90	160	160	160	160	200	147	330	21.4kg
MC-400a/R	125	200	200	200	225	250	147	400	21.4kg
MC-500a/R	147	265	265	265	265	300	280	500	60kg
MC-630a/R	190	330	330	330	330	400	280	630	60kg
MC-800a/R	220	440	440	440	440	500	280	800	60kg



Coil voltage

For MC130a/R and MC-150a/R

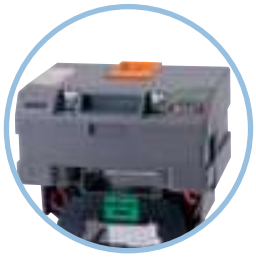
AC	50/60Hz	24, 48, 100~240, 110~120, 220~240, 300, 380~440, 500V
DC	DC	24, 48, 70~110, 100~220, 110, 220V

For MC-185a/R ~ MC-800a/R

- 24, 48, 100/200, 300, 400, 500V for types MC-185a/R, 225a/R
- 100/200, 300, 400, 500V for types MC-265a/R, 330a/R, 400a/R
- 100, 200, 300, 400, 500V for types MC-500a/R, 600a/R, 800a/R

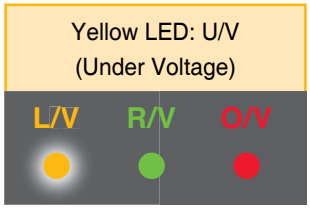
Nominal(ordering) voltage	AC 50/60Hz	DC
24V	24 - 25V	24V
48V	48 - 50V	48V
100V	100 - 127V	100 - 110V
200V	200 - 240V	200 - 220V
100/200V	100 - 240V	100 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

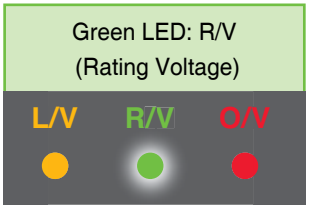
LED voltage monitoring contactor

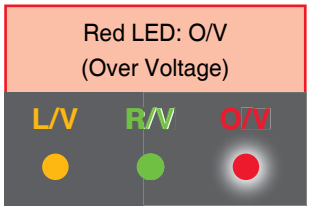


Description

- Real time observe coil voltage without additional equipments for protect coil burn out due to the under and over voltage









Coil voltage

Type	Coil voltage	LED indicate statue		
		Under Voltage (Yellow)	Rating Voltage (Green)	Over Voltage (Red)
MC-130a, MC 150a	110V	~80V	80~130V	130V~
	220V	~160V	160~260V	260V~
	400V	~330V	330~480V	280V~
MC-185a, MC-225a	110V	~80V	80~130V	130V~
	220V	~160V	160~260V	260V~
	400V	~330V	330~480V	280V~
MC-265a, MC-330a, MC-400a	110V	~80V	80~130V	130V~
	220V	~160V	160~260V	260V~
	400V	~330V	330~480V	280V~
MC-500a, MC-630a, MC-800a	110V	~80V	80~130V	130V~
	220V	~160V	160~260V	260V~

* The accuracy of LED indicate statue is ±5%

Contactor relays

Description

- 4,6,8 pole units
- AC/DC control voltage
- 16A continuous current, Ith
- DIN rail or screw mountable
- Degree of protection: IP20



MR-4



MR-6



MR-8

Selection

Poles	Composition	Types	Remarks
4 Pole	4NO 3NO1NC 2NO2NC 1NO3NC 4NC	MR-4	Additional auxiliary contacts can be mounted on the relays.
6 pole	6NO 5NO1NC 4NO2NC 3NO3NC 2NO4NC	MR-6	UA-2 is mounted on the MR-6
8 pole	8NO 7NO1NC 6NO2NC 5NO3NC 4NO4NC	MR-8	UA-4 is mounted on the MR-8.

Ratings

Rated insulation voltage (Ui)			AC 690V				
Continuous current (Ith)			16A				
Operating cycle			1800 operations/hour				
Endurance	Mechanical life time		10 mil. Operations				
	Electrical life time		0.25 mil. Operations				
Type			MR-4	MR-6		MR-8	
AC-15	Rated current (A)	120V	6	6		6	
		240V	3	3		3	
		380V	1.9	1.9		1.9	
		480V	1.5	1.5		1.5	
		500V	1.4	1.4		1.4	
		600V	1.2	1.2		1.2	
			MR-4	MR-4	UA-2	MR-4	UA-4
DC-13	Rated current (A)	125V	1.1	1.1	0.55	1.1	0.55
		250V	0.55	0.55	0.27	0.55	0.27
		400V	0.31	0.31	0.15	0.31	0.15
		500V	0.27	0.27	0.13	0.27	0.13
		600V	0.2	0.2	0.1	0.2	0.1
Minimum rated through current			DC 24V 10mA				



Coil voltage

AC	50Hz	24, 32, 42, 48, 65, 80, 100, 110, 120, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V
Voltage limit		85%~110%

Note) See page 22, 23 for details.

* Please be cautious for wiring control power cable, especially "+" (Positive) and "-" (Negative) polarity should be correctly considered.

Contactor relays for micro current

Description

- 2,4,8 pole units
- AC/DC control voltage
- 0.5 mil. operations, 2.5 mil. operations
- DIN rail or screw mountable (35mm)
- Degree of protection: IP20

* Do not place strong magnets around the main circuit.



MR-4E



MR-8E



UA-4E



UA-1E

Type

Number of poles	Contact configuration		Types	Remarks
	Normal contact	Micro contact		
2 pole	1a1b	-	UA-1E AU-100E	Auxiliary contact unit (UA-1E)
4 pole	2a 1a1b 2b	2a 1a1b 2b	MR-4E or UA-4E	Auxiliary relay (MR-4E) or Auxiliary contact unit (UA-4E)
8 pole	4a 3a1b 2a2b 1a3b 4b	4a 3a1b 2a2b 1a3b 4b	MR-8E (MR-4E + UA-4E)	Auxiliary relay (MR-4E) and Auxiliary contact unit (UA-4E)

* MR-4E UA-4E 'Negative sides of both sides' is the contact for microcurrent.
* UA-1E is applicable to 150AF or less. (Above 225AF, AU-100E is available.)
* Please note that contact welding may occur when a high voltage is applied to the contact for micro current, not the rated voltage.

Ratings

		Normal contact	Micro contact
Rated operational voltage (Ue)		AC 130V, DC 180V MAX.	AC 600V
Continuous current (Ith)		1A	16A
Switching current		0.7A MAX.	6A
Contact Capacity		10VA (10W) Less	720VA
Rated insulation voltage (Ui)		DC 200V	DC 690V
Operating cycle (AC1)		1800 operations/hour	1800 operations/hour
Endurance	Mechanical	2.5 mil. operations	10 mil. operations
	Electrical	0.5 mil. operations	0.25 mil. operations
Minimum rated through current		DC 5V 0.5mA	DC 24V 10mA
Contact reliability		1×10^{-8}	4×10^{-6}

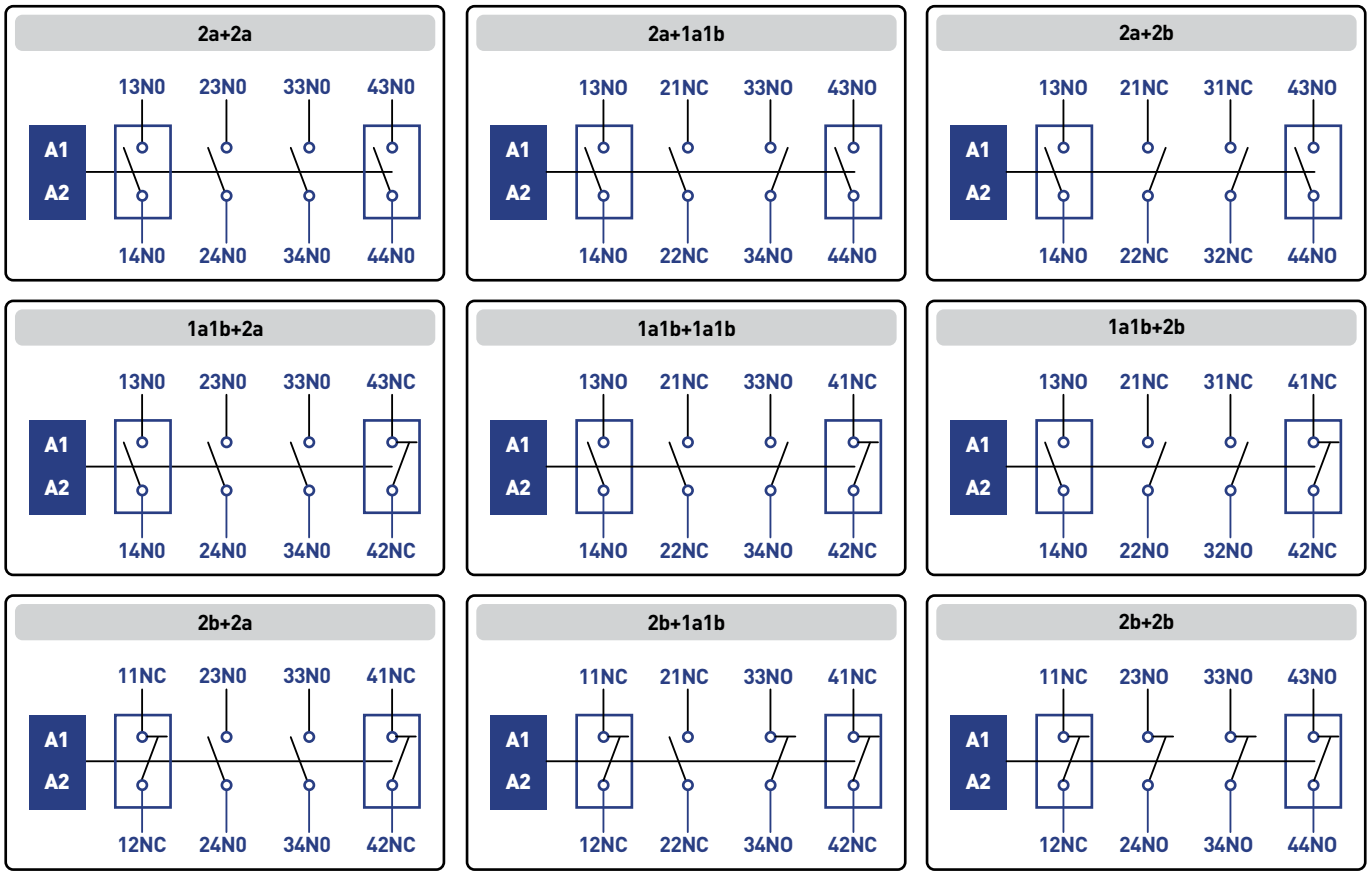


Coil voltage

AC	50Hz	24, 32, 36, 42, 48, 65, 80, 100, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
	60Hz	24, 48, 110, 120, 200, 208, 220, 230, 240, 277, 380, 440, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V
Temperature Range		85%~110%

Note) 1. Coil For detailed specifications, refer to Operating Coil Rating and Characteristics data (22AF) on Catalog 22 page.
2. For DC type products, be sure to wire the polarity of the + and - polarities when wiring the coil terminals.

Contact configuration



* □ is micro contact.

Latching contactors

9 ~ 150



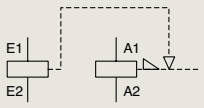


Description

- 3-pole main contact
- Mechanically latched
- Finger proof design
- DIN rail or screw mountable
- Side mountable accessories available
- Direct mountable overload relays available

Circuit diagram

MC-6a/L~150a/L



Rating

Contactor	AC3 ratings					Devices combined			
	200/240V kW	380/440V kW	500/550V kW	690V kW	1000V kW	Type	Durability (Machanical)	Operational time	Excitation time
MC-6a/L	2.2	3	3	3	-	ML-65	0.5 mil.	Less 30ms (AC) Less 50ms (DC)	More than 50ms (AC) More than 70ms (DC)
MC-9a/L	2.5	4	4	4	-				
MC-12a/L	3.5	5.5	7.5	7.5	-				
MC-18a/L	4.5	7.5	7.5	7.5	-				
MC-9b/L	2.5	4	4	4	-				
MC-12b/L	3.5	5.5	7.5	7.5	-				
MC-18b/L	4.5	7.5	7.5	7.5	-				
MC-22b/L	5.5	11	15	15	-				
MC-32a/L	11	18.5	22	22	22				
MC-40a/L	11	18.5	22	22	22				
MC-50a/L	15	22	30	30	30				
MC-65a/L	18.5	30	33	33	30				
MC-75a/L	22	37	37	37	37	ML-150	2.5 mil.		
MC-85a/L	25	45	45	45	37				
MC-100a/L	30	55	55	45	37				
MC-130a/L	37	60	60	55	75			Less 100ms	More than 130ms
MC-150a/L	45	75	75	55	75				



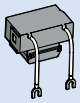
Coil voltage

AC	50/60Hz	24, 48, 100, 110, 120, 200, 220, 230, 240, 380, 400, 415, 440V
DC	DC	24, 48, 110V

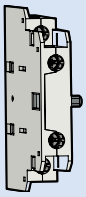
Certification

- CE(IEC)
- UL508

Accessories

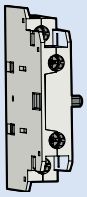


Surge unit, US
connecting to coil terminals
▶84 page



Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶77 page





Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶77 page

* It is not possible to attach " UA-1" on the right side of product for Range MC-9b~22b.



Overload relay
MT-12 ~ MT-150
▶90 page

Latching contactors

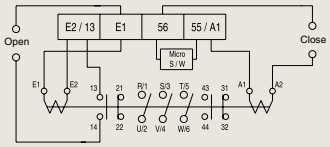
185 ~ 400



Description

- 3-pole main contact
- Mechanically interlocked
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 1NO+2NC Auxiliary contacts built-in as standard
- Top/side mountable accessories available

Circuit diagram



Rating

Contactor	AC3 ratings					Auxiliary contact (standard)
	200 ~ 240V	380 ~ 440V	500 ~ 550V	690V	1000V	
MC-185a/L	55kW 185A	90kW 185A	110kW 180A	110kW 120A	132kW 90A	2NO2NC
MC-225a/L	75kW 225A	132kW 225A	132kW 200A	140kW 150A	132kW 90A	
MC-265a/L	80kW 265A	147kW 265A	147kW 225A	160kW 185A	147kW 105A	
MC-330a/L	90kW 330A	160kW 330A	160kW 280A	200kW 225A	147kW 105A	
MC-400a/L	125kW 400A	200kW 400A	225kW 350A	250kW 300A	147kW 105A	



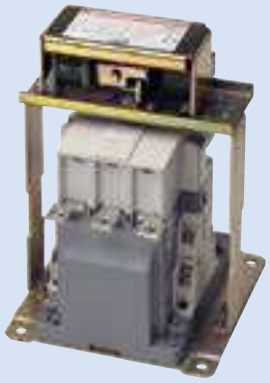
Coil voltage

Nominal(ordering) voltage	Operational voltage	Operation frequency
100V	AC100~127V	600 operations/hour
	DC100~125V	300 operations/hour
200V	AC200~240V	600 operations/hour
	DC200~220V	300 operations/hour

Accessories



AP220~400
Terminal cover
▶81 page

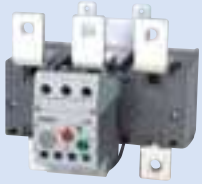




AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶78 page



AI-180
Insulation barrier
▶81 page



Overload relay
MT-225H ~ MT-400H
MT-225/3H ~ MT-400/3H
MT-225/3K ~ MT-400/3K
▶ 93 page

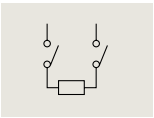
4-pole Contactors (18AF)

Description

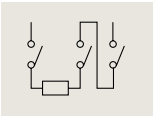
- 4-pole main contact
- Finger proof design
- DIN rail or screw mountable
- Small physical size: 44 mm wide
- Auxiliary contacts are optional
- Front/side mountable accessories
- Direct mountable overload relays available
- Degree of protection: IP20



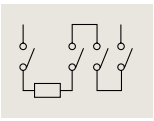
MC-18a/4



2pole for DC contact



3pole for DC contact



4pole for DC contact

Rating

Contactor type			MC-6a/4		MC-9a/4		MC-12a/4		MC-18a/4	
			kW	A	kW	A	kW	A	kW	A
AC duty	AC1	200/240V	9	25	9	25	9	25	15	40
		380/440V	17	25	17	25	17	25	27	40
		500/550V	21	25	21	25	21	25	35	40
		690V	27	25	27	25	27	25	44	40
	AC3	200/240V	2.2	9	2.5	11	3.5	13	4.5	18
		380/440V	3	7	4	9	5.5	12	7.5	18
		Ith	-	25	-	25	-	25	-	40
DC1 duty	2-pole	110V	-	6	-	6	-	10	-	13
(L/R=1ms)	in series	220V	-	3	-	3	-	7	-	8
	3-pole	110V	-	8	-	8	-	12	-	18
	in series	220V	-	8	-	8	-	12	-	18
	4-pole	110V	-	10	-	10	-	15	-	20
	in series	220V	-	10	-	10	-	12	-	20
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	3	-	3	-	5	-	7.5	-
		440/480V	5	-	5	-	7.5	-	10	-
		550/600V	7.5	-	7.5	-	10	-	15	-
		Ith	-	25	-	25	-	25	-	40
Conductor size (solid, stranded)		AWG	18~10		18~10		16~10		16~8	
(the max. number of conducts: 2)		mm²	1.5~4		1.5~10		1.5~10		1~4	
Conductor type			65/75℃ Cu-wire only							

Coil voltage

AC	50Hz	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V
	50/60Hz	24, 48, 100, 110, 120, 200, 220, 230, 240, 380, 415, 440, 500, 550V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

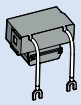
Certification

- CE(IEC)
- UL508
- CCC

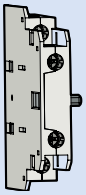
Accessories



Delayed opening device, AD-9
▶81 page

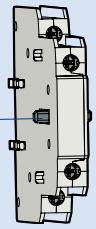


Surge unit, US
connecting to coil terminals
▶84 page

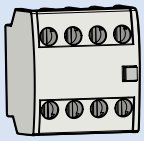


Auxiliary contact unit, UA-1
side mount, 1NO+1NC
▶77 page

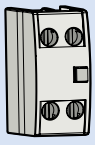




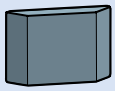
Interlock unit, UR
side mount
▶82 page



Auxiliary contact unit, UA-4
front mount, 4P
▶77 page



Auxiliary contact unit, UA-2
front mount, 2P
▶77 page



Safety cover, A7
▶87 page

4-pole Contactors (22, 40AF)

Description

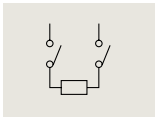
- 4-pole main contact
- Finger proof design
- DIN rail or screw mountable
- Auxiliary contacts are optional
- Front/side mountable accessories
- Direct mountable overload relays available
- Degree of protection: IP20



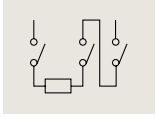
MC-22a/4



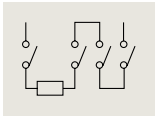
MC-40a/4



2pole for DC contact



3pole for DC contact



4pole for DC contact

Rating

Contactor type			MC-22a/4		MC-32a/4		MC-40a/4	
			kW	A	kW	A	kW	A
AC duty	AC1	200/240V	15	40	18	50	22	60
		380/440V	27	40	35	50	42	60
		500/550V	35	40	43	50	52	60
		690V	44	40	55	50	66	60
	AC3	200/240V	5.5	22	7.5	32	11	40
		380/440V	11	22	15	32	18.5	40
		Ith	-	40	-	50	-	60
DC1 duty	2-pole	110V	-	15	-	25	-	25
(L/R=1ms)	in series	220V	-	10	-	12	-	12
	3-pole	110V	-	20	-	25	-	35
	in series	220V	-	20	-	22	-	30
	4-pole	110V	-	30	-	32	-	40
	in series	220V	-	25	-	30	-	35
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	7.5	-	10	-	10	-
		440/480V	10	-	20	-	25	-
		550/600V	15	-	20	-	25	-
		Ith	-	40	-	50	-	60
Conductor size (solid, stranded)		AWG	12~8		10~6		8~6	
(the max. number of conducts: 2)		mm²	2.5~10		4~16		10~16	
Conductor type			65/75℃ Cu-wire only					

Coil voltage

AC	50Hz	24, 36, 42, 48, 65, 80, 100, 110, 220, 230, 240, 380, 400, 415, 440, 500, 550, 240(wide)V
	60Hz	24, 48, 110, 208, 220, 240, 277, 380, 440, 480, 600V
	50/60Hz	24, 48, 100, 110, 120, 200, 220, 230, 240, 380, 400, 415, 440, 500, 550V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

Certification

- CE(IEC)
- UL508
- CCC

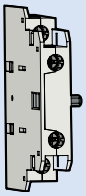
Accessories



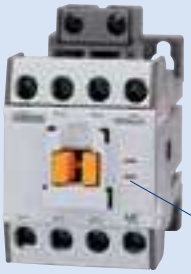
Delayed opening device, AD-9
▶81 page

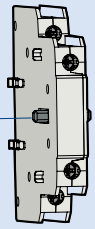


Surge unit, AS
connecting to coil terminals
▶84 page

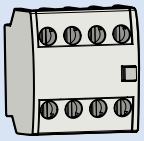


Auxiliary contact unit, AU-1 *
side mount, 1NO+1NC

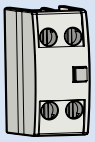




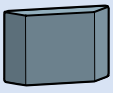
Interlock unit, AR-9 *
side mount



Auxiliary contact unit, UA-4
front mount, 4P
▶77 page



Auxiliary contact unit, UA-2
front mount, 2P
▶77 page



Safety cover, A7
▶87 page

* AU-1 and AR-9 are discontinued item. Please contact us in case of purchasing.

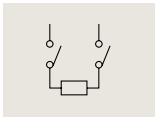
4-pole Contactors (85AF)

Description

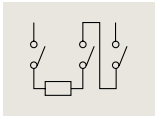
- 4-pole main contact
- Finger proof design
- DIN rail or screw mountable
- Auxiliary contacts are optional
- Front/side mountable accessories
- Direct mountable overload relays available
- Degree of protection: IP20



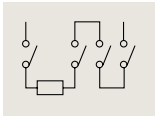
MC-85a/4



2pole for DC contact



3pole for DC contact



4pole for DC contact

Rating

Contactor type			MC-50a/4		MC-65a/4		MC-75a/4		MC-85a/4	
			kW	A	kW	A	kW	A	kW	A
AC duty	AC1	200/240V	30	80	37	100	41	110	51	135
		380/440V	56	80	70	100	76	110	95	135
		500/550V	70	80	88	100	97	110	120	135
		690V	88	80	110	100	120	110	150	135
	AC3	200/240V	15	55	18.5	65	22	75	25	85
		380/440V	22	50	30	65	37	75	45	85
		Ith	-	80	-	100	-	110	-	135
DC1 duty	2-pole	110V	-	35	-	35	-	50	-	50
(L/R=1ms)	in series	220V	-	15	-	15	-	20	-	20
	3-pole	110V	-	50	-	65	-	75	-	80
	in series	220V	-	40	-	50	-	55	-	60
	4-pole	110V	-	60	-	70	-	80	-	100
	in series	220V	-	50	-	60	-	70	-	70
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	15	-	20	-	25	-	30	-
		440/480V	30	-	40	-	50	-	50	-
		550/600V	30	-	40	-	50	-	50	-
		Ith	-	80	-	100	-	110	-	135
Conductor size (solid, stranded)		AWG	8~4		6~3		4~2		4~1	
(the max. number of conducts: 2)		mm²	10~25		16~35		25~35		25~50	
Conductor type			65/75℃ Cu-wire only							

Coil voltage

AC	50Hz	24, 48, 100, 110, 200, 220, 230, 240, 380, 400, 415, 440, 500, 550V, 240(wide)V
	60Hz	24, 48, 110, 120, 208, 220, 240, 277, 380, 440, 480, 600V
	50/60Hz	24, 48, 110, 120, 220, 230, 240, 380, 400, 415, 440, 500, 550V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

Certification

- CE(IEC)
- UL508
- CCC

Accessories

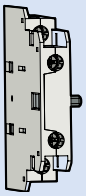


Delayed opening device, AD-50
▶81 page

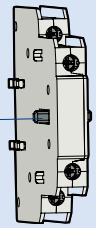


Surge unit, AS
connecting to coil terminals
▶84 page

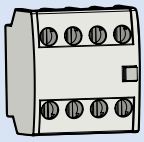




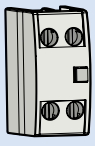
Auxiliary contact unit, AU-1 *
side mount, 1NO+1NC



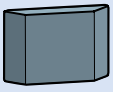
Interlock unit, AR-9 *
side mount



Auxiliary contact unit, UA-4
front mount, 4P
▶77 page



Auxiliary contact unit, UA-2
front mount, 2P
▶77 page



Safety cover, A7
▶87 page

* AU-1 and AR-9 are discontinued item. Please contact us in case of purchasing.

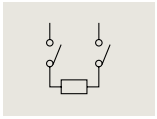
4-pole Contactors (225AF)

Description

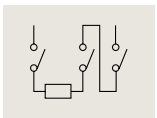
- 4-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Side mountable accessories
- Direct mountable overload relays available
- Degree of protection: IP20



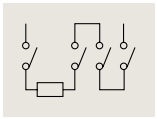
MC-225a/4



2pole for DC contact



3pole for DC contact



4-pole in series

Rating

Contactor type			MC-100a/4		MC-130a/4		MC-150a/4		MC-185a/4		MC-225a/4	
			kW	A	kW	A	kW	A	kW	A	kW	A
AC duty	AC1	200/240V	57	200	60	250	76	275	87	300	100	350
		380/440V	106	200	110	250	142	275	165	300	185	350
		500/550V	132	160	137	165	180	250	205	300	230	330
		690V	165	160	170	165	225	250	255	300	290	330
	AC3	200/240V	30	105	37	125	45	150	55	185	75	225
		380/440V	55	105	60	120	75	150	90	185	132	225
		Ith	-	200	-	250	-	275	-	300	-	350
DC1 duty	2-pole	110V	-	80	-	80	-	100	-	150	-	150
(L/R=1ms)	in series	220V	-	50	-	50	-	100	-	150	-	150
	3-pole	110V	-	100	-	100	-	150	-	180	-	220
	in series	220V	-	80	-	80	-	150	-	180	-	220
	4-pole	110V	-	100	-	100	-	150	-	180	-	220
	in series	220V	-	90	-	100	-	150	-	180	-	220
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	30	-	40	-	50	-	60	-	75	-
		440/480V	60	-	75	-	100	-	125	-	150	-
		550/600V	60	-	75	-	100	-	125	-	150	-
		Ith	-	200	-	250	-	275	-	300	-	350
Conductor size (solid, stranded)		AWG	6~00		3~00		3~4/0		1~4/0		0~300	
(the max. number of conducts: 2)		mm²	16~70		35~70		35~95		50~95		50~150	
Conductor type			75°C Cu stranded wire									

Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
24V	24 - 25V	24V
48V	48 - 50V	48V
100/200V	100 - 240V	100 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

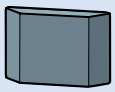
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-100
Delayed opening device
▶81 page



Safety cover, A7
▶87 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶78 page



AP220/4
Terminal cover
▶81 page



AR-180
Interlock unit
side mount
▶83 page



AI-180
Insulation barrier
▶81 page

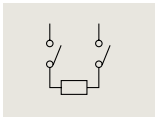
4-pole Contactors (400AF)

Description

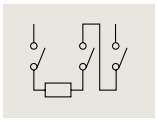
- 4-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Side mountable accessories
- Degree of protection: IP20



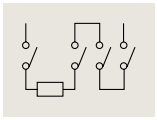
MC-400a/4



2pole for DC contact



3pole for DC contact



4pole for DC contact

Rating

Contactor type			MC-265a/4		MC-330a/4		MC-400a/4	
			kW	A	kW	A	kW	A
AC duty	AC1	200/240V	115	400	135	500	160	520
		380/440V	215	400	250	500	300	520
		500/550V	265	360	315	420	375	500
		690V	335	360	390	420	470	500
	AC3	200/240V	80	265	90	330	125	400
		380/440V	147	265	160	330	200	400
		Ith	-	400	-	500	-	520
DC1 duty	2-pole	110V	-	150	-	200	-	200
(L/R=1ms)	in series	220V	-	150	-	200	-	200
	3-pole	110V	-	220	-	300	-	400
	in series	220V	-	220	-	300	-	300
	4-pole	110V	-	220	-	300	-	400
	in series	220V	-	220	-	300	-	300
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	100	-	125	-	150	-
		440/480V	200	-	250	-	300	-
		550/600V	200	-	250	-	300	-
		Ith	-	400	-	500	-	520
Conductor size (solid, stranded)		AWG	3/0~500		4/0~500		350~700	
(the max. number of conducts: 2)		mm²	95~240		95~240		185~370	
Conductor type			75°C Cu stranded wire					

Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
100/200V	100 - 240V	100 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

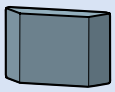
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-300
Delayed opening device
▶81 page



Safety cover, A7
▶87 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶78 page



AP-400/4
Terminal cover
▶81 page



AR-180
Interlock unit
side mount
▶83 page



AI-180
Insulation barrier
▶81 page

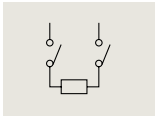
4-pole Contactors (800AF)

Description

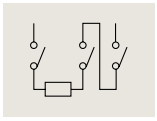
- 4-pole main contact
- AC/DC common use coil built
- Wide coil operation voltage
- Screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- Side mountable accessories
- Degree of protection: IP20



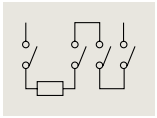
MC-800a/4



2pole for DC contact



3pole for DC contact



4pole for DC contact

Rating

Contactor type			MC-500a/4		MC-630a/4		MC-800a/4	
			kW	A	kW	A	kW	A
AC duty	AC1	200/240V	245	700	255	900	310	1050
		380/440V	450	700	470	900	570	1050
		500/550V	560	630	590	750	710	900
		690V	710	630	740	750	900	900
	AC3	200/240V	147	500	190	630	220	800
		380/440V	265	500	330	630	440	800
		Ith	-	700	-	900	-	1050
DC1 duty	2-pole	110V	-	200	-	630	-	630
(L/R=1ms)		220V	-	200	-	630	-	630
	3-pole	110V	-	400	-	630	-	800
		220V	-	300	-	630	-	800
	4-pole	110V	-	400	-	630	-	800
		220V	-	300	-	630	-	800
UL rating			**hp**	**A**	**hp**	**A**	**hp**	**A**
		220/240V	200	-	250	-	300	-
		440/480V	400	-	500	-	600	-
		550/600V	400	-	500	-	600	-
		Ith	-	700	-	900	-	1050
Conductor size (solid, stranded)		AWG	350~800		600~2000		1700~Busbar	
(the max. number of conducts: 2)		mm²	185~480		-		-	
Conductor type			75°C Cu stranded wire					

Coil voltage, AC/DC common

Nominal(ordering) voltage	AC 50/60Hz	DC
100V	100 - 127V	100 - 110V
200V	200 - 240V	200 - 220V
300V	265 - 347V	-
400V	380 - 450V	-
500V	440 - 575V	-

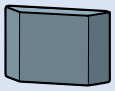
Certification

- CE(IEC)
- UL508
- CCC

Accessories



AD-600
Delayed opening device
▶81 page



Safety cover, A7
▶87 page





AU-100, 100E
Auxiliary contact unit
side mount, 1NO+1NC
▶78 page



AP-800/4
Terminal cover
▶81 page



AR-600
Interlock unit
▶83 page



AI-600
Insulation barrier
▶81 page

Enclosed motor starters, Mold enclosure







Internal device

Description

- Enclosed & full voltage type
  MW-□bM: Plastic enclosure without buttons
  MW-□bMB: Plastic enclosure with ON/OFF buttons
- With differential or non-differential type thermal overload relay
- Contactors: 3 pole with AC coil
- Screw mountable
- Degree of protection: IP40

Ratings

Type	AC3 Ratings				Continuous current Ith	Auxiliary contact (standard)	Devices combined	
	220 ~ 240V	380 ~ 440V	500 ~ 550V	690			Contactor	Overload relay
MW-9bM(B)	2.5kW 11A	4kW 9A	4kW 7A	4kW 6A	25A	1NO1NC	MC-9b	MT-32 (0.1~22A)
MW-12bM(B)	3.5kW 13A	5.5kW 12A	7.5kW 12A	7.5kW 9A	25A		MC-12b	
MW-18bM(B)	4.5kW 18A	7.5kW 18A	7.5kW 13A	7.5kW 9A	40A		MC-18b	
MW-22bM(B)	5.5kW 22A	11kW 22A	15kW 20A	15kW 18A	40A		MC-22b	

Overload relay setting range, Amp.

MT-12		MT-32		MT-63		MT-95	
0.1 - 0.16	5 - 8	0.1 - 0.16	5 - 8	4 - 6	28 - 40	7 - 10	54 - 75
0.16 - 0.25	6 - 9	0.16 - 0.25	6 - 9	5 - 8	34 - 50	9 - 13	63 - 85
0.25 - 0.4	7 - 10	0.25 - 0.4	7 - 10	6 - 9	45 - 65	12 - 18	70 - 95
0.4 - 0.63	9 - 13	0.4 - 0.63	9 - 13	7 - 10		16 - 22	80 - 100
0.63 - 1	12 - 18	0.63 - 1	12 - 18	9 - 13		18 - 25	
1 - 1.6		1 - 1.6	16 - 22	12 - 18		24 - 36	
1.6 - 2.5		1.6 - 2.5	18 - 25	16 - 22		28 - 40	
2.5 - 4		2.5 - 4	22 - 32	18 - 25		34 - 50	
4 - 6		4 - 6	28 - 40	24 - 36		45 - 65	

Coil voltage

AC	50Hz	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

Certification

- CE(IEC)
- UL508

Enclosed motor starters, Steel enclosure







Internal device

Description

- Full voltage type
  MW-□a/MW-□b: Steel enclosure without buttons
  MW-□aB/MW-□bB: Steel enclosure ON/OFF buttons
- Contactors: 3 pole with AC coil
- Overload relayWith differential or non-differential type
- Screw mountable
- Degree of protection: IP40

Ratings

Type	AC3 Ratings					Devices combined	
	220 ~ 240V	380 ~ 440V	500 ~ 550V	690V	1000V	Contactor relay	Overload
MW-6a(B)	2.2kW 9A	3kW 7A	3kW 6A	3kW 4A	-	MC-6a	
MW-9a(B)	2.5kW 11A	4kW 9A	4kW 7A	4kW 5A	-	MC-9a	MT-12
MW-12a(B)	3.5kW 13A	5.5kW 12A	7.5kW 12A	7.5kW 9A	-	MC-12a	(0.1~18A)
MW-18a(B)	4.5kW 18A	7.5kW 18A	7.5kW 13A	7.5kW 9A	-	MC-18a	
MW-9b(B)	2.5kW 11A	4kW 9A	4kW 7A	4kW 6A	-	MC-9b	
MW-12b(B)	3.5kW 13A	5.5kW 12A	7.5kW 12A	7.5kW 9A	-	MC-12b	MT-32
MW-18b(B)	4.5kW 18A	7.5kW 18A	7.5kW 13A	7.5kW 9A	-	MC-18b	(0.1~22A)
MW-22b(B)	5.5kW 22A	11kW 22A	15kW 20A	15kW 18A	-	MC-22b	
MW-32a(B)	7.5kW 32A	15kW 32A	18.5kW 28A	18.5kW 20A	22kW 17A	MC-32a	MT-32
MW-40a(B)	11kW 40A	18.5kW 40A	22kW 32A	22kW 23A	22kW 17A	MC-40a	(0.1~40A)
MW-50a(B)	15kW 55A	22kW 50A	30kW 43A	30kW 28A	30kW 23A	MC-50a	MT-63
MW-65a(B)	18.5kW 65A	30kW 65A	33kW 60A	33kW 35A	33kW 26A	MC-65a	(4~65A)
MW-75a(B)	22kW 75A	37kW 75A	37kW 64A	37kW 42A	37kW 28A	MC-75a	MT-95
MW-85a(B)	25kW 85A	45kW 85A	45kW 75A	45kW 45A	45kW 33A	MC-85a	(7~100A)
MW-100a(B)	30kW 105A	55kW 105A	55kW 85A	55kW 65A	45kW 33A	MC-100a	

Overload relay setting range, Amp.

MT-12		MT-32		MT-63		MT-95	
0.1 - 0.16	5 - 8	0.1 - 0.16	5 - 8	4 - 6	28 - 40	7 - 10	54 - 75
0.16 - 0.25	6 - 9	0.16 - 0.25	6 - 9	5 - 8	34 - 50	9 - 13	63 - 85
0.25 - 0.4	7 - 10	0.25 - 0.4	7 - 10	6 - 9	45 - 65	12 - 18	70 - 95
0.4 - 0.63	9 - 13	0.4 - 0.63	9 - 13	7 - 10		16 - 22	80 - 100
0.63 - 1	12 - 18	0.63 - 1	12 - 18	9 - 13		18 - 25	
1 - 1.6		1 - 1.6	16 - 22	12 - 18		24 - 36	
1.6 - 2.5		1.6 - 2.5	18 - 25	16 - 22		28 - 40	
2.5 - 4		2.5 - 4	22 - 32	18 - 25		34 - 50	
4 - 6		4 - 6	28 - 40	24 - 36		45 - 65	

Coil voltage

AC	50Hz	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

Certification

- CE(IEC)
- UL508

Capacitor contactors







Description

Capacitor unit(Pre-loading resistor) is connected to the terminals of the contactor to reduce the high inrush current exceeding 20×In.

- AC control coil on selection
- 3-pole(NO) main contact
- Finger proof design
- DIN rail or screw mountable
- IEC 60947-4-1 AC 6b

Type	Rating (kVar, A)			
	220~240V	400~440V	500~550V	600~690V
MC-9b with AC-9	5kVar	9.7kVar	14kVar	16kVar
	13A	14A	16A	15A
MC-12b with AC-9	6.7kVar	12.5kVar	18kVar	18kVar
	18A	18A	21A	17A
MC-18b with AC-9	8.5kVar	16.7kVar	24kVar	26kVar
	22A	24A	28A	25A
MC-22b with AC-9	10kVar	18kVar	26kVar	30kVar
	26A	26A	30A	29A
MC-32a with AC-9	15kVar	25kVar	36kVar	40kVar
	39A	36A	42A	38A
MC-40a with AC-9	20kVar	33.3kVar	48kVar	53kVar
	52A	48A	55A	51A
MC-50a with AC-50	20kVar	40kVar	58kVar	65kVar
	52A	58A	67A	63A
MC-65a with AC-50	25kVar	45.7kVar	66kVar	73kVar
	66A	66A	76A	70A
MC-75a with AC-75	29.7kVar	54kVar	78kVar	85kVar
	78A	78A	90A	82A
MC-85a with AC-75	35kVar	60kVar	92kVar	100kVar
	92A	87A	106A	96A
MC-100a with AC-75	37kVar	62kVar	94kVar	105kVar
	97A	89A	109A	101A

Type	Ith	Rating (kVar, A)	
		500~550V	600~690V
MC-9b	25	4kVar	3.5kVar
		4.6A	3.4A
MC-12b	25	4kVar	4kVar
		4.6A	3.8A
MC-18b	40	5kVar	4.5kVar
		5.8A	4.3A
MC-22b	40	8kVar	8kVar
		9.2A	7.7A
MC-32a	50	17kVar	14kVar
		20A	13A
MC-40a	60	25kVar	18kVar
		29A	17A
MC-50a	70	30kVar	25kVar
		35A	24A
MC-65a	100	40kVar	35kVar
		45A	34A
MC-75a	110	50kVar	40kVar
		58A	38A
MC-85a	135	55kVar	45kVar
		64A	43A
MC-100a	160	60kVar	48kVar
		69A	46A
MC-130a	160	70kVar	60kVar
		81A	58A
MC-150a	210	90kVar	78kVar
		104A	75A
MC-185a	230	120kVar	100kVar
		139A	96A
MC-225a	275	130kVar	105kVar
		150A	101A
MC-265a	300	140kVar	115kVar
		162A	111A
MC-330a	350	185kVar	140kVar
		214A	135A
MC-400a	450	195kVar	160kVar
		225A	154A
MC-500a	580	330kVar	290kVar
		381A	279A
MC-630a	660	340kVar	300kVar
		393A	289A
MC-800a	900	370kVar	320kVar
		427A	308A

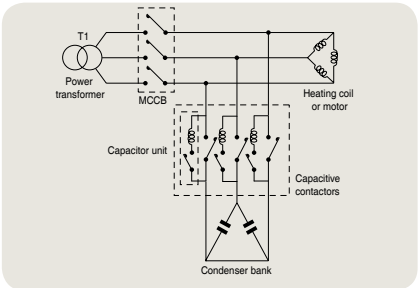
Note) 1. When the switch is closed condenser must be discharged before recharged. (Maximum residual voltage at terminals ≤ 50V)
2. To prevent short current, gG type fuse must be 1.5~2 times than rated current.

Coil voltage

AC	50Hz	24, 32, 42, 48, 80, 100, 110, 220, 230, 240, 380, 400, 500, 550V
	60Hz	24, 48, 100, 110, 120, 200, 208, 220, 230, 240, 277, 380, 480, 600V

These capacitive contactors are suitable for switching single-step or multiple-step condenser bank.
It is standardized by IEC-60947-4941, UL and CSA.
Features of capacitor unit(Pre-loading resistor)

- Damping resister that can limit the inrush current upto 60×In is connected to the mechanism that closed earlier than the main contact of the contactor
- No heat loss by the serial resistor
- Eliminate the switching surge
- Improving the performance of the capacitor system



Example

2-pole AC Contactors, Definite purpose

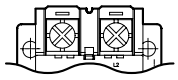


Description

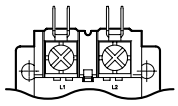
- Two normally open main contacts
- Screw clamp or fast-on connections
- Small and light design suitable for use in air conditioners, heaters, refrigerators, etc.
- Screw mountable
- No auxiliary contact
- AC control voltage



Main terminal

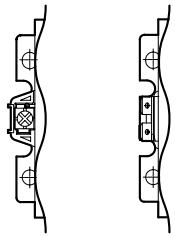


Screw clamp



Fast-on (Quick connect)

Coil terminal



Screw clamp | Fast-on (Quick connect)

Ratings

Type	AC3 ratings			Ith	Contact	
	220~240V	380~440V	500~550V		NO	NC
GMC-10P2	10A	8A	6A	20A	2	-
GMC-20P2	20A	17A	14A	30A	2	-
GMC-25P2	25A	21A	17A	35A	2	-
GMC-30P2	30A	23A	21A	40A	2	-
GMC-35P2	35A	26A	23A	45A	2	-
GMC-40P2	40A	32A	26A	50A	2	-

Insulation voltage: 690V at AC, 250V at DC

Control coil characteristics

Coil voltage	Coil consumption (VA)		Thermal dissipation (W)	Operational voltage(V)		Coil current (mA)	Operational time (ms)	
	Inrush	Holding		Pick-up	Drop-out		Closing	Opening
24V 50/60Hz	35	6.0	2	18~20	10~15	263	40	30
48V 50/60Hz	35	6.0	2	36~40	19~27	131	40	30
100V 50/60Hz	35	6.0	2	74~81	39~57	63	40	30
110V 50/60Hz	35	6.0	2	78~90	50~64	57	40	30
120V 50/60Hz	35	6.0	2	90~100	52~66	53	40	30
200V 50/60Hz	35	6.0	2	155~165	80~115	32	40	30
220V 50/60Hz	35	6.0	2	160~175	90~120	29	40	30
230V 50/60Hz	35	6.0	2	165~180	110~130	27	40	30
240V 50/60Hz	35	6.0	2	168~185	110~135	26	40	30

Note) 1. The condition of the above data: 60Hz. 20 ℃ from cold state
2. Coil consumption (VA) is average value.

Type numbering system

GMC10P2 / 20 / 00 / F7 / AC / S / E

Product (6A~2100A)	
GMC10P2	GMC-10P2
GMC20P2	GMC-20P2
GMC25P2	GMC-25PS
GMC30P2	GMC-30P2
GMC35P2	GMC-35P2
GMC40P2	GMC-40P2

Main contactor	
20	2NO0NC

Auxiliary contact	
00	None

Coil voltage
Coil code Table

Coil consumption	
AC	AC

Terminal	
S	Screw
Q	Quick Connect

Devision	
L	Local
E	Export

ex)

GMC10P2-20-00-F7-AC-S-E	GMC-10P2 AC110V 50/60Hz EXP
GMC40P2-20-00-B7-AC-Q-E	GMC-40P2 TQBQ AC24V 50/60Hz EXP

3-pole AC Contactors, Definite purpose



Description

- Three main contacts
- Screw clamp connections
- Small and light design suitable for use in air conditioners, heaters, refrigerators, etc.
- Screw mountable
- No auxiliary contact
- AC control voltage

Ratings

Type	AC3 ratings			AC4 ratings			Ith	Contact	
	220~240V	380~440V	500~550V	220~240V	380~440V	500~550V		NO	NC
GMC-20D2 GMC-20D3	20A	17A	14A	18A	13A	10A	30A	2	-
								3	-
								2	1
GMC-25D2 GMC-25D3	25A	21A	17A	20A	14A	12A	35A	2	-
								3	-
								2	1

Note) Above contact ratings are for NO contact.
Rating for NC contact: 6A@120V, 3A@240V, 1.5A@480V at AC15 duty
Rating above 24V 1A
Please use NC contact for Insulation voltage: 690V at AC, 250V at DC

Control coil characteristics

Coil voltage	Coil consumption (VA)		Thermal dissipation (W)	Operational voltage(V)		Coil current (mA)	Operational time (ms)	
	Inrush	Holding		Pick-up	Drop-out		Closing	Opening
24V 60Hz	72	9.0	3	17~19	12.5~15.5	215	30	20
48V 60Hz	72	9.0	3	32~36	26~30	125	30	20
110V 60Hz	72	9.0	3	75~85	65~75	60	30	20
220V 60Hz	72	9.0	3	150~170	120~145	41	30	20

Note) 1. The condition of the above data: 60Hz. 20 ℃ from cold state
2. Coil consumption (VA) is average value.

Type numbering system

GMC20D2 / 20 / 00 / F7 / AC / S

Product (6A~2100A)	
GMC20D2	GMC-10D2
GMC20D3	GMC-20D3
GMC25D2	GMC-25D2
GMC25P3	GMC-25D3

Main contactor	
20	2NO0NC
30	2NO1NC
21	3NO0NC

Auxiliary contact	
00	None

Coil voltage
Coil code Table

Coil consumption	
AC	AC

Terminal	
S	Screw

Single-pole AC Contactors, Definite purpose



Description

- Single-pole NO main contact
- Small and light design suitable for use in electric shock protector for AC arc welder
- DIN rail or screw mountable
- No auxiliary contact (Optional unit can be added up to 2NO2NC)
- AC control voltage

Ratings

Type	Rated current, In 110/220V	Ith	Performance	
			Making current	Breaking current
MC-65aSP	220A	500A	1.5×In	

Control coil characteristics

Coil consumption (VA)		Operational voltage		Operational time
Inrush	Holding	Pick-up	Drop-out	Closing
150VA	17VA	85% under	20% over	30ms under

Note) Coil voltage is available between AC24V and AC550V

2-pole DC Contactors

Description

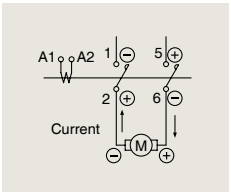


- Two normally open main contacts for DC circuit
- Positive and negative poles indicated on the front plate
- Eternal magnet installed for the effective arc extinguishing
- DIN rail or screw mountable
- 2NO+2NC Auxiliary contacts built-in as standard
- AC/DC control voltage



Selection

Contactor types	DC-2~5 ratings			DC-1 rating	Continuous current, Ith	Auxiliary contacts
	125V	250V	500V	1000V		
MD-30a	40A	30A	20A	10A	60A	2a(NO)+2b(NC)
MD-60a	80A	60A	40A	25A	100A	2a(NO)+2b(NC)
MD-100a	120A	100A	60A	35A	135A	2a(NO)+2b(NC)



Ratings of auxiliary contacts

UA-1	AC15 duty						DC13 duty				
	120V	240V	380V	480V	500V	600V	125V	250V	400V	500V	600V
Rated currents	6A	3A	1.9A	1.5A	1.4A	1.2A	1.1A	0.55A	0.31A	0.27A	0.2A
Continuous current, Ith	16A										



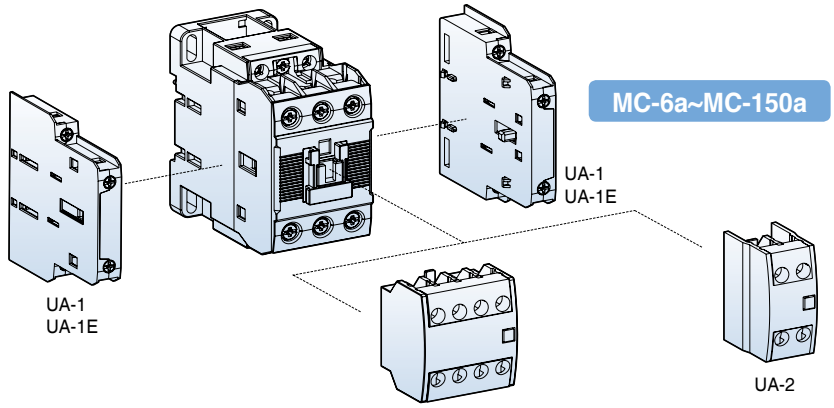
Coil voltage

AC	50/60Hz	24, 48, 100, 110, 120, 200, 220, 230, 240, 380, 400, 415, 440, 500, 550V
DC	DC	12, 20, 24, 48, 60, 80, 100, 110, 125, 200, 220, 250V

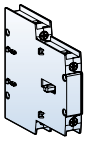
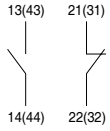
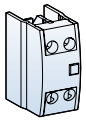
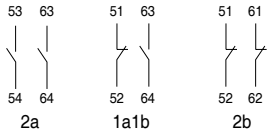
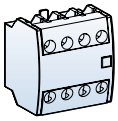
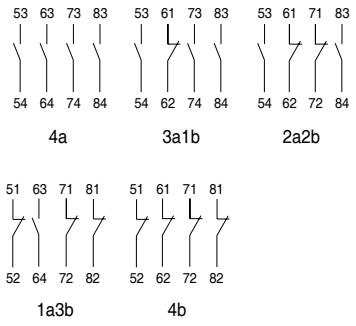
Accessory

Auxiliary contact units

- 2 and 4 pole
- Side and front mountable
- Common use for Metasol contactors from MC-6a to MC-150a
- Auxiliary contact meet the requirements for mirror contacts in IEC 60947-4-1. Annex F (A mirror contact is an auxiliary NC contact that cannot be closed simultaneously with a NO main contact)



Selection

Type	Appearance	Pole	Composition		Contact arrangement	Mount	Weight
			NO	NC			
UA-1 UA-1E		2	1	1		Side	53g
UA2		2	1 2	1		Front	28g
UA-4 UA-4E		4	4 3 2 1 -	- 1 2 3 4		Front	50g

Maximum combination of contact units

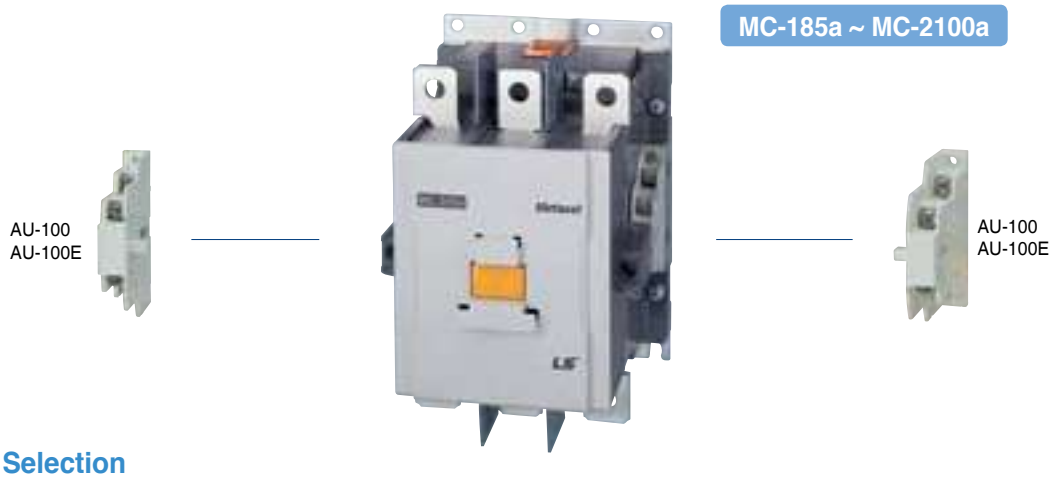
MC-6a~18a, 9b~22b (Max. 3a3b)	MC-32a~150a (Max. 4a4b)
UA-4(on front)+UA-1×1EA (1EA on one side)	UA-4(on front)+UA-1×2EA (1EA on both sides)
UA-2(on front)+UA-1×1EA (1EA on one side)	UA-2(on front)+UA-1×2EA (1EA on both sides)
UA-1×2EA (1EA on both sides)	UA-1×4EA (2EA on both sides) *For AC Coil

*To use UA-1, MC-9b~22b Type is avaliable on only left side

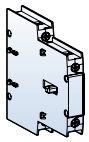
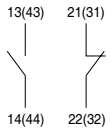
Accessory

Auxiliary contact units

- 2 pole
- Side mountable
- Common use for Metasol contactors from MC-185a to MC-2100a
- Auxiliary contact meet the requirements for mirror contacts in IEC 60947-4-1. Annex F
  (A mirror contact is an auxiliary NC contact that cannot be closed simultaneously with a NO main contact)



Selection

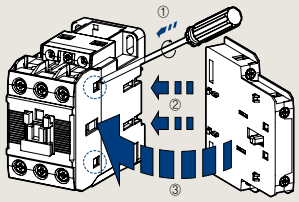
Type	Appearance	Pole	Composition		Contact arrangement	Mount	Weight
			a(NO)	b(NC)			
AU-100 AU-100E		2	1	1		Side	53g

Maximum combination of contact units

MC-185a~2100a (Max. 4a4b)
AU-100×4EA (2EA on both sides)
AU-100E×4EA (2EA on both sides)

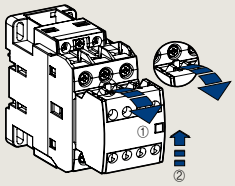
Installation

UA-1



To install side mounting unit remove the indicated part in the circle in the fig. first. And then fit each part as shown. To separate push forward and pull.

UA-2, 4



To install front mounting unit fit it on the upper part of the front of the contactor and push it down. To separate pull the lever of the unit and push the unit upward.

Accessory

Rating

Type			UA-1	UA-2, 4	AU-100
Rated operation voltage (Ue)			600V	600V	600V
Rated insulation voltage (Ui)			600V	600V	600V
Rated impulse withstand voltage (Uimp)			6kV	6kV	6kV
Rated frequency			50/60Hz	50/60Hz	50/60Hz
Contact sensibility					
	Voltage		DC 17V	DC 24V	DC 24V
	Current		DC 5mA	DC 10mA	DC 10mA
Rated thermal current (Ith, AC12 duty)			10A	16A	16A
Rated operation current					
	AC15 duty (A600)	120V	6A	6A	6A
		240V	3A	3A	3A
		380V	1.9A	1.9A	1.9A
		480V	1.5A	1.5A	1.5A
		500V	1.4A	1.4A	1.4A
		600V	1.2A	1.2A	1.2A
	DC13 duty (Q600)	125V	1.1A	1.1A	1.1A
		250V	0.55A	0.55A	0.55A
		400V	0.31A	0.31A	0.31A
		500V	0.27A	0.27A	0.27A
		600V	0.2A	0.2A	0.2A
Rated Short-time withstand current	1s		100A	-	100A
	0.5s		125A	-	120A
	0.1s		145A	-	140A
Electrical lifetime (mil. operations)					
	AC15 duty	220V	0.25	0.25	0.25
		440V	0.25	0.25	0.25
	DC13 duty	220V	0.25	0.25	0.25
		440V	0.25	0.25	0.25
Maximum operating cycles per hour			1800	1800	1800
Conductor size (Solid, stranded) (the max. number of conducts : 2)		AWG	18~10	18~10	18~10
		mm²	1~2.5	1~2.5	1~2.5

Air pressure timer

It is attached to the head-on of the magnetic contactor. It uses the principle of air pressure and it is delayed by the set time after the magnetic contactor is inserted and released and then the auxiliary contact of the air pressure timer is reversed



Air pressure timer



Combination with MC

UN – 1 N

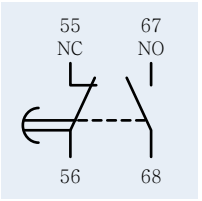
Time delay range	
1	0.1 ~ 3s
2	0.1 ~ 30s
3	10 ~ 180s

Timer Type	
N	ON delay
F	OFF delay

Note) Applicable MC: MC-6a ~ 100a, MC-9b ~ 22b, AC operation voltage

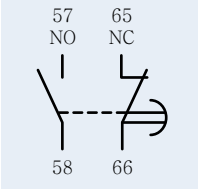
Type		MC-6a~100a / MC-9b~22b					
		UN-1N	UN-2N	UN-3N	UN-1F	UN-2F	UN-3F
Standards		IEC60947-5-1, GB14048.5					
Certification		CCC, CE					
Degree of protection		IP20					
Temperature range	Storage	-25℃ ~ +60℃					
	Operation	-5℃ ~ +40℃					
Altitude		2,000m less					
Poles		1NO + 1NC					
Rated operating voltage		690V					
Rated insulation voltage		690V					
Frequency		50/60Hz					
Rated operation current(Ie)		A600(AC-15), Q600(DC-13)					
Ith(thermal Current)		10A					
Minimum switching current		10mA					
Irms rated making capacity		10 x Ie AC-15 acc. to IEC 60947-5-1					
Non overlap time		1.5ms					
Setting time		0.1~3s	0.1~30s	10~180s	0.1~3s	0.1~30s	10~180s
Operating time tolerance		+50% less					
Mechanical durability		2.5 mil. operations					

On Delay Timer Block(UN-N)



0.1~3s
0.1~30s
10~180s

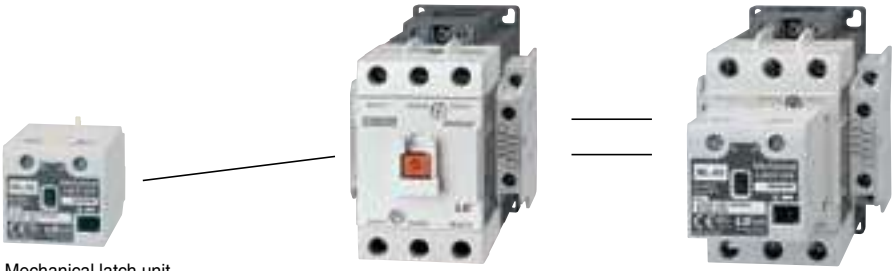
Off Delay Timer Block(UN-F)



0.1~3s
0.1~30s
10~180s

Accessory

Mechanical latch unit ML, AL



Mechanical latch unit | Contactor | Combined

Ratings

Applying contactors	Type	Coil voltage of latch unit
MC-6a/L MC-9a/L MC-12a/L MC-18a/L MC-9b/L MC-12b/L MC-18b/L MC-22b/L MC-32a/L MC-40a/L MC-50a/L MC-65a/L	ML-65	
MC-75a/L MC-85a/L MC-100a/L MC-130a/L MC-150a/L	ML-150	
MC-185a/L, 225a/L	AL-225	
MC-265a/L, 330a/L, 400a/L	AL-400	

Ordering voltage	Operation voltage
AC 24V/ DC 24V	AC 24V 50/60Hz, DC 24V
AC 48V/ DC 48V	AC 48V 50/60Hz, DC 48V
AC 100V/DC 100V	AC 100~127V 50/60Hz, DC 100V~125V
AC 200V/DC 200V	AC 200~240V 50/60Hz, DC 200V~220V
AC 400V	AC 380V~440V 50/60Hz

* Coils for 24, 48, 100, 200V are AC / DC common.

Ordering voltage	Operation voltage
AC 100V	AC 100~127V 50/60Hz, DC 100~110V
AC 200V	AC 200~240V 50/60Hz, DC 200~220V

* For using safely the product providing DC control power, Please consider that the maximum switching frequency should be less than 150 times/hour.

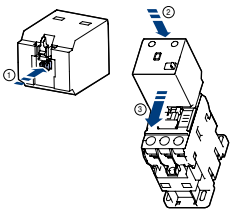
Note) Latch units for upto MC-150 are supplied seperately. From MC-185 the units are assembled at factory.



ML-65

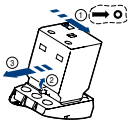


To install



1. Push the black bar
2. Position the Latch on the top of the contactor
3. Pull in the direction of the arrow until snapping

To separate



1. Pull the black lever in the direction of the arrow.
2. Pull up the yellow lever.
3. Push the latch in the direction of the arrow.

Accessory





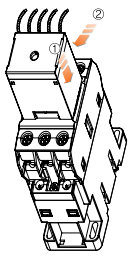
Delayed opening devices, AD

Delayed opening device prevents the inadvertent opening of a contactor in the event of a wide voltage drop or brief supply voltage interruptions of control power for 1 to 4 seconds.

Applying contactors	Type	Rated voltage	MC Rated voltage	Durability		Performance
				Mechanical	Electrical	
MC-6a~40a	AD-9	AC 110V	DC 110V	5 mil operations	1 mil operations	• Making/Breaking capacity at AC3 - Making current : min. 10xIn - Breaking current : min. 8xIn • Max. operating rate : 600 operations/hour • Delaying time : 1~4sec. • Capacitor durability 1) Operation :> 0.1 mil. operations 2) Cumulative charge life at AC 200/220V - 45℃ or less: 80,000 hours - 50℃ or less: 60,000 hours
MC-9b~22b		AC 220V	DC 220V			
MC-50a~100a	AD-50	AC 110V	DC 110V			
		AC 220V	DC 220V			
MC-130a~225a	AD-100	AC 100~240V	AC/DC 100~240V		0.5 mil operations	
MC-265a~400a	AD-300	AC 100~240V	AC/DC 100~240V			
MC-500a~800a	AD-600	AC 100~127V	AC/DC 100~127V			
		AC 200~240V	AC/DC 200~240V			

* AD shall ve used for DC type contactor





①Insult to ①way ②Push to ②way and assemble

Electronic timer unit, UT

Applying contactors		Type	Rated voltage	Performance
MC-6a~150a MC-9b~22b MR-4~8	On delay	UT-1N	AC/DC 24~48V	• Mounting: Head on • Contact: 1c • Time delay: 1 Sec~30 Sec On delay, Off delay • Frequency: AC 50/60Hz • Operation: Min. voltage 85%, Max. voltage 110% • Accuracy of setting time: ±20%
		UT-2N	AC 110~220V	
	Off delay	UT-1F	AC/DC 24~48V	
		UT-2F	AC 110~220V	



Terminal cover units for contactors, AP

Applying contactors	Type		Remarks
	Standard	Transparent	
MC-9b,12b,18b,22b (22AF)	-	AP/T-9	2 pcs included
MC-32a,40a (40AF)	-	AP/T-32	
MC-50,65a (65AF)	-	AP/T-50	
MC-75a,85a,100a (100AF)		AP/T-75	
MC-130a,150a (150AF)	-	AP/T-75	
MC-185a,225a (225AF)	AP-220	AP/T-185	
MC-265a,330a,400a (400AF)	AP-400	AP/T-265	
MC-500a,630a,800 (800AF)	AP-800	AP/T-500	





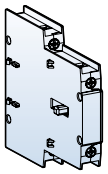
Insulation barrier units, AI

Applying contactors	Type	Remarks
MC-130a, 150a (150AF)	AI-180	4 pcs required per contactor
MC-185a, 225a (225AF)		
MC-265a, 330a, 400a (400AF)		
MC-500a, 630a, 800a, 1260a (800AF)	AI-800	
MC-1400a, 1700a, 2100a (2100AF)	AI-2100	

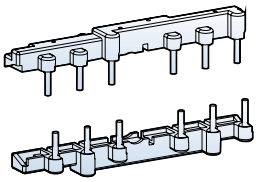


* The insulation barrier for MC-150a is available on the screw type only.

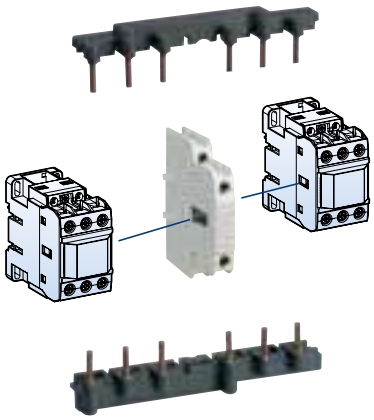
Accessory



Interlock unit



Wire kit



Interlocks

Component parts for assembling by customer

Interlock unit, UR-02 and AR-9 & Wire kit UW

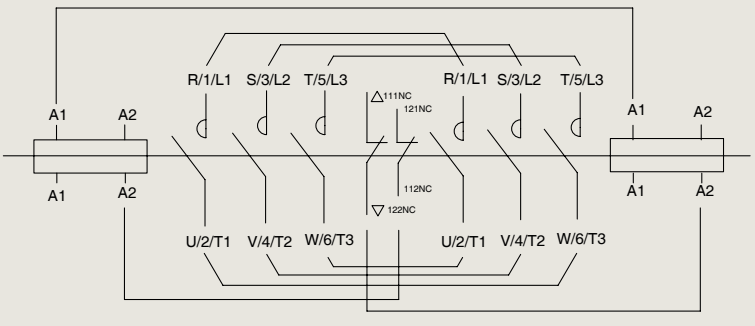
UR-02 is a mechanical interlock unit and provides 2NC contacts for use in electrical interlocking

UW is used for making reversing circuit between two contactors.

Contactor	Pole	Wire kit		Interlock unit		Aux Contact
		Type	Weight	Type	Weight	
MC-6a~18a	3P	UW-18	0.05kg	UR-02	0.06kg	2NC
	4P	-	-			
MC-9b~22b	3P	UW-22	0.04kg			-
MC-32a~40a		UW-32	0.05kg			-
MC-50a, 65a		UW-63	0.12kg			-
MC-75a~100a		UW-95	0.33kg			-
MC-130a~150a		-	-			-
MC-22a~85a	4P	-	-	AR-9	0.05kg	-

Rating of the contacts in the interlocks

Rated operation voltage			600V
Rated insulation voltage			600V
Rated frequency			50/60Hz
Rated thermal current			10A
Rated operation current			
	AC15 duty	120V	6A
	(A600)	240V	3A
		380V	1.9A
		480V	1.5A
		500V	1.4A
		600V	1.2A
	DC13 duty	125V	0.55A
	(Q300)	250V	0.27A



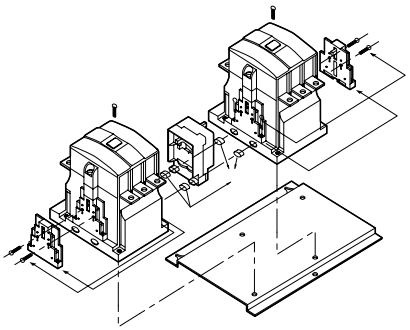
<MC-6aR ~ MC-150aR typical circuit diagram for reversing contactor>

Accessory

* The insulation barrier for MC-150a is available on the screw type only.



AR-180





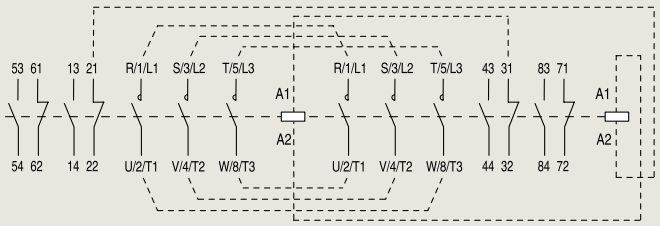
AR-600 & AR-600/V

Interlock unit, AR-180 and AR-600

AR-180 and AR-600 are mechanical interlock units in which electrical contacts for use in electrical interlocking are not included. Please use the auxiliary contacts on the sides of the contactors for that purpose.

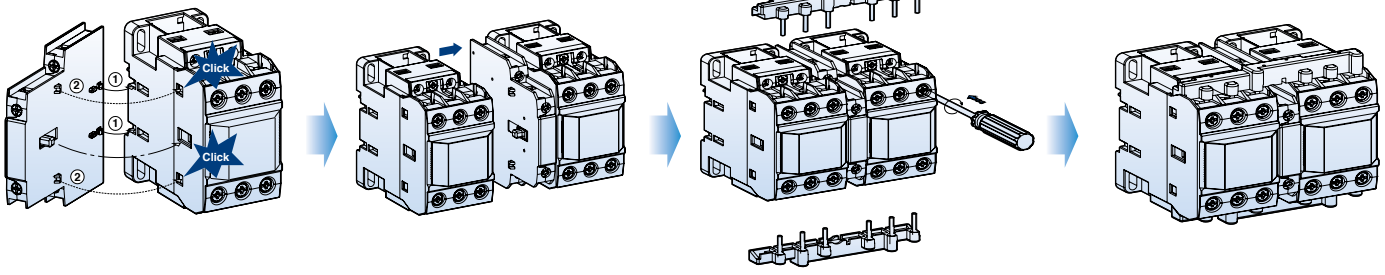
Contactor	Pole	Interlock unit	
		Type	Weight
MC-100a	4P	AR-180	0.09kg
MC-130a			
MC-150a			
MC-185a	3,4P		
MC-225a			
MC-265a			
MC-330a			
MC-400a			
MC-500a	3,4P	AR-600 AR-600/V	15.2kg
MC-630a			
MC-800a			

*AR-600 3pole product is different fro, AR-600 4pole product in size.
*The AR-600 / V is a vertical interlock unit.



<MC-180aR ~ MC-800aR typical circuit diagram for reversing contactor>

Installation of Interlock and Wire kits



Install the interlock unit on the side of a contactor first. Fit each part as indicated in the fig.

And then install the other contactor on the other side of the interlock unit as shown.

After installing interlock unit wiring kit can be assembled.
Wiring kit contains two molded wires - one for line side and the other for load side.
- In case of assembling Reversing contactor, please check the R, S, T/U, V, Wmark on the contactor and Wire kit.

Accessory



US Type



AS Type

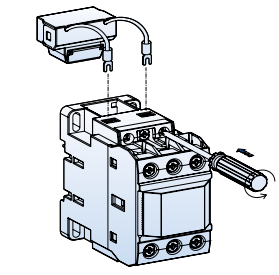
Surge unit

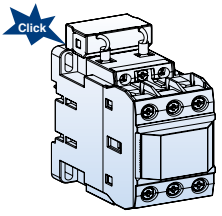
It absorbs the surge arisen out of the coil of the contactor.
It can be installed simply to the contactor.
- Rated operation voltage (Ue): AC 24 ~ 440V / DC 24 ~ 125V
- Rated insulation voltage (Ui): 1000V
- Rated impulse withstand voltage (Uimp): 8kV
- Degree of protection: IP20
- Combination with 3Pole 6a/3~150a/3, 4Pole 32a/4~85a/4

US Type

Surge unit		Rated voltage		Internal elements spec.			Contactor
Type	Composition			V	R	C	
US-1	Varistor+RC	AC	24~48V	120V	100Ω	0.1μF	3Pole MC 6a~150a 9b~22b
US-2	Varistor+RC		100~125V	270V	100Ω	0.1μF	
US-3	Varistor+RC		200~240V	470V	100Ω	0.1μF	
US-4	Varistor+RC	DC	24~48V	120V	100Ω	0.47μF	
US-5	Varistor+RC		100~125V	270V	100Ω	0.47μF	
US-6	Varistor+RC		200~220V	470V	100Ω	0.47μF	
US-11	Varistor	AC/DC common	24~48V	120V	-	-	
US-12	Varistor		100~125V	270V	-	-	
US-13	Varistor		200~240V	470V	-	-	
US-14	Varistor		380~440V	1000V	-	-	
US-22	RC	AC	100~125V	-	56Ω	1μF	

Installation (US Type)



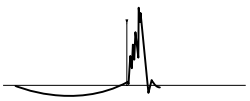


Connect the lead wires to the coil terminals of the contactor first. And then insert the body into the space of the contactor as shown above.

AS Type

Surge unit		Rated voltage		Internal elements spec.			Contactor
Type	Composition			V	R	C	
AS-1	Varistor+RC	AC	24~48V	120V	100Ω	0.1μF	3Pole MC 185a~800a 4Pole MC 22a/4~800a/4
AS-2	Varistor+RC		100~125V	270V	100Ω	0.1μF	
AS-3	Varistor+RC		200~240V	470V	100Ω	0.1μF	
AS-4	Varistor+RC	DC	24~48V	120V	100Ω	0.47μF	
AS-5	Varistor+RC		100~125V	270V	100Ω	0.47μF	
AS-6	Varistor+RC		200~220V	470V	100Ω	0.47μF	
AS-11	Varistor	AC/DC common	24~48V	120V	-	-	
AS-12	Varistor		100~125V	270V	-	-	
AS-13	Varistor		200~240V	470V	-	-	
AS-14	Varistor	AC	380~440V	1000V	-	-	

Specitication of surge unit

Without surge unit	① With Varistor unit	② With CR+Varistor unit
Surge voltage, arise when the coil is off, may provoke mis-operation and damage in the circuit.	Varistor has an effect to cut down the peak voltage	Varistor has an effect to cut down the peak voltage and high frequency wave
		

Capacitor unit



Capacitor unit is connected to the terminals of the contactor to reduce the high inrush current.
IEC 60947-4-1 AC 6b

Type				AC-9 (1b)	AC-50 (1b)	AC-75 (1b)
Rating thermal current (Ith)				16A	16A	16A
Current	AC-15	Rated Operational Current (A)	120V	6	6	6
			240V	3	3	3
			380V	1.9	1.9	1.9
			480V	1.5	1.5	1.5
			500V	1.4	1.4	1.4
			600V	1.2	1.2	1.2
	DC-13	Rated Operational Current (A)	125V	0.55	0.55	0.55
			250V	0.27	0.27	0.27
			400V	0.15	0.15	0.15
			500V	0.13	0.13	0.13
			600V	0.1	0.1	0.1

Note) - When the switch is closed condenser must be discharged before recharged. (Maximum residual voltage at terminals ≤ 50V)
- To prevent short current, gG type fuse must be 1.5~2 times than rated current.

Features of capacitor unit (Pre-loading resistor)

- Damping resister that can limit the inrush current upto 60×In is connected to the mechanism that closed earlier than the main contact of the contactor
- No heat loss by the serial resistor
- Eliminate the switching surge
- Improving the performance of the capacitor system

Operation sequence

Capacitor unit: OFF
Contactor: OFF

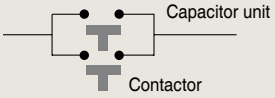


Fig.1

Capacitor unit: ON
Contactor: OFF

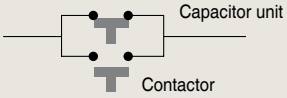


Fig.2

Capacitor unit: OFF
Contactor: ON

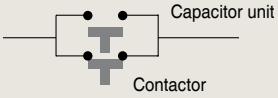


Fig.3

Note) Closing sequence: Fig.1 => Fig.2 => Fig.3
Opening sequence: Fig.3 => Fig.1

Accessory

Replacement coils

To replace the existing coil to change voltage or repair.





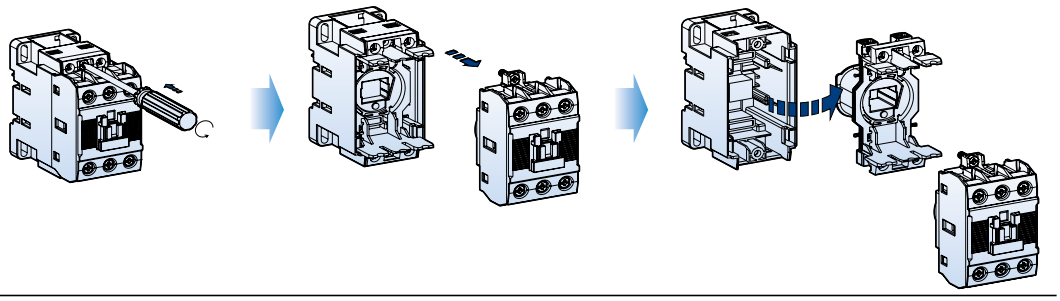






Contactor	Coil voltage							
	AC coil						DC coil	
	AC 60Hz (V)		AC 50Hz (V)		AC 50/60Hz (V)		DC (V)	
MC-6a MC-9a MC-12a MC-18a MR-4, 6, 8	24 48 100 110 120 200 208 220	230 240 277 380 440 480 600	24 36 42 48 80 100 110 220	230 240 380 400 415 440 500 550	24 48 100 110 120 200 220 230	240 380 415 440 500 550	-	
MC-9b MC-12b MC-18b MC-22b	24 48 100 110 120 200 208 220	230 240 277 380 440 480 600	24 36 42 48 80 100 110 220	230 240 380 400 415 440 500 550	24 48 100 110 120 200 220 230	240 380 415 440 500 550	-	
MC-32a MC-40a	24 48 100 110 120 200 208 220	230 240 277 380 440 480 600	24 36 42 48 80 100 110 220	230 240 380 400 415 440 500 550	24 48 100 110 120 200 220 230	240 380 415 440 500 550	12 20 24 48 60 80 100	110 125 200 220 250
MC-50a MC-65a	24 48 100 110 120 200 208 220	230 240 277 380 440 480 600	24 36 42 48 80 100 110 220	230 240 380 400 415 440 500 550	24 48 100 110 120 200 220 230	240 380 415 440 500 550	12 20 24 48 60 80 100	110 125 200 220 250
MC-75a MC-85a MC-100a	24 48 100 110 120 200 208 220	230 240 277 380 440 480 600	24 36 42 48 80 100 110 220	230 240 380 400 415 440 500 550	24 48 100 110 120 200 220 230	240 380 415 440 500 550	12 20 24 48 60 80 100	110 125 200 220 250
MC-130a MC-150a	-		-		24 48 24~48 100~240 110~120	220~240 300 380~440 500	24 48 24~48 70~110 100~220	110 220

Replacement process





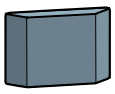
To replace the existing coil to change voltage or repair.

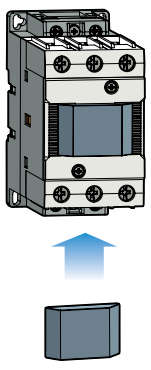
Contactor	Coil voltage	
	AC/DC common coil (V)	AC coil (V)
MC-185a MC-225a	24 48 100~240/100~220	300 400 500
MC-300a MC-400a	100~240/100~220	400 500
MC-500a MC-630a MC-800a	100/100 200/200	300 400 500
MC-1260a	100~240/100~220	-
MC-1400a MC-1700a MC-2100a	100~240/100~220	-

Safety cover (Option)

To prevent arc from coming out from the contactor
To protect from unintended manual operation
To prevent dust from entering into the inside of the contactor

Type	Contactor
AP-9	MC-6a~150a
AP-185	MC-185a~800a





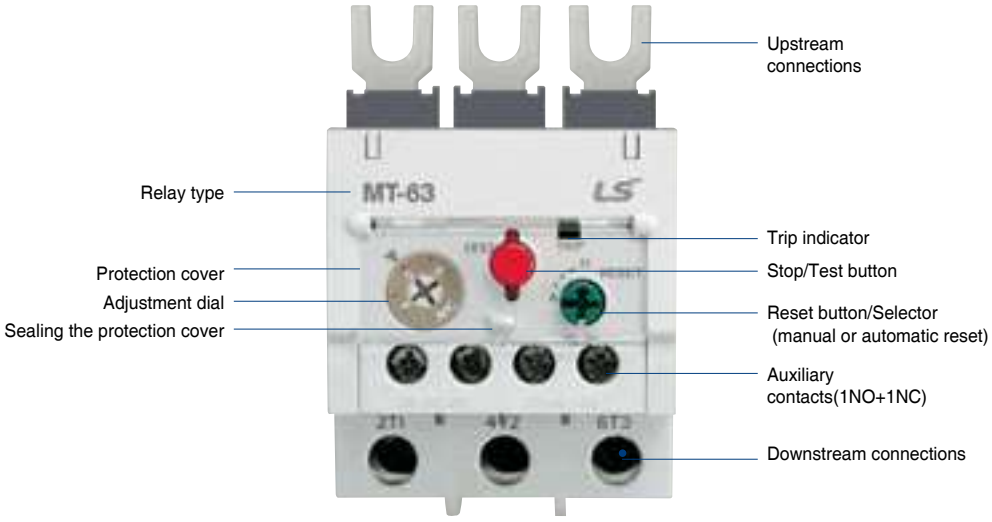
Overload relays

General descriptions

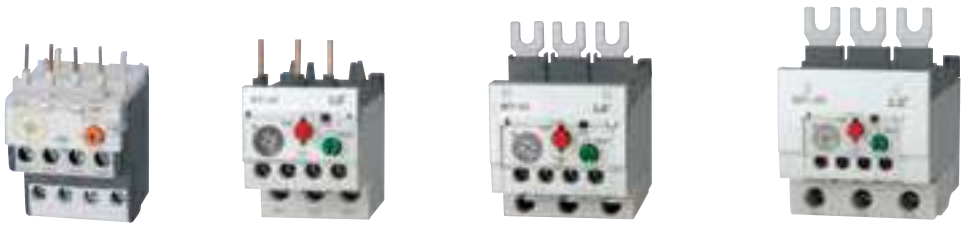
Type MT, bimetal-style, overload relays are designed to protect AC circuits and motors against overloads, phase failure, long starting times and prolonged stalling of the motor.

Four versions are available according to the protection function and trip class as below.
1. Non-differential: 3pole-2bimetal(heater) and trip class 10A
2. Non-differential: 3pole-3bimetal(heater) and trip class 10A
3. Differential: 3pole-3bimetal(heater) and trip class 10A
4. Differential: 3pole-3bimetal(heater) and trip class 20

Configuration of the front

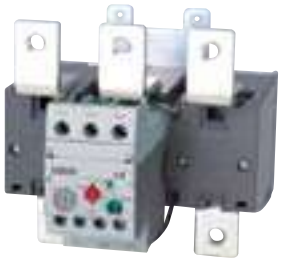
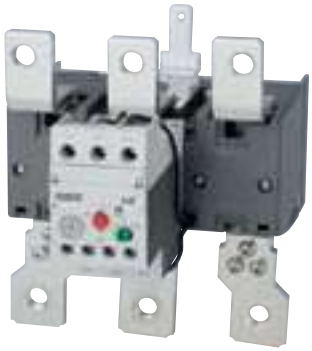


MT-12	MT-32	MT-63	MT-95



Environment and auxiliary circuit

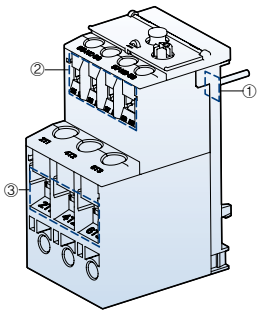
Environment			MT-12~800
Standards			IEC/EN 60947-1, IEC/EN 60947-4-1
Certifications			CE, CSA, UL, CCC
Rated operation voltage			Max. 690V
Rated insulation voltage			690V
Rated frequency			50/60Hz
Degree of protection (Conforming to IEC 60 529)			IP 20
Ambient air temperature		Storage	-30 ~ +65℃
		Operation	-25 ~ +60℃ (MT refer to Graph)
Mounting position			Vertical plane
Shock resistance (Conforming to IEC 68-2-7)			15gn - 11ms
Vibration resistance (Conforming to IEC68-2-6)			6G
Insulation strength (Conforming to IEC 255-5)			6kV
Rated impulse withstand voltage (Conforming to IEC 801-5)			6kV
Auxiliary contacts characteristics			
Composition			1NO +1NC
Rated thermal current			5A
Rated operation current			C600, R300
	AC15 duty	120V	1.5A
	(C600)	240V	0.75A
		380V	0.47A
		480V	0.375A
		500V	0.35A
		600V	0.3A
	DC13 duty	120V	0.22A
	(R300)	240V	0.1A
Connecting conductor		Size	18AWG /1㎟
Connection to screw clamp terminals		Type	65/75℃ Cu-wire

MT-150	MT-225	MT-400	MT-800
			

Overload relay specifications

MT type
Thermal Overload Relays





① Main terminals/upstream
② Main terminals/downstream
③ Auxiliary terminals

Frame size		18AF	
Trip class		10A	20
Type	Non-differential(3P-2 heater)	MT-12/2H□S	-
	Non-differential(3P-3 heater)	MT-12/3H□S	-
	Differential type	MT-12/3K□S	MT-12/3D□S
Terminals	Main-upstream	Screw clamp	
	Main-downstream	Screw clamp	
	Auxiliary	Screw clamp	
Number of poles		3	
Rated operational voltage (Ue)		690V	
Rated insulation voltage (Ui)		Up to 690V	
Rated impulse voltage (Uimp)		6kV	
Degree of protection (IEC 60 529)		IP 20	
Temperature compensation (℃)		-5~+40℃	
Functions	Trip indicating	■	
	Stop	-	
	Test	■	
	Manual/Automatic Reset	■	
Setting range(A)		0.1~18A	1~18A
Nominal current rating	Wire size mm²	Wire size AWG	
0.14	1	18	0.1~0.16
0.21			0.16~0.25
0.33			0.25~0.4
0.52			0.4~0.63
0.82			0.63~1
1.3			1~1.6
2.1			1.6~2.5
3.3			2.5~4
5			4~6
6.5			5~8
7.5	1~1.5	18~16	6~9
8.5			7~10
11	1.5~2.5	16~14	9~13
15	2.5	14~12	12~18
19	2.5~4	12~10	
21.5			
27	4~6	10	
30	4~10	10~8	
34	6~10	10~8	
42	10	8	
54			
65	16~25	6~4	
74			
83	25~35	4~3	
90			
Applied contactors		MC-6a, 9a, 12a, 18a	
Separate mounting unit		UZ-12	

Note : Screw clamp and Lug types are available for the main terminals of 65AF, 100AF and 150AF relays.
The other AF relays provide Screw clamp types only.



40AF	
10A	20
MT-32/2H□S	-
MT-32/3H□S	-
MT-32/3K□S	MT-32/3D□S
Screw clamp	
Screw clamp	
Screw clamp	
3	
690V	
Up to 690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
0.1~40A	1~40A
0.1~0.16	
0.16~0.25	
0.25~0.4	
0.4~0.63	
0.63~1	
1~1.6	1~1.6
1.6~2.5	1.6~2.5
2.5~4	2.5~4
4~6	4~6
5~8	5~8
6~9	6~9
7~10	7~10
9~13	9~13
12~18	12~18
16~22	16~22
18~25	18~25
22~32	22~32
-	-
28~40	28~40
MC-9b, 12b, 18b, 22b, 32a, 40a	
UZ-32	



65AF	
10A	20
MT-63/2H□S/L	-
MT-63/3H□S/L	-
MT-63/3K□S/L	MT-63/3D□S/L
Screw/Lug clamp	
Screw/Lug clamp	
Screw clamp	
3	
690V	
Up to 690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
4~65A	
4~6	4~6
5~8	5~8
6~9	6~9
7~10	7~10
9~13	9~13
12~18	12~18
16~22	16~22
18~25	18~25
-	-
24~36	24~36
28~40	28~40
34~50	34~50
45~65	45~65
MC-50a, 65a	
UZ-63S/L	

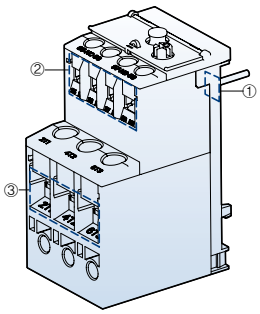


100AF	
10A	20
MT-95/2H□S/L	-
MT-95/3H□S/L	-
MT-95/3K□S/L	MT-95/3D□S/L
Screw/Lug clamp	
Screw/Lug clamp	
Screw clamp	
3	
690V	
Up to 690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
7~100A	
7~10	7~10
9~13	9~13
12~18	12~18
16~22	16~22
18~25	18~25
-	-
24~36	24~36
28~40	28~40
34~50	34~50
45~65	45~65
54~75	54~75
63~85	63~85
70~95	70~95
80~100	80~100
MC-75a, 85a, 100a	
UZ-95S/L	

Overload relay specifications

MT type
Thermal Overload Relays

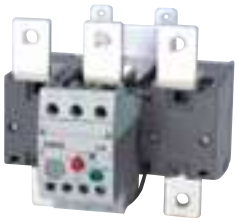


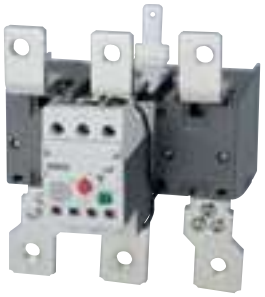


① Main terminals/upstream
② Main terminals/downstream
③ Auxiliary terminals

Frame size		150AF	
Trip class		10A	20
Type	Non-differential(3P-2 heater)	MT-150/2H□S/L	-
	Non-differential(3P-3 heater)	MT-150/3H□S/L	-
	Differential type	MT-150/3K□S/L	MT-150/3D□S/L
Terminals	Main-upstream	Screw/Lug clamp	
	Main-downstream	Screw/Lug clamp	
	Auxiliary	Screw clamp	
Number of poles		3	
Rated operational voltage (Ue)		690V	
Rated insulation voltage (Ui)		690V	
Rated impulse voltage (Uimp)		6kV	
Degree of protection (IEC 60 529)		IP 20	
Temperature compensation (℃)		-5~+40℃	
Functions	Trip indicating	■	
	Stop	■	
	Test	■	
	Manual/Automatic Reset	■	
Setting range(A)		34~150A	34~150A
Nominal current rating	Wire size mm²	AWG	
42	10	8	34~50
55	16	6	45~65
65	25	4	54~75
74	25	4	63~85
80	35	3	-
93	35	2	80~105
107	50	1	-
113	50	1	95~130
130	50	0	110~150
130	70	00	
153	95	000	
200	120	250	
265	185	350	
350	240	500	
515	185×2n	350×2n	
660	240×2n	300×3n	
Applied contactors		MC-130a, 150a	
Separate mounting unit		UZ-150S/L	

Note : Screw clamp and Lug types are available for the main terminals of 65AF, 100AF and 150AF relays.
The other AF relays provide Screw clamp types only.





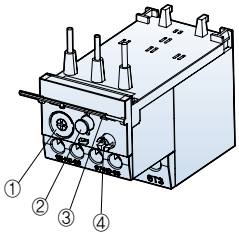


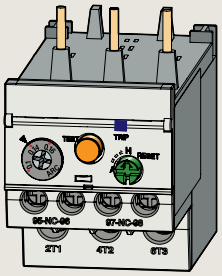
225AF	
10A	20
MT-225/2H□S	-
MT-225/3H□S	-
MT-225/3K□S	MT-225/3D□S
Screw clamp	
Screw clamp	
Screw clamp	
3	
690V	
690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
65~100	
-	
85~125	
-	
-	
100~160	
120~185	
160~240	
MC-185a, 225a	
-	

400AF	
10A	20
MT-400/2H□S	-
MT-400/3H□S	-
MT-400/3K□S	MT-400/3D□S
Screw clamp	
Screw clamp	
Screw clamp	
3	
690V	
690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
85~125	
-	
-	
100~160	
120~185	
160~240	
200~330	
260~400	
MC-265a, 330a, 400a	
-	

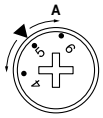
800AF	
10A	20
MT-800/2H□S	-
MT-800/3H□S	-
MT-800/3K□S	MT-800/3D□S
Screw clamp	
Screw clamp	
Screw clamp	
3	
690V	
690V	
6kV	
IP 20	
-5~+40℃	
■	
■	
■	
■	
200~800A	200~800A
200~300	
260~400	
400~600	
520~800	
MC-500a, 630a, 800a	
-	

Overload relay operation





1. Adjustment dial



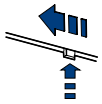
Before adjusting the dial open the protection cover.
Current setting can be done easily by using (+) or (-) driver and have to be set by dotting or graduations standard.
Do not rotate the dial out of the setting range.

2. Stop/Test button



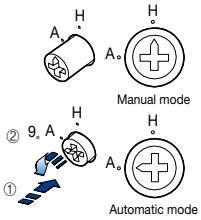
STOP function is executed by pushing the button, which causes the next sequence.
In case of operation test pull this button.

3.Trip indicator



If relay is tripped it comes out.

4. Reset button/Selector



Using a driver the reset mode can be set.
In case of Manual mode(H) push the button to reset the relay.
To change to Automatic mode(A) from Manual mode push the button and rotate as shown in the fig.

5.Auxiliary contact operation

Terminal no.	Normal	STOP	TEST/TRIP	RESET
NC 95-96				
NO 97-98				

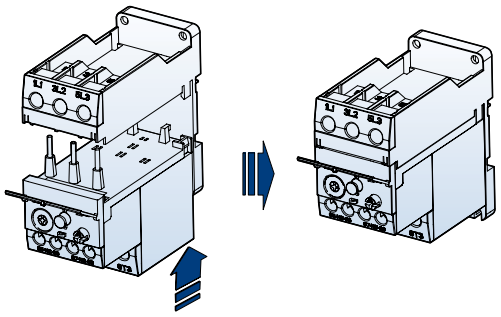
Accessories for overload relay

Separate mounting units

These accessories are used to mount the relays separately from contactors.
If a relay is combined with a unit, it can be mounted on DIN rail or panel by fixing screws.



Relay	Unit	
	Type	Weight
MT-12/2H□ MT-12/3H□ MT-12/3K□ MT-12/3D□	UZ-12	35g
MT-32/2H□ MT-32/3H□ MT-32/3K□ MT-32/3D□	UZ-32	68g
MT-63/2H□ MT-63/3H□ MT-63/3K□ MT-63/3D□	UZ-63S/L	134g
MT-95/2H□ MT-95/3H□ MT-95/3K□ MT-95/3D□	UZ-95S/L	230g
MT-150/2H□ MT-150/3H□ MT-150/3K□ MT-150/3D□	UZ-150S/L	284g



Terminal cover units for overload relays



Relay	Remarks
MT-225	2 pcs
MT-400	

Accessories for overload relay

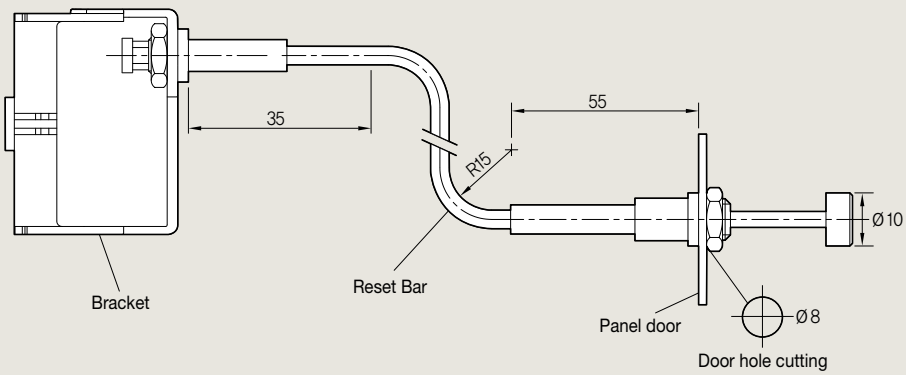
Remote reset units



These accessories are used to reset the relays on the panel door.

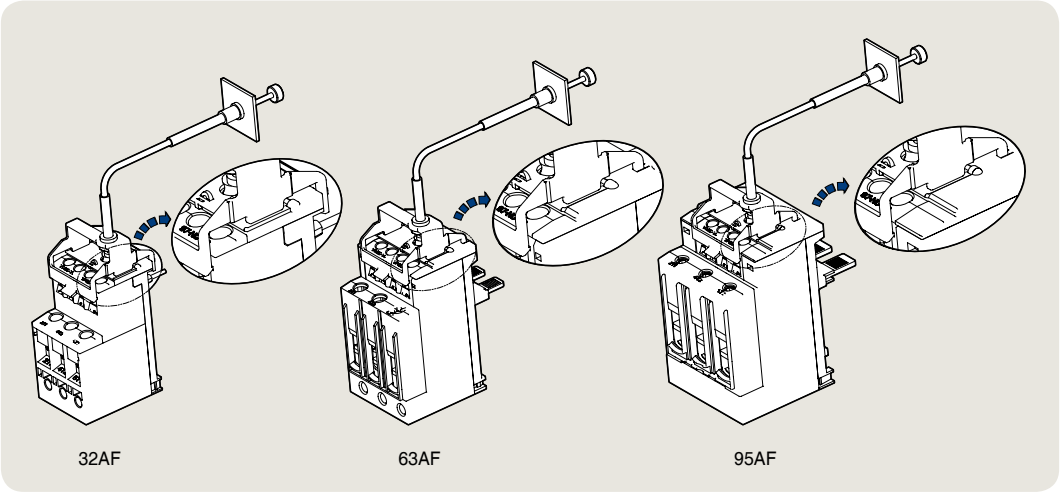
Type	Cable length (L)
UM-4R	400 mm
UM-5R	500 mm
UM-6R	600 mm

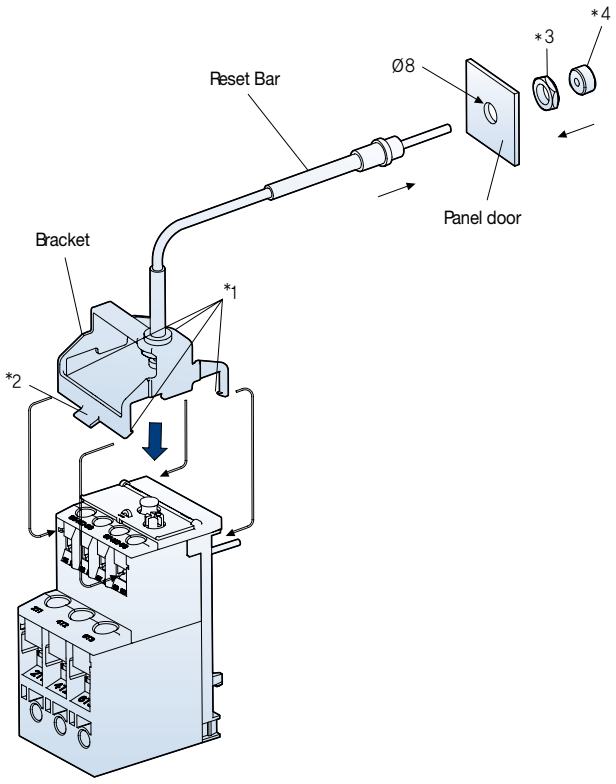
1. Make the reset bar straight at least 55mm from the panel door and 35mm from the bracket.
2. The bending radius of the flexible reset bar should not be less than 15mm.



Installation

1. Fit the bracket with the relay as indicated(*1) below.
2. Separate Nut(*3) and Head cap(*4) from the reset bar first.
   Insert the reset bar into the panel hole and then fix it with the Nut and Head.
3. To separate the bracket from the relay lift the *2 part as shown in the fig.

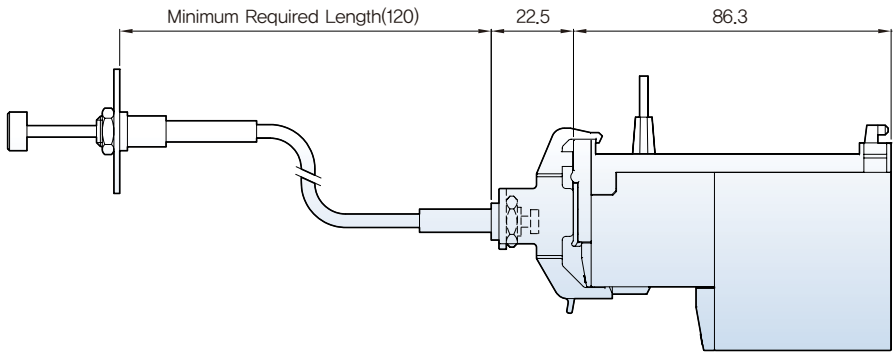




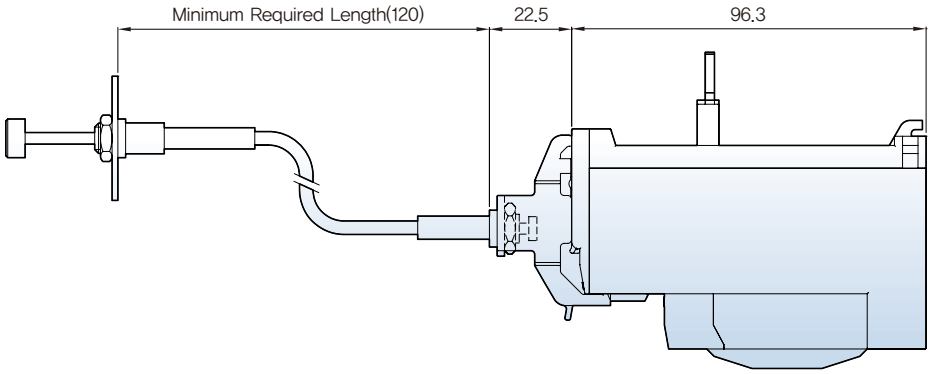
Accessories for overload relay

Remote reset units

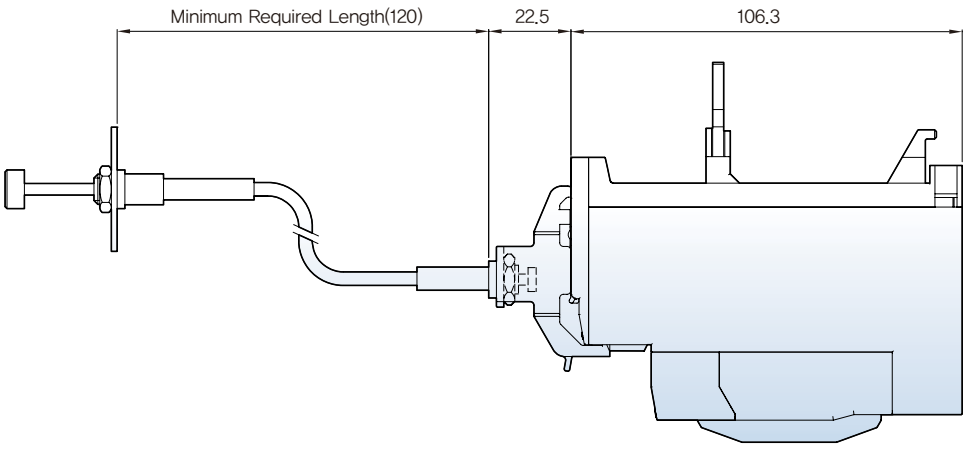
MT-32



MT-63

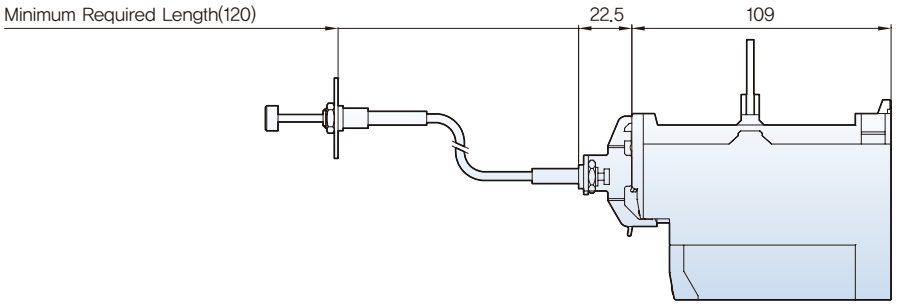


MT-95

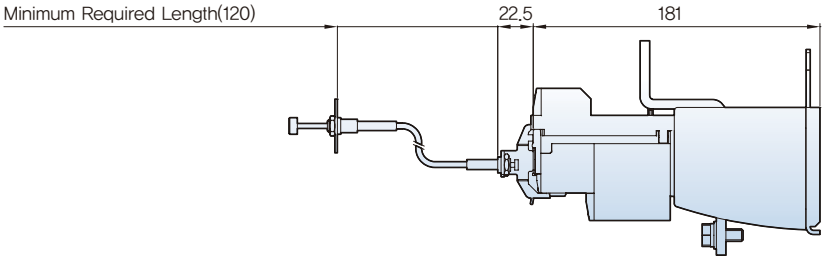


Remote reset units

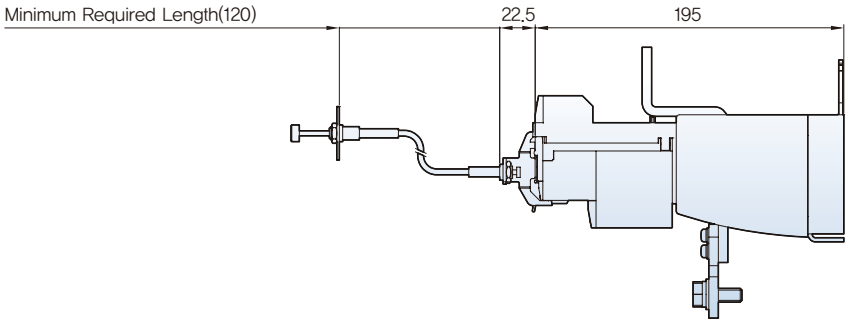
MT-150



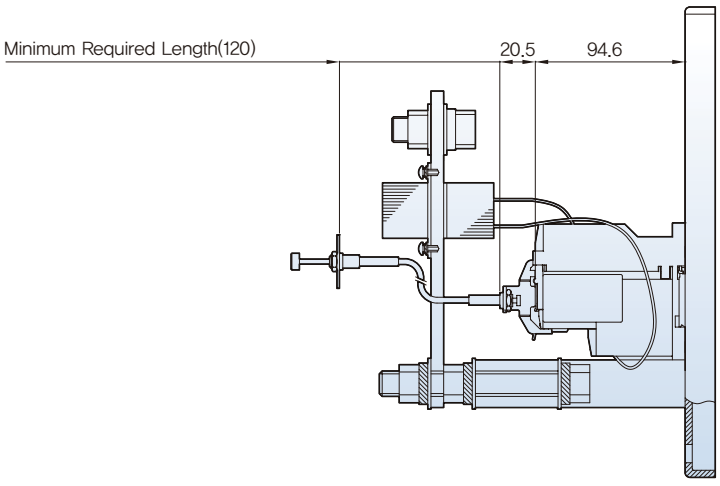
MT-225



MT-400



MT-800



Overload relay ordering types

Trip class 10A

Terminal type: Screw

Degree of protection: IP20

MT-12



MT-12/□

Setting range (A)	Ordering type			Contactor	Separate mounting unit
	Non-differential (3P-2 heater)	Non-differential (3P-3 heater)	Differential		
0.1~0.16	MT-12/2H0.14	MT-12/3H0.14	MT-12/3K0.14	MC-6a MC-9a MC-12a MC-18a	UZ-12
0.16~0.25	MT-12/2H0.21	MT-12/3H0.21	MT-12/3K0.21		
0.25~0.4	MT-12/2H0.33	MT-12/3H0.33	MT-12/3K0.33		
0.4~0.63	MT-12/2H0.52	MT-12/3H0.52	MT-12/3K0.52		
0.63~1	MT-12/2H0.82	MT-12/3H0.82	MT-12/3K0.82		
1~1.6	MT-12/2H1.3	MT-12/3H1.3	MT-12/3K1.3		
1.6~2.5	MT-12/2H2.1	MT-12/3H2.1	MT-12/3K2.1		
2.5~4	MT-12/2H3.3	MT-12/3H3.3	MT-12/3K3.3		
4~6	MT-12/2H5	MT-12/3H5	MT-12/3K5		
5~8	MT-12/2H6.5	MT-12/3H6.5	MT-12/3K6.5		
6~9	MT-12/2H7.5	MT-12/3H7.5	MT-12/3K7.5		
7~10	MT-12/2H8.5	MT-12/3H8.5	MT-12/3K8.5		
9~13	MT-12/2H11	MT-12/3H11	MT-12/3K11		
12~18	MT-12/2H15	MT-12/3H15	MT-12/3K15		

MT-32



MT-32/□

Setting range (A)	Ordering type			Contactor	Separate mounting unit
	Non-differential (3P-2 heater)	Non-differential (3P-3 heater)	Differential		
0.1~0.16	MT-32/2H0.14	MT-32/3H0.14	MT-32/3K0.14	MC-9b MC-12b MC-18b MC-22b MC-32a MC-40a	UZ-32
0.16~0.25	MT-32/2H0.21	MT-32/3H0.21	MT-32/3K0.21		
0.25~0.4	MT-32/2H0.33	MT-32/3H0.33	MT-32/3K0.33		
0.4~0.63	MT-32/2H0.52	MT-32/3H0.52	MT-32/3K0.52		
0.63~1	MT-32/2H0.82	MT-32/3H0.82	MT-32/3K0.82		
1~1.6	MT-32/2H1.3	MT-32/3H1.3	MT-32/3K1.3		
1.6~2.5	MT-32/2H2.1	MT-32/3H2.1	MT-32/3K2.1		
2.5~4	MT-32/2H3.3	MT-32/3H3.3	MT-32/3K3.3		
4~6	MT-32/2H5	MT-32/3H5	MT-32/3K5		
5~8	MT-32/2H6.5	MT-32/3H6.5	MT-32/3K6.5		
6~9	MT-32/2H7.5	MT-32/3H7.5	MT-32/3K7.5		
7~10	MT-32/2H8.5	MT-32/3H8.5	MT-32/3K8.5		
9~13	MT-32/2H11	MT-32/3H11	MT-32/3K11		
12~18	MT-32/2H15	MT-32/3H15	MT-32/3K15		
16~22	MT-32/2H19	MT-32/3H19	MT-32/3K19		
18~25	MT-32/2H21.5	MT-32/3H21.5	MT-32/3K21.5		
22~32	MT-32/2H27	MT-32/3H27	MT-32/3K27		
28~40	MT-32/2H34	MT-32/3H34	MT-32/3K34		UZ-32(34A)

Trip class 10A

Terminal type: Screw(S), Lug(L)

Degree of protection: IP20

MT-63



MT-63/□S



MT-63/□L

Setting range (A)	Ordering type			Contactor	Separate mounting unit
	Non-differential (3P-2 heater)	Non-differential (3P-3 heater)	Differential		
4~6	MT-63/2H5S/L	MT-63/3H5S/L	MT-63/3K5S/L	MC-50a MC-65a	UZ-63S/L
5~8	MT-63/2H6.5S/L	MT-63/3H6.5S/L	MT-63/3K6.5S/L		
6~9	MT-63/2H7.5S/L	MT-63/3H7.5S/L	MT-63/3K7.5S/L		
7~10	MT-63/2H8.5S/L	MT-63/3H8.5S/L	MT-63/3K8.5S/L		
9~13	MT-63/2H11S/L	MT-63/3H11S/L	MT-63/3K11S/L		
12~18	MT-63/2H15S/L	MT-63/3H15S/L	MT-63/3K15S/L		
16~22	MT-63/2H19S/L	MT-63/3H19S/L	MT-63/3K19S/L		
18~25	MT-63/2H21.5S/L	MT-63/3H21.5S/L	MT-63/3K21.5S/L		
24~36	MT-63/2H30S/L	MT-63/3H30S/L	MT-63/3K30S/L		
28~40	MT-63/2H34S/L	MT-63/3H34S/L	MT-63/3K34S/L		
34~50	MT-63/2H42S/L	MT-63/3H42S/L	MT-63/3K42S/L		
45~65	MT-63/2H55S/L	MT-63/3H55S/L	MT-63/3K55S/L		

MT-95



MT-95/□S



MT-95/□L

Setting range (A)	Ordering type			Contactor	Separate mounting unit
	Non-differential (3P-2 heater)	Non-differential (3P-3 heater)	Differential		
7~10	MT-95/2H8.5S/L	MT-95/3H8.5S/L	MT-95/3K8.5S/L	MC-75a MC-85a MC-100a	UZ-95S/L
9~13	MT-95/2H11S/L	MT-95/3H11S/L	MT-95/3K11S/L		
12~18	MT-95/2H15S/L	MT-95/3H15S/L	MT-95/3K15S/L		
16~22	MT-95/2H19S/L	MT-95/3H19S/L	MT-95/3K19S/L		
18~25	MT-95/2H21.5S/L	MT-95/3H21.5S/L	MT-95/3K21.5S/L		
24~36	MT-95/2H30S/L	MT-95/3H30S/L	MT-95/3K30S/L		
28~40	MT-95/2H34S/L	MT-95/3H34S/L	MT-95/3K34S/L		
34~50	MT-95/2H42S/L	MT-95/3H42S/L	MT-95/3K42S/L		
45~65	MT-95/2H55S/L	MT-95/3H55S/L	MT-95/3K55S/L		
54~75	MT-95/2H65S/L	MT-95/3H65S/L	MT-95/3K65S/L		
63~85	MT-95/2H74S/L	MT-95/3H74S/L	MT-95/3K74S/L		
70~95	MT-95/2H83S/L	MT-95/3H83S/L	MT-95/3K83S/L		
80~100	MT-95/2H90S/L	MT-95/3H90S/L	MT-95/3K90S/L		

MT-150



MT-150/□S

Setting range (A)	Ordering type			Contactor	Separate mounting unit
	Non-differential		Differential		
	(3P-2 heater)	(3P-3 heater)			
34~50	MT-150/2H42S/L	MT-150/3H42S/L	MT-150/3K42S/L	MC-130a MC-150a	UZ-150S/L
45~65	MT-150/2H55S/L	MT-150/3H55S/L	MT-150/3K55S/L		
54~75	MT-150/2H65S/L	MT-150/3H65S/L	MT-150/3K65S/L		
63~85	MT-150/2H74S/L	MT-150/3H74S/L	MT-150/3K74S/L		
80~105	MT-150/2H93S/L	MT-150/3H93S/L	MT-150/3K93S/L		
95~130	MT-150/2H113S/L	MT-150/3H113S/L	MT-150/3K113S/L		
110~150	MT-150/2H130S/L	MT-150/3H130S/L	MT-150/3K130S/L		

Overload relay ordering types

Trip class 10A

Terminal type: Screw

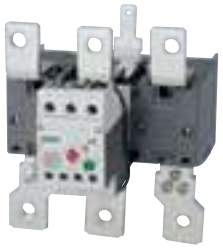
Degree of protection: IP20



MT-225/☐

MT-225

Setting range (A)	Ordering type			Contactor	Remarks
	Non-differential		Differential		
	(3P-2 heater)	(3P-3 heater)			
65~100	MT-225/2H80	MT-225/3H80	MT-225/3K80	MC-185a MC-225a	-
85~125	MT-225/2H107	MT-225/3H107	MT-225/3K107		
100~160	MT-225/2H130	MT-225/3H130	MT-225/3K130		
120~185	MT-225/2H153	MT-225/3H153	MT-225/3K153		
160~240	MT-225/2H200	MT-225/3H200	MT-225/3K200		



MT-400/☐

MT-400

Setting range (A)	Ordering type			Contactor	Remarks
	Non-differential		Differential		
	(3P-2 heater)	(3P-3 heater)			
85~125	MT-400/2H107	MT-400/3H107	MT-400/3K107	MC-265a MC-330a MC-400a	-
100~160	MT-400/2H130	MT-400/3H130	MT-400/3K130		
120~185	MT-400/2H153	MT-400/3H153	MT-400/3K153		
160~240	MT-400/2H200	MT-400/3H200	MT-400/3K200		
200~330	MT-400/2H265	MT-400/3H265	MT-400/3K265		
260~400	MT-400/2H350	MT-400/3H350	MT-400/3K350		



MT-800/☐

MT-800

Setting range (A)	Ordering type			Contactor	Remarks
	Non-differential		Differential		
	(3P-2 heater)	(3P-3 heater)			
200~330	MT-800/2H265	MT-800/3H265	MT-800/3K265	MC-500a MC-630a MC-800a	-
260~400	MT-800/2H350	MT-800/3H350	MT-800/3K350		
400~630	MT-800/2H515	MT-800/3H515	MT-800/3K515		
520~800	MT-800/2H660	MT-800/3H660	MT-800/3K660		

Trip class 20

Terminal type: Screw

Degree of protection: IP20

MT-12



MT-12/3D□

Setting range (A)	Ordering type	Contactor	Separate mounting unit
	Differential		
1~1.6	MT-12/3D1.3	MC-6a MC-9a MC-12a MC-18a	UZ-12
1.6~2.5	MT-12/3D2.1		
2.5~4	MT-12/3D3.3		
4~6	MT-12/3D5		
5~8	MT-12/3D6.5		
6~9	MT-12/3D7.5		
7~10	MT-12/3D8.5		
9~13	MT-12/3D11		
12~18	MT-12/3D15		

MT-32



MT-32/3D□

Setting range (A)	Ordering type	Contactor	Separate mounting unit
	Differential		
1~1.6	MT-32/3D1.3	MC-9b MC-12b MC-18b MC-22b MC-32a MC-40a	UZ-32
1.6~2.5	MT-32/3D2.1		
2.5~4	MT-32/3D3.3		
4~6	MT-32/3D5		
5~8	MT-32/3D6.5		
6~9	MT-32/3D7.5		
7~10	MT-32/3D8.5		
9~13	MT-32/3D11		
12~18	MT-32/3D15		
16~22	MT-32/3D19		
18~25	MT-32/3D21.5		
22~32	MT-32/3D27		
28~40	MT-32/3D34		UZ-32(34A)

Overload relay ordering types

Trip class 20

Terminal type: Screw(S), Lug(L)
Degree of protection: IP20

MT-63


MT-63/ S


MT-63/ L

Setting range (A)	Ordering type	Contactor	Separate mounting unit
	Differential		
4~6	MT-63/3D5S/L	MC-50a MC-65a	UZ-63S/L
5~8	MT-63/3D6.5S/L		
6~9	MT-63/3D7.5S/L		
7~10	MT-63/3D8.5S/L		
9~13	MT-63/3D11S/L		
12~18	MT-63/3D15S/L		
16~22	MT-63/3D19S/L		
18~25	MT-63/3D21.5S/L		
24~36	MT-63/3D30S/L		
28~40	MT-63/3D34S/L		
34~50	MT-63/3D42S/L		
45~65	MT-63/3D55S/L		

MT-95


MT-95/ S


MT-95/ L

Setting range (A)	Ordering type	Contactor	Separate mounting unit
	Differential		
7~10	MT-95/3D8.5S/L	MC-75a MC-85a MC-100a	UZ-95S/L
9~13	MT-95/3D11S/L		
12~18	MT-95/3DK15S/L		
16~22	MT-95/3D19S/L		
18~25	MT-95/3D21.5S/L		
24~36	MT-95/3D30S/L		
28~40	MT-95/3D34S/L		
34~50	MT-95/3D42S/L		
45~65	MT-95/3D55S/L		
54~75	MT-95/3D65S/L		
63~85	MT-95/3D74S/L		
70~95	MT-95/3D83S/L		
80~100	MT-95/3D90S/L		

MT-150


MT-150/ S

Setting range (A)	Ordering type	Contactor	Separate mounting unit
	Differential		
34~50	MT-150/3D42S/L	MC-130a MC-150a	UZ-150S/L
45~65	MT-150/3D55S/L		
54~75	MT-150/3D65S/L		
63~85	MT-150/3D74S/L		
80~105	MT-150/3D93S/L		
95~130	MT-150/3D113S/L		
110~150	MT-150/3D130S/L		

Trip class 20

Terminal type: Screw
Degree of protection: IP20

MT-225



MT-225/3D

Setting range (A)	Ordering type	Contactor	Remarks
	Differential		
65~100	MT-225/3D80	MC-185a MC-225a	-
85~125	MT-225/3D107		
100~160	MT-225/3D130		
120~185	MT-225/3D153		
160~240	MT-225/3D200		

MT-400



MT-400/3D

Setting range (A)	Ordering type	Contactor	Remarks
	Differential		
85~125	MT-400/3D107	MC-265a MC-330a MC-400a	-
100~160	MT-400/3D130		
120~185	MT-400/3D153		
160~240	MT-400/3D200		
200~330	MT-400/3D265		
260~400	MT-400/3D350		

MT-800



MT-800/3D

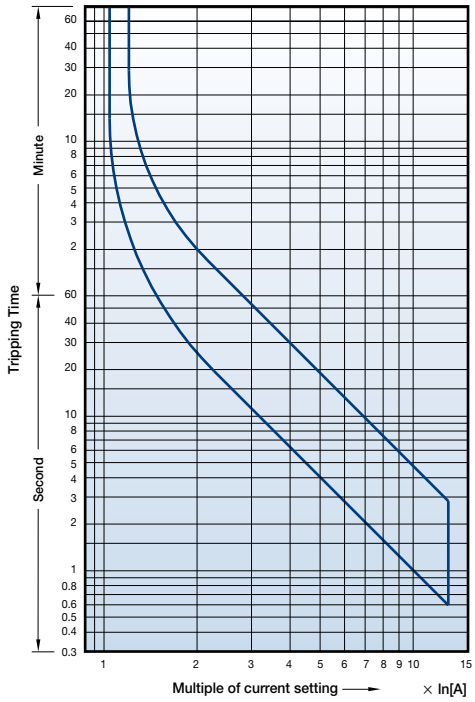
Setting range (A)	Ordering type	Contactor	Remarks
	Differential		
200~330	MT-800/3D265	MC-500a MC-630a MC-800a	-
260~400	MT-800/3D350		
400~630	MT-800/3D515		
520~800	MT-800/3D660		

Overload relay trip curves

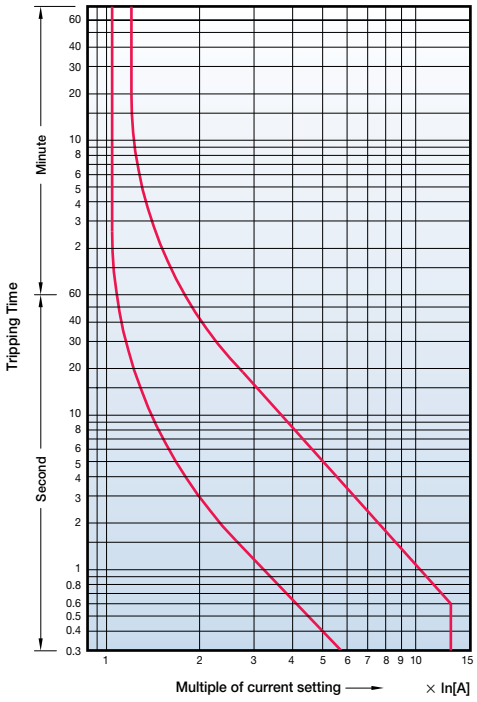
Class 10A, 18AF

MT-12/2H
MT-12/3H
MT-12/3K

Cold starting



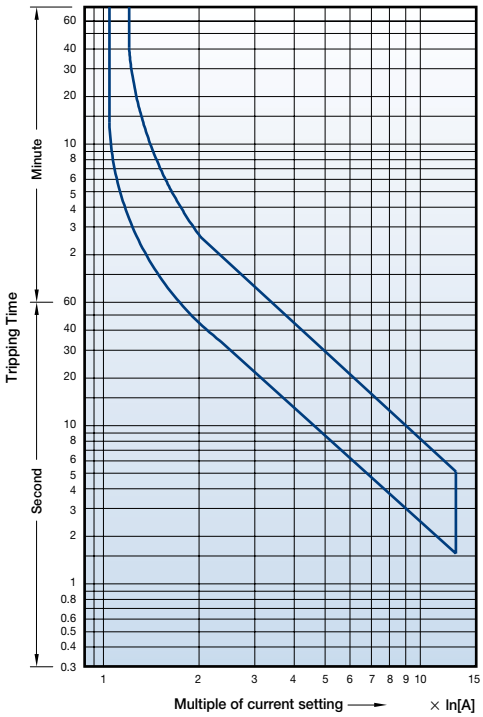
Hot starting



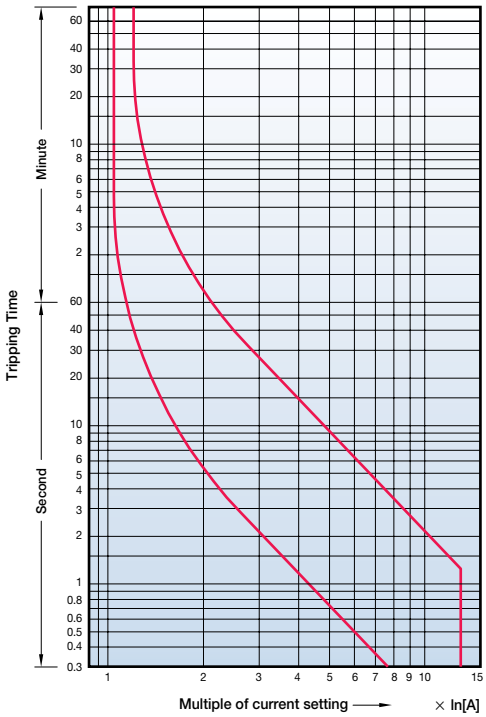
Class 20, 18AF

MT-12/3D

Cold starting



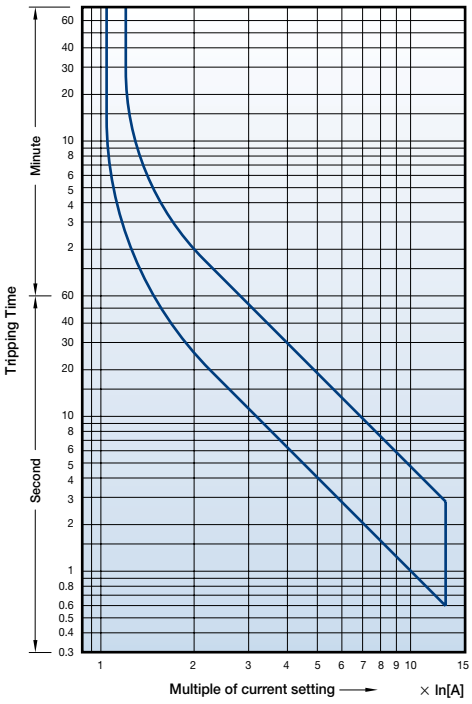
Hot starting



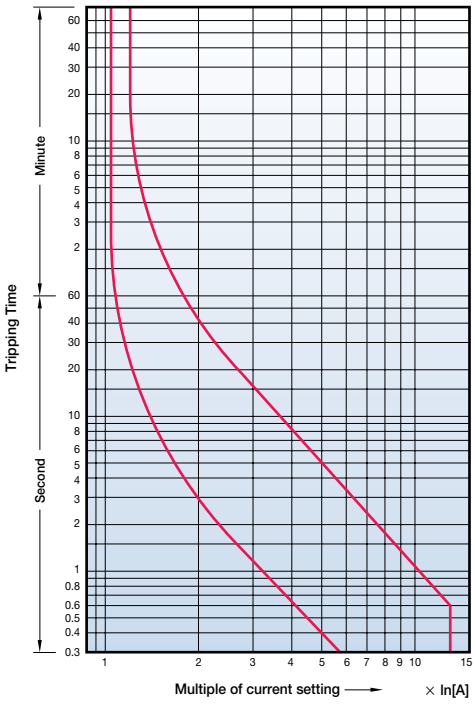
Class 10A, 40AF

MT-32/2H
MT-32/3H
MT-32/3K

Cold starting



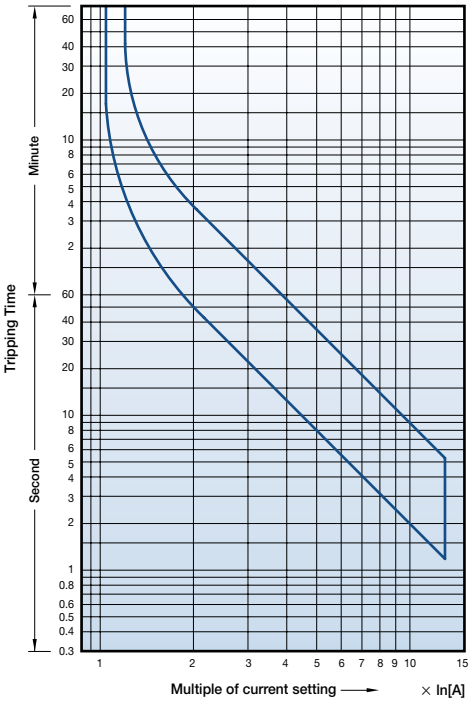
Hot starting



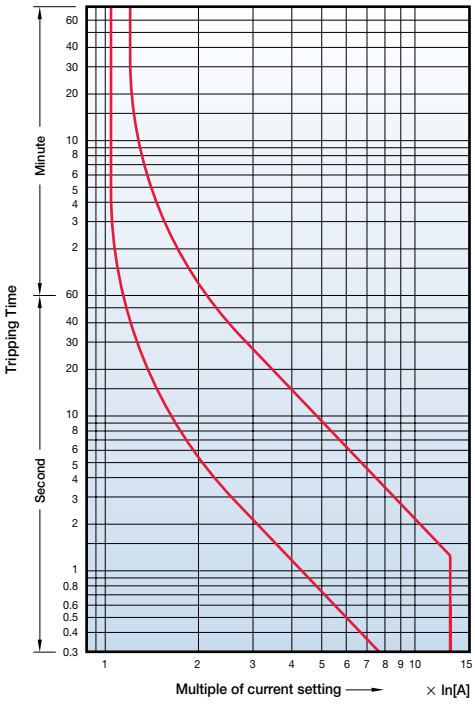
Class 20, 40AF

MT-32/3D

Cold starting



Hot starting

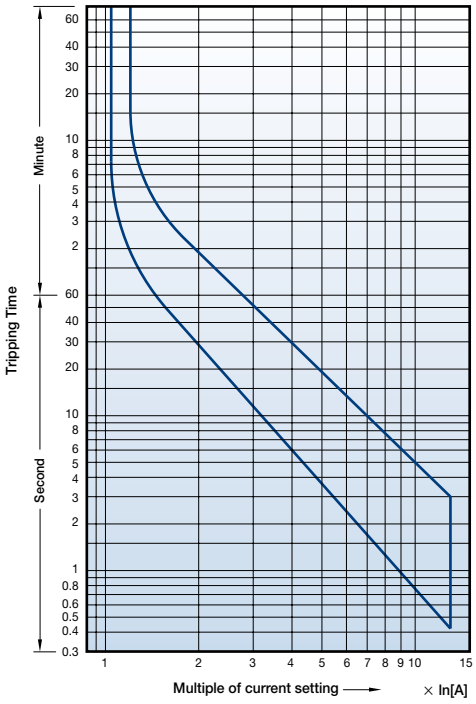


Overload relay trip curves

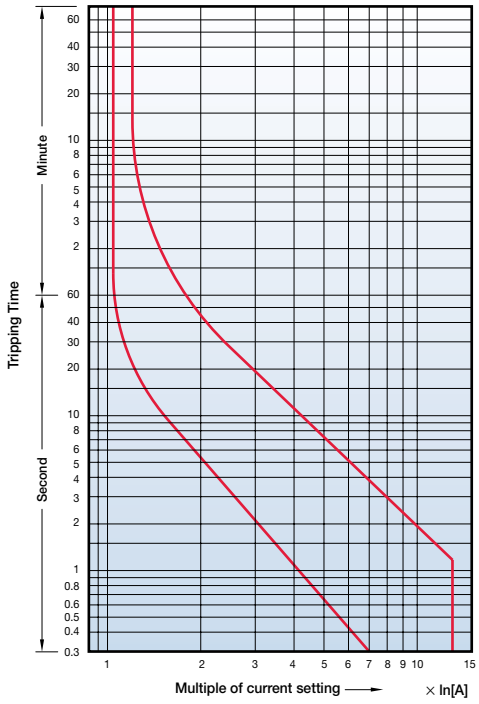
Class 10A, 65AF

MT-63/2H
MT-63/3H
MT-63/3K

Cold starting



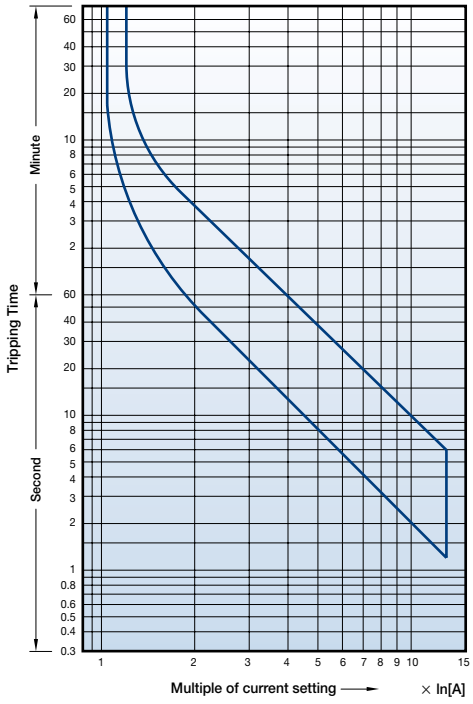
Hot starting



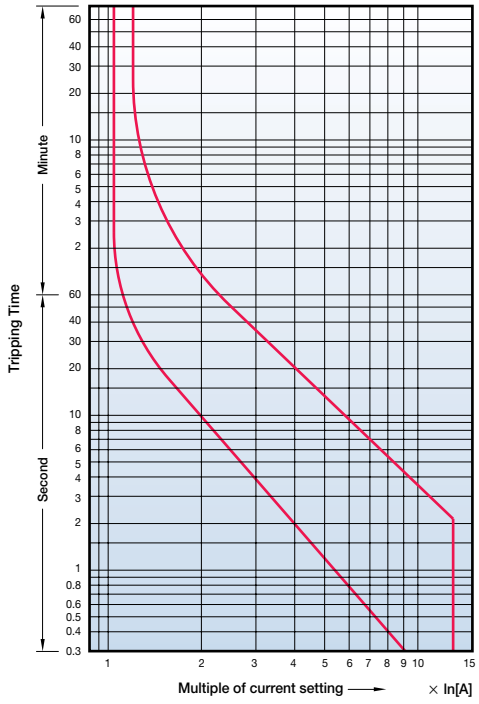
Class 20, 65AF

MT-63/3D

Cold starting



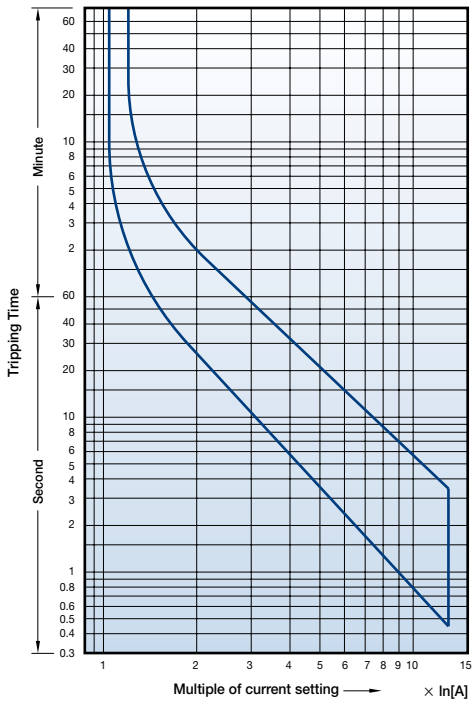
Hot starting



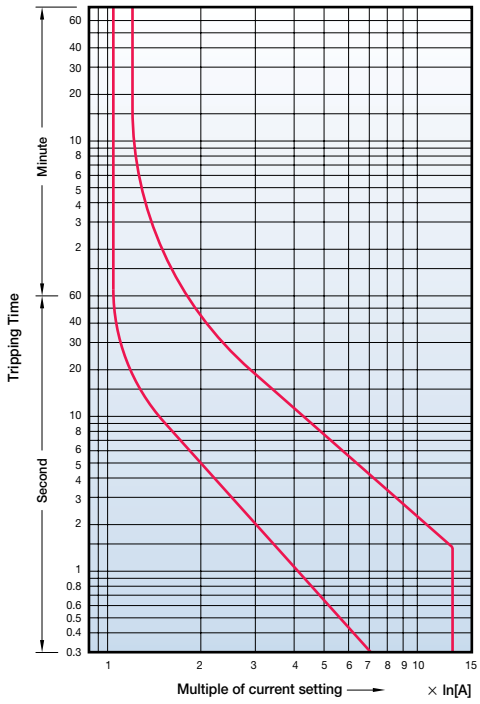
Class 10A, 100AF

MT-95/2H
MT-95/3H
MT-95/3K

Cold starting



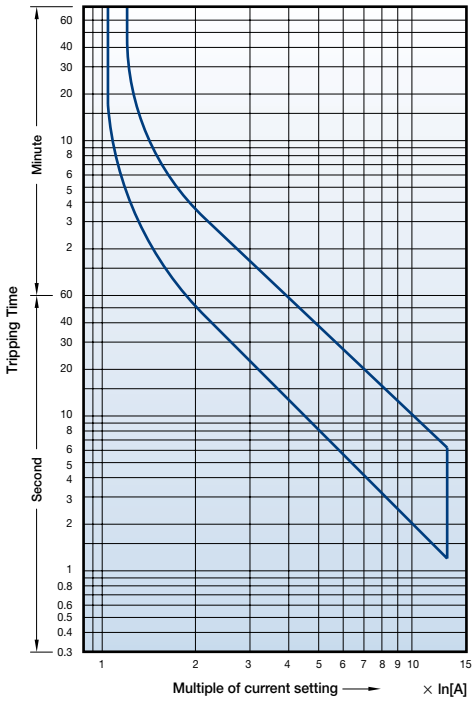
Hot starting



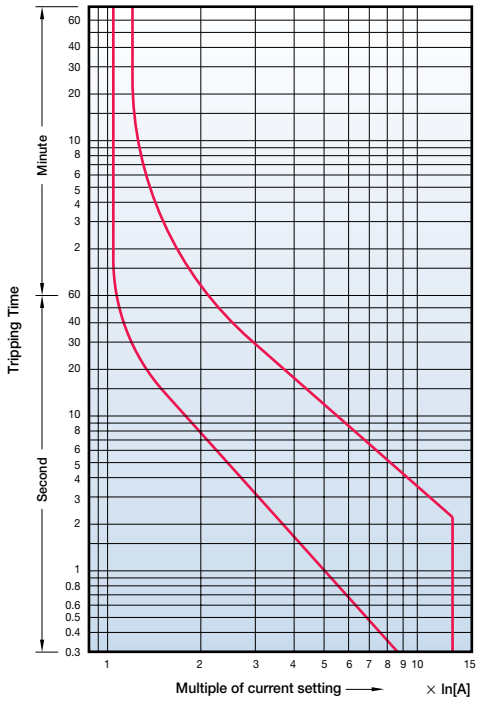
Class 20, 100AF

MT-95/3D

Cold starting



Hot starting

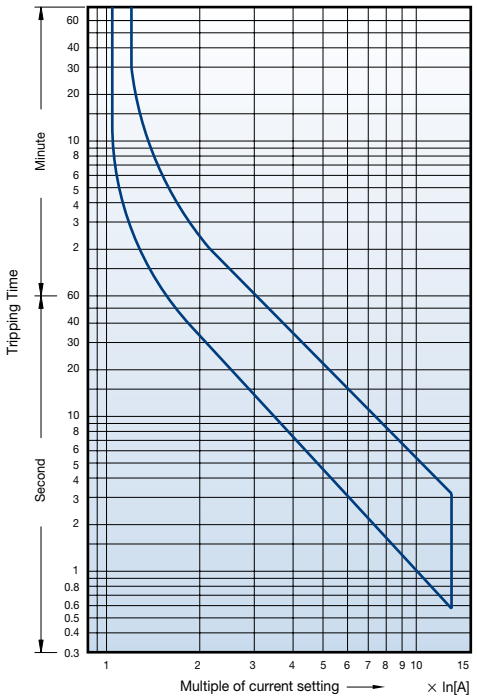


Overload relay trip curves

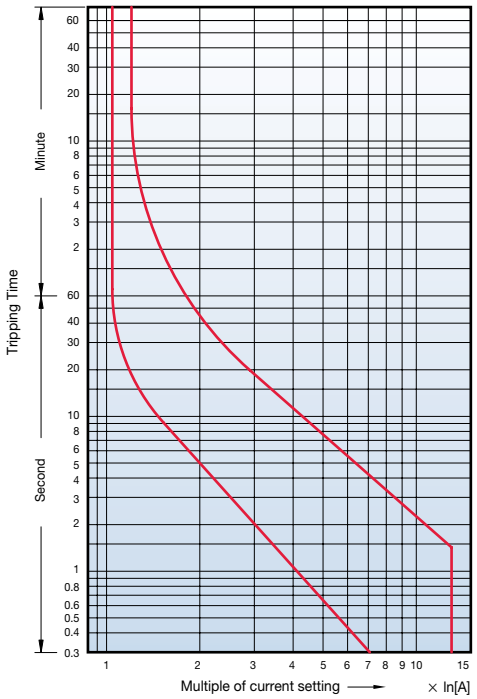
Class 10A, 150AF

MT-150/2H
MT-150/3H
MT-150/3K

Cold starting



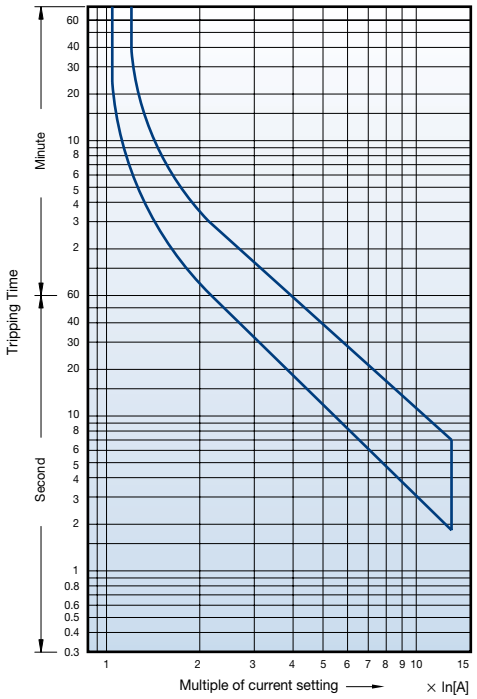
Hot starting



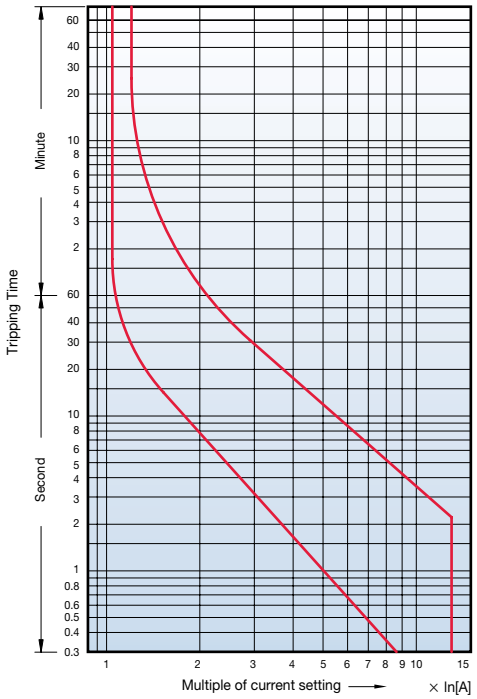
Class 20, 150AF

MT-150/3D

Cold starting



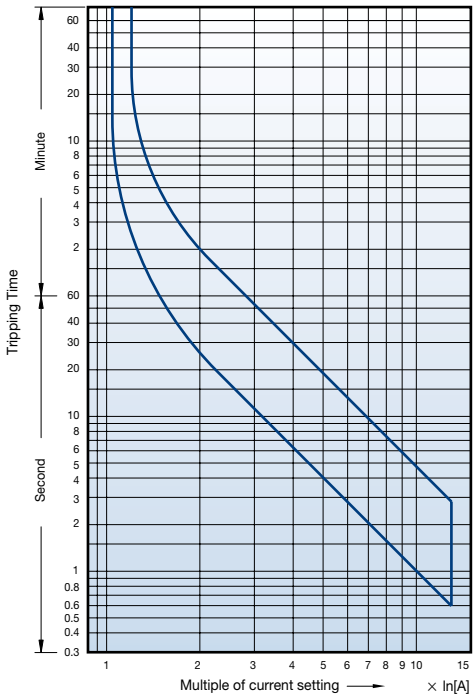
Hot starting



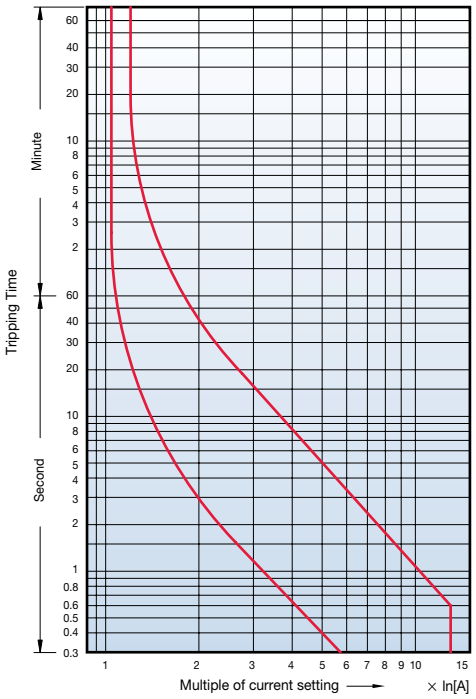
Class 10A, 225AF

MT-225/2H
MT-225/3H
MT-225/3K

Cold starting



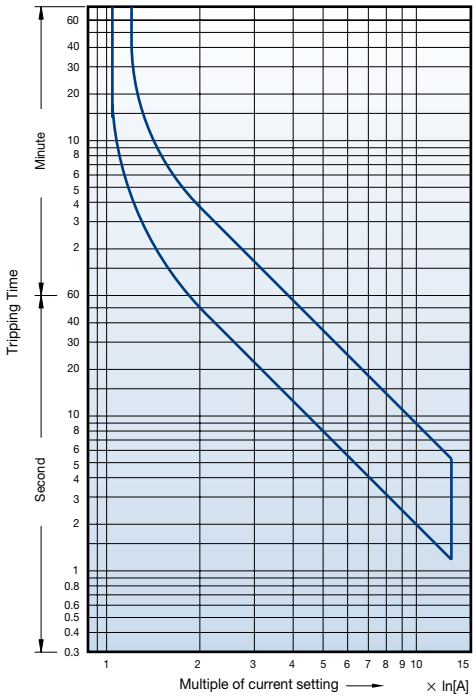
Hot starting



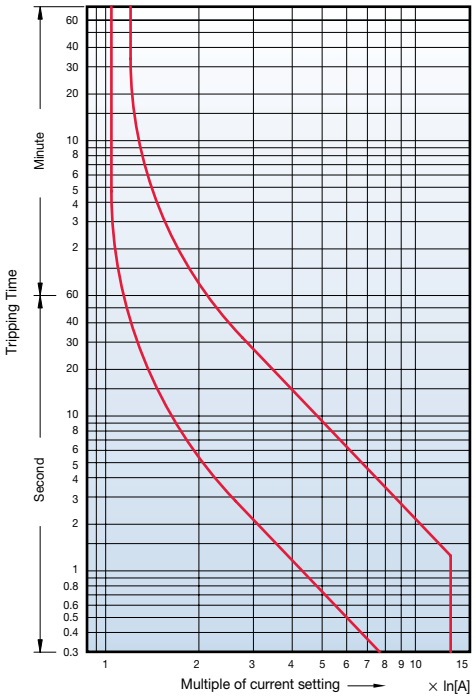
Class 20, 225AF

MT-225/3D

Cold starting



Hot starting

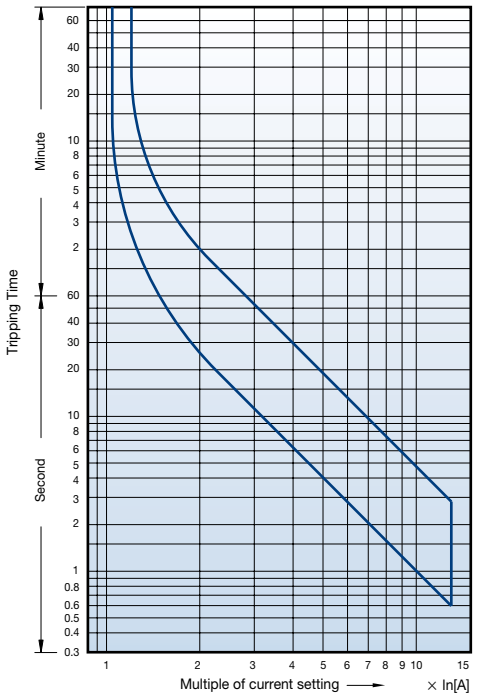


Overload relay trip curves

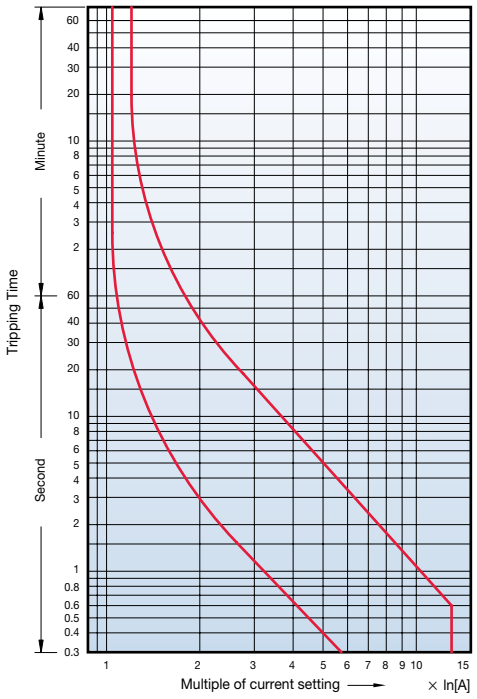
Class 10A, 400AF

MT-400/2H
MT-400/3H
MT-400/3K

Cold starting



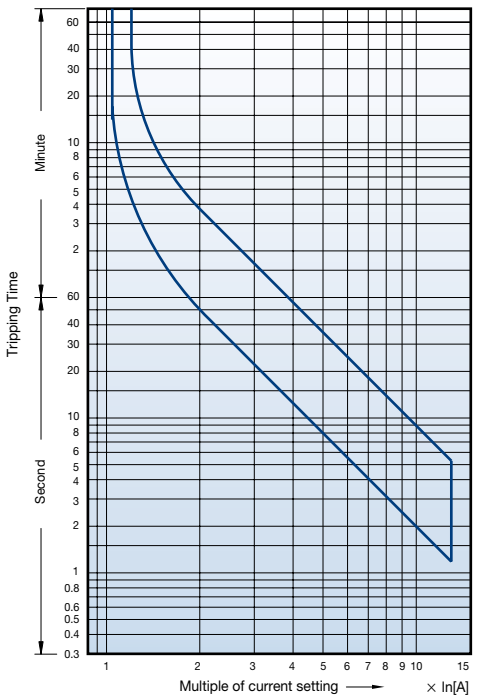
Hot starting



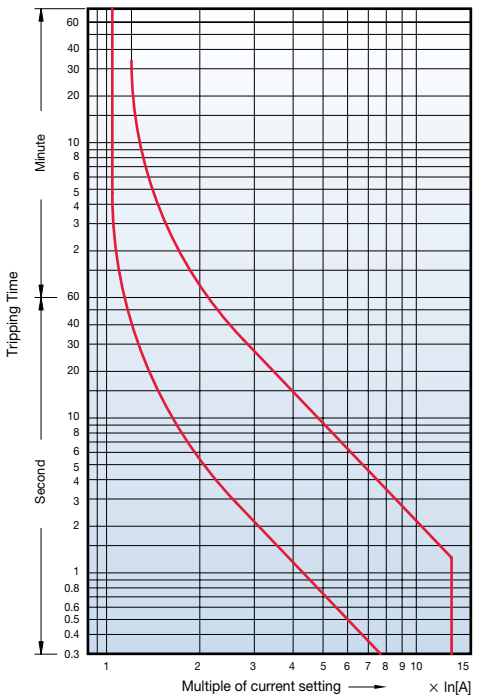
Class 20, 400AF

MT-400/3D

Cold starting



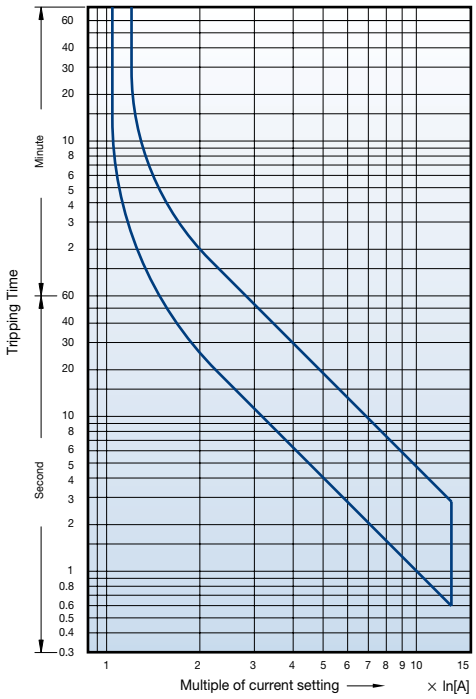
Hot starting



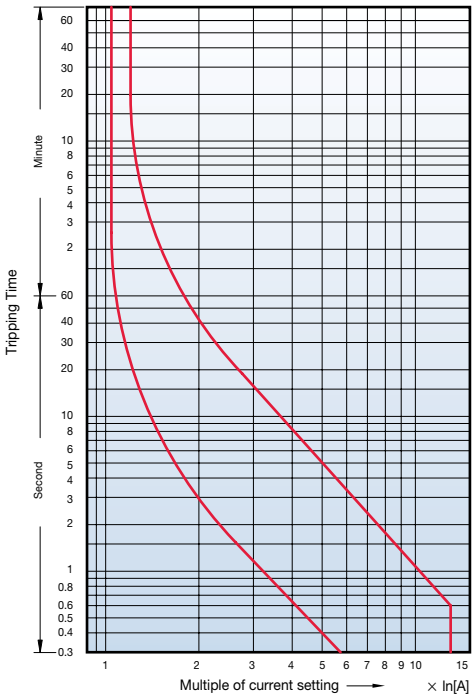
Class 10A, 800AF

MT-800/2H
MT-800/3H
MT-800/3K

Cold starting



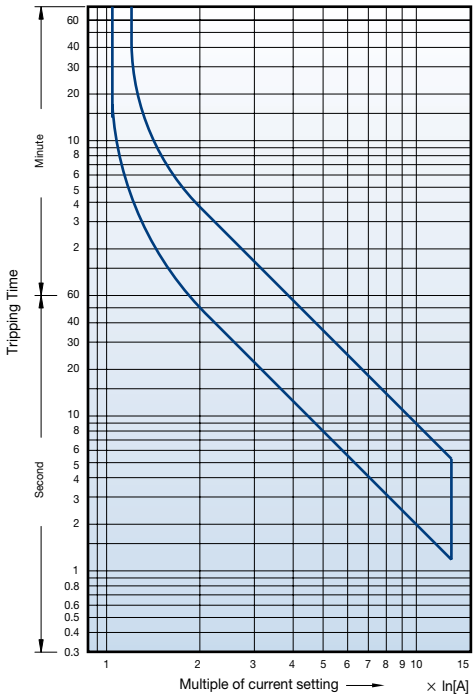
Hot starting



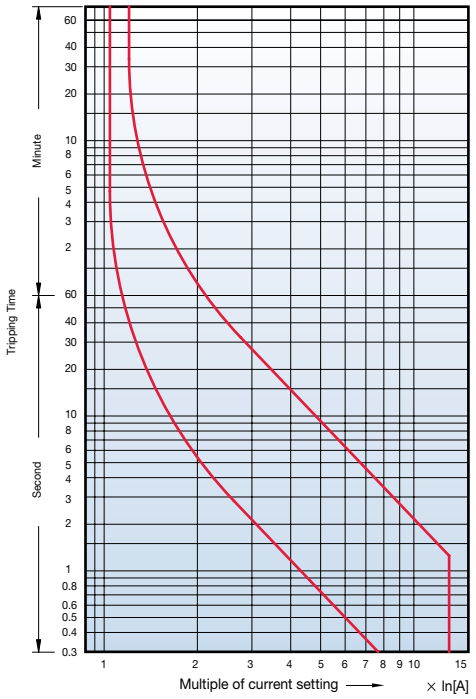
Class 20, 800AF

MT-800/3D

Cold starting



Hot starting



Type 2 Coordination (380/415V)

415V

Susol MCCB

Rated Operational Power				Circuit Breaker		Contactor	Thermal Overload Relays		Short Circuit Breaking Capacity	
kW	Current [A]			Type	Rating [A]	Type	Type	Setting Range [A]	Ir [kA]	Iq [kA]
	380V	400V	415V							
0.06	0.21	0.20	0.19	TS100 (1)	1.6	MC-6a	MT-12	0.16~0.25	1	50
0.09	0.32	0.30	0.29	TS100 (1)	1.6	MC-6a	MT-12	0.25~0.4	1	50
0.12	0.46	0.44	0.42	TS100 (1)	1.6	MC-6a	MT-12	0.4~0.63	1	50
0.18	0.63	0.60	0.58	TS100 (1)	1.6	MC-6a	MT-12	0.4~0.63	1	50
0.25	0.89	0.85	0.82	TS100 (1)	1.6	MC-6a	MT-12	0.63~1	1	50
0.37	1.16	1.10	1.06	TS100 (1)	1.6	MC-12a,12b	MT-12	1~1.6	1	50
0.55	1.6	1.5	1.4	TS100 (1)	1.6	MC-12a,12b	MT-12	1~1.6	1	50
0.75	2.0	1.9	1.8	TS100 (1)	1.6	MC-12a,12b	MT-12	1.6~2.5	1	50
1.1	2.8	2.7	2.6	TS100 (1)	3.2	MC-22b	MT-32	2.5~4	1	50
1.5	3.8	3.6	3.5	TS100 (1)	3.2	MC-22b	MT-32	2.5~4	1	50
2.2	5.2	4.9	4.7	TS100 (1)	3.2	MC-22b	MT-32	4~6	1	50
3.0	6.8	6.5	6.3	TS100 (1)	6.3	MC-40a	MT-32	5~8	1	50
4.0	8.9	8.5	8.2	TS100 (1)	6.3	MC-40a	MT-32	6~9	1	50
5.5	12.1	11.5	11.1	TS100 (1)	12	MC-40a	MT-32	9~13	3	50
7.5	16.3	15.5	14.9	TS100 (1)	12	MC-40a	MT-32	12~18	3	50
11.0	23.2	22.0	21.2	TD100	25	MC-50a	MT-63	18~25	3	70
15.0	31	29	28	TD100	32	MC-50a	MT-63	24~36	3	70
18.5	37	35	34	TD100	40	MC-50a	MT-63	28~40	3	70
22	43	41	40	TD100	50	MC-50a	MT-63	34~50	3	70
30	58	55	53	TD100	63	MC-65a	MT-63	45~65	3	70
37	69	66	64	TD100	80	MC-75a	MT-95	54~75	5	70
45	84	80	77	TD100	100	MC-85a	MT-95	63~85	5	70
55	-	-	93	TD100	100	MC-100a	MT-95	70~95	5	70

Note) (1) This marks admit for motor protection MTU trip unit only.

Type 2 Coordination (440V)

440V

Susol MCCB N-type

Rated Operational Power		Circuit Breaker		Contactor	Thermal Overload Relays		Short Circuit Breaking Capacity	
kW	Current [A]	Type	Rating [A]	Type	Type	Setting Range [A]		
	440V						Ir [kA]	Iq [kA]
1.1	2.37	TS100N	16	MC-12b	MT-32	2.5~4	1	50
1.5	3.06	TS100N	16	MC-18b	MT-32	2.5~4	1	50
2.2	4.42	TS100N	16	MC-22b	MT-32	4~6	1	50
3	5.77	TS100N	16	MC-22b	MT-32	5~8	1	50
4	7.90	TS100N	16	MC-32a	MT-32	6~9	1	50
5.5	10.40	TS100N	16	MC-32a	MT-32	9~13	3	50
7.5	13.70	TS100N	20	MC-32a	MT-32	12~18	3	50
11	20.10	TS100N	32	MC-40a	MT-32	18~25	3	50
15	26.50	TS100N	40	MC-40a	MT-32	22~32	3	50
18.5	32.80	TS100N	40	MC-50a	MT-63	28~40	3	50
22	39.00	TS100N	50	MC-50a	MT-63	34~50	3	50
30	51.50	TS100N	63	MC-65a	MT-63	45~65	3	50
37	64.00	TS100N	80	MC-75a	MT-95	54~75	5	50
45	76.00	TS100N	100	MC-85a	MT-95	63~85	5	50
55	90.00	TS100N	100	MC-100a	MT-95	70~95	5	50
59	97.00	TS160N	100	MC-130a	MT-150	80~105	5	50
75	125	TS160N	160	MC-150a	MT-150	110~150	10	50
90	146	TS160N	160	MC-185a	MT-225	120~185	10	50
110	178	TS250N	200	MC-185a	MT-225	160~240	10	50
132	215	TS250N	250	MC-225a	MT-225	160~240	10	65
160	256	TS400N	300	MC-400a	MT-400	200~330	18	65
200	330	TS400N	400	MC-400a	MT-400	260~400	18	65
220	353	TS400N	400	MC-400a	MT-400	260~400	18	65
250	401	TS630N	500	MC-630a	MT-800	260~400	18	65
300	481	TS630N	500	MC-630a	MT-800	400~630	18	65

Type 2 Coordination (440V)

440V

Susol MCCB H-type

Rated Operational Power		Circuit Breaker		Contactor	Thermal Overload Relays		Short Circuit Breaking Capacity	
kW	Current [A]	Type	Rating [A]	Type	Type	Setting Range [A]	Ir [kA]	Iq [kA]
	440V							
1.1	2.37	TS100H	16	MC-12b	MT-32	2.5~4	1	50
1.5	3.06	TS100H	16	MC-18b	MT-32	2.5~4	1	50
2.2	4.42	TS100H	16	MC-22b	MT-32	4~6	1	50
3	5.77	TS100H	16	MC-22b	MT-32	5~8	1	50
4	7.90	TS100H	16	MC-32a	MT-32	6~9	1	50
5.5	10.40	TS100H	16	MC-32a	MT-32	9~13	3	50
7.5	13.70	TS100H	20	MC-32a	MT-32	12~18	3	70
11	20.10	TS100H	32	MC-40a	MT-32	18~25	3	70
15	26.50	TS100H	40	MC-40a	MT-32	22~32	3	70
18.5	32.80	TS100H	40	MC-50a	MT-63	28~40	3	70
22	39.00	TS100H	50	MC-50a	MT-63	34~50	3	70
30	51.50	TS100H	63	MC-65a	MT-63	45~65	3	70
37	64.00	TS100H	80	MC-75a	MT-95	54~75	5	70
45	76.00	TS100H	100	MC-85a	MT-95	63~85	5	70
55	90.00	TS100H	100	MC-100a	MT-95	70~95	5	70
59	97.00	TS160H	100	MC-130a	MT-150	80~105	5	70
75	125	TS160H	160	MC-150a	MT-150	110~150	10	70
90	146	TS160H	160	MC-185a	MT-225	120~185	10	70
110	178	TS250H	200	MC-185a	MT-225	160~240	10	70
132	215	TS250H	250	MC-225a	MT-225	160~240	10	70
160	256	TS400H	300	MC-400a	MT-400	200~330	18	85
200	330	TS400H	400	MC-400a	MT-400	260~400	18	85
220	353	TS400H	400	MC-400a	MT-400	260~400	18	85
250	401	TS630H	500	MC-630a	MT-800	260~400	18	85
300	481	TS630H	500	MC-630a	MT-800	400~630	18	100

440V

Susol MCCB L-type

Rated Operational Power		Circuit Breaker		Contactor	Thermal Overload Relays		Short Circuit Breaking Capacity	
kW	Current [A]	Type	Rating [A]	Type	Type	Setting Range [A]	Ir [kA]	Iq [kA]
	440V							
1.1	2.37	TS100L	16	MC-12b	MT-32	2.5~4	1	50
1.5	3.06	TS100L	16	MC-18b	MT-32	2.5~4	1	50
2.2	4.42	TS100L	16	MC-22b	MT-32	4~6	1	50
3	5.77	TS100L	16	MC-22b	MT-32	5~8	1	50
4	7.90	TS100L	16	MC-32a	MT-32	6~9	1	50
5.5	10.40	TS100L	16	MC-32a	MT-32	9~13	3	50
7.5	13.70	TS100L	20	MC-32a	MT-32	12~18	3	100
11	20.10	TS100L	32	MC-40a	MT-32	18~25	3	100
15	26.50	TS100L	40	MC-40a	MT-32	22~32	3	100
18.5	32.80	TS100L	40	MC-50a	MT-63	28~40	3	130
22	39.00	TS100L	50	MC-50a	MT-63	34~50	3	130
30	51.50	TS100L	63	MC-65a	MT-63	45~65	3	130
37	64.00	TS100L	80	MC-75a	MT-95	54~75	5	130
45	76.00	TS100L	100	MC-85a	MT-95	63~85	5	130
55	90.00	TS100L	100	MC-100a	MT-95	70~95	5	130
59	97.00	TS160L	100	MC-130a	MT-150	80~105	5	130
75	125	TS160L	160	MC-150a	MT-150	110~150	10	130
90	146	TS160L	160	MC-185a	MT-225	120~185	10	130
110	178	TS250L	200	MC-185a	MT-225	160~240	10	130
132	215	TS250L	250	MC-225a	MT-225	160~240	10	130
160	256	TS400L	300	MC-400a	MT-400	200~330	18	130
200	330	TS400L	400	MC-400a	MT-400	260~400	18	130
220	353	TS400L	400	MC-400a	MT-400	260~400	18	130
250	401	TS630L	500	MC-630a	MT-800	260~400	18	130
300	481	TS630L	500	MC-630a	MT-800	400~630	18	130

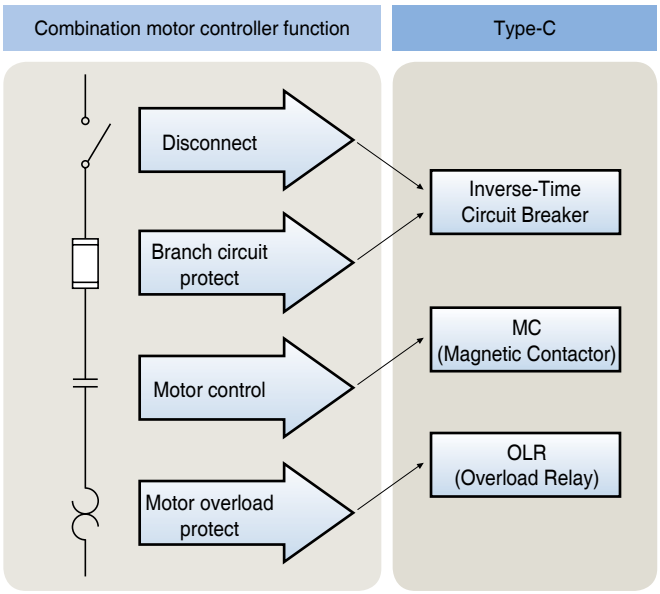
Type 2 Coordination (440V)

440V

Metasol MCCB N-type

Rated Operational Power		Circuit Breaker		Contactor	Thermal Overload Relays		Short Circuit Breaking Capacity	
kW	Current [A]	Type	Rating [A]	Type	Type	Setting Range [A]	Ir [kA]	Iq [kA]
	440V							
1.1	2.37	ABN53c	15	MC-12b	MT-32	2.5~4	1	14
1.5	3.06	ABN53c	15	MC-18b	MT-32	2.5~4	1	14
2.2	4.42	ABN53c	15	MC-22b	MT-32	4~6	1	14
3.0	5.77	ABN53c	15	MC-22b	MT-32	5~8	1	14
4.0	7.90	ABN53c	15	MC-32a	MT-32	6~9	1	14
5.5	10.40	ABN53c	15	MC-32a	MT-32	9~13	3	14
7.5	13.70	ABN53c	20	MC-32a	MT-32	12~18	3	14
11.0	20.10	ABN103c	30	MC-40a	MT-32	18~25	3	18
15.0	26.50	ABN103c	40	MC-40a	MT-32	22~32	3	18
18.5	32.80	ABN103c	40	MC-50a	MT-63	28~40	3	18
22.0	39.00	ABN103c	50	MC-50a	MT-63	34~50	3	18
30.0	51.50	ABN103c	75	MC-65a	MT-63	45~65	3	18
37.0	64.00	ABN103c	75	MC-75a	MT-95	54~75	5	18
45.0	76.00	ABN103c	100	MC-85a	MT-95	63~85	5	18
55.0	90.00	ABN103c	100	MC-100a	MT-95	70~95	5	18
59.0	97.00	ABN103c	100	MC-130a	MT-150	80~105	5	18
75.0	125	ABN203c	150	MC-150a	MT-150	110~150	10	26
90.0	146	ABN203c	200	MC-185a	MT-225	120~185	10	26
110.0	178	ABN203c	200	MC-185a	MT-225	160~240	10	26
132.0	215	ABN203c	250	MC-225a	MT-225	160~240	10	26
160.0	256	ABN403c	300	MC-400a	MT-400	200~330	18	37
200.0	330	ABN403c	400	MC-400a	MT-400	260~400	18	37
220.0	353	ABN403c	400	MC-400a	MT-400	260~400	18	37
250.0	401	ABN803c	500	MC-630a	MT-800	260~400	18	37
300.0	481	ABN803c	500	MC-630a	MT-800	400~630	18	37

Rated UL 508 Combination motor controller type-C



240V

Susol UL MCCB C-type

CB	MC	OLR	kA@240V	Maximum HP	Maximum NEC FLC [A]
UTE100E	MC-6a	MT-12 0.1~10A	50	3	9.6
UTE100N			65	3	
UTE100E	MC-9a	MT-12 0.1~10A	50	3	
UTE100N			65	3	
UTE100E	MC-12a	MT-12 0.1~18A	50	5	15.2
UTE100N			65	5	
UTE100E	MC-18a	MT-12 0.1~18A	50	7.5	22
UTE100N			65	7.5	
UTS150N			65	7.5	
UTS150L			100	7.5	
UTE100E	MC-9b	MT-32 0.1~10A	50	3	9.6
UTE100N			65	3	
UTE100E	MC-12b	MT-32 0.1~18A	50	5	15.2
UTE100N			65	5	
UTE100E	MC-18b	MT-32 0.1~25A	50	7.5	22
UTE100N			65	7.5	
UTS150N			65	7.5	
UTS150L			100	7.5	
UTE100E	MC-22b	MT-32 0.1~32A	50	10	28
UTE100N			65	10	
UTS150N			65	10	
UTS150L			100	10	
UTE100E	MC-32a	MT-32 0.1~32A	50	10	28
UTE100N			65	10	
UTS150N			65	10	
UTS150L			100	10	
UTE100E	MC-40a	MT-32 0.1~40A	50	15	42
UTE100N			65	15	
UTS150N			65	15	
UTS150L			100	15	

Rated UL 508 Combination motor controller type-C

240V

Susol UL MCCB C-type

CB	MC	OLR	kA@240V	Maximum HP	Maximum NEC FLC [A]
UTE100E	MC-6a	MT-12 0.1~10A	50	3	9.6
UTE100N			65	3	
UTE100E	MC-9a	MT-12 0.1~10A	50	3	
UTE100N			65	3	
UTE100E	MC-12a	MT-12 0.1~18A	50	5	15.2
UTE100N			65	5	
UTE100E	MC-18a	MT-12 0.1~18A	50	7.5	22
UTE100N			65	7.5	
UTS150N			65	7.5	
UTS150L			100	7.5	
UTE100E	MC-9b	MT-32 0.1~10A	50	3	9.6
UTE100N			65	3	
UTE100E	MC-12b	MT-32 0.1~18A	50	5	15.2
UTE100N			65	5	
UTE100E	MC-18b	MT-32 0.1~25A	50	7.5	22
UTE100N			65	7.5	
UTS150N			65	7.5	
UTS150L			100	7.5	
UTE100E	MC-22b	MT-32 0.1~32A	50	10	28
UTE100N			65	10	
UTS150N			65	10	
UTS150L			100	10	
UTE100E	MC-32a	MT-32 0.1~32A	50	10	28
UTE100N			65	10	
UTS150N			65	10	
UTS150L			100	10	
UTE100E	MC-40a	MT-32 0.1~40A	50	15	42
UTE100N			65	15	
UTS150N			65	15	
UTS150L			100	15	
UTE100E	MC-50a	MT-63 4~65A	50	25	65
UTE100N			65	25	
UTS150N			65	25	
UTS150L			100	25	
UTE100E	MC-65a	MT-63 4~65A	50	30	80
UTE100N			65	30	
UTS150N			65	30	
UTS150L			100	30	
UTS150N	MC-75a	MT-95 7~95A	65	30	80
UTS150L			100	30	
UTS150N	MC-85a	MT-95 7~100A	65	40	104
UTS150L			100	40	
UTS150N	MC-100a	MT-95 7~100A	65	40	104
UTS150L			100	40	
UTS150N	MC-130a	MT-150 34~130A	65	40	104
UTS150L			100	40	
UTS150N	MC-150a	MT-150 34~130A	65	50	130
UTS150L			100	50	
UTS250N			65	50	
UTS250L			100	50	
UTS250N	MC-185a	MT-225 65~185A	65	60	154
UTS250L			100	60	
UTS400N			65	60	
UTS400L			100	60	
UTS400N	MC-225a	MT-225 65~240A	65	75	192
UTS400L			100	75	
UTS250N			65	75	
UTS250L			100	75	
UTS400N	MC-265a	MT-400 85~330A	65	100	248
UTS400L			100	100	
UTS400N	MC-330a	MT-400 85~330A	65	100	248
UTS400L			100	100	
UTS600N	MC-400a	MT-400 85~400A	65	150	360
UTS600L			100	150	
UTS800L	MC-500a	MT-800 200~600A	100	200	480
UTS800L	MC-630a	MT-800 200~800A	100	250	604

Rated UL 508 Combination motor controller type-C

480V

Susol UL MCCB C-type

CB	MC	OLR	Voltage (V)	Short Circuit Rating (kA) (4)	Maximum HP	Maximum NEC FLC [A]
UTE100E	MC-6a	MT-12 0.1~10A	480	25	5	7.6
UTE100N			480	35	5	
UTE100E	MC-9a	MT-12 0.1~10A	480	25	5	
UTE100N			480	35	5	
UTE100E	MC-12a	MT-12 0.1~13A	480	25	7.5	11
UTE100N			480	35	7.5	
UTE100E	MC-18a	MT-12 0.1~18A	480	25	10	14
UTE100N			480	35	10	
UTS150N			480	35	10	
UTS150H			480	65	10	
UTS150L			480	100	10	
UTE100E	MC-9b	MT-32 0.1~10A	480	25	5	7.6
UTE100N			480	35	5	
UTE100E	MC-12b	MT-32 0.1~13A	480	25	7.5	11
UTE100N			480	35	7.5	
UTE100E	MC-18b	MT-32 0.1~18A	480	25	10	14
UTE100N			480	35	10	
UTE100E	MC-22b	MT-32 0.1~25A	480	25	15	21
UTE100N			480	35	15	
UTS150N			480	35	15	
UTS150H			480	65	15	
UTS150L			480	100	15	
UTE100E	MC-32a	MT-32 0.1~32A	480	25	20	27
UTE100N			480	35	20	
UTS150N			480	35	20	
UTS150H			480	65	20	
UTS150L			480	100	20	
UTE100E	MC-40a	MT-32 0.1~40A	480	25	30	40
UTE100N			480	35	30	
UTS150N			480	35	30	
UTS150H			480	65	30	
UTS150L			480	100	30	
UTE100E	MC-50a	MT-63 4~65A	480	25	40	52
UTE100N			480	35	40	
UTS150N			480	35	40	
UTS150H			480	65	40	
UTS150L			480	100	40	
UTE100E	MC-65a	MT-63 4~65A	480	25	50	65
UTE100N			480	35	50	
UTS150N			480	35	50	
UTS150H			480	65	50	
UTS150L			480	100	50	
UTS150N	MC-75a	MT-95 7~75A	480	35	50	65
UTS150H			480	65	50	
UTS150L			480	100	50	
UTS150N	MC-85a	MT-95 7~85A	480	35	60	77
UTS150H			480	65	60	
UTS150L			480	100	60	
UTS150N	MC-100a	MT-95 7~100A	480	35	75	96
UTS150H			480	65	75	
UTS150L			480	100	75	
UTS150N	MC-130a	MT-150 34~130A	480	35	75	96
UTS150H			480	65	75	
UTS150L			480	100	75	
UTS150N	MC-150a	MT-150 34~130A	480	35	100	124
UTS150H			480	65	100	
UTS150L			480	100	100	
UTS250N			480	35	100	
UTS250H			480	65	100	
UTS250L			480	100	100	
UTS250N	MC-185a	MT-225 65~185A	480	35	125	156
UTS250H			480	65	125	
UTS250L			480	100	125	
UTS400N			480	35	150	
UTS400H			480	65	150	
UTS400L			480	100	150	
UTS250N	MC-225a	MT-225 65~240A	480	35	125	180
UTS250H			480	65	125	
UTS250L			480	100	125	
UTS400N			480	35	150	
UTS400H			480	65	150	
UTS400L			480	100	150	
UTS400N	MC-265a	MT-400 85~330A	480	35	200	240
UTS400H			480	65	200	
UTS400L			480	100	200	
UTS400N	MC-330a	MT-400 85~330A	480	35	200	302
UTS400H			480	65	200	
UTS400L			480	100	200	
UTS600N	MC-400a	MT-400 85~400A	480	35	300	361
UTS600H			480	65	300	
UTS600L			480	100	300	
UTS800H	MC-500a	MT-800 200~600A	480	65	400	477
UTS800L			480	100	400	
UTS800H	MC-630a	MT-800 200~800A	480	65	500	590
UTS800L	MC-630a		480	100	500	

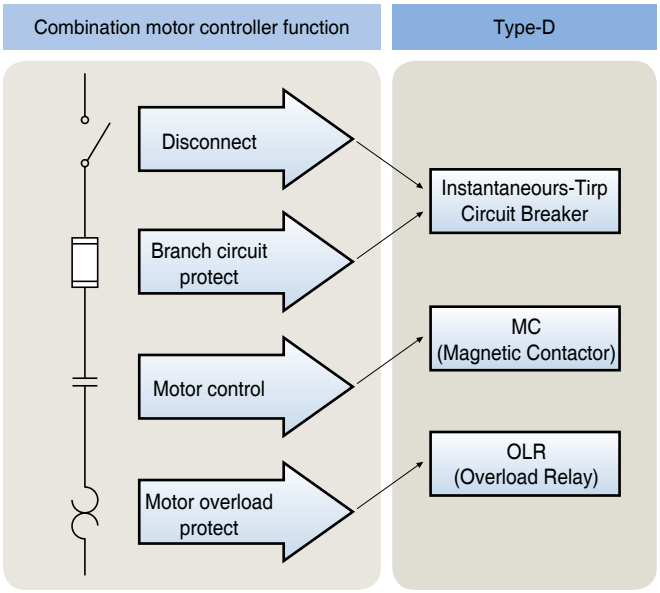
Rated UL 508 Combination motor controller type-C

600V

Susol UL MCCB C-type

CB	MC	OLR	Voltage (V)	Short Circuit Rating (kA) (4)	Maximum HP	Maximum NEC FLC [A]
UTE100E	MC-6a	MT-12 0.1~10A	600Y/347	14	7.5	9
UTE100N			600Y/347	18	7.5	
UTE100E	MC-9a	MT-12 0.1~10A	600Y/347	14	7.5	
UTE100N			600Y/347	18	7.5	
UTE100E	MC-12a	MT-12 0.1~13A	600Y/347	14	10	11
UTE100N			600Y/347	18	10	
UTE100E	MC-18a	MT-12 0.1~18A	600Y/347	14	15	17
UTE100N			600Y/347	18	15	
UTS150N			600	18	15	
UTS150H			600	35	15	
UTS150L			600	50	15	
UTE100E	MC-9b	MT-32 0.1~10A	600Y/347	14	7.5	9
UTE100N			600Y/347	18	7.5	
UTE100E	MC-12b	MT-32 0.1~13A	600Y/347	14	10	11
UTE100N			600Y/347	18	10	
UTE100E	MC-18b	MT-32 0.1~18A	600Y/347	14	15	17
UTE100N			600Y/347	18	15	
UTE100E	MC-22b	MT-32 0.1~25A	600Y/347	14	20	22
UTE100N			600Y/347	18	20	
UTS150N			600	18	20	
UTS150H			600	35	20	
UTS150L			600	50	20	
UTE100E	MC-32a	MT-32 0.1~32A	600Y/347	14	25	27
UTE100N			600Y/347	18	25	
UTS150N			600	18	25	
UTS150H			600	35	25	
UTS150L			600	50	25	
UTE100E	MC-40a	MT-32 0.1~32A	600Y/347	14	30	32
UTE100N			600Y/347	18	30	
UTS150N			600	18	30	
UTS150H			600	35	30	
UTS150L			600	50	30	
UTE100E	MC-50a	MT-63 4~65A	600Y/347	14	50	52
UTE100N			600Y/347	18	50	
UTS150N			600	18	50	
UTS150H			600	35	50	
UTS150L			600	50	50	
UTE100E	MC-65a	MT-63 4~65A	600Y/347	14	60	62
UTE100N			600Y/347	18	60	
UTS150N			600	18	60	
UTS150H			600	35	60	
UTS150L			600	50	60	
UTS150N	MC-75a	MT-95 7~75A	600	18	60	62
UTS150H			600	35	60	
UTS150L			600	50	60	
UTS150N	MC-85a	MT-95 7~85A	600	18	75	77
UTS150H			600	35	75	
UTS150L			600	50	75	
UTS150N	MC-100a	MT-95 7~85A	600	18	75	77
UTS150H			600	35	75	
UTS150L			600	50	75	
UTS150N	MC-130a	MT-150 34~85A	600	18	75	77
UTS150H			600	35	75	
UTS150L			600	50	75	
UTS150N	MC-150a	MT-150 34~85A	600	18	75	77
UTS150H			600	35	75	
UTS150L			600	50	75	
UTS250N	MC-185a	MT-150 65~185A	600	18	125	125
UTS250H			600	35	125	
UTS250L			600	50	125	
UTS400N			600	18	150	
UTS400H			600	35	150	
UTS400L			600	50	150	
UTS250N	MC-225a	MT-225 65~185A	600	18	125	144
UTS250H			600	35	125	
UTS250L			600	50	125	
UTS400N			600	18	150	
UTS400H			600	35	150	
UTS400L			600	50	150	
UTS400N	MC-265a	MT-400 85~240A	600	18	200	192
UTS400H			600	35	200	
UTS400L			600	50	200	
UTS400N	MC-330a	MT-400 85~330A	600	18	200	242
UTS400H			600	35	200	
UTS400L			600	50	200	
UTS600N	MC-400a	MT-400 85~400A	600	18	300	289
UTS600H			600	35	300	
UTS600L			600	50	300	
UTS800H	MC-500a	MT-800 200~400A	600	35	400	382
UTS800L			600	50	400	
UTS800H	MC-630a	MT-800 200~600A	600	35	500	472
UTS800L			600	50	500	

Rated UL 508 Combination motor controller type-D



240V

Susol UL MCCB D-type

CB	MC	OLR	kA@240V	Maximum HP	Maximum NEC FLC [A]
UTS150L MCP	MC-6a	MT-12 0.1~10A	100	3	9.6
UTS150L MCP	MC-9a	MT-12 0.1~10A	100	3	9.6
UTS150L MCP	MC-12a	MT-12 0.1~18A	100	5	15.2
UTS150L MCP	MC-18a	MT-12 0.1~18A	100	7.5	22
UTS150L MCP	MC-9b	MT-32 0.1~10A	100	3	9.6
UTS150L MCP	MC-12b	MT-32 0.1~18A	100	5	15.2
UTS150L MCP	MC-18b	MT-32 0.1~25A	100	7.5	22
UTS150L MCP	MC-22b	MT-32 0.1~32A	100	10	28
UTS150L MCP	MC-32a	MT-32 0.1~32A	100	10	28
UTS150L MCP	MC-40a	MT-32 0.1~40A	100	15	42
UTS150L MCP	MC-50a	MT-63 4~65A	100	25	65
UTS150L MCP	MC-65a	MT-63 4~65A	100	30	80
UTS150L MCP	MC-75a	MT-95 7~95A	100	30	80
UTS150L MCP	MC-85a	MT-95 7~100A	100	40	104
UTS150L MCP	MC-100a	MT-95 7~100A	100	40	104
UTS150L MCP	MC-130a	MT-150 34~130A	100	40	104
UTS250L MCP	MC-150a	MT-150 34~130A	100	50	130
UTS250L MCP	MC-185a	MT-225 65~160A	100	60	154
UTS400L MCP	MC-185a	MT-225 65~185A	100	60	154
UTS250L MCP	MC-225a	MT-225 65~240A	100	75	192
UTS400L MCP	MC-225a	MT-225 65~240A	100	75	192
UTS400L MCP	MC-265a	MT-400 85~330A	100	100	248
UTS400L MCP	MC-330a	MT-400 85~330A	100	100	248
UTS600L MCP	MC-400a	MT-400 85~400A	100	150	360
UTS800L MCP	MC-500a	MT-800 200~600A	100	200	480
UTS800L MCP	MC-630a	MT-800 200~800A	100	250	604

Rated UL 508 Combination motor controller type-D

480V

Susol UL MCCB D-type

CB	MC	OLR	kA@400V	Maximum HP	Maximum NEC FLC [A]
UTS150L MCP	MC-6a	MT-12 0.1~10A	65	5	7.6
UTS150L MCP	MC-9a	MT-12 0.1~10A	65	5	7.6
UTS150L MCP	MC-12a	MT-12 0.1~13A	65	7.5	11
UTS150L MCP	MC-18a	MT-12 0.1~18A	65	10	14
UTS150L MCP	MC-9b	MT-32 0.1~10A	65	5	7.6
UTS150L MCP	MC-12b	MT-32 0.1~13A	65	7.5	11
UTS150L MCP	MC-18b	MT-32 0.1~18A	65	10	14
UTS150L MCP	MC-22b	MT-32 0.1~25A	65	15	21
UTS150L MCP	MC-32a	MT-32 0.1~32A	65	20	27
UTS150L MCP	MC-40a	MT-32 0.1~40A	65	20	40
UTS150H MCP	MC-50a	MT-63 4~65A	65	40	52
UTS150L MCP	MC-50a	MT-63 4~65A	100	40	52
UTS150H MCP	MC-65a	MT-63 4~65A	65	50	65
UTS150L MCP	MC-65a	MT-63 4~65A	100	50	65
UTS150H MCP	MC-75a	MT-95 7~75A	65	50	65
UTS150L MCP	MC-75a	MT-95 7~75A	100	50	65
UTS150H MCP	MC-85a	MT-95 7~85A	65	60	77
UTS150L MCP	MC-85a	MT-95 7~85A	100	60	77
UTS150H MCP	MC-100a	MT-95 7~100A	65	75	96
UTS150L MCP	MC-100a	MT-95 7~100A	100	75	96
UTS150H MCP	MC-130a	MT-150 34~130A	65	75	96
UTS150L MCP	MC-130a	MT-150 34~130A	100	75	96
UTS250H MCP	MC-150a	MT-150 34~130A	65	100	124
UTS250L MCP	MC-150a	MT-150 34~130A	100	100	124
UTS250H MCP	MC-185a	MT-225 65~185A	65	125	156
UTS250L MCP	MC-185a	MT-225 65~185A	85	125	156
UTS400H MCP	MC-185a	MT-225 65~185A	65	125	156
UTS400L MCP	MC-185a	MT-225 65~185A	100	125	156
UTS250H MCP	MC-225a	MT-225 65~240A	65	150	180
UTS250L MCP	MC-225a	MT-225 65~240A	85	150	180
UTS400H MCP	MC-225a	MT-225 65~240A	65	150	180
UTS400L MCP	MC-225a	MT-225 65~240A	100	150	180
UTS400H MCP	MC-265a	MT-400 85~330A	65	200	240
UTS400L MCP	MC-265a	MT-400 85~330A	100	200	240
UTS400H MCP	MC-330a	MT-400 85~330A	65	200	240
UTS400L MCP	MC-330a	MT-400 85~330A	100	200	240
UTS600H MCP	MC-400a	MT-400 85~400A	65	300	361
UTS600L MCP	MC-400a	MT-400 85~400A	100	300	361
UTS800L MCP	MC-500a	MT-800 200~600A	100	400	477
UTS800L MCP	MC-630a	MT-800 200~800A	100	500	590

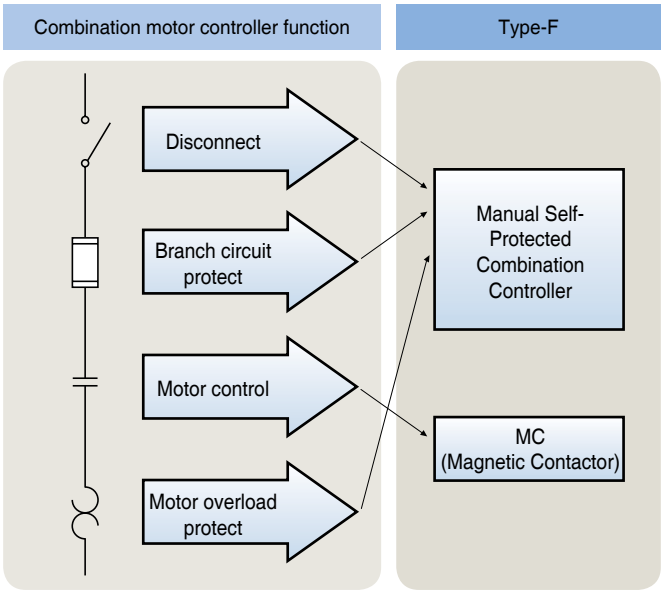
Rated UL 508 Combination motor controller type-D

600V

Susol UL MCCB D-type

CB	MC	OLR	kA@600V	Maximum HP	Maximum NEC FLC [A]
UTS150L MCP	MC-6a	MT-12 0.1~10A	25	7.5	9
UTS150L MCP	MC-9a	MT-12 0.1~10A	25	7.5	9
UTS150L MCP	MC-12a	MT-12 0.1~13A	25	10	11
UTS150L MCP	MC-18a	MT-12 0.1~18A	25	15	17
UTS150L MCP	MC-9b	MT-32 0.1~10A	25	7.5	9
UTS150L MCP	MC-12b	MT-32 0.1~13A	25	10	11
UTS150L MCP	MC-18b	MT-32 0.1~18A	25	15	17
UTS150L MCP	MC-22b	MT-32 0.1~25A	25	20	22
UTS150L MCP	MC-32a	MT-32 0.1~32A	25	25	27
UTS150L MCP	MC-40a	MT-32 0.1~32A	25	25	32
UTS150H MCP	MC-50a	MT-63 4~65A	25	50	52
UTS150L MCP	MC-50a	MT-63 4~65A	50	50	62
UTS150H MCP	MC-65a	MT-63 4~65A	25	60	62
UTS150L MCP	MC-65a	MT-63 4~65A	50	60	62
UTS150H MCP	MC-75a	MT-95 7~75A	25	60	77
UTS150L MCP	MC-75a	MT-95 7~75A	50	60	77
UTS150H MCP	MC-85a	MT-95 7~85A	25	75	77
UTS150L MCP	MC-85a	MT-95 7~85A	50	75	77
UTS150H MCP	MC-100a	MT-95 7~85A	25	75	77
UTS150L MCP	MC-100a	MT-95 7~85A	50	75	77
UTS150H MCP	MC-130a	MT-150 65~185A	25	75	77
UTS150L MCP	MC-130a	MT-150 65~185A	50	75	77
UTS150H MCP	MC-150a	MT-150 65~185A	25	75	77
UTS150L MCP	MC-150a	MT-150 65~185A	50	75	77
UTS250H MCP	MC-185a	MT-225 65~185A	25	125	125
UTS250L MCP	MC-185a	MT-225 65~185A	50	125	125
UTS400H MCP	MC-185a	MT-225 65~185A	25	125	125
UTS400L MCP	MC-185a	MT-225 65~185A	50	125	125
UTS250H MCP	MC-225a	MT-225 65~185A	25	150	144
UTS250L MCP	MC-225a	MT-225 65~185A	50	150	144
UTS400H MCP	MC-225a	MT-225 65~185A	25	150	144
UTS400L MCP	MC-225a	MT-225 65~185A	50	150	144
UTS400H MCP	MC-265a	MT-400 85~240A	25	200	192
UTS400L MCP	MC-265a	MT-400 85~240A	50	200	192
UTS400H MCP	MC-330a	MT-400 85~330A	25	200	242
UTS400L MCP	MC-330a	MT-400 85~330A	50	200	242
UTS600H MCP	MC-400a	MT-400 85~400A	25	300	289
UTS600L MCP	MC-400a	MT-400 85~400A	50	300	289
UTS800L MCP	MC-500a	MT-800 200~400A	50	400	382
UTS800L MCP	MC-630a	MT-800 200~600A	50	500	472

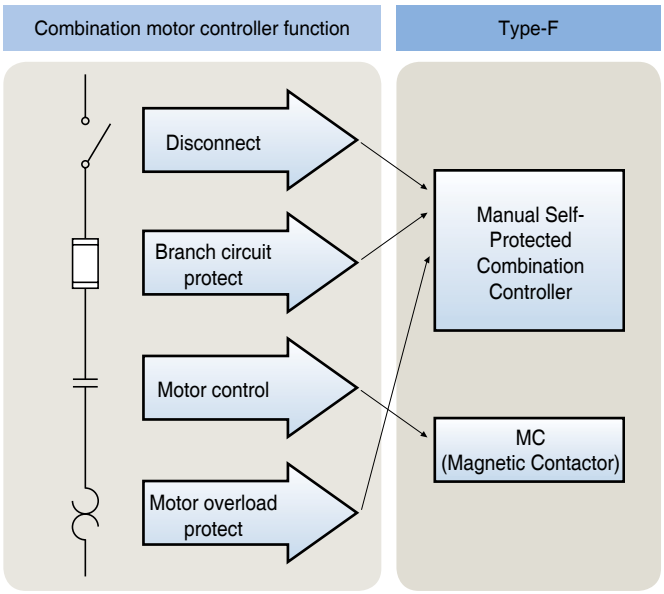
Rated UL 508 combination motor controller Type-F



32AF Rotary operation, High interruption

Manual Self-Protected	Adjustment range (A)	Short circuit rating (kA)			MC (Applied contactor)
		240V	480Y/277V	600Y/347V	Metasol
MMS-32H-0.16	0.1~0.16	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-0.25	0.16~0.25	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-0.4	0.25~0.4	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-0.63	0.4~0.63	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-1	0.63~1	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-1.6	1~1.6	100	65	25	MC-6a/MC-9a/MC-9b
MMS-32H-2.5	1.6~2.5	100	65	25	MC-12a/MC-12b
MMS-32H-4	2.5~4	100	65	25	MC-12a/MC-12b
MMS-32H-6	4~6	100	65	25	MC-18a/MC-18b
MMS-32H-8	5~8	100	65	10	MC-18a/MC-18b
MMS-32H-10	6~10	100	65	10	MC-18a/MC-18b
MMS-32H-13	9~13	100	65	10	MC-22b/MC-25a
MMS-32H-17	11~17	100	30	10	MC-22b/MC-25a
MMS-32H-22	14~22	100	30	10	MC-22b/MC-25a
MMS-32H-26	18~26	100	30	10	MC-32a
MMS-32H-32	22~32	100	30	10	MC-32a
MMS-32H-40	28~40	100	30	10	MC-35a/MC-40a

Rated UL 508 combination motor controller Type-F



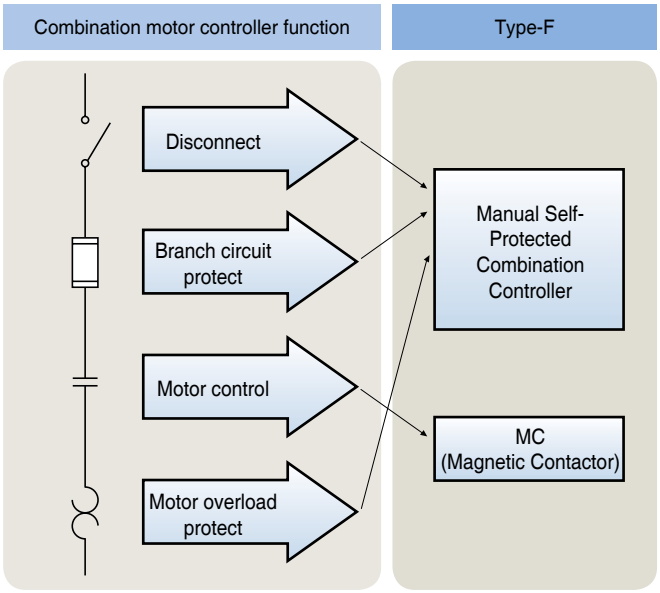
63AF Rotary operation, standard interruption

Manual Self-Protected	Adjustment range (A)	Short circuit rating (kA)			MC (Applied contactor)
		240V	480Y/277V	600Y/347V	Metasol
MMS-63S-10	6~10	100	50	10	MC-50a
MMS-63S-13	9~13	100	50	10	MC-50a
MMS-63S-17	11~17	100	40	10	MC-50a
MMS-63S-22	14~22	100	40	10	MC-50a
MMS-63S-26	18~26	100	40	10	MC-50a
MMS-63S-32	22~32	100	40	10	MC-50a
MMS-63S-40	28~40	100	40	10	MC-50a
MMS-63S-40	28~40	100	40	10	MC-50a
MMS-63S-50	34~50	100	40	10	MC-50a
MMS-63S-63	45~63	100	40	10	MC-65a
MMS-63S-65	47~65	100	40	10	MC-65a

63AF Rotary operation, High interruption

Manual Self-Protected	Adjustment range (A)	Short circuit rating (kA)			MC (Applied contactor)
		240V	480Y/277V	600Y/347V	Metasol
MMS-63H-10	6~10	100	65	25	MC-50a
MMS-63H-13	9~13	100	65	25	MC-50a
MMS-63H-17	11~17	100	50	10	MC-50a
MMS-63H-22	14~22	100	50	10	MC-50a
MMS-63H-26	18~26	100	50	10	MC-50a
MMS-63H-32	22~32	100	50	10	MC-50a
MMS-63H-40	28~40	100	50	10	MC-50a
MMS-63H-50	34~50	100	50	10	MC-50a
MMS-63H-63	45~63	100	50	10	MC-50a
MMS-63H-65	47~65	100	40	10	MC-65a

Rated UL 508 combination motor controller Type-F



100AF Rotary operation, Standard interruption

Manual Self-Protected	Adjustment range (A)	Short circuit rating (kA)			MC (Applied contactor)
		240V	480Y/277V	600Y/347V	Metasol
MMS-100S-17	11~17	100	50	10	MC-75a
MMS-100S-22	14~22	100	50	10	MC-75a
MMS-100S-26	18~26	100	50	10	MC-75a
MMS-100S-32	22~32	100	50	10	MC-75a
MMS-100S-40	28~40	100	50	10	MC-75a
MMS-100S-50	34~50	100	50	10	MC-75a
MMS-100S-63	45~63	100	40	10	MC-75a
MMS-100S-75	55~75	100	40	10	MC-75a
MMS-100S-90	70~90	100	40	10	MC-85a
MMS-100S-100	80~100	100	40	10	MC-100a

100AF Rotary operation, High interruption

Manual Self-Protected	Adjustment range (A)	Short circuit rating (kA)			MC (Applied contactor)
		240V	480Y/277V	600Y/347V	Metasol
MMS-100H-17	11~17	100	65	25	MC-75a
MMS-100H-22	14~22	100	65	20	MC-75a
MMS-100H-26	18~26	100	65	20	MC-75a
MMS-100H-32	22~32	100	65	20	MC-75a
MMS-100H-40	28~40	100	65	20	MC-75a
MMS-100H-50	34~50	100	65	20	MC-75a
MMS-100H-63	45~63	100	50	10	MC-75a
MMS-100H-75	55~75	100	50	10	MC-75a
MMS-100H-90	70~90	100	50	10	MC-85a
MMS-100H-100	80~100	100	50	10	MC-100a

Ready for IE3

LS contactor ready for IE3-rated motors



What is the IE3?

Motor is rated from IE1 through IE4 depends on its energy efficiency which means IE3 is more efficient compared to IE1 and IE2-rated motors. IE3-rated motor is not only more efficient than IE2-rated motor but it also has lower rated current and higher starting current & inrush current.



WHY IE3?

As part of the EU Enviromental policy "20/30-20-20" strategy, the EU is committed to achieve the followings by 2020

- Reduce greenhouse gas emissions by up to 30 %
- Increase the share of renewable energy to 20 % and
- Boost energy efficiency by 20% in general.IEC 60034-30 standard specifies IE classes for motors in accordance with the above requirements.
- IEC 60034-30-1 standard
  - IE1 Standard Efficiency
  - IE2 High Efficiency
  - IE3 Premium Efficiency
  - IE4 Super Premium Efficiency





IE3 motors can be associated with standard starter solutions, like contactor, manual motor starter.
During the starting the IE3 motor, Inrush and starting current are increased compared with IE2 motor.
The increased inrush and starting current can affect the selection of the electrical device.
In order to secure the highest efficiency in your electrical system, it is strongly recommended to use the more efficient IE3 class product.
LSIS contactors are already compliant with IE3.





Ready for IE3






Frame size			18AF				22AF				40AF		65AF	
Type			MC-6a	MC-9a	MC-12a	MC-18a	MC-9b	MC-12b	MC-18b	MC-22b	MC-32a	MC-40a	MC-50a	MC-65a
AC-3	200/240V	kW	1.1	1.5	2	2.5	1.5	2	2.5	3	4.5	5.5	7.5	9
		A	5	6	7.5	10	6	7.5	10	12.5	18	23	31.5	37
	380/440V	kW	2	2.2	3.5	5	2.2	3.5	5	6.5	9	11	15	18.5
		A	4	5	7	10	5	7	10	12.5	18	23	28.5	37
	500/550V	kW	2	2.2	4	4.5	2.2	4	4.5	7.5	10	11	15	22
		A	3.5	4	7	7.5	4	7	7.5	11.5	16	18	24.5	34
	690V	kW	2	2.2	4	4	3	4	4	9	10	11	15	18.5
		A	2.3	3	5	5	3.5	5	5	10	11.5	13	16	20
	1000V	kW	-	-	-	-	-	-	-	-	11	11	16	16
		A	-	-	-	-	-	-	-	-	9.7	9.7	13	13

*Thermal Overload Relay" Products are available regardless of de-rating






Frame size			100AF			150AF		225AF		400AF		
Type			MC-75a	MC-85a	MC-100a	MC-130a	MC-150a	MC-185a	MC-225a	MC-265a	MC-330a	MC-400a
AC-3	200/240V	kW	11	15	18.5	22	25	30	37	45	55	75
		A	43	49	60	74	86	106	126	151	189	229
	380/440V	kW	22	25	30	40	45	55	75	85	110	132
		A	43	49	60	74	86	106	129	151	189	229
	500/550V	kW	27	30	35	37	40	70	85	90	100	140
		A	37	43	49	51	57	103	114	129	143	200
	690V	kW	22	25	30	30	30	65	80	100	130	160
		A	24	26	37	34	34	69	86	106	126	171
	1000V	kW	20	20	20	37	37	70	70	80	80	80
		A	16	16	16	30	30	51	51	60	60	60

*Thermal Overload Relay" Products are available regardless of de-rating




Frame size			800AF			2100AF		
Type			MC-500a	MC-630a	MC-800a	MC-1400a	MC-1700a	MC-2100a
AC-3	200/240V	kW	80	110	132	147	180	-
		A	286	360	457	491	600	-
	380/440V	kW	160	200	250	265	355	500
		A	286	360	457	491	600	829
	500/550V	kW	160	200	280	-	-	-
		A	229	286	411	-	-	-
	690V	kW	200	230	300	450	500	-
		A	217	240	360	491	543	-
	1000V	kW	160	160	160	-	-	-
		A	126	126	126	-	-	-

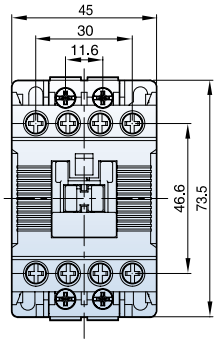
*Thermal Overload Relay" Products are available regardless of de-rating

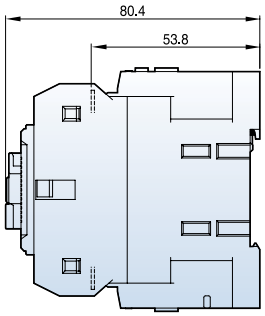
3-pole contactors dimension (18AF/22AF)

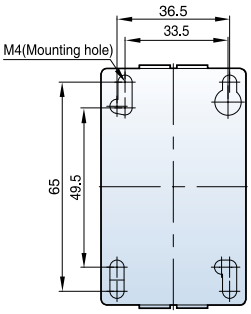
[mm]

MC-6a~18a AC

1NO or 1NC



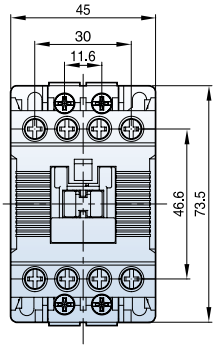


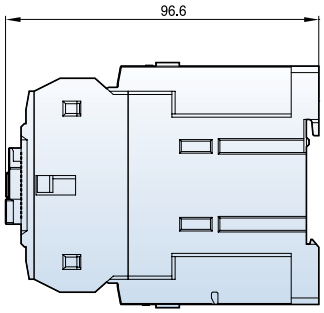


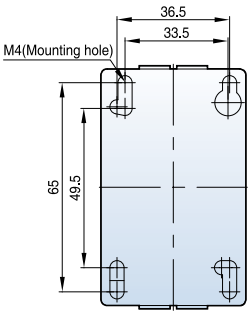
0.3kg

MC-6a~18a DC

1NO or 1NC



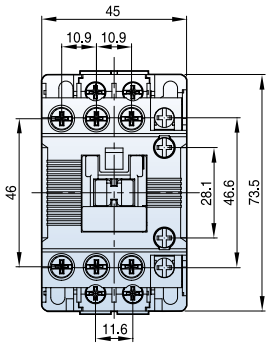


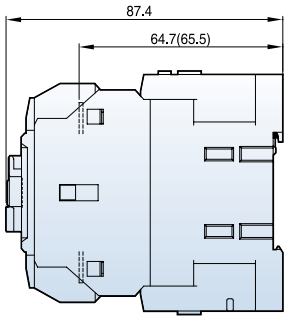


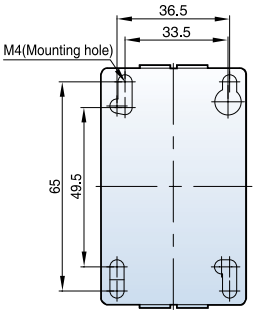
0.5kg

MC-9b~22b AC

1NO1NC





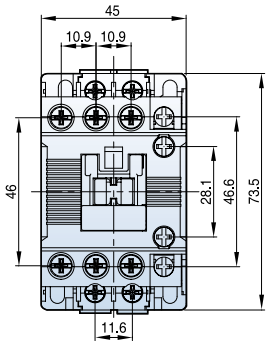


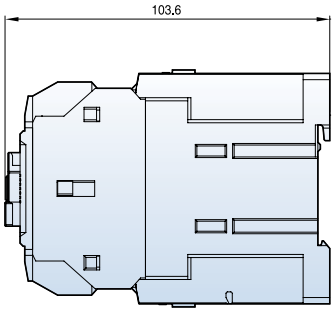
0.3kg

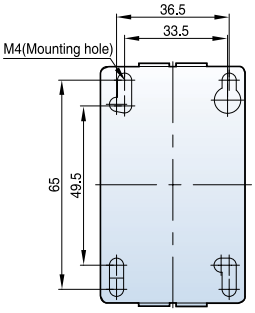
* ( ) MC-18b, 22b

MC-9b~22b DC

1NO1NC





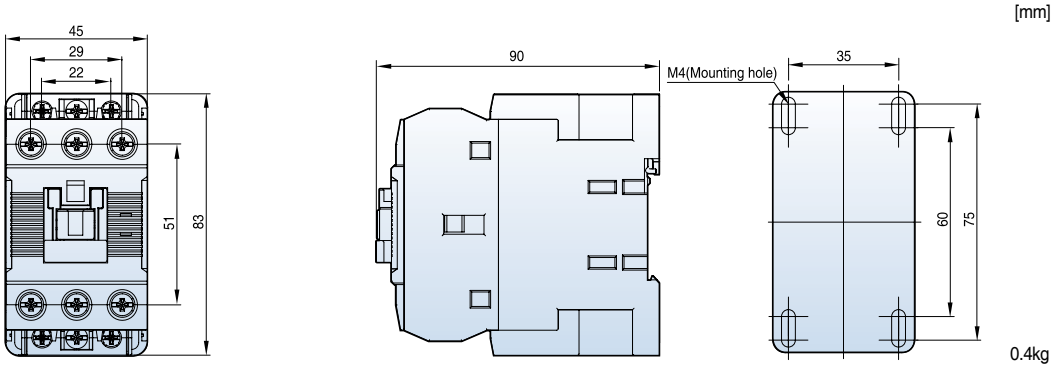


0.5kg

3-pole contactors dimension (40AF)

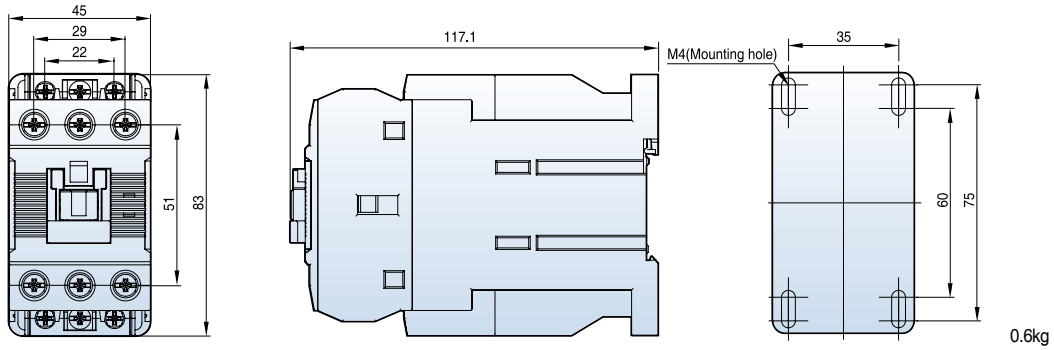
[mm]

MC-32a~40a AC



0.4kg

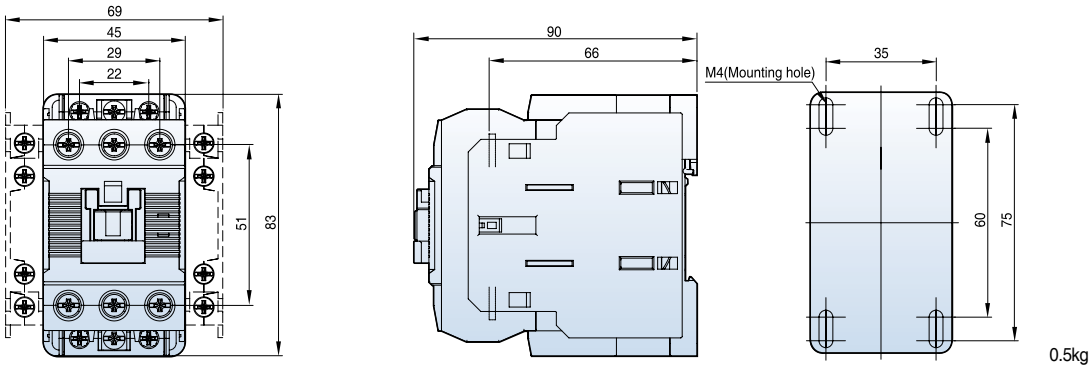
MC-32a~40a DC



0.6kg

MC-32a~40a AC

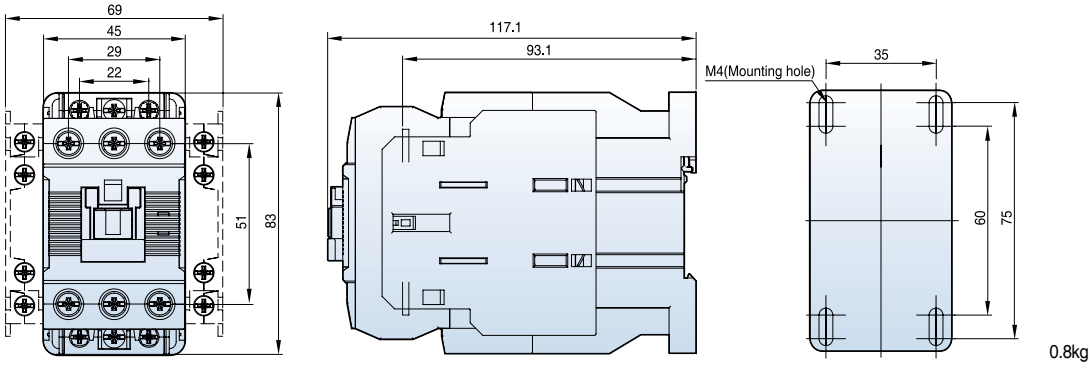
2NO2NC



0.5kg

MC-32a~40a DC

2NO2NC

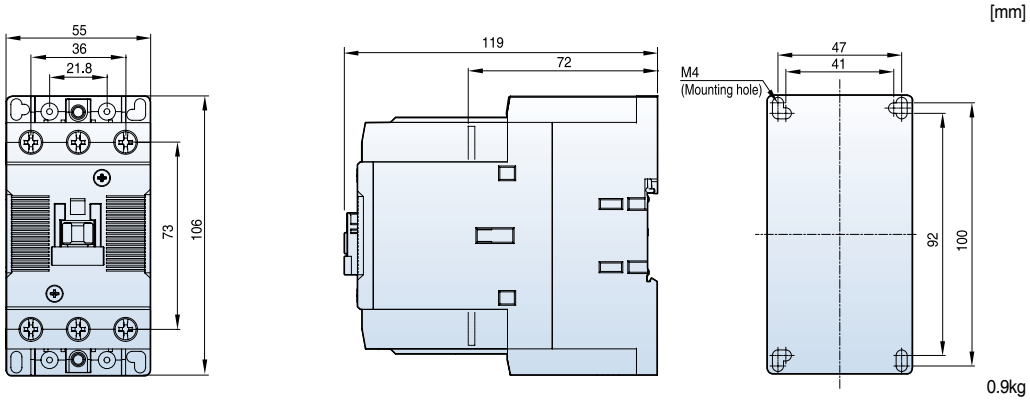


0.8kg

3-pole contactors dimension (65AF)

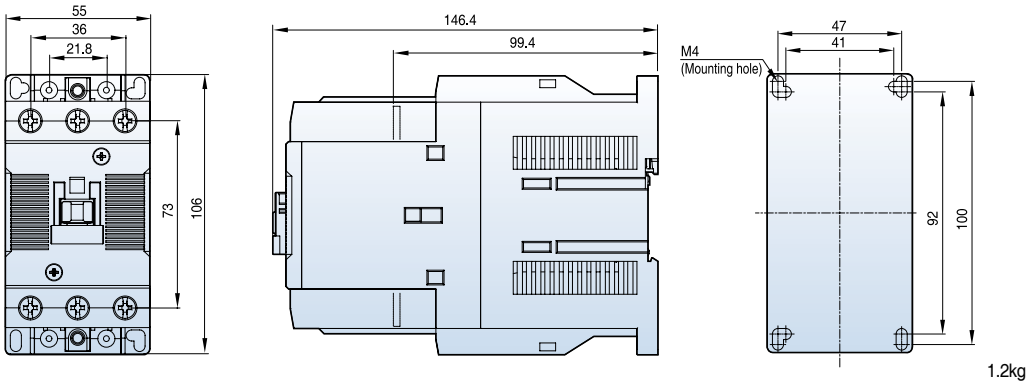
[mm]

MC-50a~65a AC



0.9kg

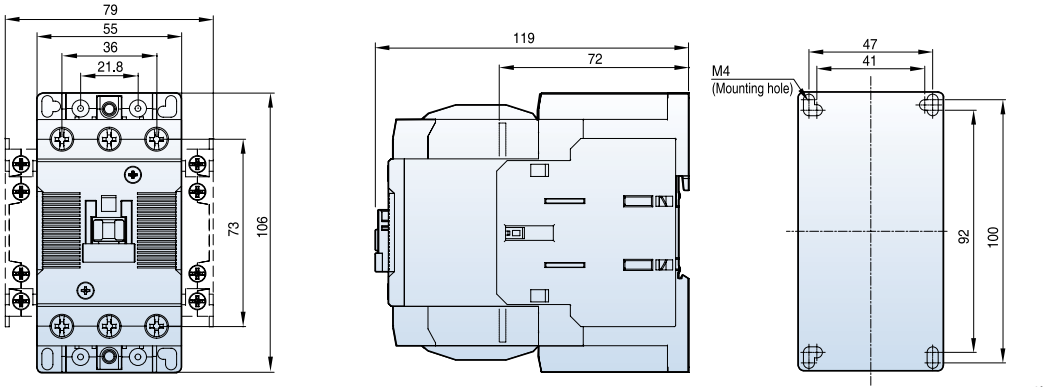
MC-50a~65a DC



1.2kg

MC-50a~65a AC

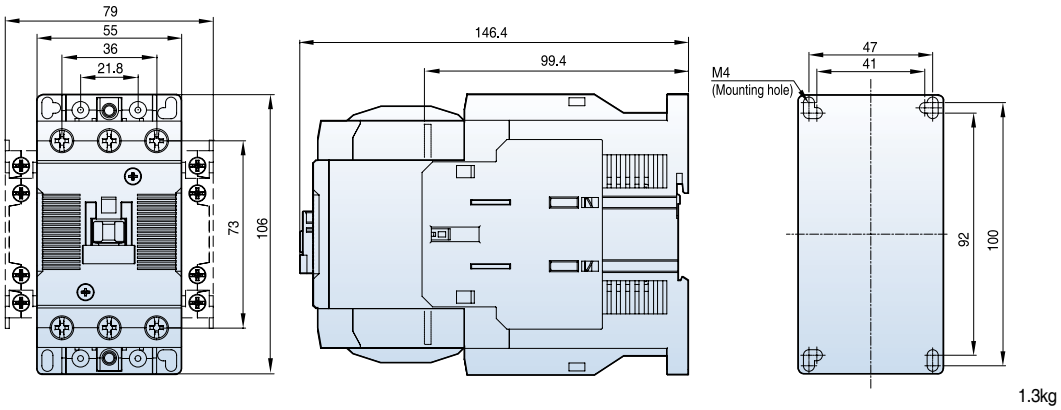
2NO2NC



1kg

MC-50a~65a DC

2NO2NC

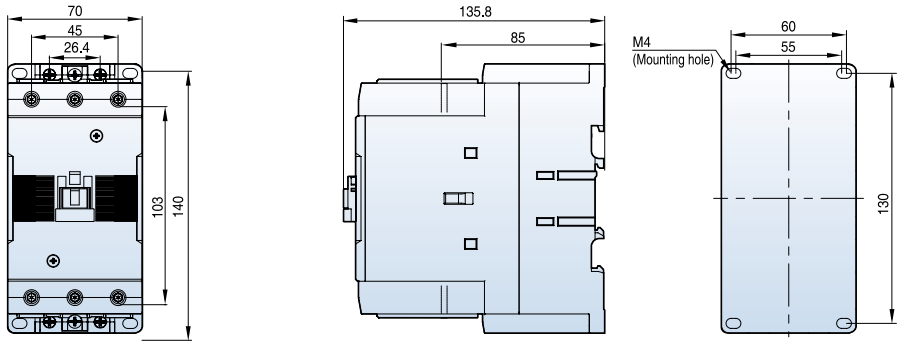


1.3kg

3-pole contactors dimension (100AF)

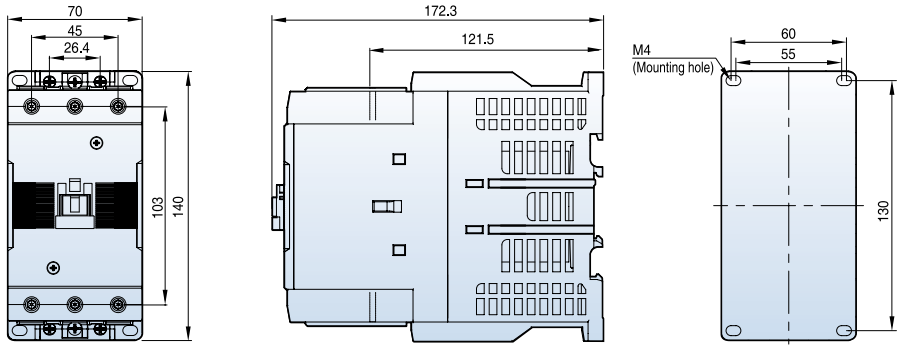
[mm]

MC-75a~100a AC



1.8kg

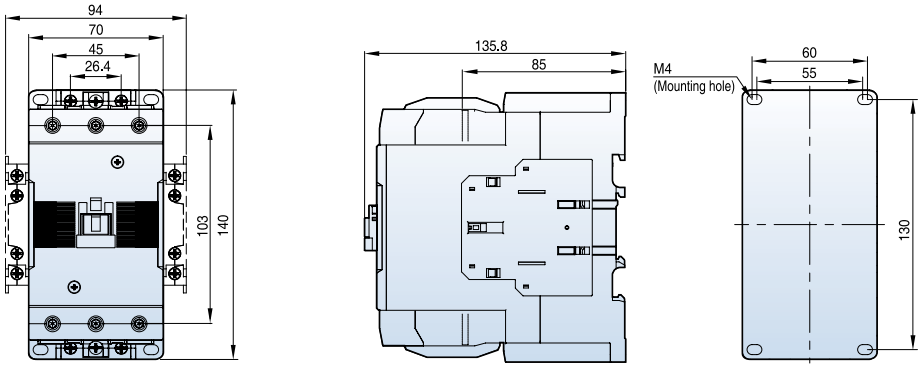
MC-75a~100a DC



2.7kg

MC-75a~100a AC

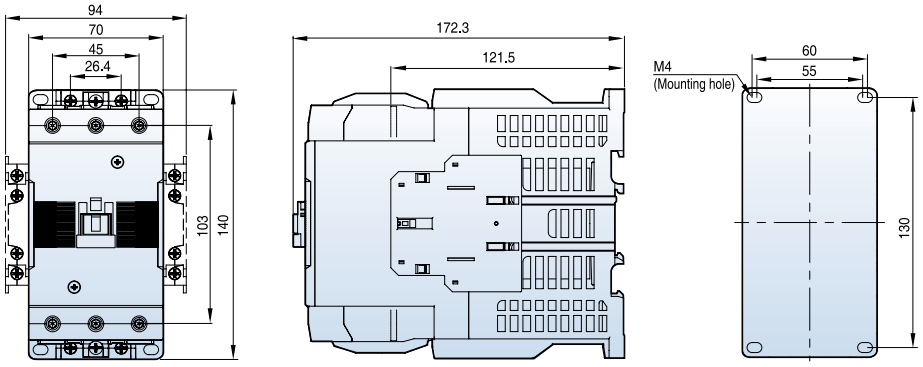
2NO2NC



1.9kg

MC-75a~100a DC

2NO2NC

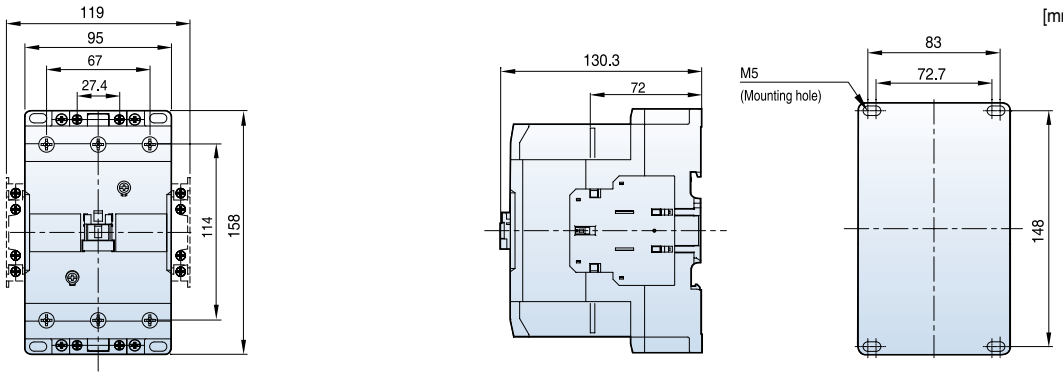


2.8kg

3-pole contactors dimension (150AF~400AF)

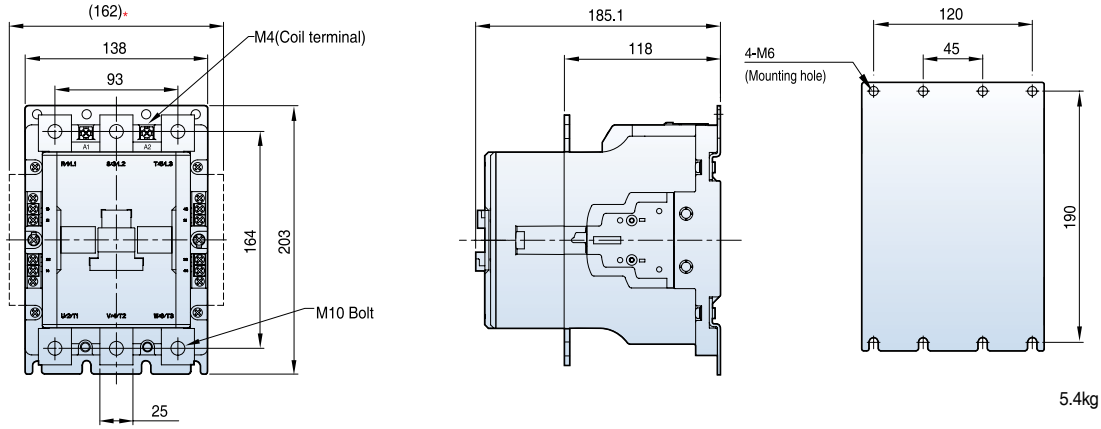
[mm]

MC-130a
MC-150a



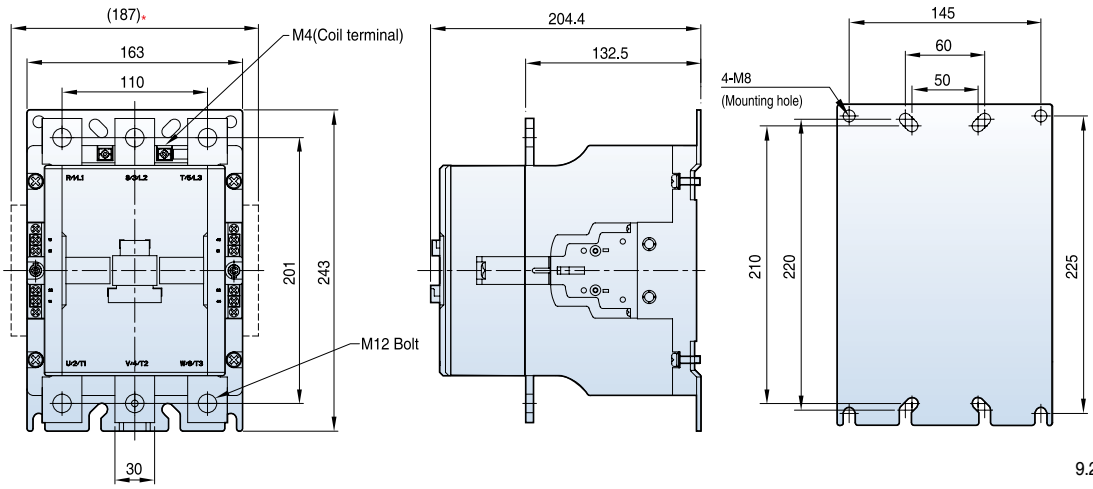
2.4kg

MC-185a
MC-225a



5.4kg

MC-265a
MC-330a
MC-400a



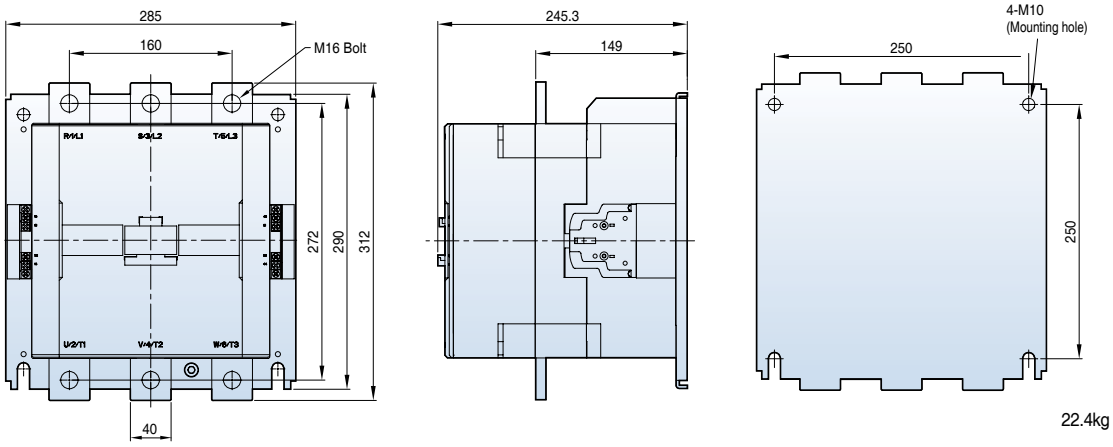
9.2kg

* Optional auxiliary contact unit (side mount type)

3-pole contactors dimension (800AF~2100AF)

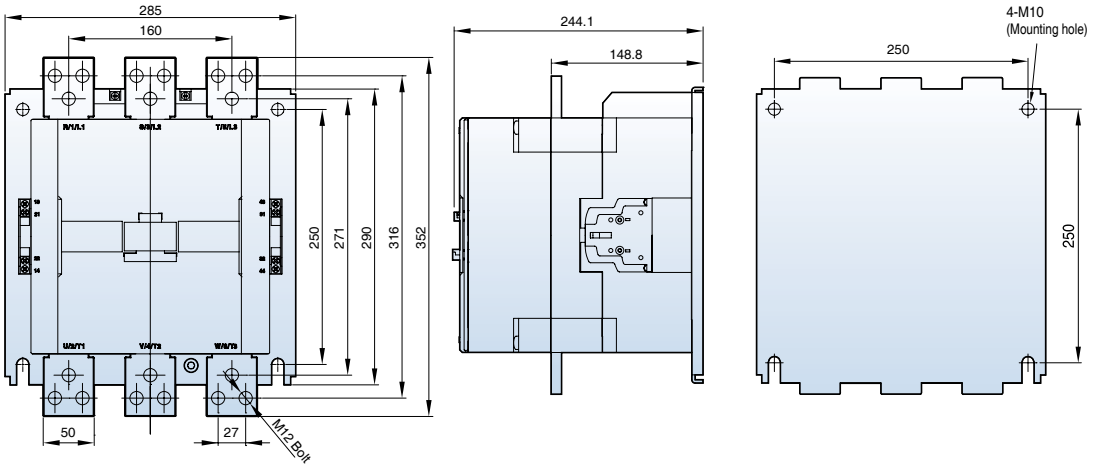
[mm]

MC-500a
MC-630a
MC-800a

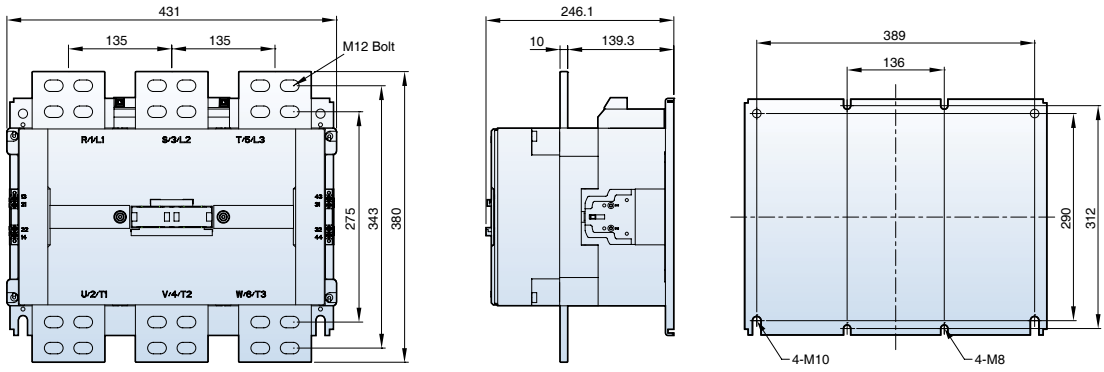


22.4kg

MC-1260a



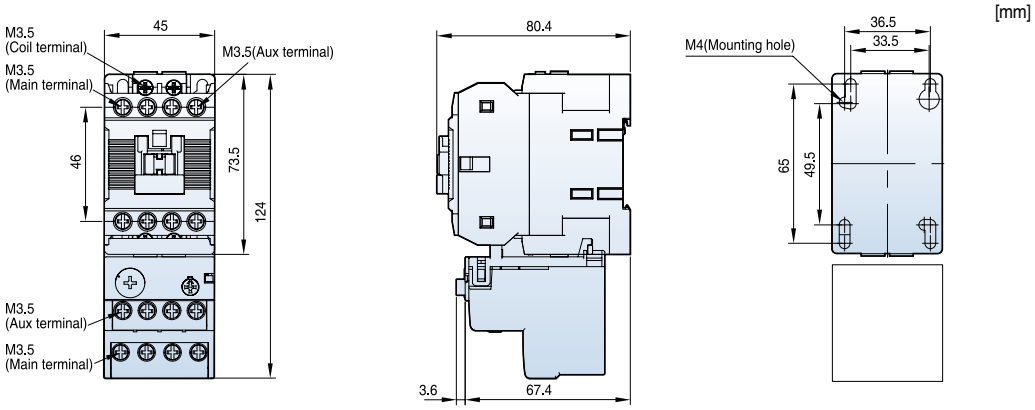
MC-1400a
MC-1700a
MC-2100a



Motor starters dimension (18AF~40AF)

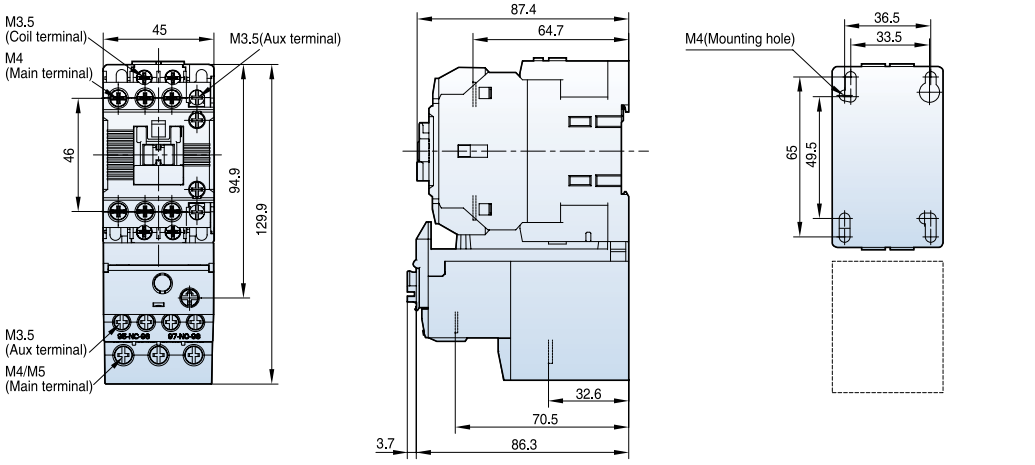
[mm]

MS-6a~18a



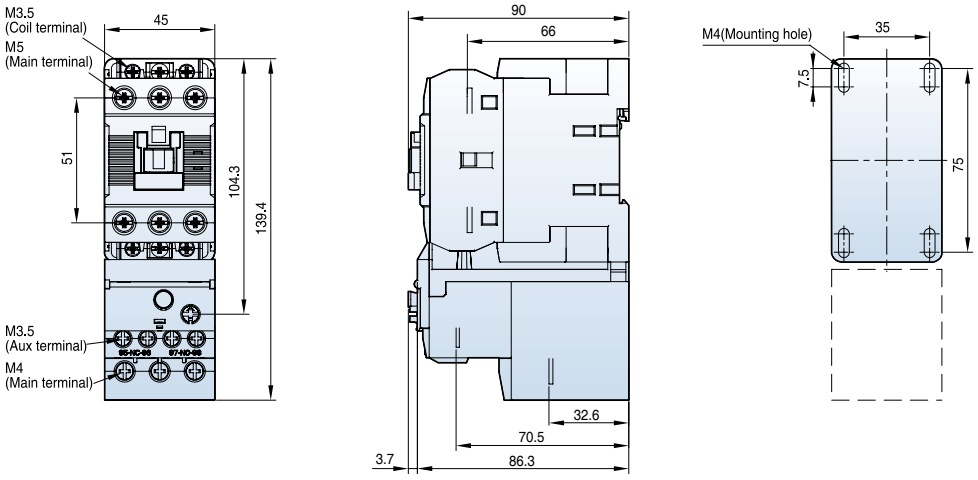
0.5kg

MS-9b~22b



0.5kg

MS-32a~40a

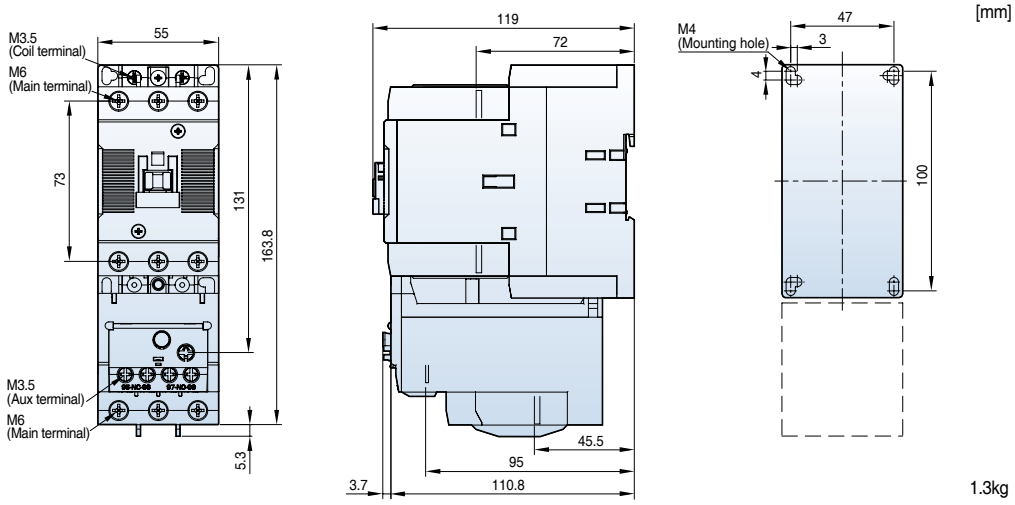


0.6kg

Motor starters dimension (65AF~150AF)

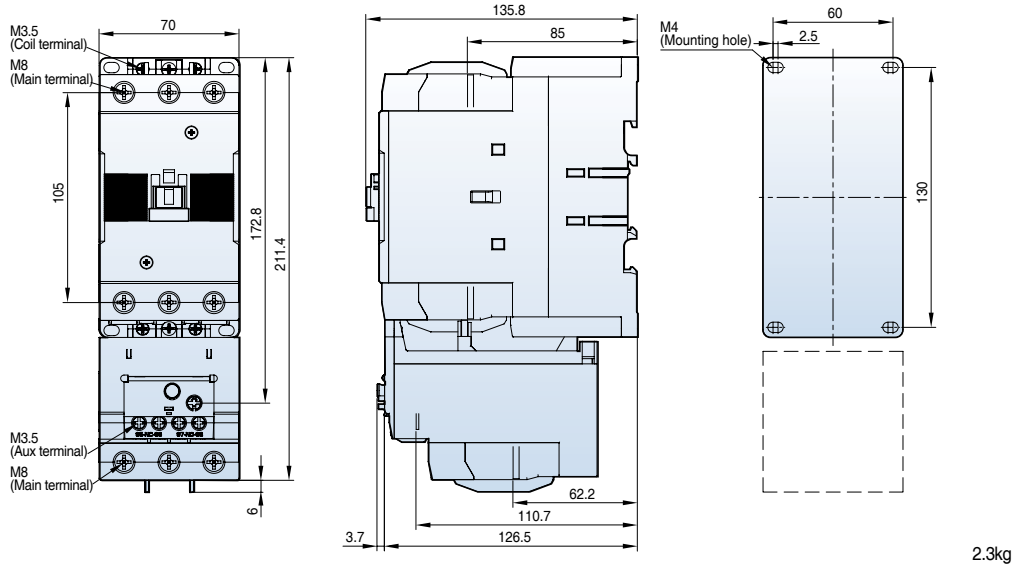
[mm]

MS-50a~65a



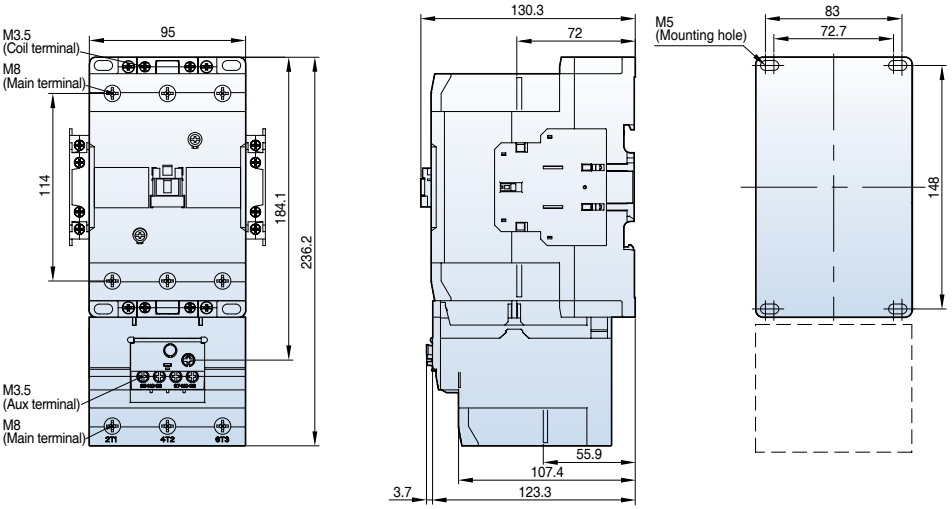
1.3kg

MS-75a~100a



2.3kg

MS-130a
MS-150a

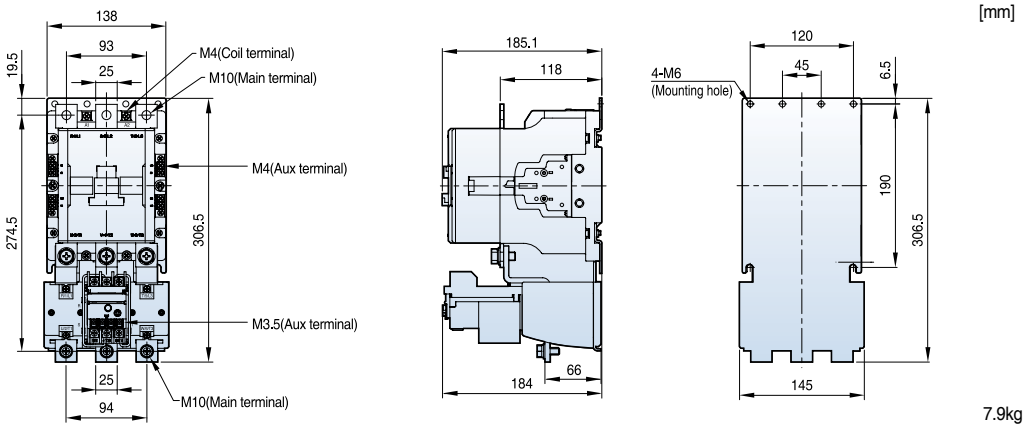


2.9kg

Motor starters dimension (185AF~800AF)

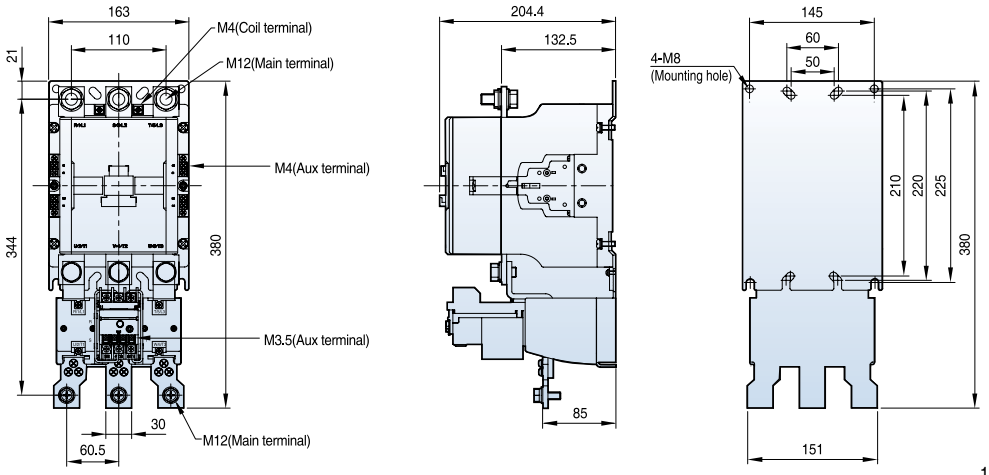
[mm]

MS-185a
MS-225a



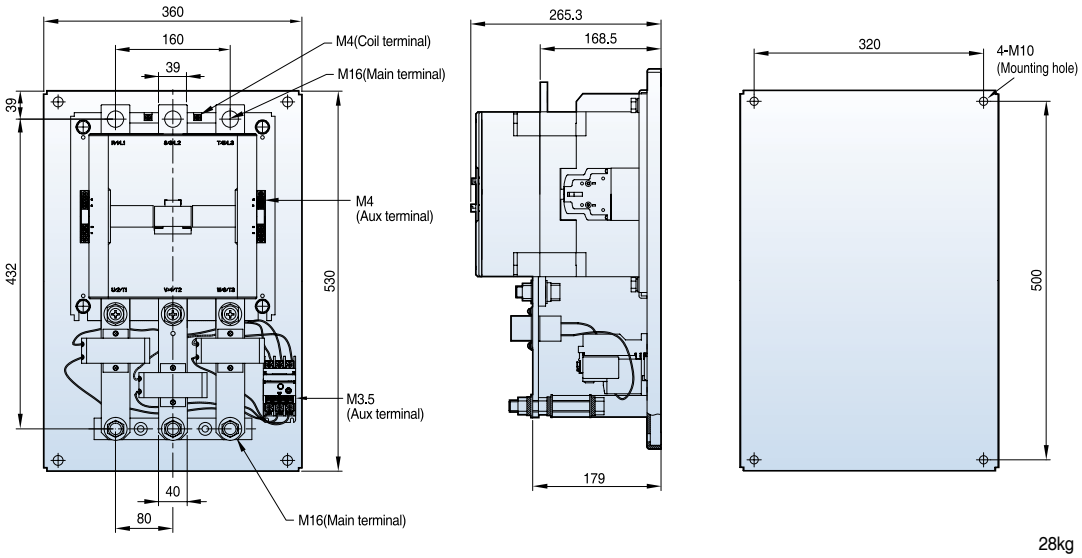
7.9kg

MS-265a
MS-330a
MS-400a



11.8kg

MS-500a
MS-630a
MS-800a

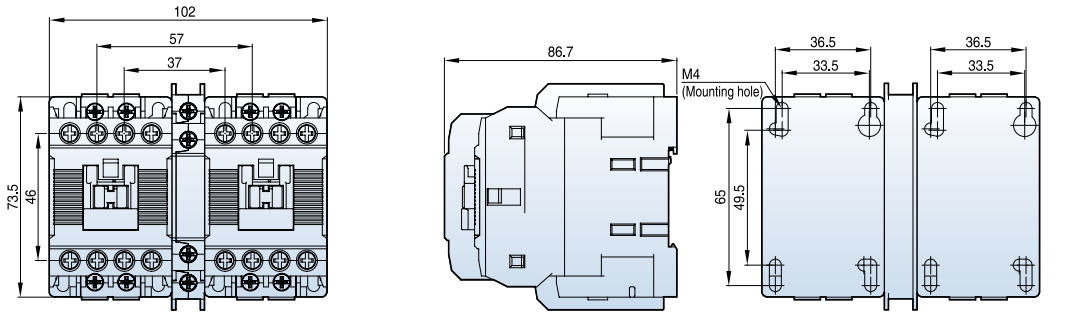


28kg

Reversing contactors and starters dimension (18AF)

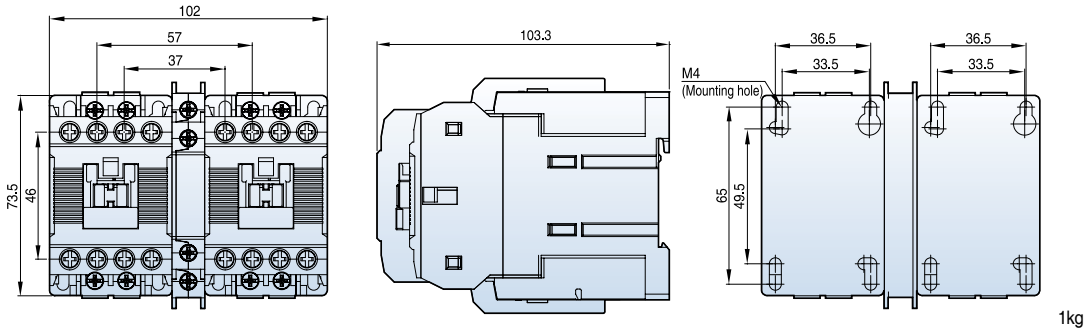
[mm]

MC-6aR~18aR AC



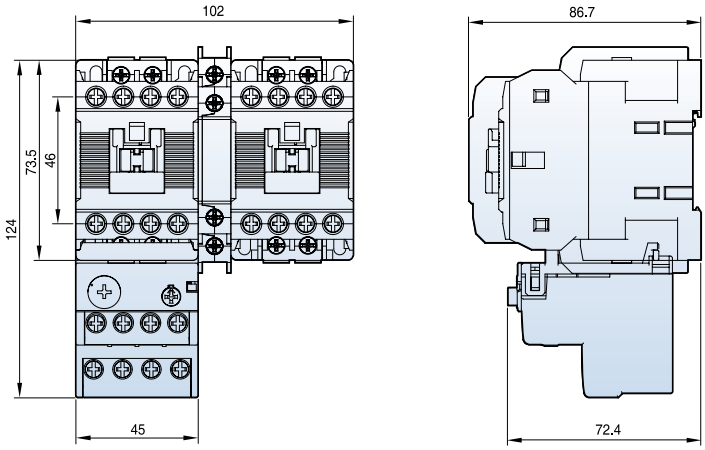
0.7kg

MC-6aR~18aR DC



1kg

MS-6aR~18aR AC

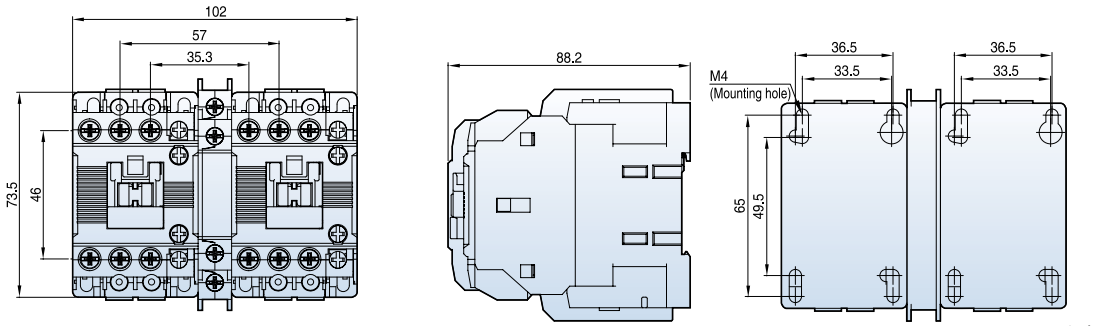


0.9kg

Reversing contactors and starters dimension (22AF)

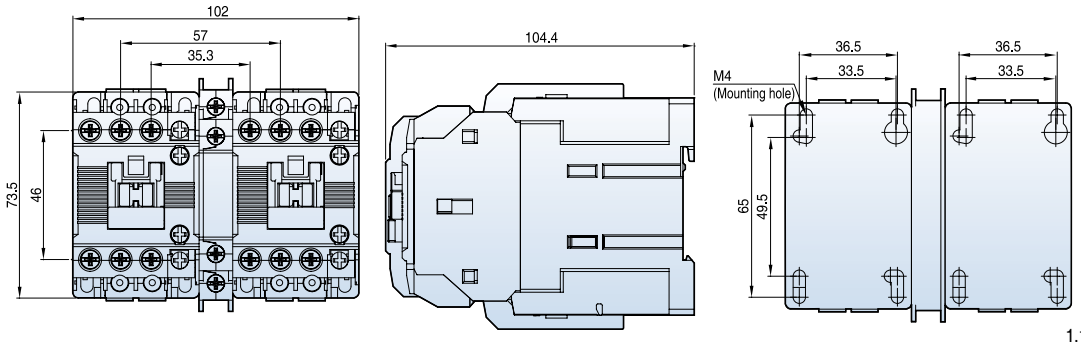
[mm]

MC-9bR~22bR AC



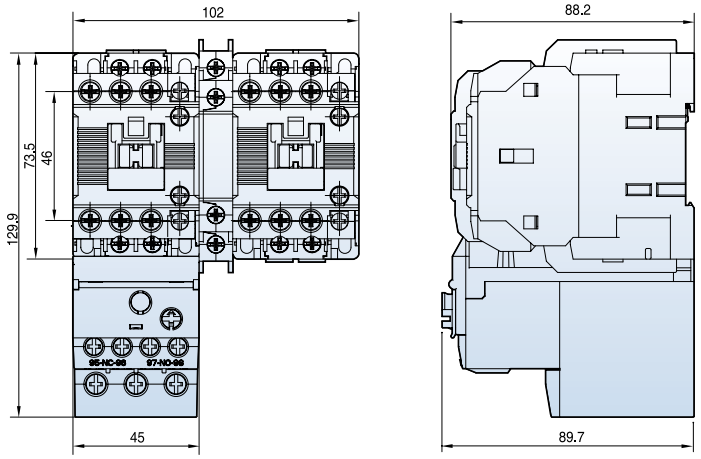
0.7kg

MC-9bR~22bR DC



1.1kg

MS-9bR~22bR AC

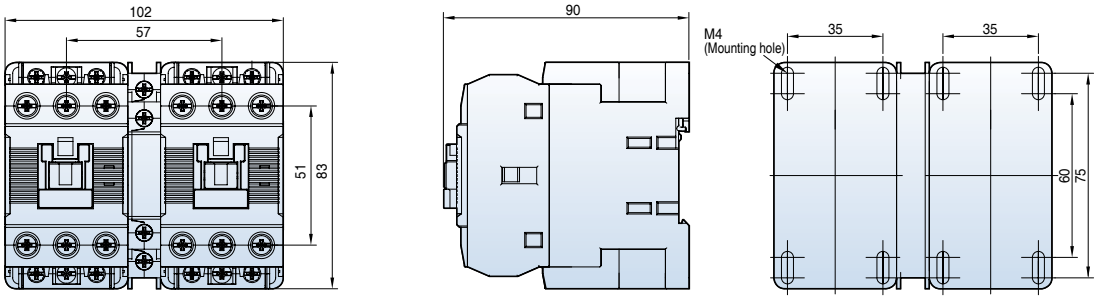


1.3kg

Reversing contactors and starters dimension (40AF)

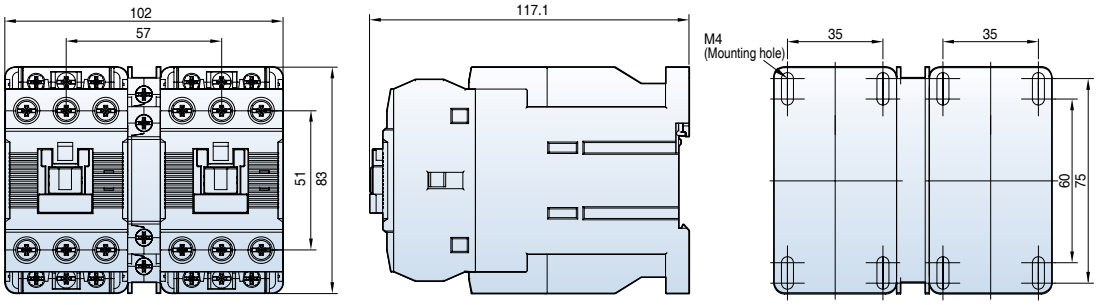
[mm]

MC-32aR~40aR AC



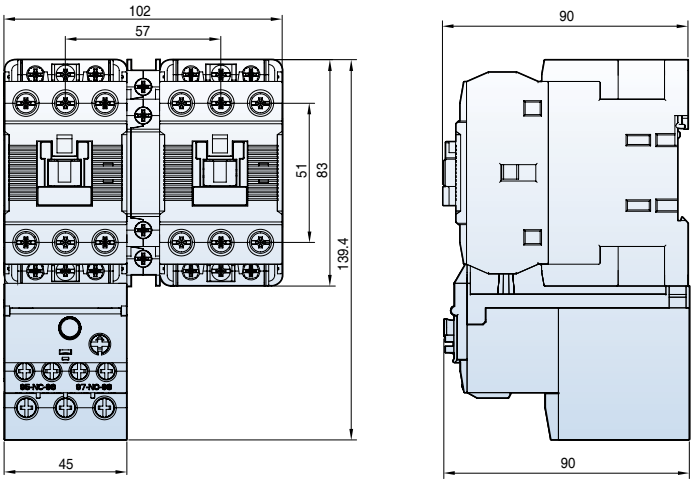
0.9kg

MC-32aR~40aR DC



1.4kg

MS-32aR~40aR AC

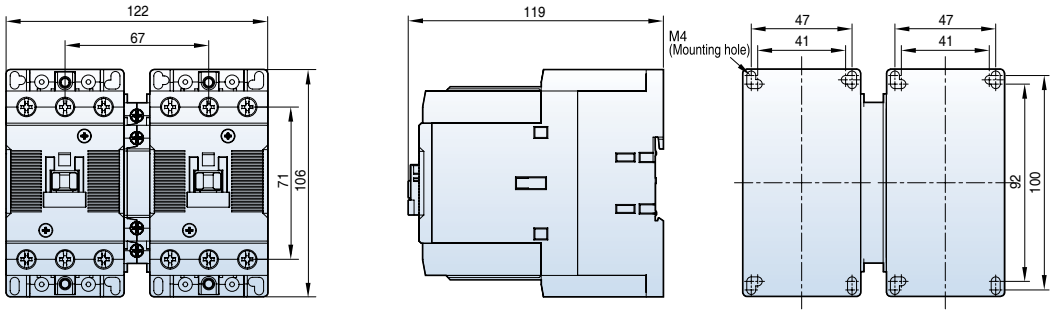


1.1kg

Reversing contactors and starters dimension (65AF)

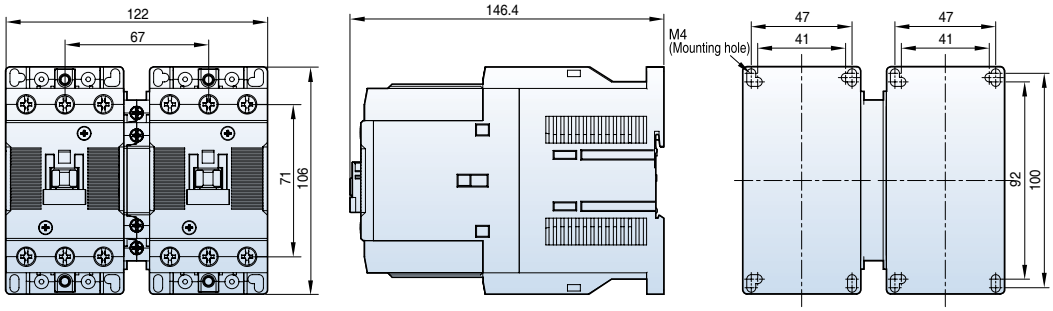
[mm]

MC-50aR~65aR AC



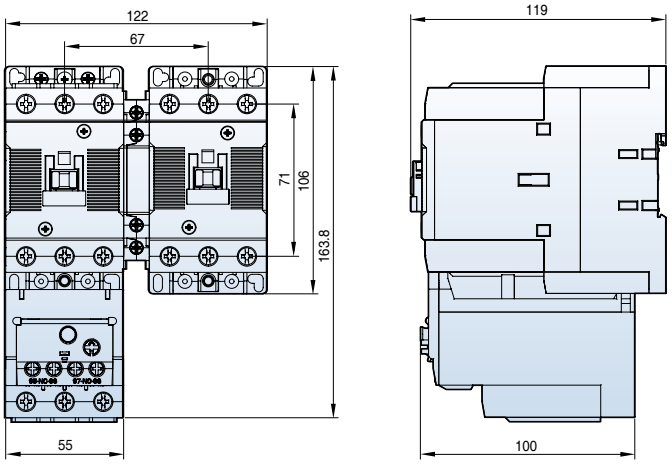
1.8kg

MC-50aR~65aR DC



2.4kg

MS-50aR~65aR AC

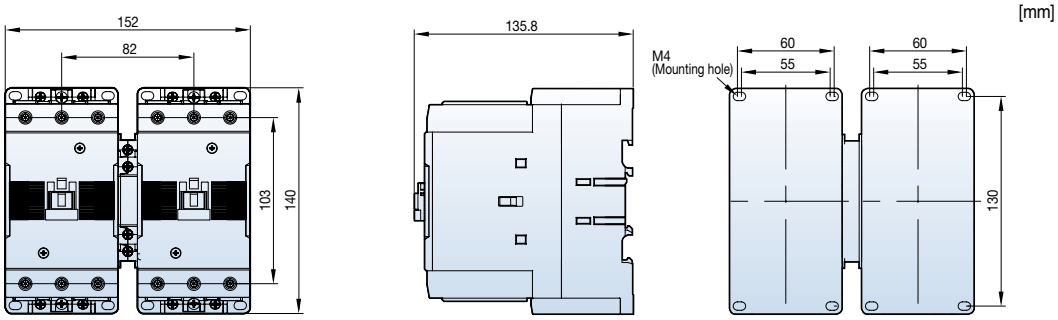


1.9kg

Reversing contactors and starters dimension (100AF)

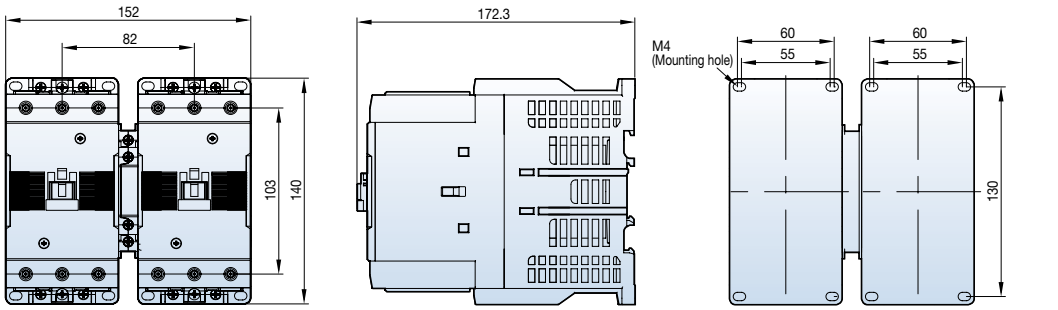
[mm]

MC-75aR~100aR AC



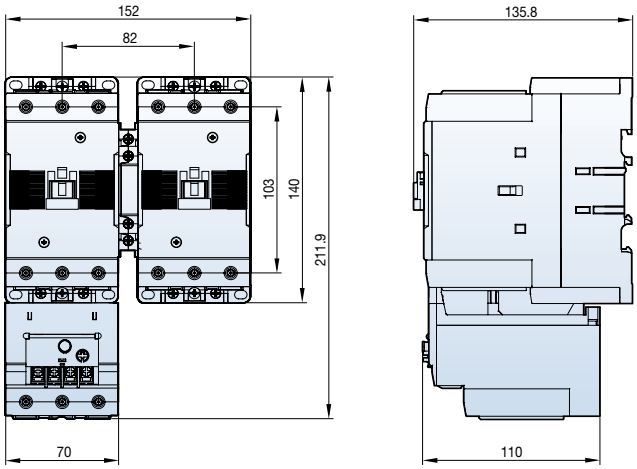
3.3kg

MC-75aR~100aR DC



5.2kg

MS-75aR~100aR AC

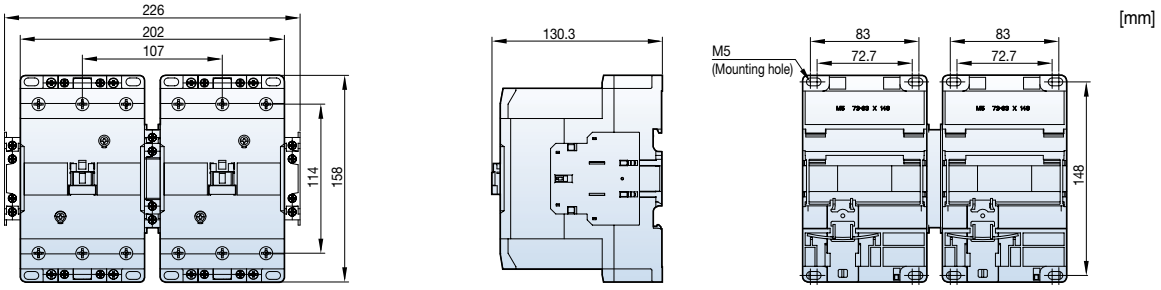


3.8kg

Reversing contactors and starters dimension (150~800AF)

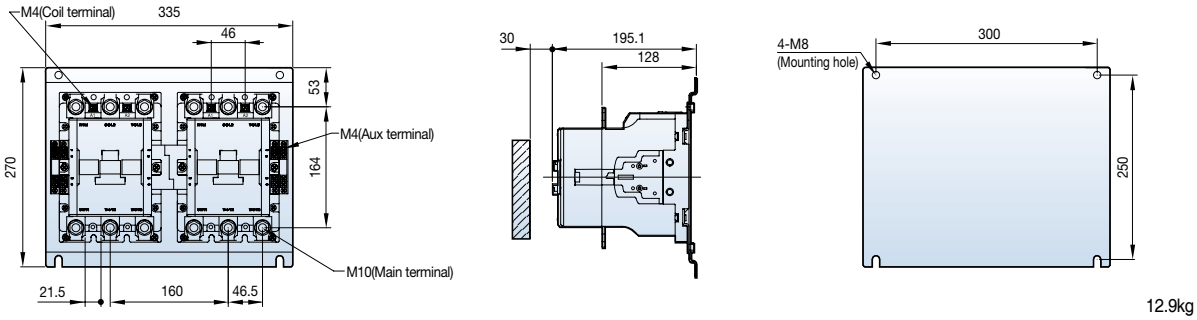
[mm]

MC-130aR
MC-150aR



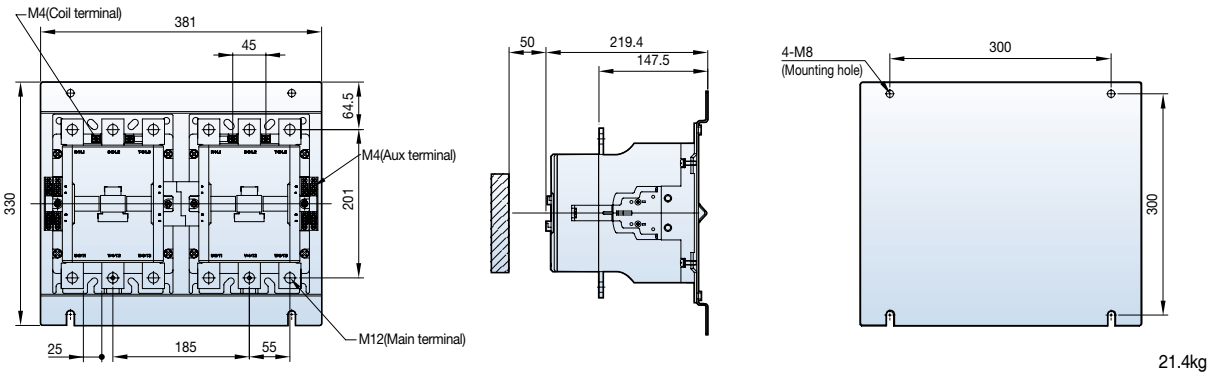
4.3kg

MC-185aR
MC-225aR



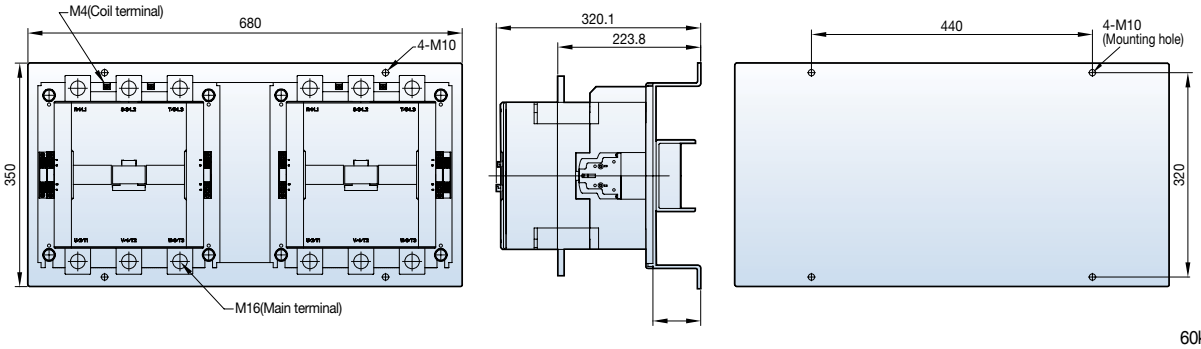
12.9kg

MC-265aR
MC-330aR
MC-400aR



21.4kg

MC-500aR
MC-630aR
MC-800aR

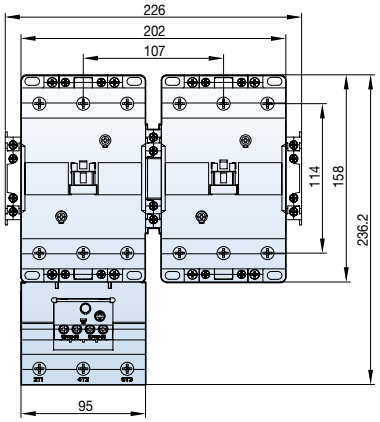


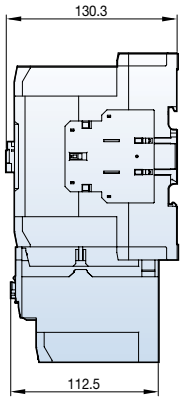
60kg

Reversing contactors and starters dimension (150~400AF)

[mm]

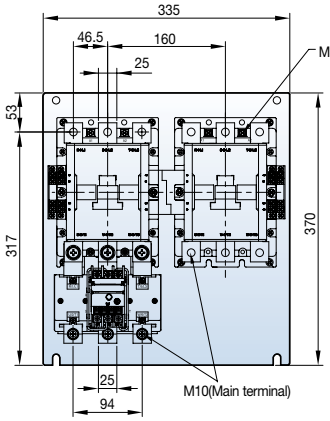
MS-130aR AC
MS-150aR AC

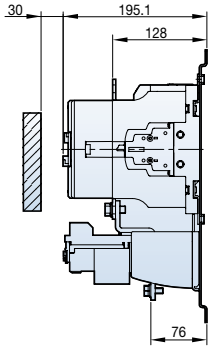


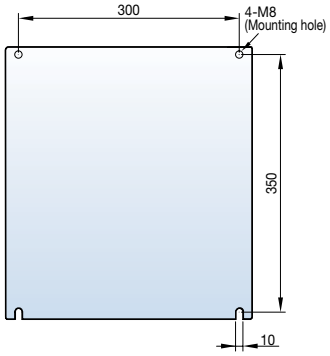


5kg

MS-185aR AC
MS-225aR AC

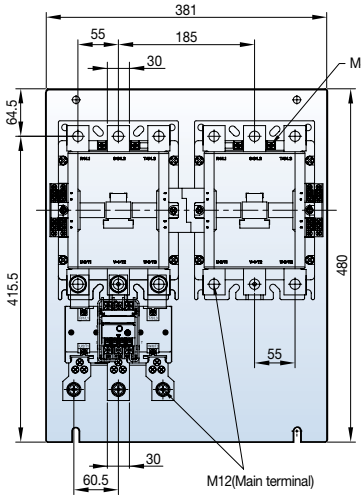


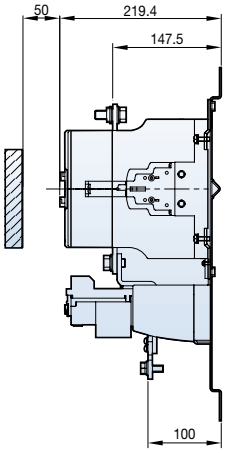


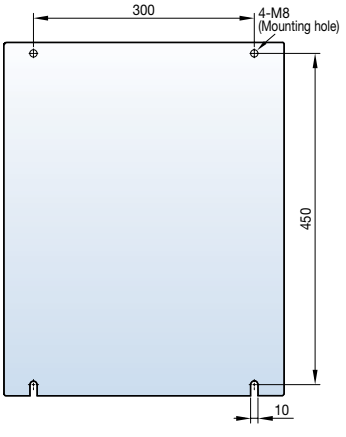


15.5kg

MS-265aR DC
MS-330aR DC
MS-400aR DC





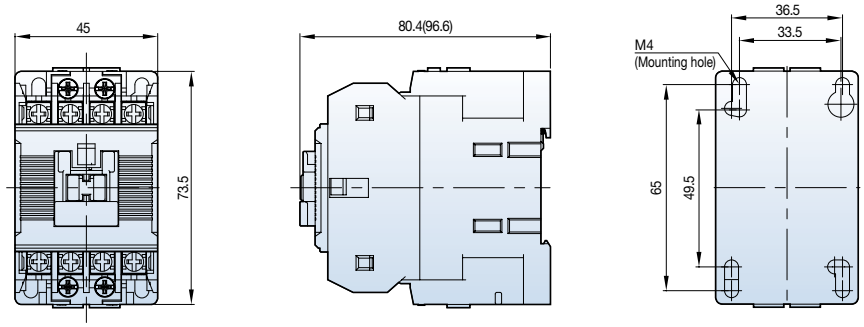


23kg

Contactor relays dimension

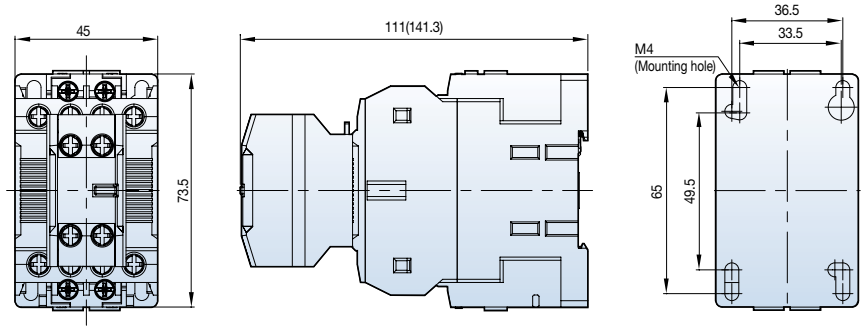
[mm]

MR-4 (D)



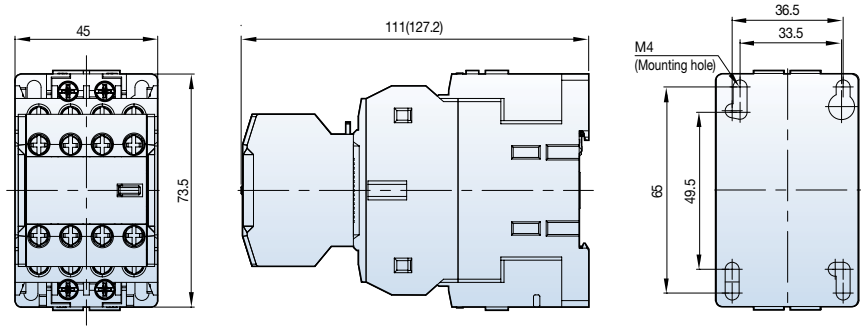
284g

MR-6 (D)



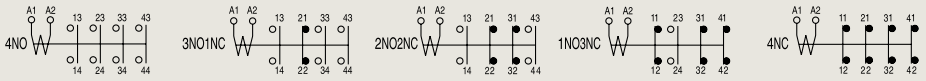
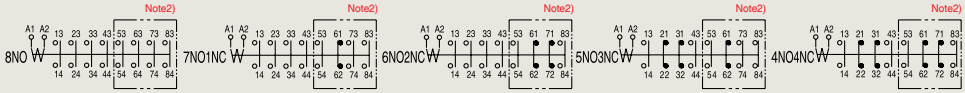
315g

MR-8 (D)



339g

Composition

MR-4	
MR-6	
MR-8	

Note) 1. UA-2: Auxiliary contact units 2. UA-4: Auxiliary contact units

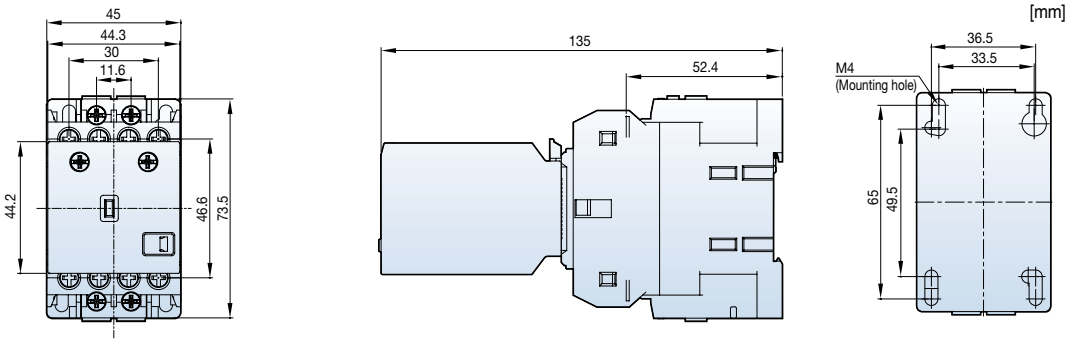
- Please refer to the contact diagram represented on catalog when contactor relays MR-4, 4D and auxiliary contact units UA-2,4 are purchased separately.
- Be sure to order the auxiliary contact units separately except for the above contact units. • ( ) : Dimension for DC control contactor relays

Latching contactors dimension (18AF/22AF)

[mm]

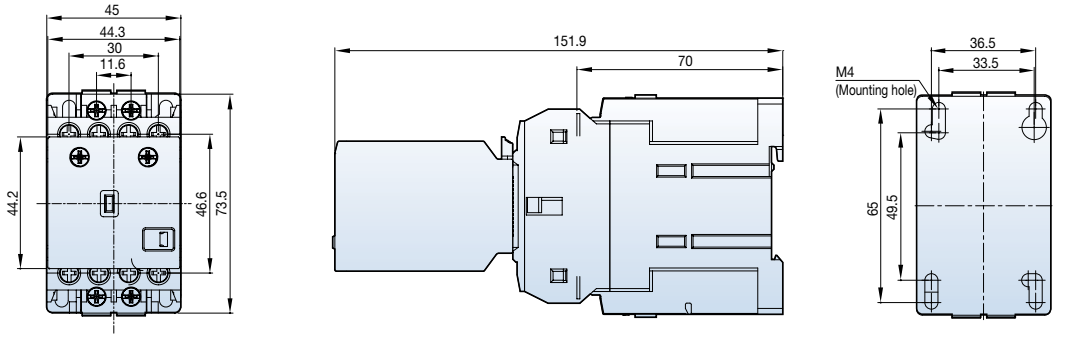
MC-6a~18a AC

1NO or 1NC + ML-65



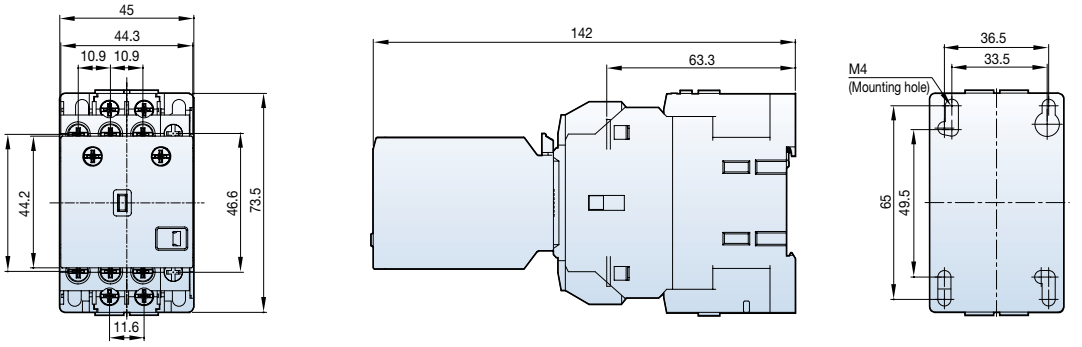
MC-6a~18a DC

1NO or 1NC + ML-65



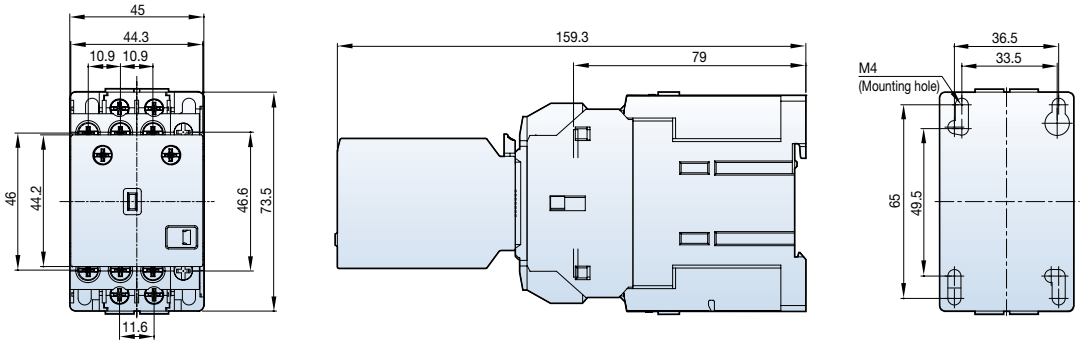
MC-9b~22b AC

1NO1NC + ML-65



MC-9b~22b DC

1NO1NC + ML-65

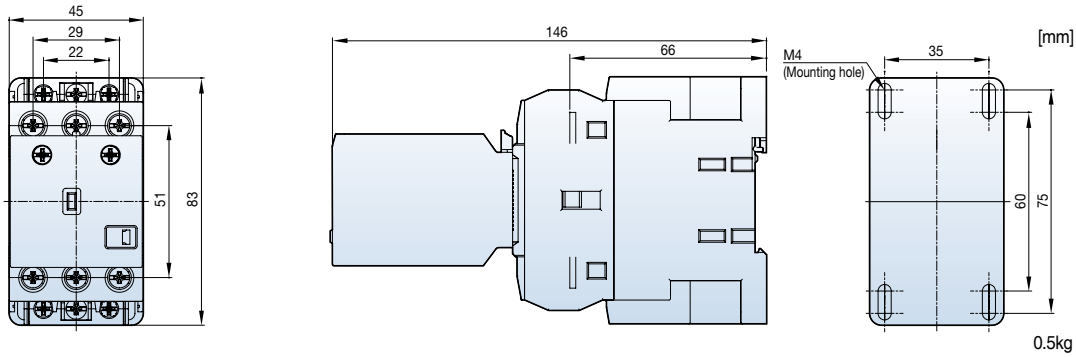


Latching contactors dimension (40AF)

[mm]

MC-32a~40a AC

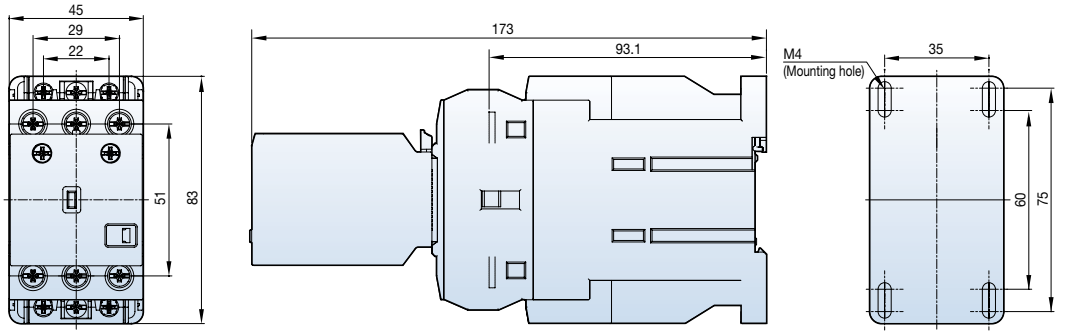
+ML-65



0.5kg

MC-32a~40a DC

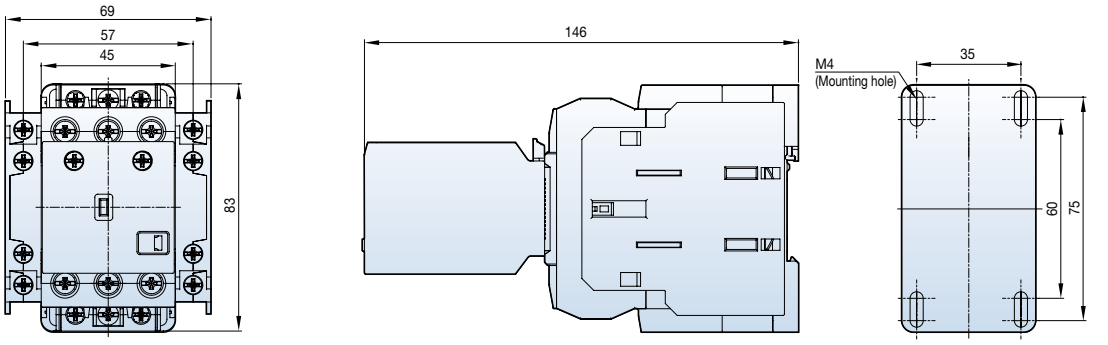
+ ML-65



0.8kg

MC-32a~40a AC

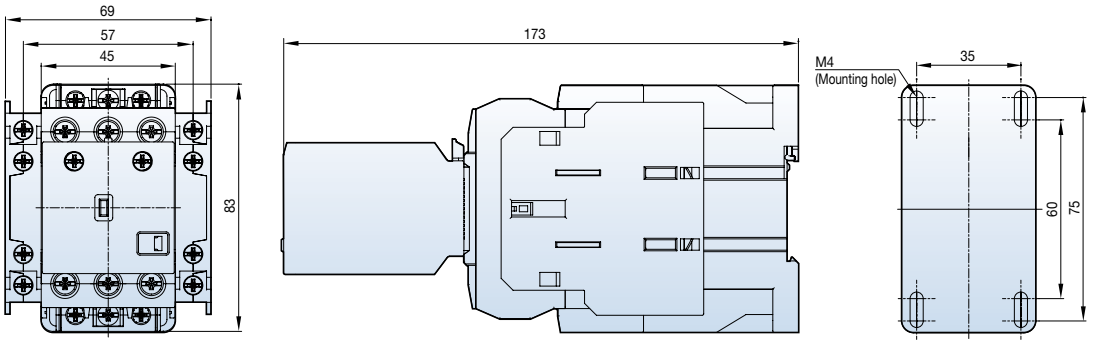
2NO2NC + ML-65



1kg

MC-32a~40a DC

2NO2NC + ML-65



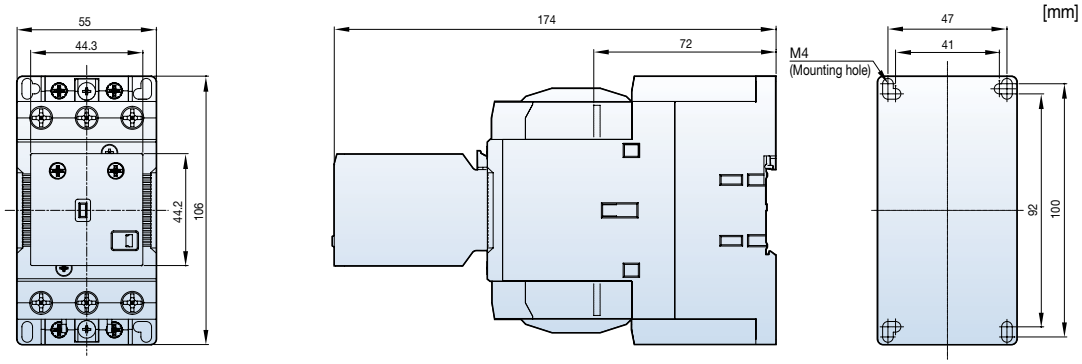
1.3kg

Latching contactors dimension (65AF)

[mm]

MC-50a~65a AC

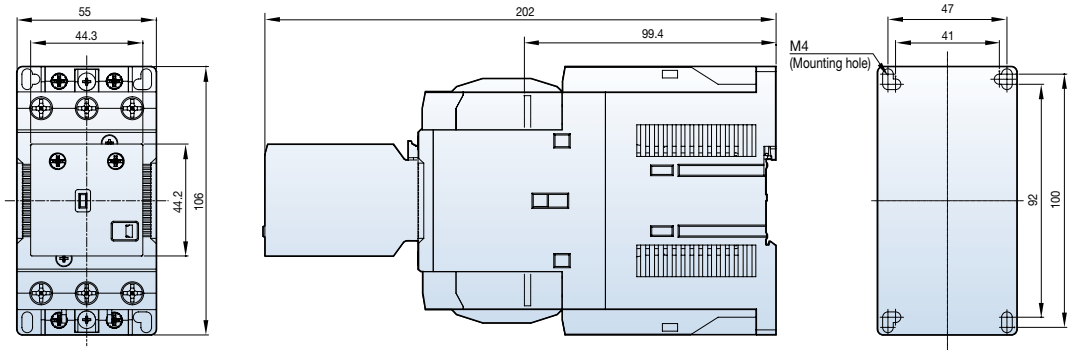
+ ML-65



1.9kg

MC-50a~65a DC

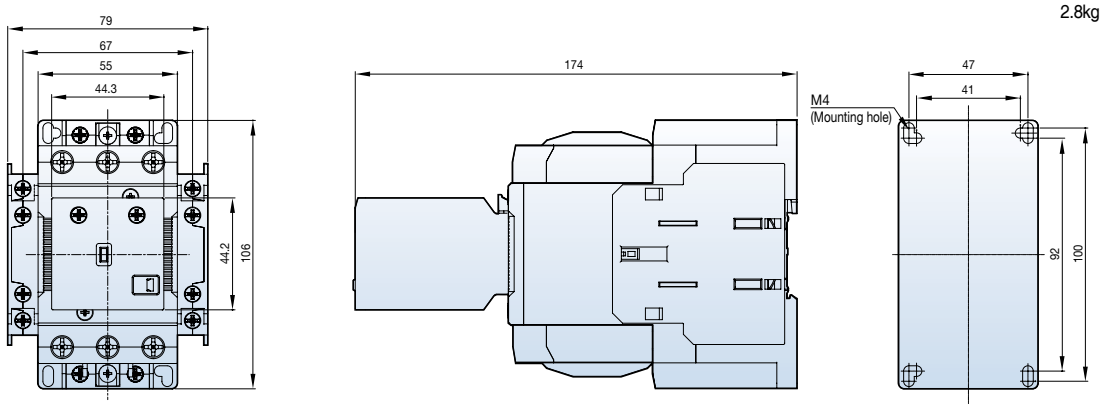
+ ML-65



2.8kg

MC-50a~65a AC

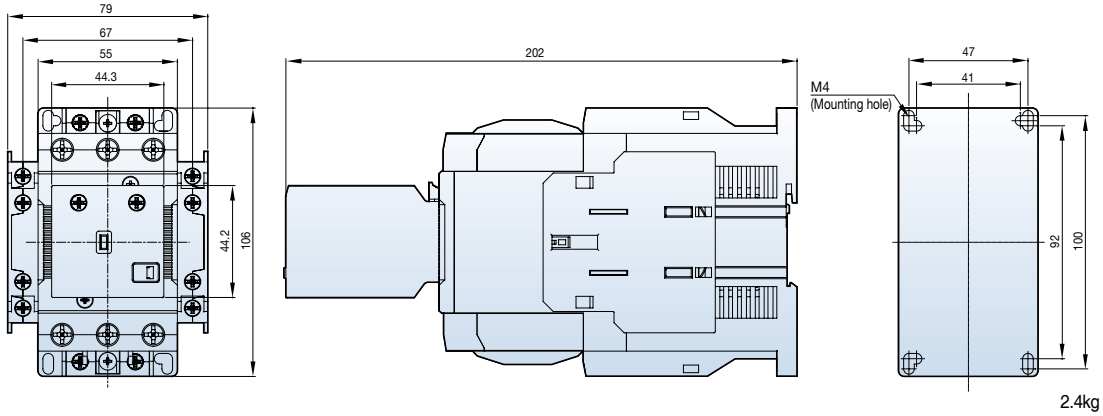
2NO2NC + ML-65



2.4kg

MC-50a~65a DC

2NO2NC + ML-65



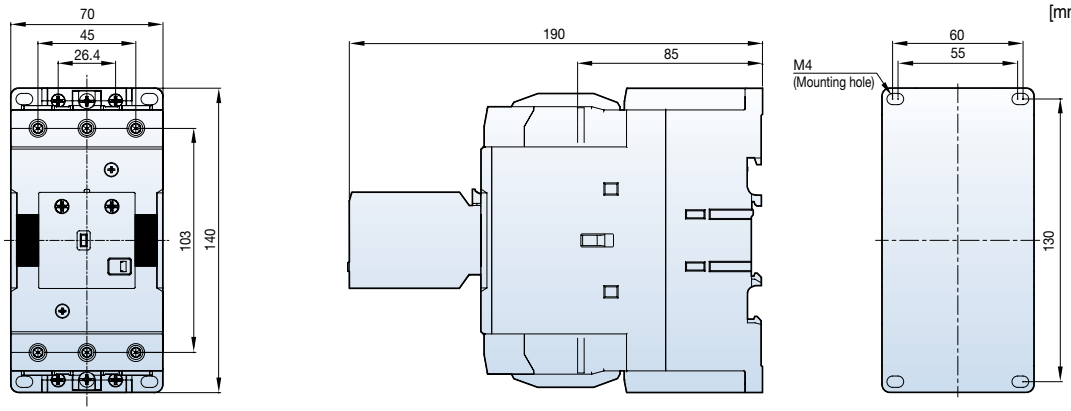
2.4kg

Latching contactors dimension (100AF)

[mm]

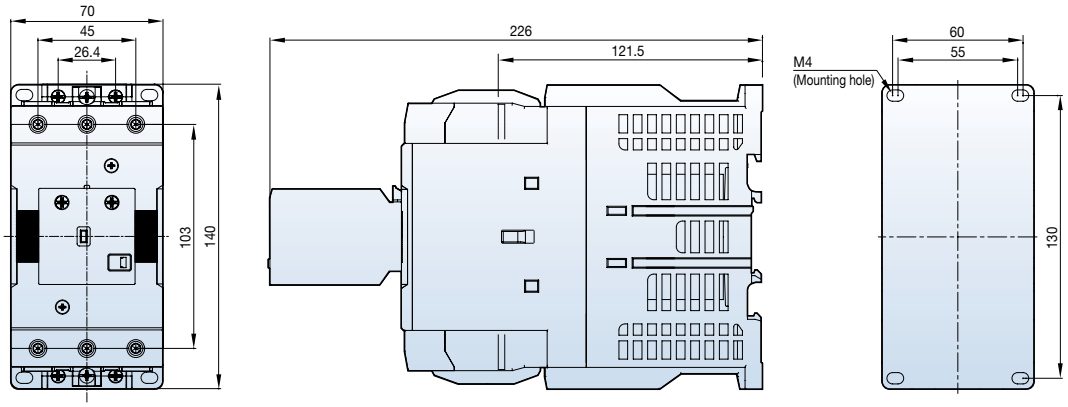
MC-75a~100a AC

+ ML-150



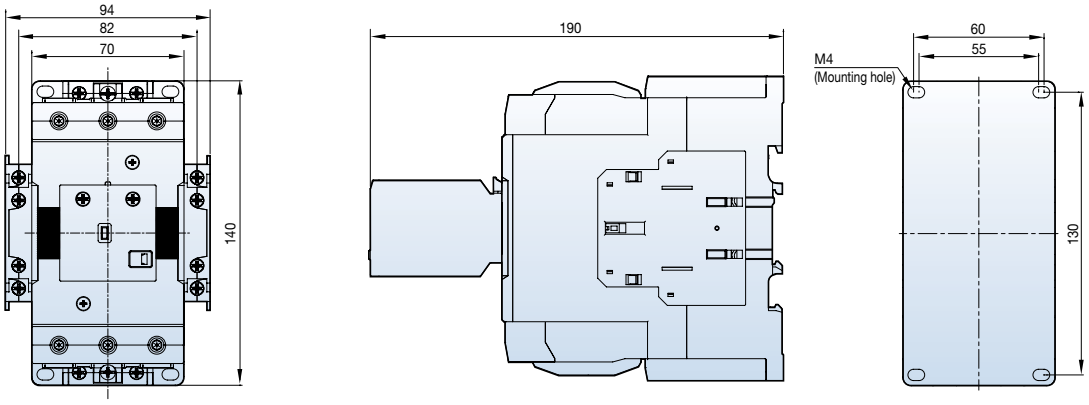
MC-75a~100a DC

+ ML-150



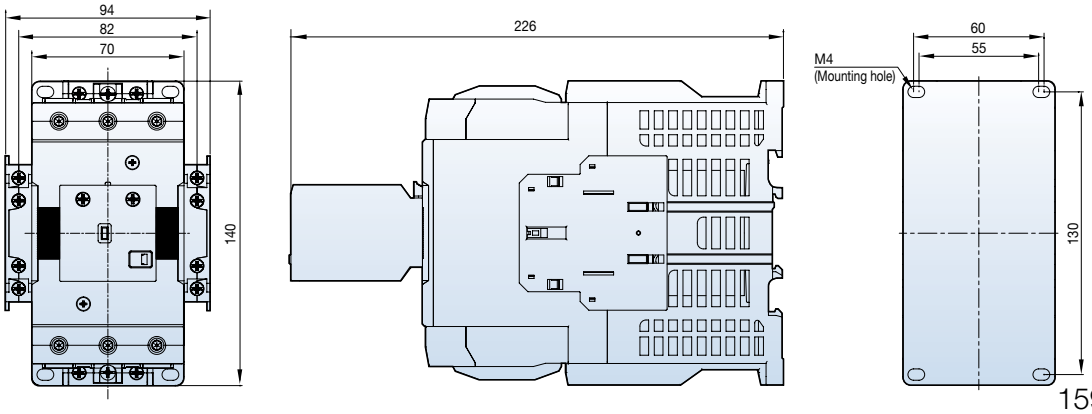
MC-75a~100a AC

2NO2NC + ML-150



MC-75a~100a DC

2NO2NC + ML-150

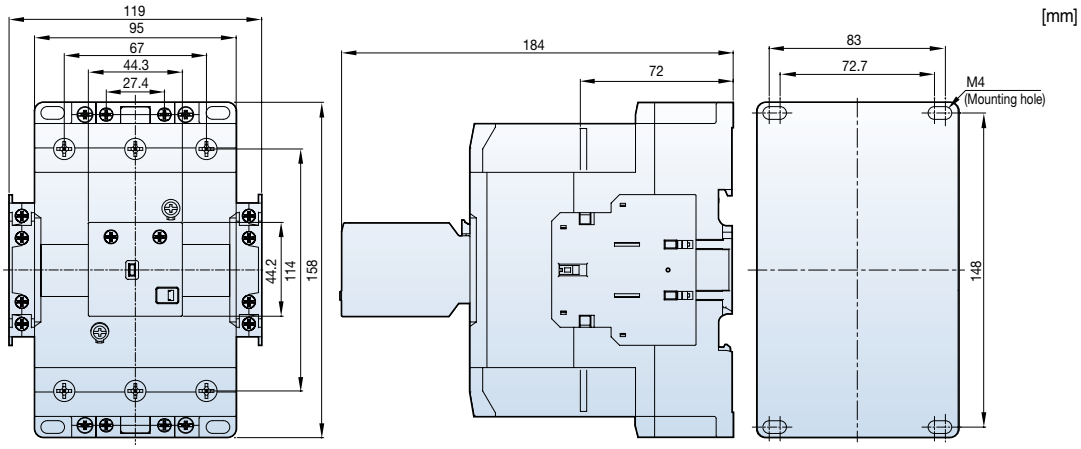


Latching contactors dimension (150AF~400AF)

[mm]

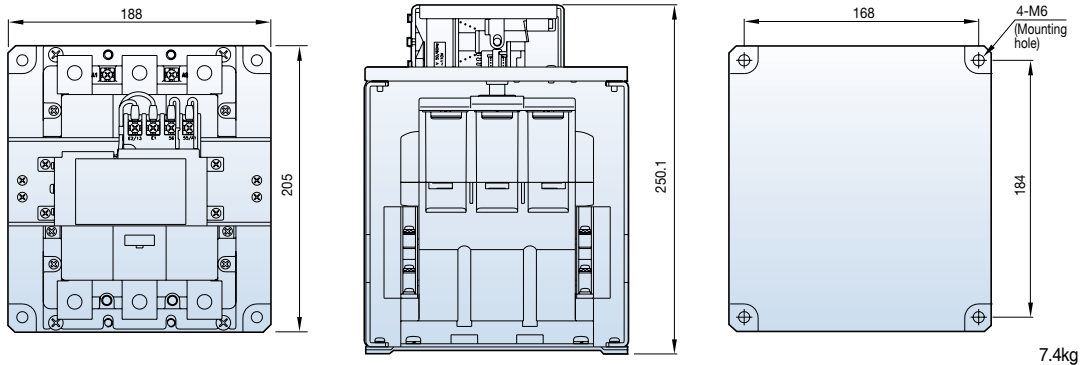
MC-130a~150a

2NO2NC + ML-150



MC-185a

MC-225a

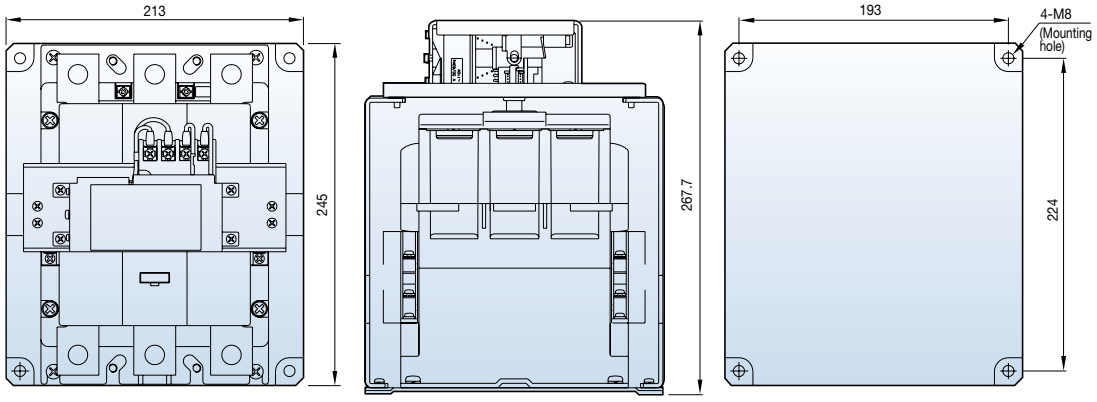


7.4kg

MC-265a

MC-330a

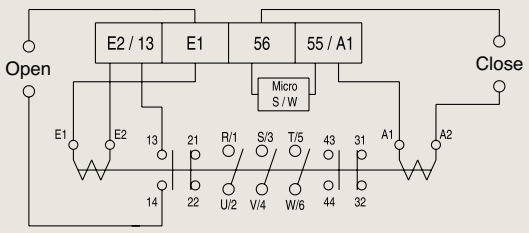
MC-400a



11.6kg

Composition

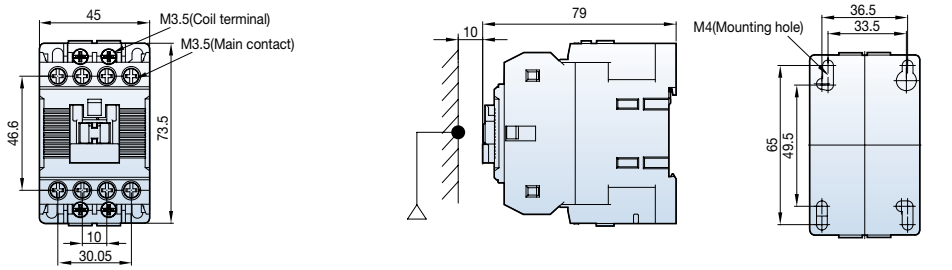
MC-185a/L~400a/L



4-pole contactors dimension (MC-6a/4~22a/4)

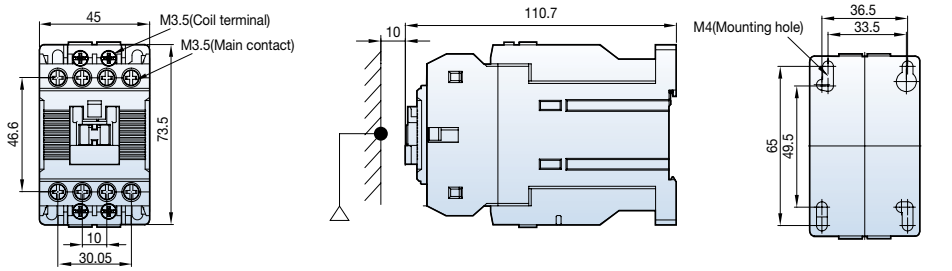
[mm]

MC-6a/4~18a/4 AC



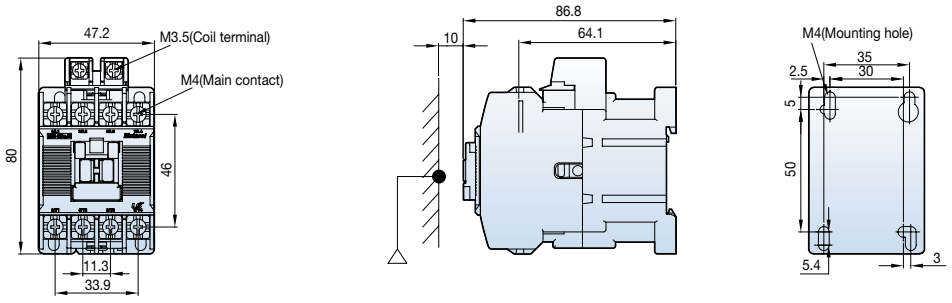
0.3kg

MC-6a/4~18a/4 DC



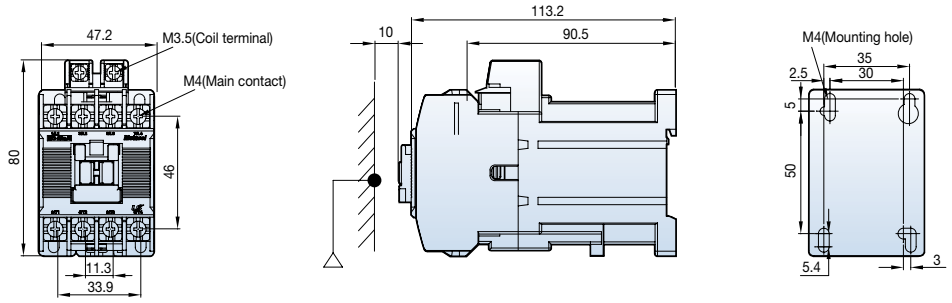
0.45kg

MC-22a/4 AC



0.4kg

MC-22a/4 DC

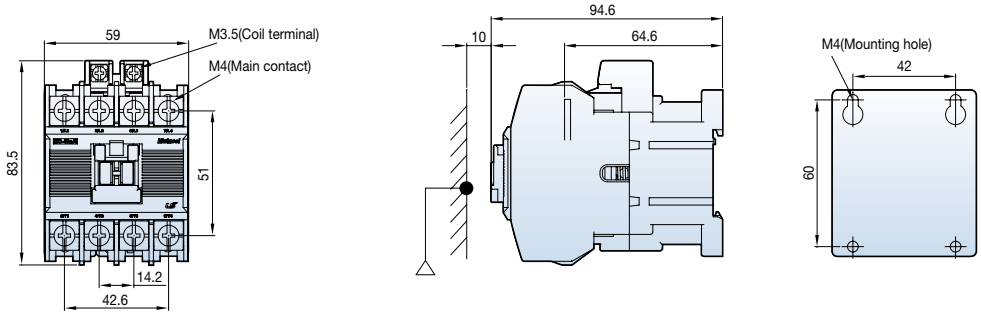


0.5kg

4-pole contactors dimension (MC-32a/4~85a/4)

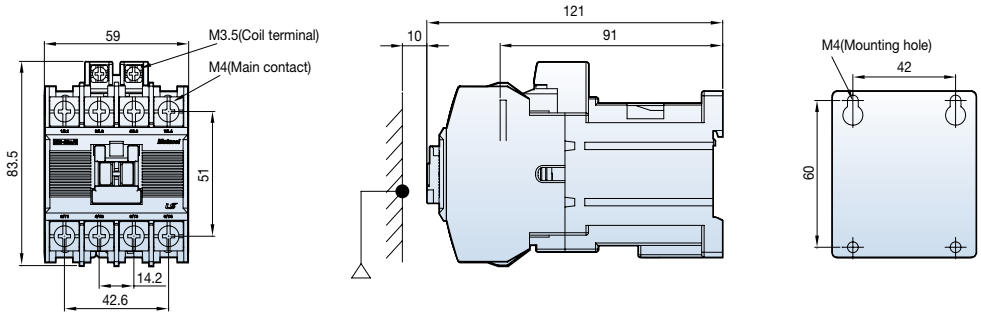
[mm]

MC-32a/4~40a/4 AC



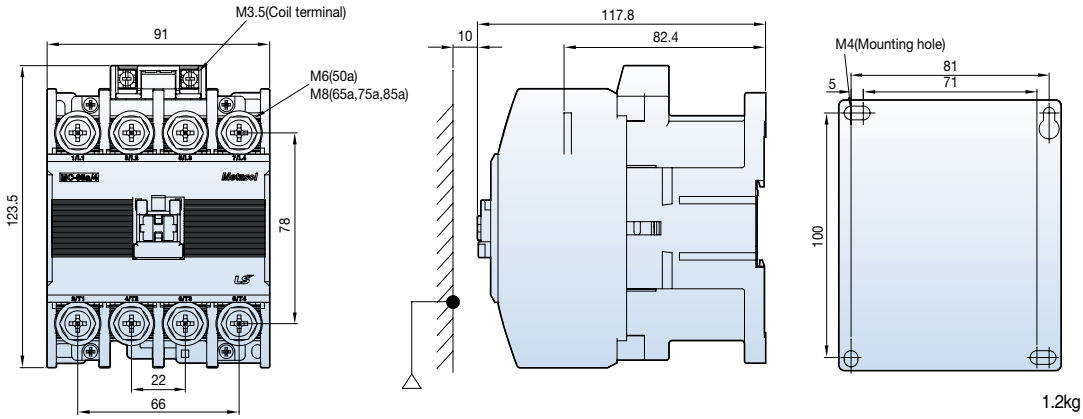
0.59kg

MC-32a/4~40a/4 DC



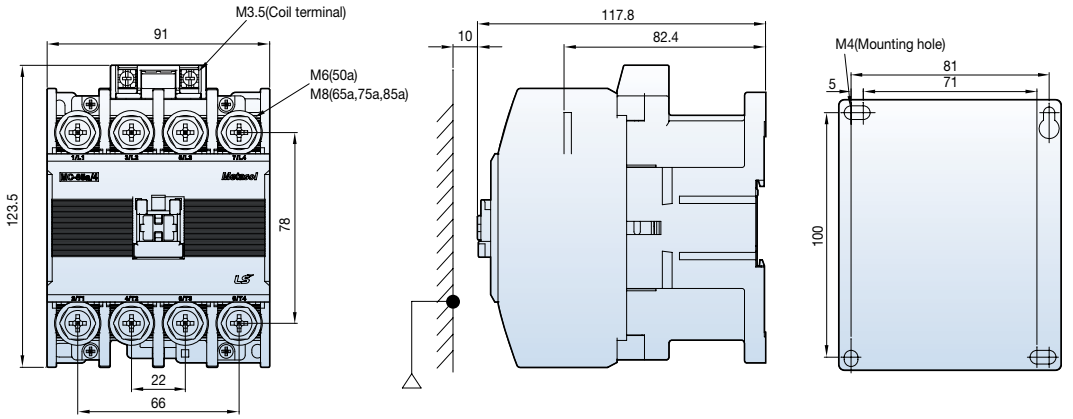
0.7kg

MC-50a/4~85a/4 AC



1.2kg

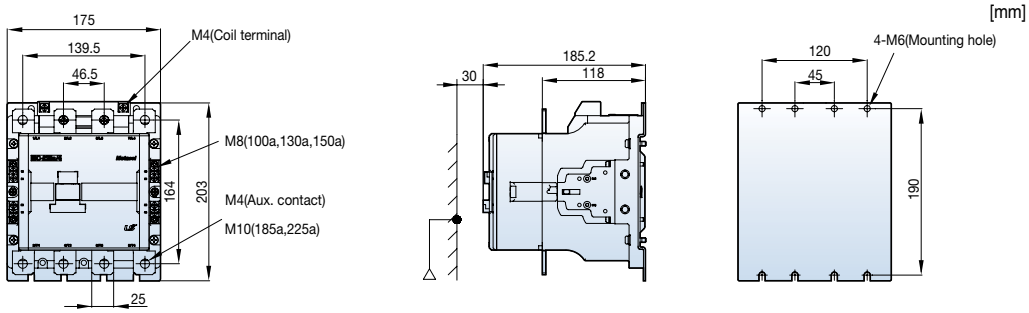
MC-50a/4~85a/4 DC



4-pole contactors dimension (MC-100a/4~800a/4)

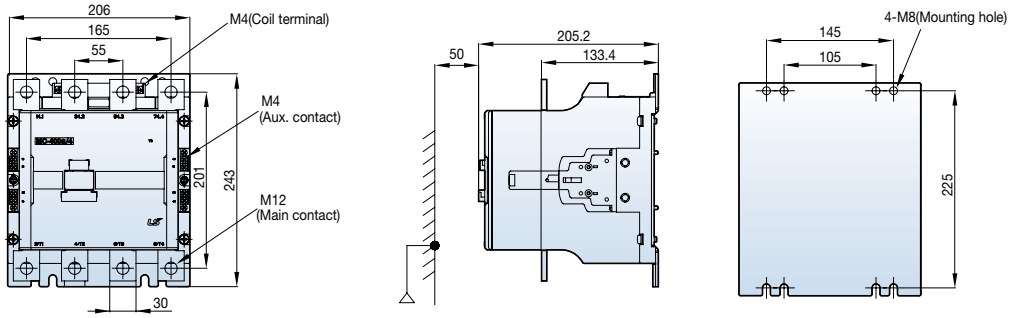
[mm]

MC-100a/4
MC-130a/4
MC-150a/4
MC-185a/4
MC-225a/4



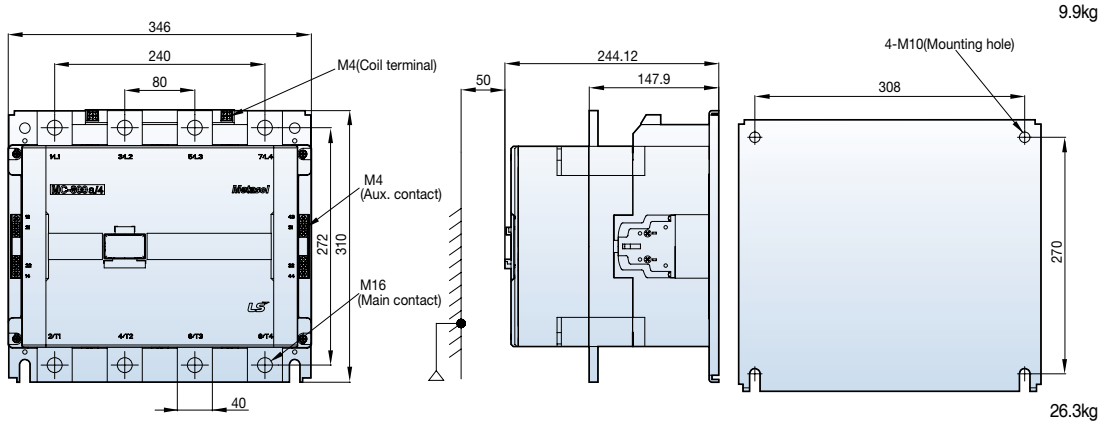
5.6kg

MC-265a/4
MC-330a/4
MC-400a/4



9.9kg

MC-500a/4
MC-630a/4
MC-800a/4

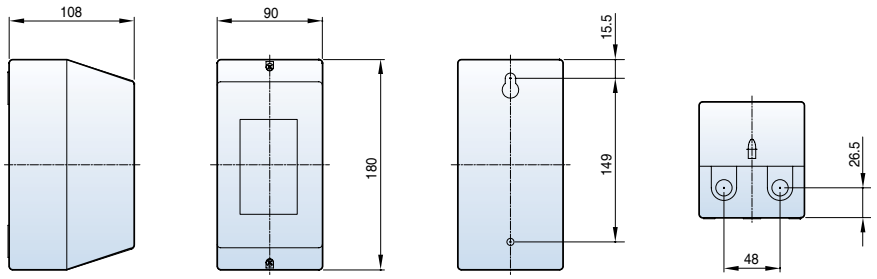


26.3kg

Enclosed motor starters, Mold enclosure dimension

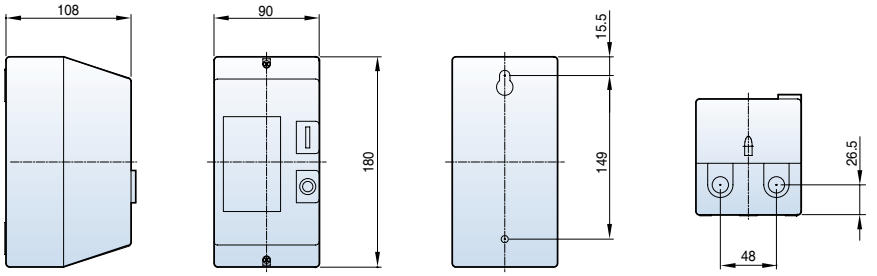
MW-9bM~22bM

[mm]



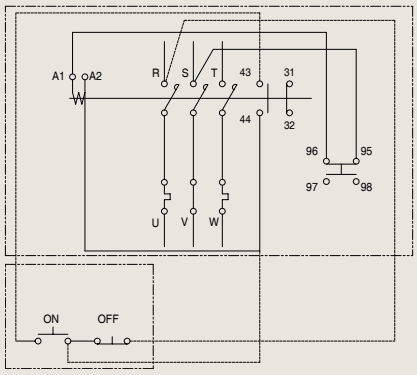
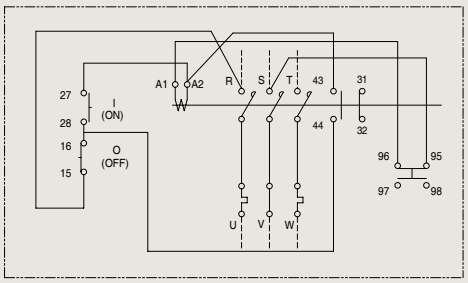
0.8kg

MW-9bMB~22bMB



0.8kg

Circuit diagram

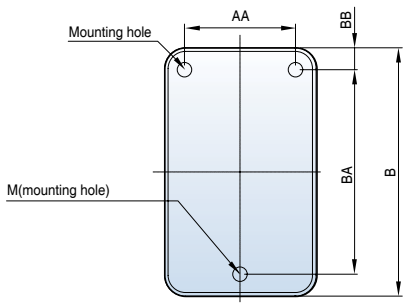
MW-9bM~22bM	MW-9bMB~22bMB
	

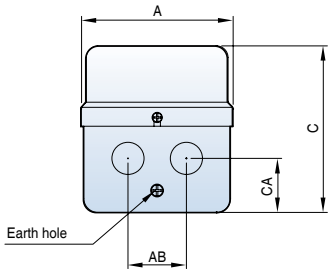
Enclosed motor starters, Steel enclosure dimension

MW-6a~100a

MW-9b~22b

[mm]

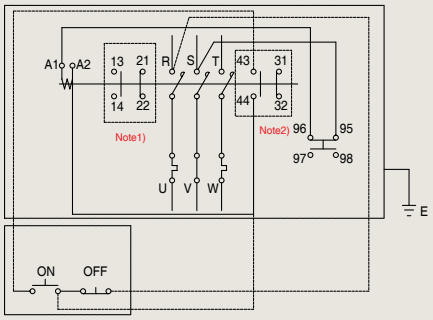




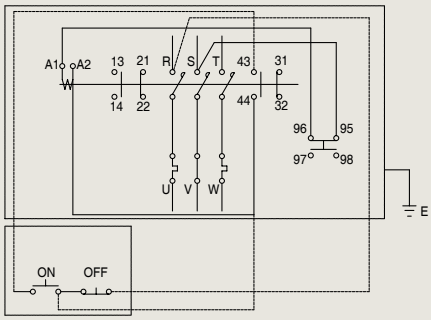
Type	A	AA	AB	B	BA	BB	C	CA	M	N	Weight(kg)
MW-9b~22b, 6a~18a	104	76	40	170	140	15	114	41	M5	Ø22	1.1
MW-32a, 40a	145	95	55	245	165	40	130	50	M5	Ø35	1.6
MW-50a, 65a	160	120	74	290	220	35	141	50	M5	Ø35	2.7
MW-75a, 85a, 100a	160	120	74	290	220	35	161	67	M5	Ø35	3.6

Circuit diagram

MW-6a~18a, MW-9b~22b



MW-32a~100a



Note) 1. In case of adding the auxiliary contact unit(1NO1NC).
2. See below for MC-6a~18a(1NO)

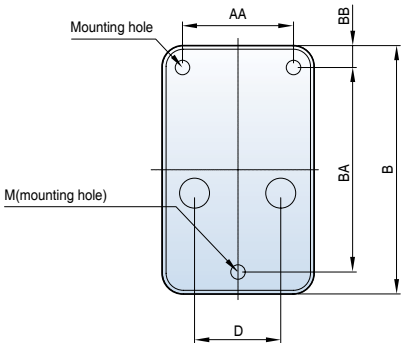


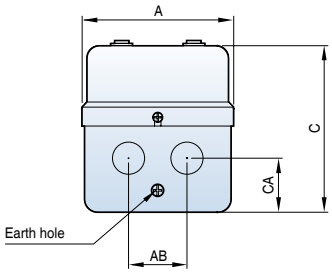
Enclosed motor starters, Steel enclosure dimension

MW-6aB~100aB

MW-9bB~22bB

[mm]

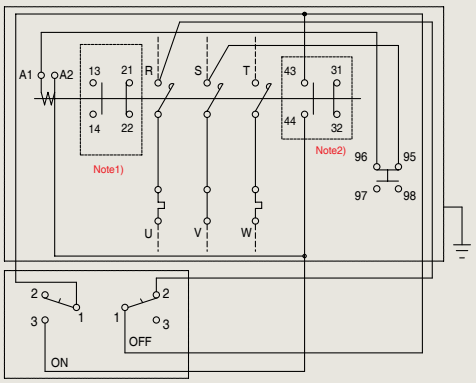




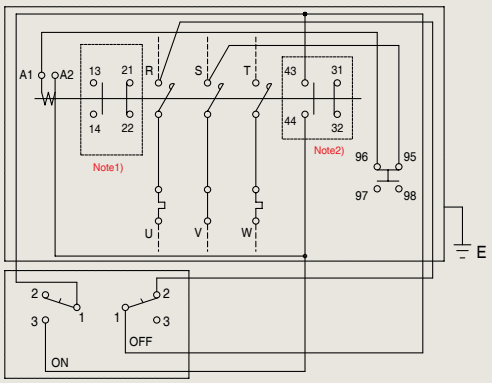
Type	A	AA	AB	B	BA	BB	C	CA	D	M	N	Weight(kg)
MW-9bB~22bB, 6aB~18aB	104	76	40	170	140	15	114	41	59	M5	Ø22	1.1
MW-32aB, 40aB	145	95	55	245	165	40	130	50	97	M5	Ø35	1.6
MW-50aB, 65aB	160	120	74	290	220	35	141	50	114	M5	Ø35	2.7
MW-75aB, 85aB, 100aB	160	120	74	290	220	35	161	67	114	M5	Ø35	3.6

Circuit diagram

MW-6aB~18aB, MW-9bB~22bB



MW-32aB~100aB



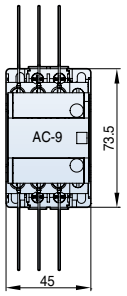
Note) 1. In case of adding the auxiliary contact unit(1NO1NC).
2. See below for MC-6a~18a(1NO)

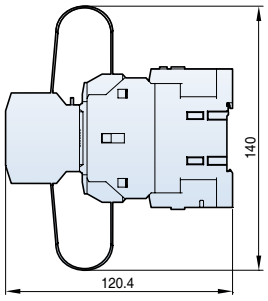


Capacitor contactors (MC-9b~40a with AC-9, MC 50a~100a with AC-50)

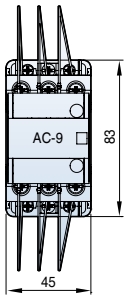
[mm]

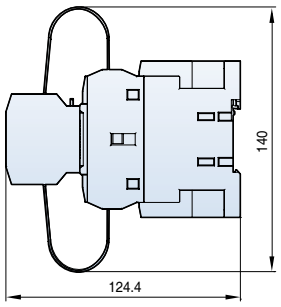
MC-9b~22b with AC-9



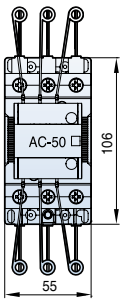


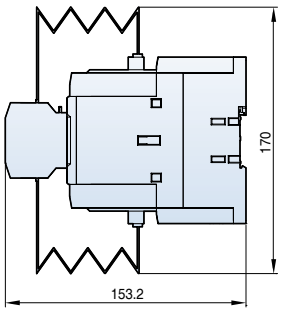
MC-32a,40a with AC-9



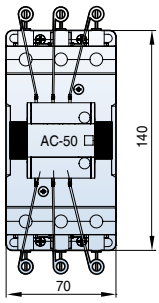


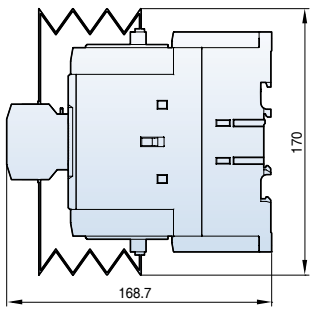
MC-50a,65a with AC-50



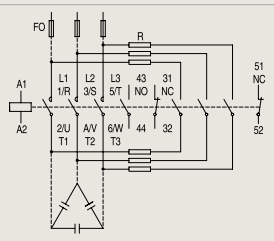
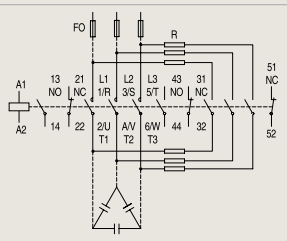


MC-75a~100a with AC-50





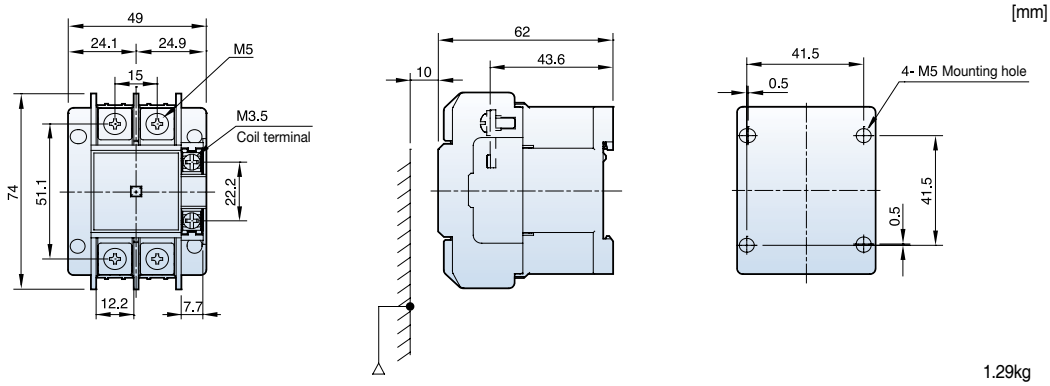
Circuit diagram

MC-9b~22b with AC-9	MC-32a, 40a with AC-9 MC 50a~100a with AC-50
	

2-pole AC Contactors dimension (GMC-10P2~40P2)

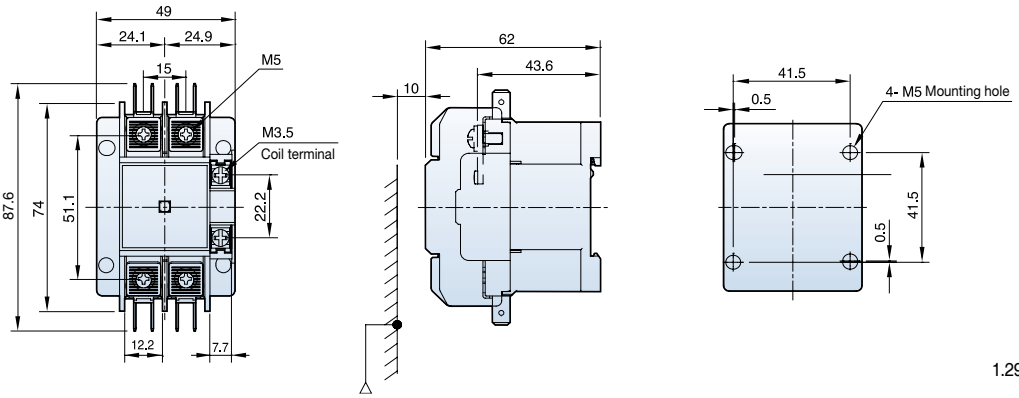
GMC-10P2
GMC-20P2
GMC-25P2
GMC-30P2
GMC-35P2
GMC-40P2
(TSBS Type)

[mm]



1.29kg

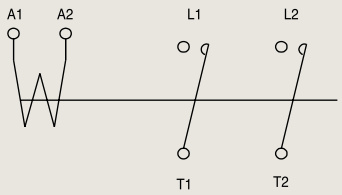
GMC-10P2
GMC-20P2
GMC-25P2
GMC-30P2
GMC-35P2
GMC-40P2
(TQBS Type)



1.29kg

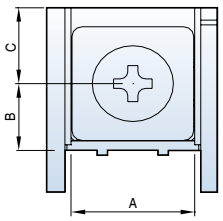
Composition

GMC - 10P2~40P2



Terminal

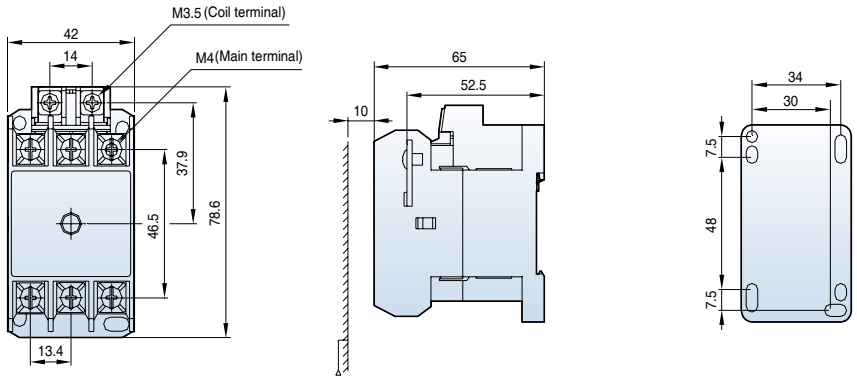
Type	Terminals	Screw	Wire (mm²)	Torque (kgf·cm)	A×B×C (mm)
GMC-10P2~40P2	Main	M5	1-16	26	12.2×7×8
	Coil	M3.5	1-2.5	12	7.7×7.5×4.5



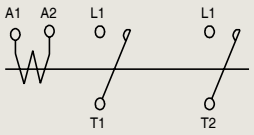
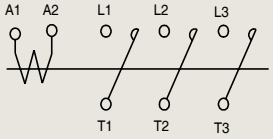
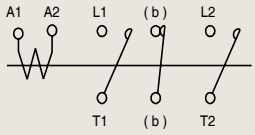
3-pole AC Contactors dimension (GMC-20D, 25D)

GMC-20D
GMC-25D

[mm]

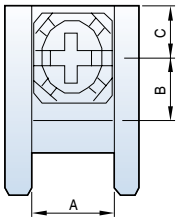


Contact arrangement

2NO	3NO	2NO1NC
		

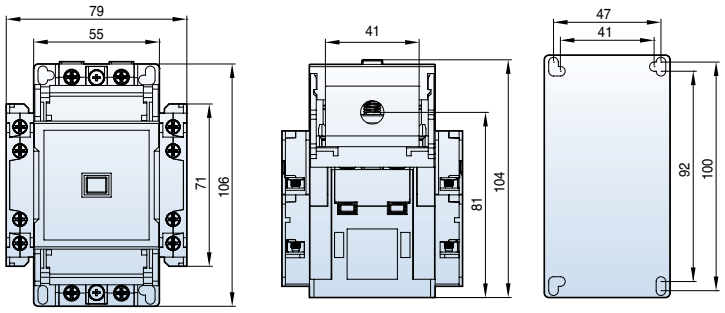
Terminal

Type	Screw	Wire (mm²)	Torque (kgf·cm)	A×B×C (mm)
GMC-20D~25D	M4	1-10	15	9.4×5×85.1
Coil~	M3.5	1.25~2	12	8×5×4.9

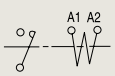
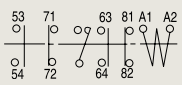


Single-pole AC Contactors (MC-65aSP)

MC-65aSP



Composition

without auxiliary contact	
with auxiliary contacts	

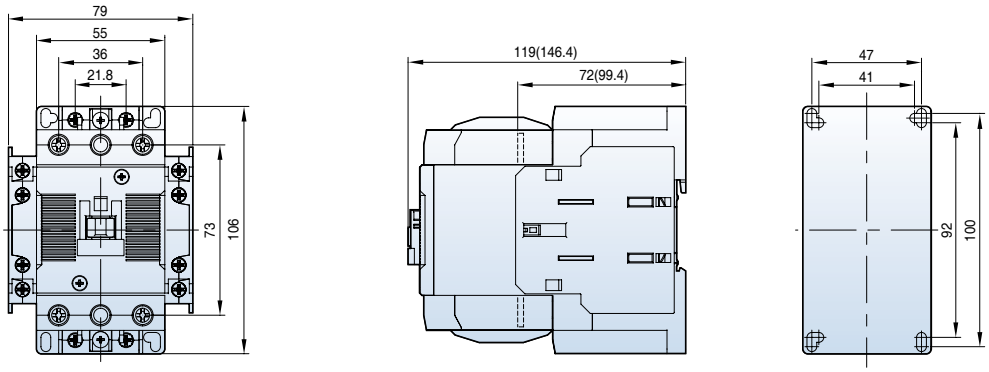
0.65kg

DC Contactors dimension (MD-30a, 60a,100a)

[mm]

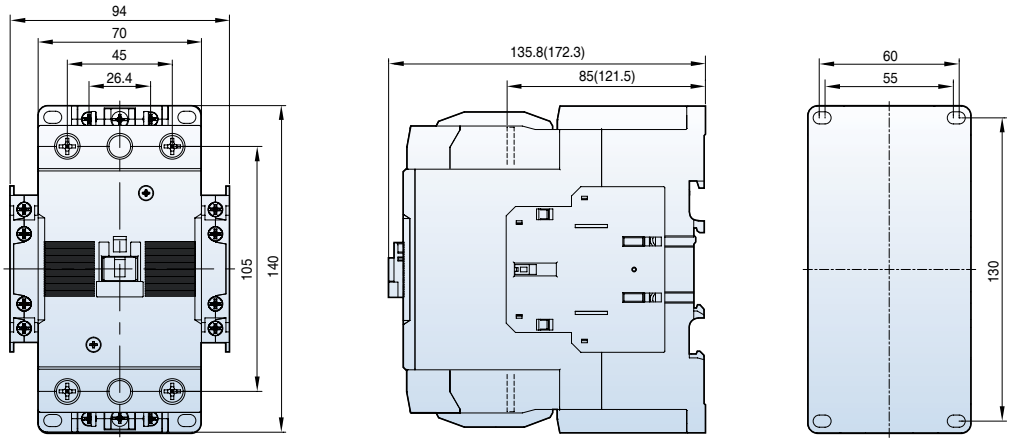
MD-30a AC(DC)

2NO2NC



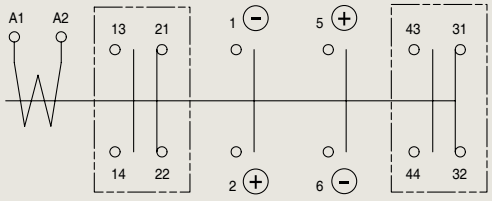
MD-60a,100a AC(DC)

2NO2NC

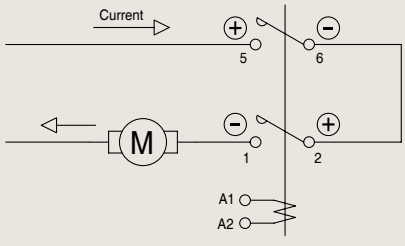


Composition

MD-30a, 60a, 100a



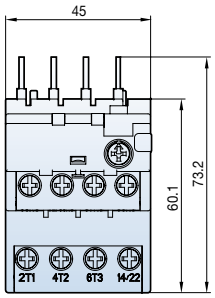
Contact arrangement

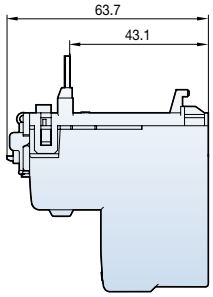


Overload relays dimension

[mm]

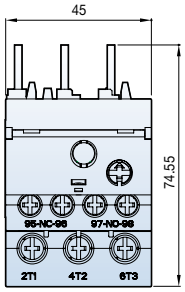
MT-12

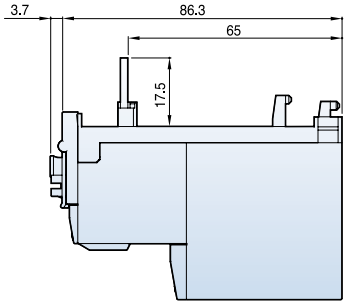




100g

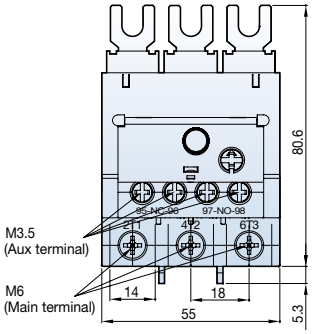
MT-32

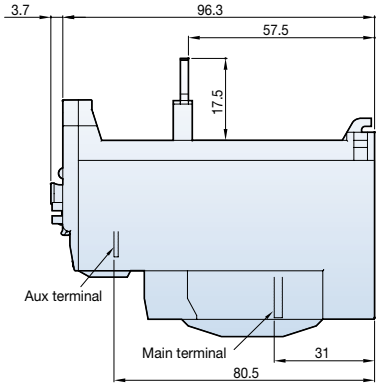




167g

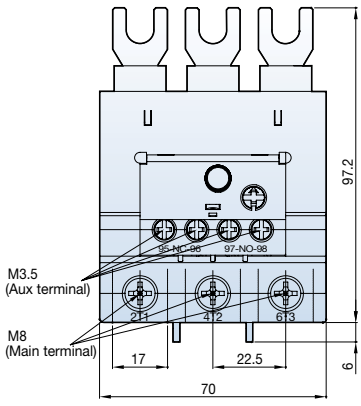
MT-63

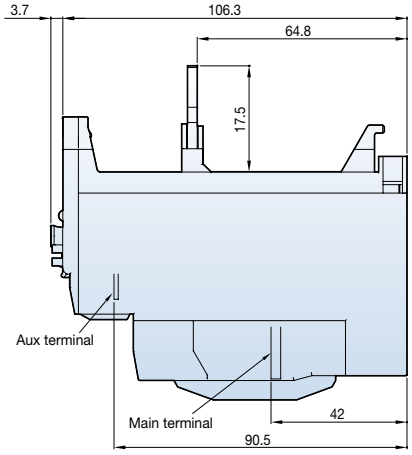




303g

MT-95



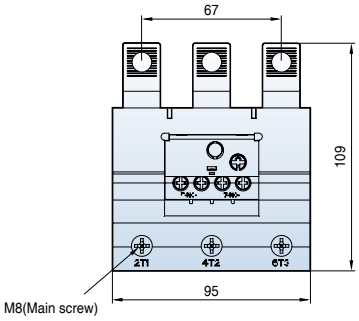


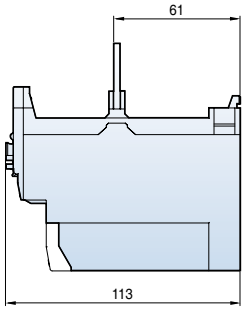
474g

Overload relays dimension

[mm]

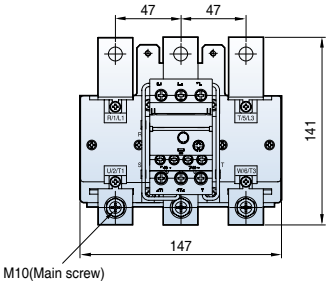
MT-150

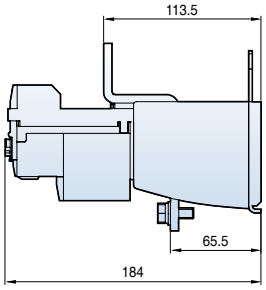




0.7kg

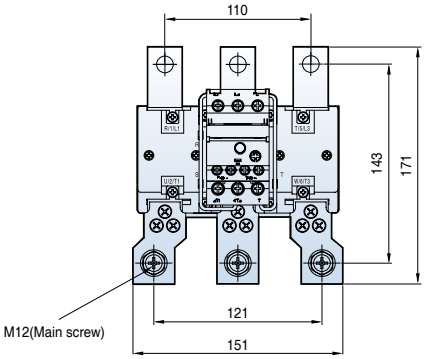
MT-225

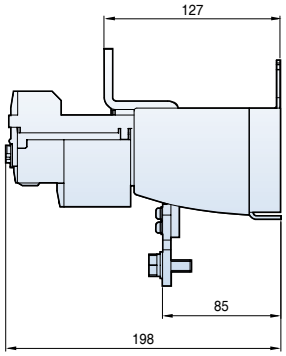




2.5kg

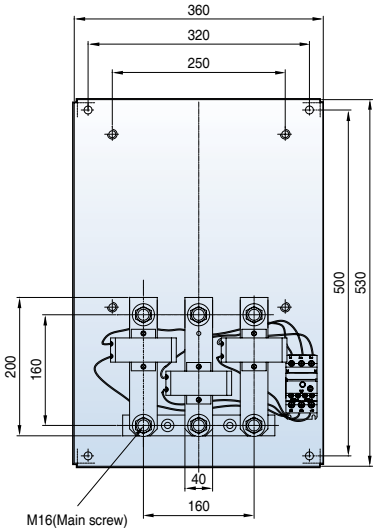
MT-400

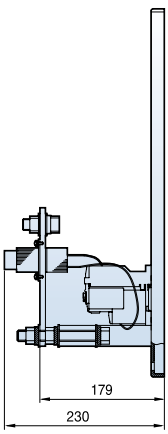




2.6kg

MT-800



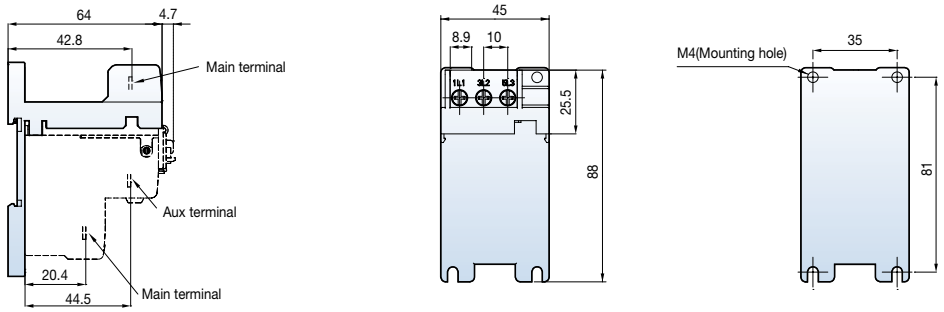


11.5kg

Overload relays dimension (Separate mount)

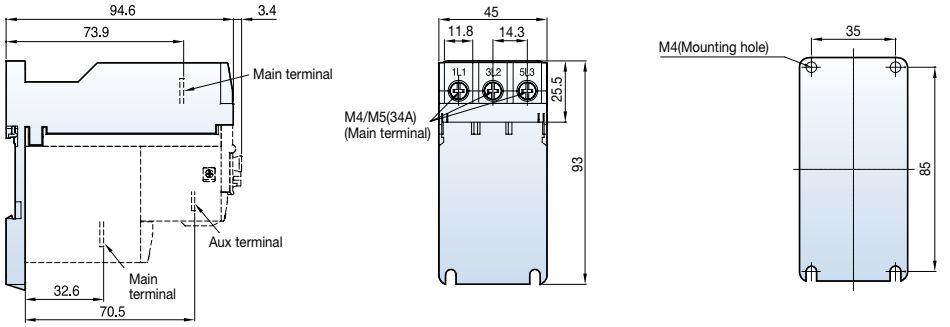
[mm]

UZ-12



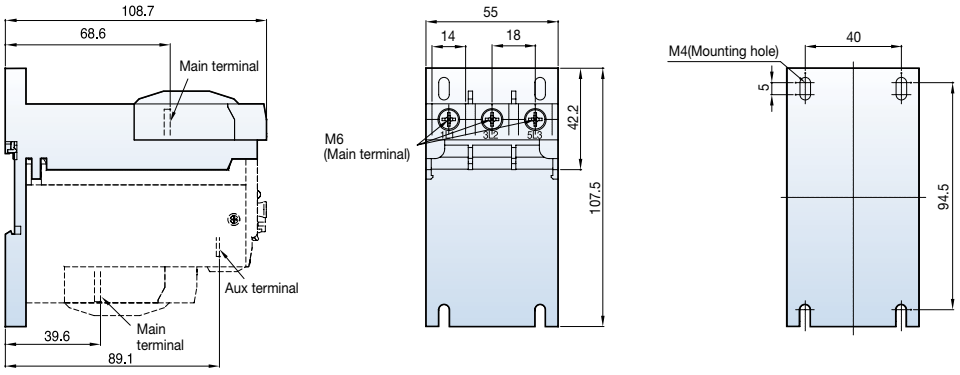
35g

UZ-32



68g

UZ-63

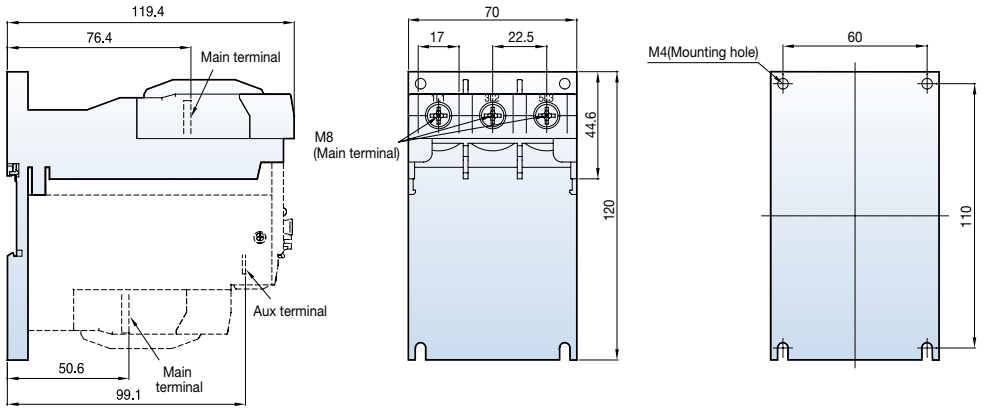


134g

Overload relays dimension (Separate mount)

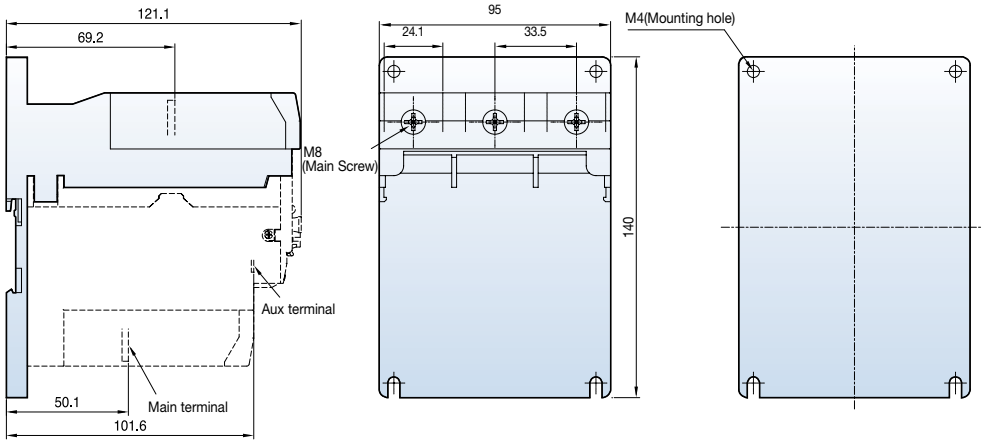
[mm]

UZ-95



230g

UZ-150

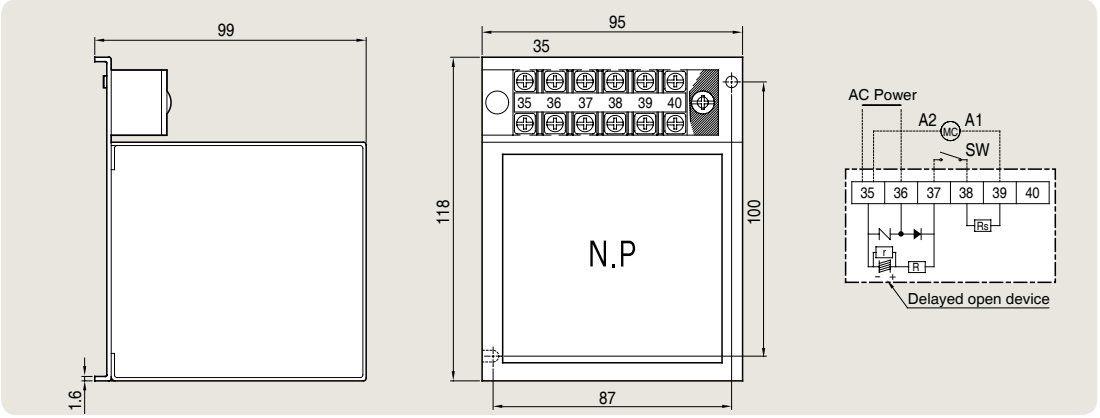


284g

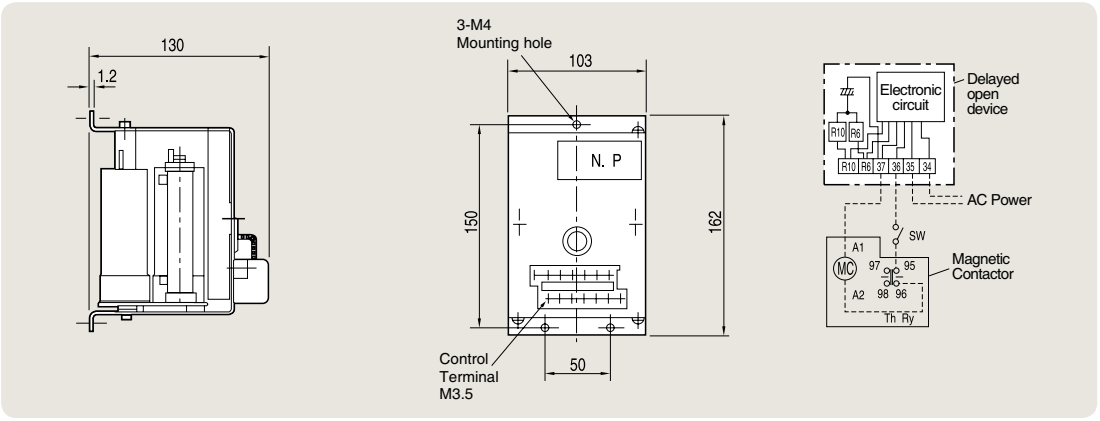
Delayed Opening Device

Dimension

AD - 9

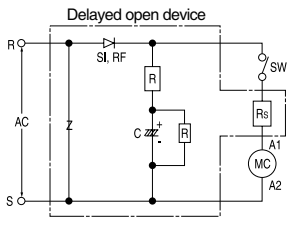
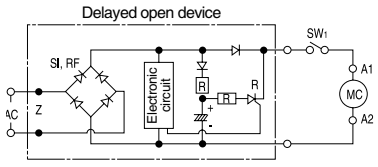


AD - 50
AD - 100
AD - 300
AD - 600



0.5kg

Control Circuit Diagram

AD – 9	AD – 50~600
	

SI, RF : Rectifier
C : Condenser
Rs : Resistance in series

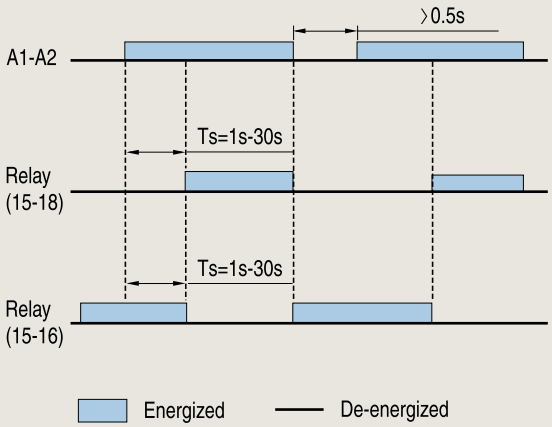
SI, RF : Rectifier
C : Condenser
Rs : Resistance in series

Electronic timer unit

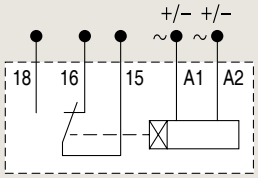
Electronic timer unit

ON delay timer

On time(Close MC) delay of magnetic contactors 1~30sec

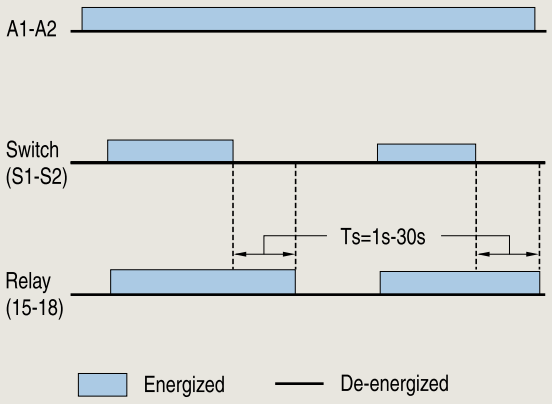


Circuit diagram

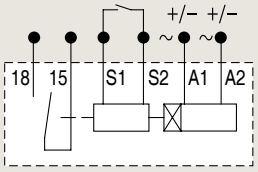


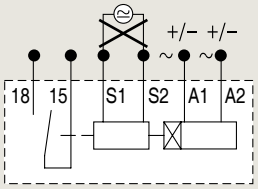
OFF delay timer

Off time(Open MC) delay of magnetic contactors 1~30sec

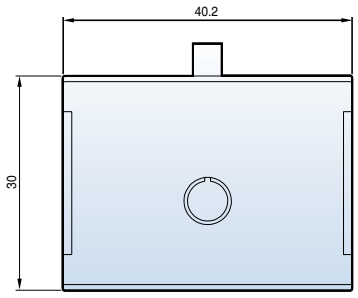


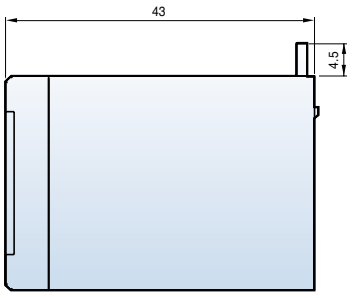
Circuit diagram





Dimension



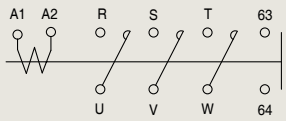


Contact arrangement

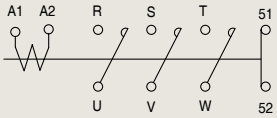
Contactors

MC-6a~18a

1NO

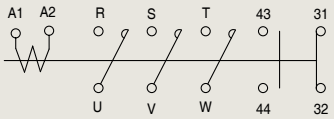


1NC



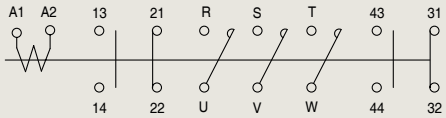
MC-9b~22b

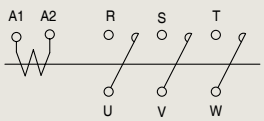
1NO1NC



MC-32a~150a

2NO2NC

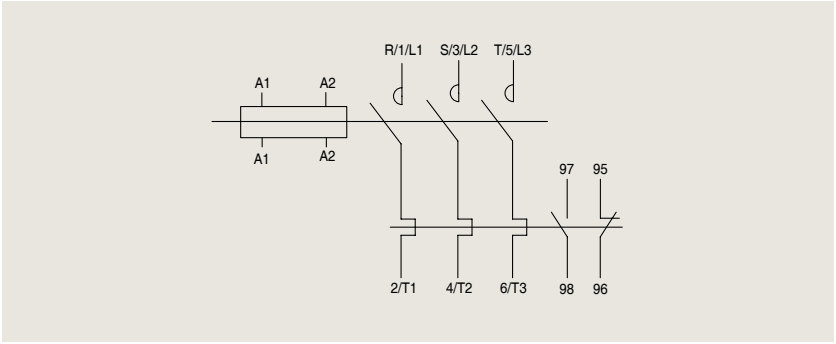


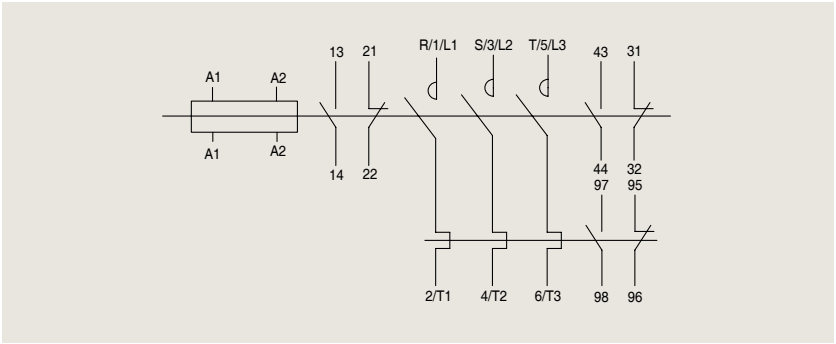


Contact arrangement

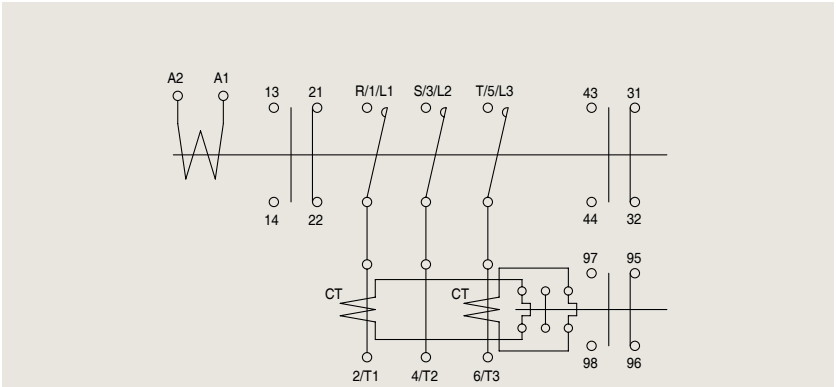
Motor starters

MS-6a~150a



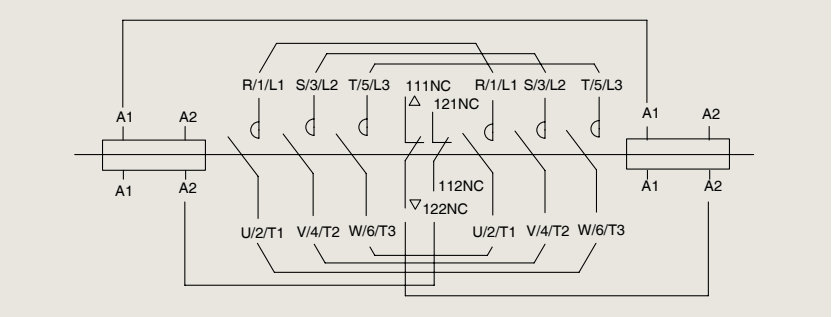


MS-185a~800a

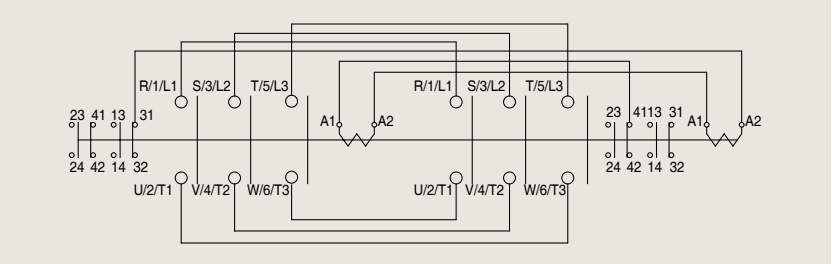


Reversing contactors

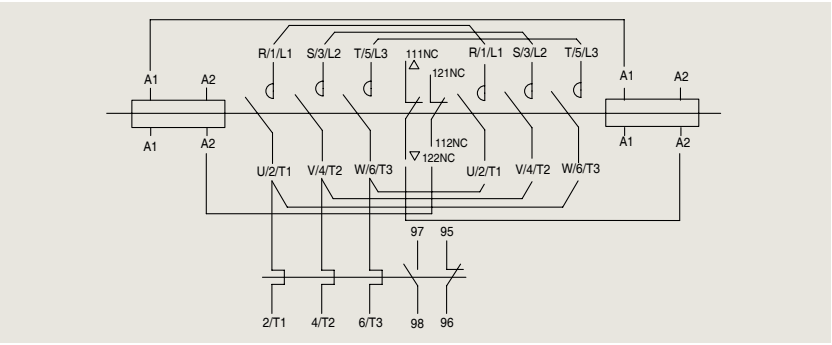
MC-6aR~150aR



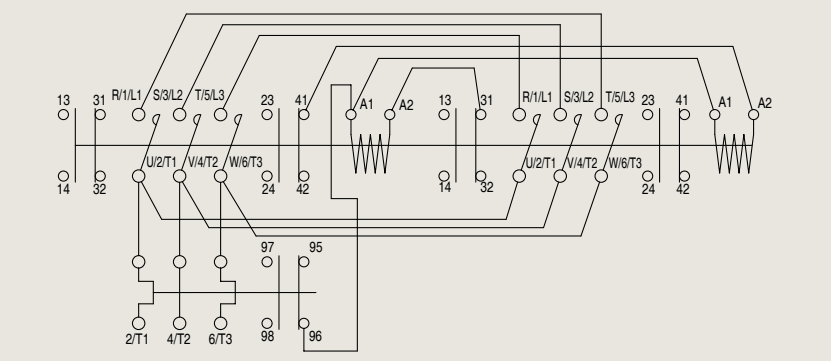
MC-185aR~800aR



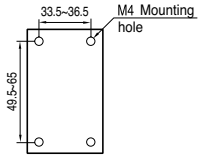
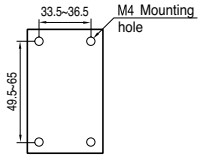
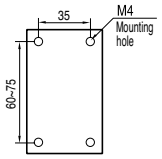
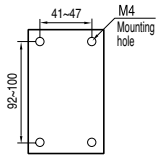
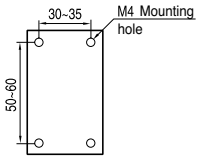
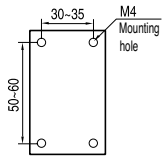
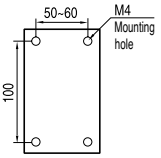
MS-6aR~150aR



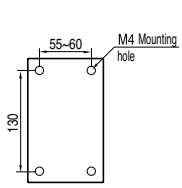
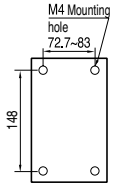
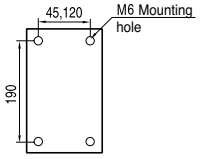
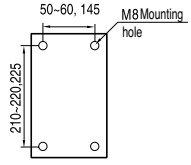
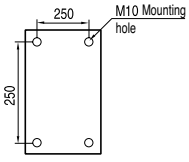
MS-185aR~400aR

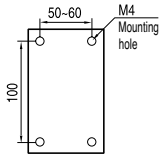
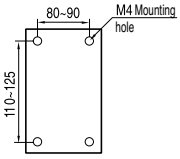
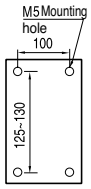
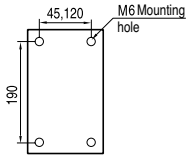
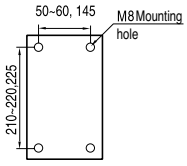
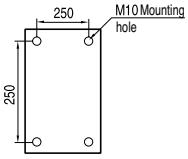


Meta-MEC and Metasol MC comparison table

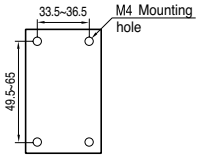
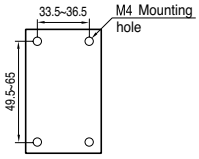
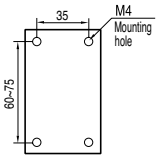
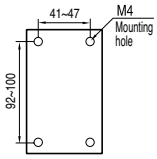
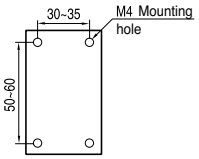
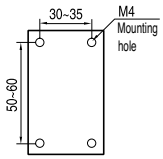
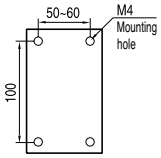
Type				MC-6a	MC-9a	MC-12a	MC-18a	MC-9b	MC-12b	MC-18b	MC-22b	MC-32a	MC-40a	MC-50a	MC-65a
Manufactured products	Metasol MC	Current and power AC3	200~240V	2.2kW 9A	2.5kW 11A	3.5kW 13A	4.5kW 18A	2.5kW 11A	3.5kW 13A	4.5kW 18A	5.5kW 22A	7.5kW 32A	11kW 40A	15kW 55A	18.5kW 65A
			380~440V	3kW 7A	4kW 9A	5.5kW 12A	7.5kW 18A	4kW 9A	5.5kW 12A	7.5kW 18A	11kW 22A	15kW 32A	18.5kW 40A	22kW 50A	30kW 65A
			500~550V	3kW 6A	4kW 7A	7.5kW 12A	7.5kW 13A	4kW 7A	7.5kW 12A	7.5kW 13A	15kW 20A	18.5kW 28A	22kW 32A	30kW 43A	33kW 60A
		Auxiliary (Standard)		1NO or 1NC				1NO1NC				-		-	
		Mounting size													
		Weight (kg)		0.33				0.34				0.4		0.9	
		Demension (W×H×D)		45×73.5×80.4				45×73.5×87.4				45×83×90		55×106×119	
Type				-				**GMC-9**	**GMC-12**	**GMC-18**	**GMC-22**	**GMC-32**	**GMC-40**	**GMC-50**	**GMC-65**
Discontinued priducts	Meta-MEC MC	Current and power AC3	200~220V	-				2.5kW 11A	3.5kW 13A	4.5kW 18A	5.5kW 22A	7.5kW 32A	11kW 40A	15kW 55A	18.5kW 65A
			380~440V	-				4kW 9A	5.5kW 12A	7.5kW 18A	11kW 22A	15kW 32A	18.5kW 40A	22kW 50A	30kW 65A
			500~550V	-				4kW7.5kW 7A	7.5kW 12A	15kW 13A	18.5kW 22A	22kW 28A	30kW 32A	33kW 43A	60A
		Auxiliary (Standard)		-				1NO1NC	1NO1NC	1NO1NC	1NO1NC	2NO2NC	2NO2NC	2NO2NC	2NO2NC
		Mounting size		-											
		Weight (kg)		-				0.33		0.37		0.45		1	
		Demension (W×H×D)		-				44×80×86.8				68×82×94.6		94×122×115.7	
	Compatibility with Metasol			-				○				▲		▲	

Note) Marks of compatibility
○: can be compatible
▲: compatible when applied the bracket
×: can't be compatible(have to use bracket)

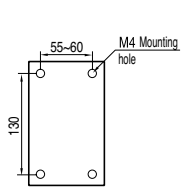
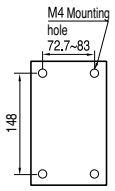
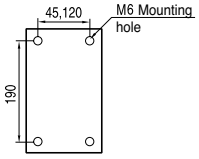
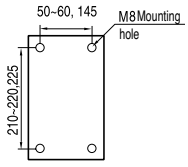
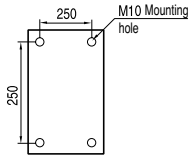
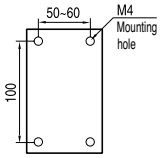
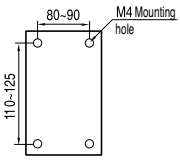
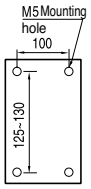
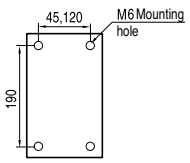
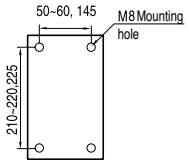
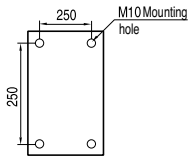
MC-75a	MC-85a	MC-100a	MC-130a	MC-150a	MC-185a	MC-225a	MC-265a	MC-330a	MC-400a	MC-500a	MC-630a	MC-800a
22 kW	25kW	30kW	37kW	45kW	55kW	75kW	80kW	90kW	125kW	147kW	190kW	220kW
75A	85A	105A	130A	150A	185A	225A	265A	330A	400A	500A	630A	800A
37 kW	45kW	55kW	60kW	75kW	90kW	132kW	147kW	160kW	200kW	265kW	330kW	440kW
75A	85A	105A	130A	150A	185A	225A	265A	330A	400A	500A	630A	800A
37 kW	45 kW	55kW	60kW	70kW	110kW	132kW	147kW	160kW	225kW	265kW	330kW	500kW
64A	75A	85A	90A	100A	180A	200A	225A	280A	350A	400A	500A	720A
-			-		2NO2NC		2NO2NC			2NO2NC		
												
1.6			2.4		5.4		9.2			22.4		
70×140×135.8			95×158×130.3		138×203×185.1		163×243×204.4			285×312×245.3		

GMC-75	GMC-85	GMC-100	GMC-125	GMC-150	GMC-180	GMC-220	-	GMC-300	GMC-400	-	GMC-600	GMC-800
22kW	25kW	30kW	37kW	45kW	55kW	75kW	-	90kW	125kW	-	190kW	220kW
75A	85A	105A	125A	150A	180A	250A		300A	400A		630A	800A
37kW	45kW	55kW	60kW	75kW	90kW	132kW	-	160kW	200kW	-	330kW	440kW
75A	85A	105A	120A	150A	180A	250A		300A	400A		630A	800A
37kW	45kW	55kW	60kW	90kW	110kW	132kW	-	160kW	225kW	-	330kW	500kW
64A	75A	85A	90A	140A	180A	200A		250A	350A		500A	720A
2NO2NC	2NO2NC	2NO2b	2NO2NC	2NO2NC	2NO2NC	2NO2NC	-	2NO2NC	2NO2NC	-	2NO2NC	2NO2NC
												
1		2.9		3.4	5.4		-	9.2		-	22	
94×122×115.7		100×157.4×146.5		120×166×157	138×203×181		-	163×243×198		-	285×290×242	
▲		×		×	○		-	○		-	○	

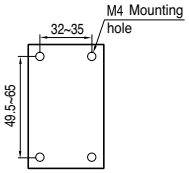
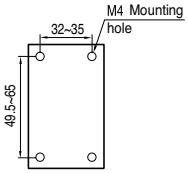
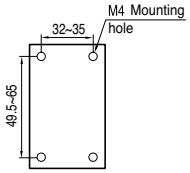
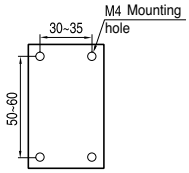
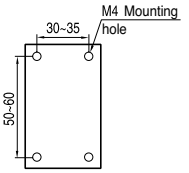
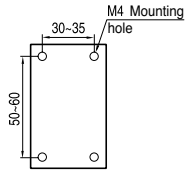
Meta-MEC and Metasol MS comparison table

	Type			MS-6a	MS-9a	MS-12a	MS-18a	MS-9b	MS-12b	MS-18b	MS-22b	MS-32a	MS-40a	MS-50a	MS-65a
Manufactured products	Metasol MC	Current and power AC3	200~240V	2.2kW 9A	2.5kW 11A	3.5kW 13A	4.5kW 18A	2.5kW 11A	3.5kW 13A	4.5kW 18A	5.5kW 22A	7.5kW 32A	11kW 40A	15kW 55A	18.5kW 65A
			380~440V	3kW 7A	4kW 9A	5.5kW 12A	7.5kW 18A	4kW 9A	5.5kW 12A	7.5kW 18A	11kW 22A	15kW 32A	18.5kW 40A	22kW 50A	30kW 65A
			500~550V	3kW 6A	4kW 7A	7.5kW 12A	7.5kW 13A	4kW 7A	7.5kW 12A	7.5kW 13A	15kW 20A	18.5kW 28A	22kW 32A	30kW 43A	33kW 60A
		Auxiliary (Standard)		1NO or 1NC				1NO1NC				-		-	
		Mounting size													
		Weight (kg)		0.5				0.5				0.7		1.5	
		Demension (W×H×D)		45×124×80.4				45×129.9×87.4				45×139.4×90		55×163.8×119	
	Type			-				GMS-9	GMS-12	GMS-18	GMS-22	GMS-32	GMS-40	GMS-50	GMS-65
Discontinued priducts	Meta-MEC MC	Current and power AC3	200~220V	-				2.5kW 11A	3.5kW 13A	4.5kW 18A	5.5kW 22A	7.5kW 32A	11kW 40A	15kW 55A	18.5kW 65A
			380~440V	-				4kW 9A	5.5kW 12A	7.5kW 18A	11kW 22A	15kW 32A	18.5kW 40A	22kW 50A	30kW 65A
			500~550V	-				4kW 7A	7.5kW 12A	7.5kW 13A	15kW 22A	18.5kW 28A	22kW 32A	30kW 43A	33kW 60A
		Auxiliary (Standard)		-				1NO1NC	1NO1NC	1NO1NC	1NO1NC	2NO2NC	2NO2NC	2NO2NC	2NO2NC
		Mounting size		-											
		Weight (kg)		-				0.46		0.5		0.45		1.12	
		Demension (W×H×D)		-				44×128.9×86.8				68×136.5×94.6		94×181.5×115.7	
	Compatibility with Metasol			-				○				×		×	

Note) Marks of compatibility
○: can be compatible
×: can't be compatible(have to use bracket)

MS-75a	MS-85a	MS-100a	MS-130a	MS-150a	MS-185a	MS-225a	MS-265a	MS-330a	MS-400a	MS-500a	MS-630a	MS-800a
22 kW	25kW	30kW	37kW	45kW	55kW	75kW	80kW	90kW	125kW	147kW	190kW	220kW
75A	85A	105A	130A	150A	185A	225A	265A	330A	400A	500A	630A	800A
37 kW	45kW	55kW	60kW	75kW	90kW	132kW	147kW	160kW	200kW	265kW	330kW	440kW
75A	85A	105A	130A	150A	185A	225A	265A	330A	400A	500A	630A	800A
37 kW	45 kW	55kW	60kW	70kW	110kW	132kW	147kW	160kW	225kW	265kW	330kW	500kW
64A	75A	85A	90A	100A	180A	200A	225A	280A	350A	400A	500A	720A
-			-		2NO2NC		2NO2NC			2NO2NC		
												
2.4			3		7.9		11.8			28		
70×211.4×135.8			95×236.2×130.3		138×306.5×185.1		163×380×204.4			360×530×245.3		
GMC-75	**GMC-85**	**GMC-100**	**GMC-125**	**GMC-150**	**GMC-180**	**GMC-220**	**-**	**GMC-300**	**GMC-400**	**-**	**GMC-600**	**GMC-800**
22kW	25kW	30kW	37kW	45kW	55kW	75kW	-	90kW	125kW	-	190kW	220kW
75A	85A	105A	125A	150A	180A	250A		300A	400A		630A	800A
37kW	45kW	55kW	60kW	75kW	90kW	132kW	-	160kW	200kW	-	330kW	440kW
75A	85A	105A	120A	150A	180A	250A		300A	400A		630A	800A
37kW	45kW	55kW	60kW	90kW	110kW	132kW	-	160kW	225kW	-	330kW	500kW
64A	75A	85A	90A	140A	180A	200A		250A	350A		500A	720A
2NO2NC	2NO2NC	2NO2NC	2NO2NC	2NO2NC	2NO2NC	2NO2NC	-	2NO2NC	2NO2NC	-	2NO2NC	2NO2NC
													
1.12		3.4		4	7.9		-	11.8		-	28	
94×181.5×115.7		114×231.4×146.5		120×206.4×157	138×306.54×181		-	163×380×198		-	360×530×262	
×			×		○		-	○		-	○	

Meta-MEC and Metasol relays comparison table

Type				MR-4	MR-6	MR-8
Manufactured products	Metasol MR	Auxiliary	(kW)	4NO, 3NO1NC, 2NO2NC 1NO3NC, 4NC	6NO, 5NO1NC, 4NO2NC 3NO3NC, 2NO4NC	8NO, 7NO1NC, 6NO2NC 5NO3NC, 4NO4NC
		Rated current (A) AC15	120 V	6	6	6
			240 V	3	3	3
		Mounting size				
		Demension (W×H×D)		45×73.5×79(96.6)	45×73.5×109.6(127.2)	45×73.5×109.6(127.2)
		Weight (kg)		0.284	0.315	0.339
Type				**GMR-4(D)**	**GMR-6(D)**	**GMR-8(D)**
Discontinued products	GMR	Auxiliary	(kW)	4NO, 3NO1NC, 2NO2NC 1NO3NC, 4NC	6NO, 5NO1NC, 4NO2NC 3NO3NC, 2NO4NC	8NO, 7NO1NC, 6NO2NC 5NO3NC, 4NO4NC
		Rated current (A) AC11	AC110V	6	6	6
			AC220V	5	5	5
		Mounting size				
		Demension (W×H×D)		44×80×86.8(113.2)	44×80×118.6(145)	44×80×118.6(145)
		Weight (kg)		0.34(0.56)	0.37(0.59)	0.39(0.61)

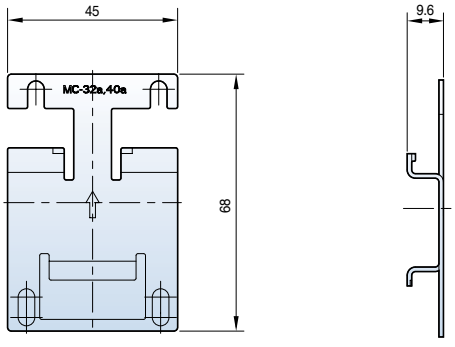
Meta-MEC and Metasol TOR comparison table

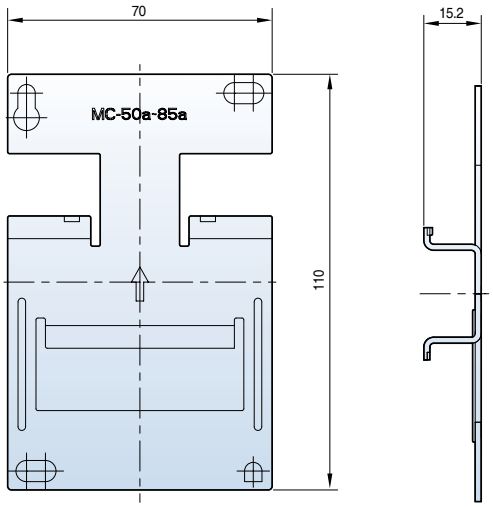
Manufactured products		Discontinued priducts	
MT		GTH (K)	
MT-12	0.14(0.1~0.16), 0.21(0.16~0.25), 0.33(0.25~0.4), 0.52(0.4~0.63), 0.82(0.63~1), 1.3(1~1.6), 2.1(1.6~2.5), 3.3(2.5~4), 5(4~6), 6.5(5~8), 7.5(6~9), 8.5(7~10), 11(9~13), 15(12~18), 19(16~22)	GTH(K)-22	0.14(0.1~0.16), 0.21(0.16~0.25), 0.33(0.25~0.4), 0.52(0.4~0.63), 0.82(0.63~1), 1.3(1~1.6), 2.1(1.6~2.5), 3.3(2.5~4), 5(4~6), 6.5(5~8), 7.5(6~9), 8.5(7~10), 11(9~13), 15(12~18), 19(16~22)
MT-32	0.14(0.1~0.16), 0.21(0.16~0.25), 0.33(0.25~0.4), 0.52(0.4~0.63), 0.82(0.63~1), 5(4~6), 6.5(5~8), 7.5(6~9), 8.5(7~10), 11(9~13), 15(12~18), 19(16~22), 22(18~26), 34(28~40)	GTH(K)-40	5(4~6), 6.5(5~8), 7.5(6~9), 8.5(7~10), 11(9~13), 15(12~18), 19(16~22), 22(18~26), 30(24~36), 34(28~40)
MT-63	5(4~6), 6.5(5~8), 7.5(6~9), 8.5(7~10), 11(9~13), 15(12~18), 19(16~22), 22(18~26), 30(24~36), 34(28~40), 42(34~50), 54(45~65)	GTH(K)-85	8.5(7~10), 11(9~13), 15(12~18), 19(16~22), 22(18~26), 30(24~36), 34(28~40), 42(34~50), 55(45~65), 65(54~75), 74(63~85)
MT-95	8.5(7~10), 11(9~13), 15(12~18), 19(16~22), 22(18~26), 30(24~36), 34(28~40), 42(34~50), 54(45~65), 65(54~75), 74(63~85), 83(70~95), 90(80~100)	GTH(K)-100	41(34~50), 48(39~57), 56(43~65), 67(54~80), 80(65~100), 107(85~125)
MT-150	42(34~50), 55(45~65), 65(54~75), 74(63~85), 93(80~105), 113(95~130), 130(110~150)	GTH(K)-150	41(34~50), 48(39~57), 56(43~65), 67(54~80), 80(65~100), 107(85~125), 130(100~150)
MT-225	80(65~100), 107(85~125), 130(100~160), 153(120~185), 200(160~240)	GTH(K)-220	80(65~100), 107(85~125), 130(100~160), 150(120~180), 200(160~240)
MT-400	107(85~125), 130(100~160), 153(120~185), 200(160~240), 265(200~330), 350(260~400)	GTH(K)-400	107(85~125), 130(100~160), 150(120~180), 200(160~240), 250(200~300), 350(260~400)
MT-800	265(200~300), 350(260~400), 515(400~630), 660(520~800)	GTH(K)-600	250(200~300), 350(260~400), 500(400~600), 660(520~800)

Compatibility bracket

- Mounting Metasol MC on Din-Rail of braket after install the bracket on panel
- Please use compatibility bracket when replace from Meta-MEC to Metasol MC without to change of panel size

	MP-40	MP-85
Manufactured products	MC-32a~40a	MC-50a~85a
Discontinued priducts	GMC-32~40	GMC-50~85





Certifications

Type		Approvals					Certificates		
Cerficate		-	IEC	UL	CSA	GB	IEC		KEPIC
Mark and name			CE	c UL us			KEMA	DEMKO	
Type		Korea	Europe	U.S.A & Canada	Canada	China	Netherlands	Denmark	Korea
Metasol Contactors	MC-6a	●	●	●	●	●	●		●
	MC-9a	●	●	●	●	●	●		●
	MC-12a	●	●	●	●	●	●		●
	MC-18a	●	●	●	●	●	●		●
	MC-9b	●	●	●	●	●	●		●
	MC-12b	●	●	●	●	●	●		●
	MC-18b	●	●	●	●	●	●		●
	MC-22b	●	●	●	●	●	●		●
	MC-32a	●	●	●	●	●	●		●
	MC-40a	●	●	●	●	●	●		●
	MC-50a	●	●	●	●	●	●		●
	MC-65a	●	●	●	●	●	●		●
	MC-75a	●	●	●	●	●	●		●
	MC-85a	●	●	●	●	●	●		●
	MC-100a	●	●	●	●	●	●		●
	MC-130a	●	●	●		●	●		●
	MC-150a	●	●	●		●	●		●
	MC-185a	●	●	●		●	●		●
	MC-225a	●	●	●		●	●		●
	MC-265a	●	●	●		●	●		●
	MC-330a		●	●		●	●		●
	MC-400a		●	●		●	●		●
	MC-500a		●	●		●	●		●
	MC-630a		●	●		●	●		●
	MC-800a		●	●		●	●		●
	MC-1260a		●	●	●			●	
	MC-1400a		●	●	●	●	●		
	MC-1700a		●	●	●	●	●		
	MC-2100a		●	●	●	●	●		
Thermal Overload Relays Nondifferential (2-heater)	MT-12/2H		●	●			●		●
	MT-32/2H		●	●	●	●	●		●
	MT-63/2H		●	●	●	●	●		●
	MT-95/2H		●	●	●	●	●		●
	MT-150/2H		●	●			●		●
	MT-225/2H		●	●			●		●
	MT-400/2H		●	●			●		●
	MT-800/2H		●	●			●		●
Thermal Overload Relays Nondifferential (3-heater)	MT-12/3H		●	●			●		●
	MT-32/3H		●	●	●	●	●		●
	MT-63/3H		●	●	●	●	●		●
	MT-95/3H		●	●	●	●	●		●
	MT-150/3H		●	●			●		●
	MT-225/3H		●	●			●		●
	MT-400/3H		●	●			●		●
	MT-800/3H		●	●			●		●
Thermal Overload Relays Differential	MT-12/3K		●	●			●		●
	MT-32/3K		●	●	●	●	●		●
	MT-63/3K		●	●	●	●	●		●
	MT-95/3K		●	●	●	●	●		●
	MT-150/3K		●	●			●		●
	MT-225/3K		●	●			●		●
	MT-400/3K		●	●			●		●
	MT-800/3K		●	●			●		●
Thermal Overload Relays Class 20	MT-12/3D		●	●			●		
	MT-32/3D		●	●	●	●	●		
	MT-63/3D		●	●	●	●	●		
	MT-95/3D		●	●	●	●	●		
	MT-150/3D		●	●			●		
	MT-225/3D		●	●			●		
	MT-400/3D		●	●			●		
	MT-800/3D		●	●			●		

Note) ● : Completion

Marine certifications

Type	Mark and name	Approvals							
		KR	LR	BV	GL	DNV	ABS	NK	RINA
		Korea	U.K	France	Germany	Norway	U.S.A	Japan	Italy
Metasol Contactors	MC-6a	●	●	●	●	●	●	●	●
	MC-9a	●	●	●	●	●	●	●	●
	MC-12a	●	●	●	●	●	●	●	●
	MC-18a	●	●	●	●	●	●	●	●
	MC-9b	●	●	●	●	●	●	●	●
	MC-12b	●	●	●	●	●	●	●	●
	MC-18b	●	●	●	●	●	●	●	●
	MC-22b	●	●	●	●	●	●	●	●
	MC-32a	●	●	●	●	●	●	●	●
	MC-40a	●	●	●	●	●	●	●	●
	MC-50a	●	●	●	●	●	●	●	●
	MC-65a	●	●	●	●	●	●	●	●
	MC-75a	●	●	●	●	●	●	●	●
	MC-85a	●	●	●	●	●	●	●	●
	MC-100a	●	●	●	●	●	●	●	●
	MC-130a	●	●	●	●	●	●	●	●
	MC-150a	●	●	●	●	●	●	●	●
	MC-185a	●	●	●	●	●	●	●	●
	MC-225a	●	●	●	●	●	●	●	●
	MC-265a	●	●	●	●	●	●	●	●
	MC-330a	●	●	●	●	●	●	●	●
	MC-400a	●	●	●	●	●	●	●	●
	MC-500a	●	●	●	●	●	●	●	●
	MC-630a	●	●	●	●	●	●	●	●
	MC-800a	●	●	●	●	●	●	●	●
Thermal Overload Relays Nondifferential (2-heater)	MT-12/2H	●	●	●	●	●	●	●	●
	MT-32/2H	●	●	●	●	●	●	●	●
	MT-63/2H	●	●	●	●	●	●	●	●
	MT-95/2H	●	●	●	●	●	●	●	●
	MT-150/2H	●	●	●	●	●	●	●	●
	MT-225/2H	●	●	●	●	●	●	●	●
	MT-400/2H	●	●	●	●	●	●	●	●
	MT-800/2H	●	●	●	●	●	●	●	●
Thermal Overload Relays Nondifferential (3-heater)	MT-12/3H	●	●	●	●	●	●	●	●
	MT-32/3H	●	●	●	●	●	●	●	●
	MT-63/3H	●	●	●	●	●	●	●	●
	MT-95/3H	●	●	●	●	●	●	●	●
	MT-150/3H	●	●	●	●	●	●	●	●
	MT-225/3H	●	●	●	●	●	●	●	●
	MT-400/3H	●	●	●	●	●	●	●	●
	MT-800/3H	●	●	●	●	●	●	●	●
Thermal Overload Relays Differential	MT-12/3K	●	●	●	●	●	●	●	●
	MT-32/3K	●	●	●	●	●	●	●	●
	MT-63/3K	●	●	●	●	●	●	●	●
	MT-95/3K	●	●	●	●	●	●	●	●
	MT-150/3K	●	●	●	●	●	●	●	●
	MT-225/3K	●	●	●	●	●	●	●	●
	MT-400/3K	●	●	●	●	●	●	●	●
	MT-800/3K	●	●	●	●	●	●	●	●
Thermal Overload Relays Class 20	MT-12/3D	●	●	●	●	●	●	●	●
	MT-32/3D	●	●	●	●	●	●	●	●
	MT-63/3D	●	●	●	●	●	●	●	●
	MT-95/3D	●	●	●	●	●	●	●	●
	MT-150/3D	●	●	●	●	●	●	●	●
	MT-225/3D	●	●	●	●	●	●	●	●
	MT-400/3D	●	●	●	●	●	●	●	●
	MT-800/3D	●	●	●	●	●	●	●	●

Note) ●: Completion

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LSMS

This application is selection guide of motor starter, overload relays and MCCBs

<Android> <IOS>

Safety Instructions

- For your safety, please read user's manual thoroughly before operating.
- Contact the nearest authorized service facility for examination, repair, or adjustment.
- Please contact qualified service technician when you need maintenance.
  Do not disassemble or repair by yourself!
- Any maintenance and inspection shall be performed by the personnel having expertise concerned.

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